

FOUNDATIONS OF EXERCISE MECHANICS

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TABLE OF CONTENT

1 FOUNDATIONS OF EXERCISE MECHANICS

1.1 Introduction to the Foundations of Exercise Mechanics	5
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2 FUNDAMENTOS DE LAS FUERZAS Y PROPIEDADES DE LAS CARGAS

2.1 What is force?	11
2.2 Forces and vectors	12
2.3 External and internal forces. That is not the point.	15
2.4 Forces: different names depending on the context	15
2.5 Creating resistances	22
2.6 Types of load	23
2.7 The human body responds to external forces...but it also creates reactive inertial responses through intention	40

3 LEVER SYSTEMS

3.1 Creating a lever system. Linkages.	42
3.2 Linkages in the human body (Arthrokinematics)	46
3.3 Arthrokinematics in Physical Exercise and health	56
3.4 Anatomical directions	61
3.5 Axes and planes of movements	62
3.6 The kinematics of lever systems (linkages)	74
3.7 Instructional progression of lever systems for Physical Activity	78
3.8 Resistance profile and lever systems	103
3.9 Velocity and introduction to dynamic mechanics	105

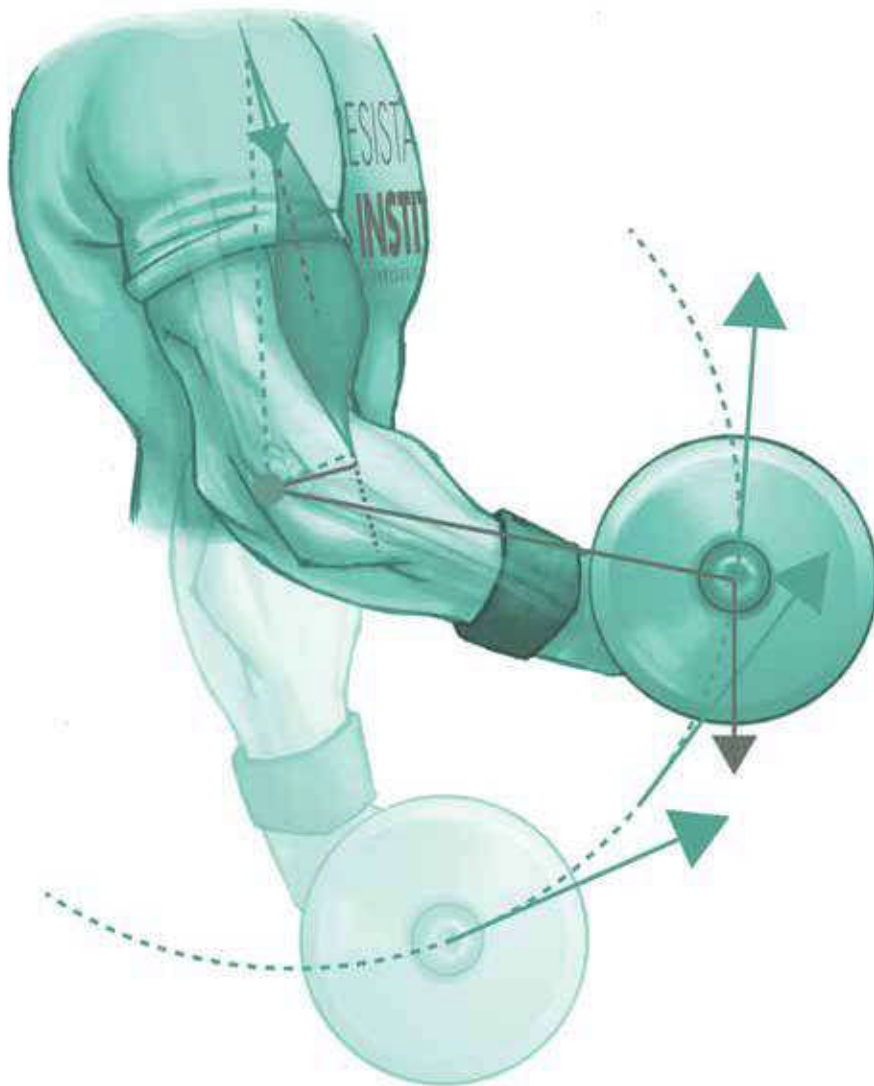
4 FUNCIÓN MUSCULAR Y CONTROL ARTICULAR

4.1 Creating internal forces	113
4.2 Skeletal muscle: structure and function	113
4.3 Structure and contractile ability	114
4.4 Neurological control	117
4.5 Force and mechanical characteristics	122
4.6 Negotiating with forces and muscle participation	147
4.7 Traditional kinesiology and muscle participation	165
4.8 Stability and resistance training: muscle participation	170
4.9 Muscle control	175

Bibliography	181
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1

FOUNDATIONS OF EXERCISE MECHANICS



1.1 INTRODUCTION TO THE FOUNDATIONS OF EXERCISE MECHANICS

'My age being what it is, universal acceptance of what we are now doing may not come within my lifetime; but it will come, because what we are doing is clearly established by simple laws of basic physics that cannot be denied forever.'

Arthur Jones

'In 2001, I started looking at exercise through renewed eyes, getting away from the traditional focus on basic physical abilities and muscle group-oriented training. That is when my journey towards understanding human motion-applied Physics and Engineering began. I then realized that exercise and Engineering were very much related, and a new perspective on Physical Activity and Sports Science was coming about.'

Lucas Leal

This fascinating journey towards Exercise Mechanics started over a decade ago. The author will now go on to explain his vision on physical conditioning and will offer guidance to help professionals better apply their knowledge.

LOOKING AT EXERCISE WITH NEW EYES

When thinking about exercise, many people in our field choose to disregard the evidence of Physics when considering exercise. They often also choose to ignore the challenge of understanding the most complex engineering of all: Human Engineering. Understanding these fields is a must if we want to get deeper into the least explored branch of Biomechanics: Exercise Mechanics. It is essential that we achieve a sound knowledge of forces as applied, generated and tolerated by the body system.

The reward to this effort will be worthwhile. A profound comprehension of Exercise Mechanics will make it possible for the exercise specialist to gain a whole new perspective. They will then see through renewed eyes, and this will enable them to analyze 'the invisible'; that which is not seen but is, indeed, present in every movement, every method, every piece of equipment.

The movie Matrix constitutes a good example of what does

happen when you acquire deeper understanding, deeper knowledge. Even though it may not be easy to visualize forces, these are always present and they are responsible for any change our body experiences. Exercise professionals will be on their way to understanding Exercise Mechanics when they:

- can visualize the applied forces and be able to manipulate them,
- know what constitutes an appropriate neuromuscular response to those forces, are able to assess the characteristics and effects of forces,
- understand a structure's tolerance to those forces,
- know human structures and their force control mechanisms.

Professionals will then be able to understand exercise beyond well-established myths. Their new eyes will give me better judgment to analyze the foundations of physical activity.

FORCES AND EXERCISE MECHANICS

The human body doesn't know a thing about methods, or brand names. It only cares about controlling and moving the musculoskeletal system, depending on the challenges/stimuli that it encounters. It deals with forces, in order to preserve its structure and achieve the necessary mobility.

Exercise professionals should start off by asking themselves the following question:

What is exercise?

As exercise professionals, trying to reply from the standpoint of Physics will allow us to generate a paradigm shift, creating a new vision over our work.

The answer to this question may seem simple and obvious (there are many definitions available), but it can still be simplified further:

- **Exercise: Intended force applied to a structure, within a specific scenario, with the purpose of causing an adaptation.**

By fully understanding the magnitude of this definition, exercise will be looked at differently.

Forces are involved in every kind of exercise. These forces affect our body and that of our client and are always applied in the hope of generating an adaptation.

This focus is based on the following: exercise relates to forces, so that there is no exercise without forces.

Force should therefore be considered as the sole instrument to centralize the input to generate adaptations.

Before we continue, it is important to make clear that all efforts to classify exercise based upon its specific goal (i.e. aerobic exercise, strength training, explosive move, etc.), is just the manifestation of force in different scenarios.

Many things will look different when a sound understanding of Newton's laws is achieved and we are able to truly apply the principles of Mechanics. Exercises will no longer fall into categories, such as flexibility/stretching, endurance or proprioception, but a scenario of forces will be visualized.

Whoever manages to understand and assimilate the above-mentioned definition will gain a new vision of exercise. This implies being able to manipulate all the mechanical variables that affect the stimulus, obtaining greater specificity and better results. If done this way, the exercise professional will be able to create exercises and better adjust the applied resistances to their clients' needs. There is only one way to get there: principles and reasoning have to be applied and information needs to be critically approached (it is often contradictory and poorly grounded). No book or paper should establish how we act, as if it were a dogma.

Foundations are needed to create appropriate stimuli. Each stimulus has to do with forces, so that exercise professionals have a 'moral duty' to learn as much about them as they possibly can. So... let's get started!

Is it possible to 'see' forces while working with clients?

Whatever training scenario you may have created up until now can be mechanically set into pieces, so that the forces which generate the stimulus can be 'seen'.

Understanding and analyzing the different force-scenarios constitutes the first step towards reaching reality. It is hardly possible to optimise our work and offer a good service to

our clients if we don't understand and are able to control the mechanical variables affecting the move.

As said, step number one is to analyze any given exercise from a Physics standpoint. After that, but most importantly, we will have to be able to manipulate the variables, in order to create the perfect setting for our goals.

Looking at exercise with new eyes implies not giving in to categories or marketing names. It means being able to autonomously assess the work and making decisions. And by doing so, a whole new array of possibilities comes along, for you to excel in your job and to enjoy it.

Have you ever thought of this while reading or listening to the indications or contraindications of a given exercise or piece of equipment?

If you thought of exercise as forces that are applied within a given scenario, would it seem easy to say that something is good or bad? –In spite of it being common practice, it is not appropriate to name the benefits or dangers of an exercise. Mel C Siff (former professor at the Mechanical Engineering school of Witwatersrand University in Johannesburg. He is also known for having published, together with Dr. Verkoshansky, the book called Supertraining) once published this statement:

'... we must keep away from making frivolous and unspecific generalizations about the superiority or inferiority, the safety or the dangers of a given exercise, because every effect of an exercise depends on the situation, the context, the time and the individual.'

Common as it is, making a study of any given exercise and generally stating its benefits is frivolous and unspecific. It is often disregarded that generalizing is totally unspecific and that the particular context always needs to be taken into account.

One of the most important factors which should be taken into consideration is the situation of the individual.

When we are able to fully grasp this, it becomes clear that the concept behind the term 'functional training' doesn't quite exist. If an exercise were to be functional, it would have to be so within a given context, for a given individual, and so on. This is not a critic to 'functional exercise' but it is indeed useful to realize that it merely a marketing term. Functional means nothing per se, just like saying 'magical exercise' would not make an exercise magical.)

When we don't understand this, we don't understand that

in every exercise there are always forces.

Every generic concept needs of a specific context to be analyzed within. This is the specific context:

Exercise = intended force applied to a structure.

And can be also put this other way:

Exercise = intended stimulus that leads to adaptation.

Which are the factors that determine the stimuli?

- Amount and type of force that reach each of the joint axes.
- Articular ROM (range of motion) and ROM modification during the application of stimuli on each of the joint axes.
- Duration of the stimulus.
- Tempo of applied stimulus, relative to each of the joint axes.
- Frequency of stimulus application.

This stimulus relates then to the individual:

- Genotype (specially affecting the musculoskeletal system)
- Phenotype (specially affecting the musculoskeletal system)
- Neuromuscular and proprioceptive function
- Diseases, tissue damage, age
- Mental and psychological situation

What needs to be noted is that it is the dosage that matters, and that it will necessarily be different for each individual. In fact, even for one given individual, the context will change and the dosage will need to vary accordingly.

CONTEMPLATING EXERCISE AS A VACCINE

We should start considering exercise as a vaccine: an inadequate dosage will give rise to the disease.

Exercise = intended stimulus that causes adaptation

It is forces that are responsible for the stimulus, but they too could create the disease. Unless forces adequately relate to the individual's tolerance levels, it can cause damage to the human structure.

Forces act like a 'virus', our body adapts to them and increases its tolerance to the 'virus'. That is why the main goal for any exercise and health professional is to apply an adequate stimulus, in order to create adaptations that will

Taking all of this into consideration, some common statements become meaningless. For instance:

- A specific movement being considered as 'functional', without taking into consideration all of the other factors mentioned above.
- A piece of equipment being 'good/bad'. Take the leg extension. You couldn't generally state that it is dangerous because you lack sufficient information.
- An exercise being better than another. It is impossible to determine that solely by the name of the exercise.
- More information on the stimulus and the individual are required.
- The advantages of a specific technique or method without specific data on the scenario, the stimulus and the individual. For example: 'isometric exercise is good for you.'

Conclusion:

In order to begin to understand the complexity of exercise, we need to go back to the foundations of the stimuli (study forces) and the individual (study the human body).

enable the system to better tolerate the 'virus'.

So, what causes an exercise to be 'good' or 'bad'?

The Risk-Benefit Ratio

The potential risk of a given exercise depends on the relationship between the application of a force on the structure and how the control mechanisms deal with it.

There is no such thing as a risk-free exercise. The key is to analyze its variables and to be able to manipulate them, in order to minimize the risks.

FORCE AND MUSCLES

What do muscles really do?

They generate force. Their only intention is to respond to the external or internal forces that reach your joints.

Although we share with physiologists the way force is generated, exercise professionals should, nevertheless, bear in mind that the whole process of muscular tension generation has one main and very clear purpose: to generate forces that will deal with those other forces that are altering our homeostasis. And by doing so, making sure that our body preserves its integrity.

The remaining functions of musculature are 'secondary' and

derive from its use to be able to generate force in different scenarios (hypertrophy, aerobic training, endurance, etc.).

'Physiology could be thought of as a response to physics'

Mitch Simon, Dr of Physiology

In Resistance Institute, we strongly believe that this perspective is not yet clear to everybody. We therefore strive to further explain some of the concepts that are still being misunderstood in the field of exercise and health.

FORCE AND BASIC PHYSICAL ABILITIES

When talking about physical abilities, force should be considered the sole centralizing ability.

In the absence of forces, exercise cannot take place.

Schools and colleges continue to teach how to train these abilities, without taking into account that they are all actually manifestations of force in varying scenarios.

By doing so, people tend to try to understand and work these abilities from a self-contained perspective. Instead, it is advisable to get deeper into studying the underlying forces.

Most exercise educational programs continue to look at the development of the basic physical abilities as if they were somehow independent from strength training.

This is why in Resistance Institute, we often feel there is a considerable gap between our views and those of many other fitness professionals. For us, it is essential that force is considered as the only and centralizing tool to generate adaptations.

Everything changes from the time you start visualizing forces in every movement. You acquire a whole new perspective, and you begin to see exercise differently.

The first thing that an Exercise Mechanic realizes is that no distinctions can be made between strength, endurance and flexibility training.

There is no way that any of the above abilities can be trained. There is no movement, technique or brand name which does not require forces as their cornerstone to create stimuli.

There is no other option but to go back to the foundations of Exercise Mechanics, if we want to begin to understand the complexity in the word exercise.

'We tend to classify things in order to better understand ourselves. But then, we lose perspective and insight in understanding the way our body works.'

(Lucas Leal)

JUST HOW MUCH FORCE: THE DOSAGE

Exercise, and any other dosage of force alike, should be thought of as a prescription drug. You would open the case, read the package insert and find the indications. If you continued reading, you would find the contraindications, interactions and side effects.

It is a well known thing that marketing strategies and commercial interests tend to emphasize the benefits. But exercise experts need to bear in mind that every stimulus can cause hazardous side effects, depending on the context and the particular individual.

Exercise is highly invasive.

Why is it so simple and logical to expect that forces will produce positive effects beyond our skin (i.e. hypertrophy, improvement in metabolism, etc) but we find it hard to imagine the damaging effects that those same forces can cause to our system?

Anyone who applies forces to their clients (trainers, physical therapists, osteopaths, fitness instructors, etc.) should hold themselves responsible for the forces applied for the sake of exercise. Just like surgeons hold themselves responsible for any damage inflicted to tissue, exercise professionals should be responsible for any tissue damage derived from the application of forces.

FOUNDATIONS OF EXERCISE MECHANICS

In order to fully understand exercise mechanics, three basic pillars should be considered:

- Forces
- Structure of the musculoskeletal system
- Control mechanisms (the nervous system, in particular)

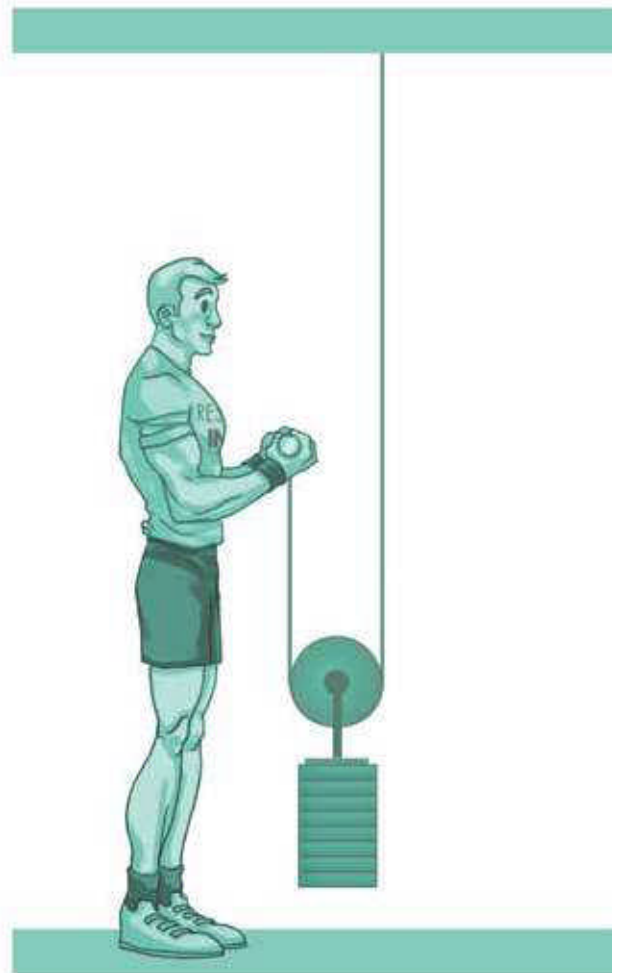
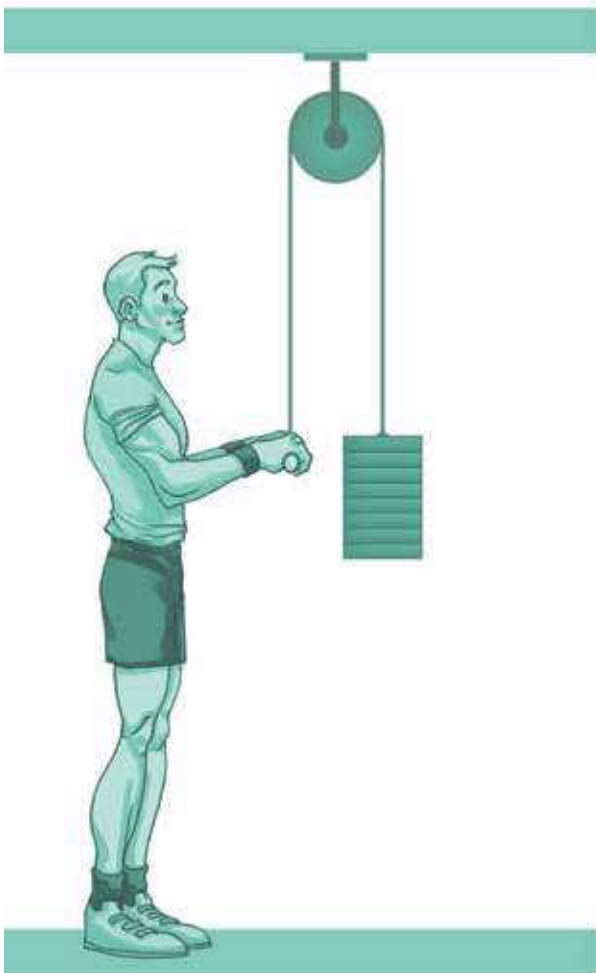
Professionals will highly benefit from expanding their knowledge within these three fields. This will enable them to become better at applying forces to the human body, whether they work in personal training, physical therapy, as sports coaches, in rehabilitation, or other.

This book summarizes 20 years of work at Resistance Institute. Its goal is to go deeper into the foundations of force as it affects the human body (musculoskeletal structure). Our hope is that it will contribute to the study of forces, joints and muscle mechanics, since these are the pillars of Exercise Mechanics.

I hope you enjoy it.

Lucas Leal
Barcelona, June 2020.

2 FOUNDATIONS OF FORCES AND PROPERTIES OF LOAD



This chapter addresses one of the great pillars in exercise mechanics: force. It is essential to fully understand this concept, in order to be able to integrate the contents explained in the following chapters. You will now discover what exactly forces are, how they act and in which contexts they may be present. An analysis of the various types of existing loads and their characteristics when applied to the human body will be presented at the end of the chapter.

2.1 WHAT IS FORCE?

Force is anything that can cause change in a structure. You could say that force is an agent of change. From the standpoint of Exercise Mechanics, we are talking about changes in the state of movement (not to those of shape, temperature, etc.)

In order to quantify force, it first needs to contact something. That is to say, the force needs to pull or push an object and thus try to cause a change upon it. This thought leads to the equation that allows to determine how much force is affecting a given object: Force = mass * acceleration. Should there be no object involved, we would then be talking about energy.

Forces are commonly broken down into a mass and an acceleration, but people rarely stop to think that there are also forces involved in an object that is standing still on the ground, with no resulting acceleration.

This equation never intended to break down the force. We also see this when we clear the equation and get:

$$m = F/a$$

Would it then be possible to state that, because m can be solved for, it is made of a force that relates to an acceleration? –The answer is NO. Mass is the amount of matter that makes up an object and is measured in mass units. Force is an agent of change and is measured in units of force.

The above stated equation determines an actual relationship between these three measurements: force, mass and acceleration. Similarly, Ohm's law defines the relationship between voltage, current and electrical resistance.

$$F = m * a$$

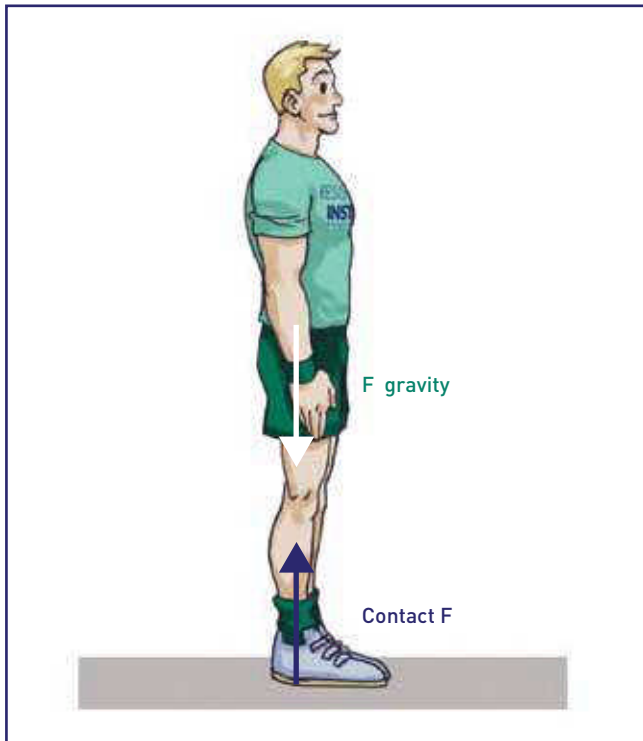
What this means is that, thanks to this formula, the amount of force can be known if we have the data on acceleration and the mass that it affects. Similarly, you can solve for mass, if you know the data for force and acceleration. Or you can

find out about the acceleration, if you know the amount of mass and force.

Do note that it all becomes a bit more complicated when the object receives more than one force. In this case, acceleration will result from adding up all forces acting upon the object (net force). But even then, we would still have to consider how each specific force affects the object/body and thus tries to cause it to accelerate. That is the reason why this equation constitutes one of the foundations of Newtonian physics.

Forces never act on their own, but always come in pairs. This does not mean that they counteract each other, though. In order for two forces to counteract each other, they need to be acting on the same body, with equal magnitudes, on the same line but in opposite directions (picture 2.1.1). Understanding this is important, since it can otherwise lead to misunderstanding when analyzing the balance or imbalance of two forces acting upon a given object.

For instance, if body 'x' applies 20-kp force on a ball that weighs 500 grams, the ball will respond against that body with 20 kp. The ball would receive an unbalancing force of 20 kp and would get some acceleration, depending on its mass. Body 'x' would receive 20 kp of force. Forces act in pairs, but they do not necessarily do so on one single object.



Picture 2.1.1

This picture shows how the contact force counteract the force of gravity. Both forces act upon the body and keep it balanced.

FORCE UNITS OF MEASUREMENT

The International System of Units determines that units of force be measured in Newtons (N).

One Newton (N) equals the force that is required to accelerate a mass of 1 kg by 1 m/s².

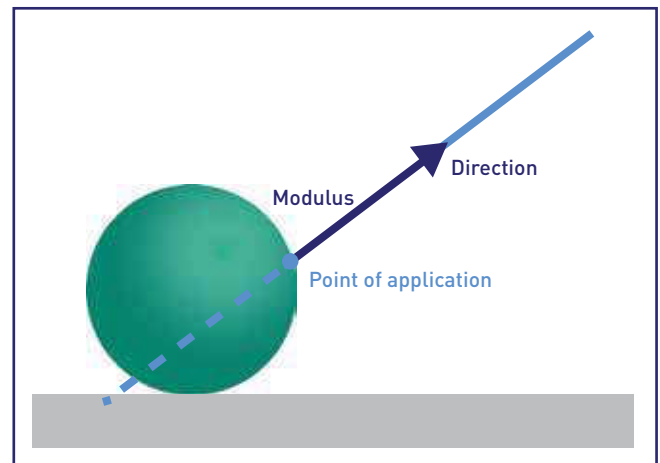
However, most people use kiloponds (kp). 1 kp is equal to the force that accelerates a mass of 1 kg to 9.8 m/s². Usually, this is a more convenient unit to use. One kilogram that is affected by the gravitational acceleration (9.8 m/s²) is thus a kilopond. By doing this, a direct equivalent is reached, between a measure of mass and a measure of force. And this is the reason why household dynamometers use kilograms as a unit of force, although they should be using kp.

2.2 FORCES AND VECTORS

Because forces cannot actually be seen, they are represented by force vectors. Vectors are graphic representations of force and consist of a line and an arrow head. Magnitudes such as temperature, mass or volume – known as scalar quantities or scalars– can simply be represented by a number. As opposed to that, force is a vector quantity that needs to be represented by a vector. In other words, in order to depict a force, you cannot just use a number (magnitude), but you also need to use other components so as to exactly determine it.

A vector depicts:

- The point of application of the force on an object.
- The direction of the force.
- The magnitude of the force. The length of the line shows the magnitude or amount of force that is exerted, relative to all other forces applying (other vectors). Alternatively, the magnitude of the force can be shown by placing a number over the line/arrow.



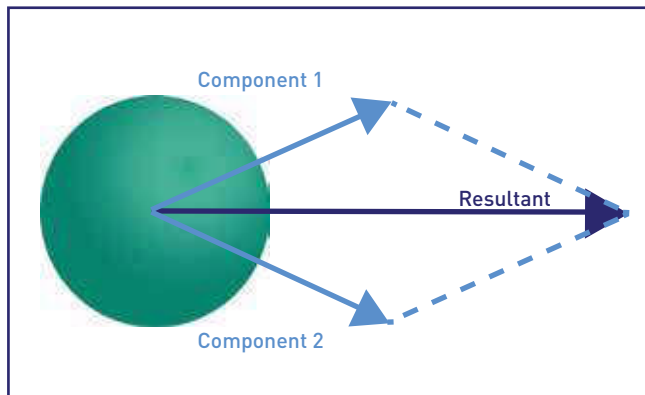
Picture 2.2.1

Picture showing a force vector and its components.

FORCE SYSTEMS

A force system is a collection of forces, all acting on the same body.

- Each of the forces in the system is called component.
- The line that represents the equivalent to all other lines and replaces them in the system is called resultant. The resultant takes both the magnitude and the direction of all the other forces).



Picture 2.2.1

Force system. The resultant force can be calculated from its two component forces.

ANALYSING A FORCE SYSTEM:

1.- FORCE COMPOSITION

2.- DECOMPOSITION OF FORCES

1. FORCE COMPOSITION

Two or more forces acting on the same mass, at the same point of contact, can be substituted by a single force that produces an equal effect. We use force composition to find a resultant force in a system where the individual components are known.

Three different situations may come about:

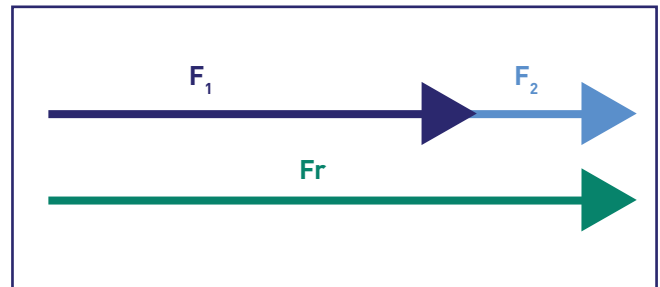
- Composing forces with the same direction.
- Composing concurrent forces acting on the same point (an angle is formed between them).
- Composing parallel forces.

Forces that lie on the same plane are called coplanar, and they are the ones that we will be covering in this book.

A. COMPOSITION OF FORCES IN THE SAME DIRECTION

A.1. FORCES ON THE SAME LINE AND SAME DIRECTION

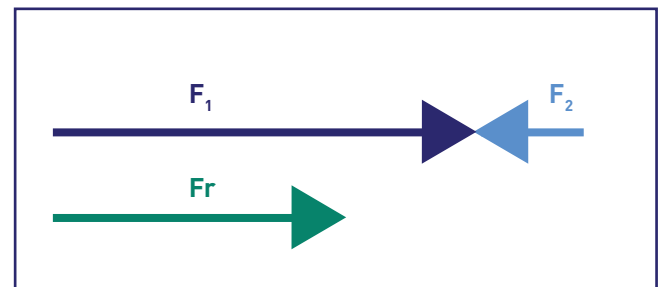
The lengths of force 1 (F_1) and force 2 (F_2) are added together. What results from applying both forces on the same point is called the resultant force (F_r). Picture 2.2.3a.



Picture 2.2.3a

A.2. FORCES ON THE SAME LINE AND OPPOSITE DIRECTIONS

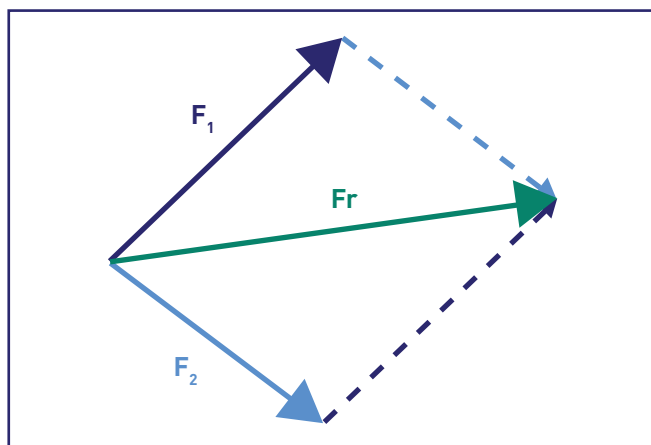
When this is the case, the length of force 2 (F_2) is subtracted from force 1 (F_1). The resultant force (F_r) is consequently reduced. Picture 2.2.3b.



Picture 2.2.3b

B. COMPOSITION OF CONCURRENT FORCES (they form an angle):

When the vectors to be added are not on the same line, the parallelogram law of forces is used. In order to find out the resultant force, a parallelogram is constructed, with vectors F_1 and F_2 acting as its sides, and F_r coming up as a diagonal resulting vector. Picture 2.2.3c.



Picture 2.2.3c

C. COMPOSITION OF PARALELL FORCES

These do not come within the scope of this book and will therefore not be discussed.

2- DECOMPOSITION OF FORCES:

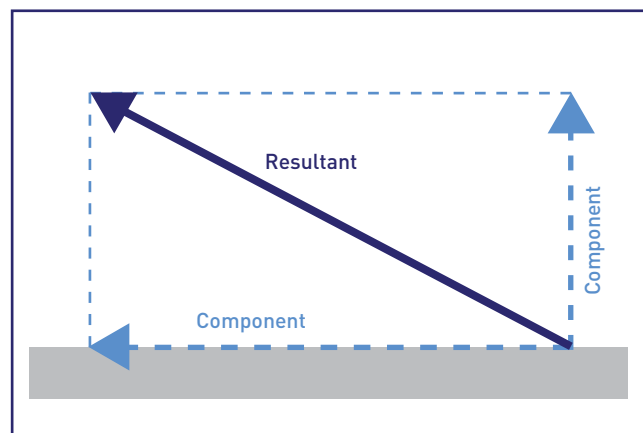
As mentioned earlier, a substitutive resultant force can be calculated out of two or more forces. However, it may at times be preferred to do the contrary. That is, to replace one sole force by two or more forces that produce the same effect as the resultant (by creating a force system). These forces are called components. By playing around with the amount of components, their magnitude and their directions, a force can be resolved into an infinite number of combinations of components.

RECTANGULAR COMPONENTS OF A FORCE

This section will help the reader understand the components of an applied force, and it will set up the basics in order to understand lever systems and muscle function later on.

Professionals should necessarily learn to decompose a force into its two perpendicular components, in order for them to correctly apply loads

Picture 2.2.4 shows the rectangle resulting from decomposing a force.



Picture 2.2.4

This diagram shows vector resolution into rectangular components.

RESOLUTION OR FORCE COMPOSITION METHODS

- Graphic method: forces, in the form of vectors, are represented on a vector diagram. This is the method as shown in the previous pictures.
- Algebraic method: algebraic equations and concepts of trigonometry are used.

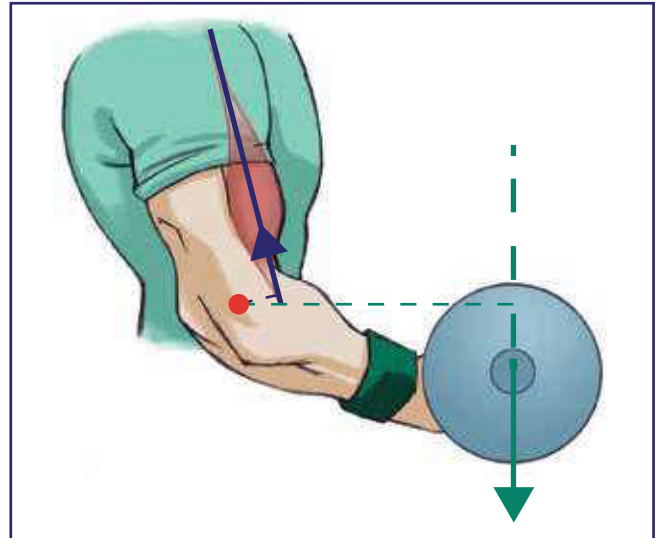
The different systems to resolve a force will not be looked at in detail. You will find out more information on this topic in trigonometry and basic algebra manuals/books. This should help you better understand the basics of exercise mechanics foundations.

2.3 EXTERNAL AND INTERNAL FORCES. THAT IS NOT THE POINT

Forces that come from outside the body, such as free weights, bands and balls are usually labelled as external forces. Those originated inside –basically muscle force– are known as internal forces. In order to eventually understand the effects of inertial loads, it is important to consider the gravitational force as an external one, even when it is your own body's.

In order to carry on a mechanical assessment, what really matters is to be able to analyze how these forces act on the human body. No matter where they originate, they all end up causing an effect on the joints. And it is this that should concern the exercise professional when analysing the effects on the articular levers.

Forces may originate inside or outside of the body. But what really matters is being able to calculate their effects on the joints, and other parts of the connective tissue, such as bones, tendons, ligaments, etc.



Picture 2.3.1

This picture shows an example of internal forces (biceps brachii) and external forces (weight). Both the external and the internal forces end up affecting the musculoskeletal system.

2.4 FORCES: DIFFERENT NAMES DEPENDING ON THE CONTEXT

All the terms that are about to be explained share force as a common denominator, though expressed in different situations and contexts.

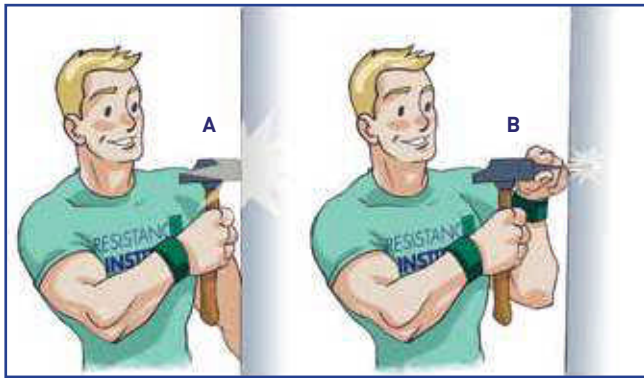
MECHANICAL STRESS

Mechanical stress is the ratio of force that is being applied on a structure, relative to the contact area. As per the International System of Units, it is measured in N/m². Its equation is as follows:

$$\text{Mechanical stress} = \text{Force (N)} / \text{surface (m}^2\text{)}$$

From a mechanical point of view, the term mechanical stress simply refers to a quotient between a force and the area over which it is applied. Taking this into consideration, a mechanical stress cannot be labelled as good or bad. Only the context in which it is being applied can make it to have positive or negative effects. If a mechanical stress were

excessive, it could cause damage to the structure. If the stress was too little, it could cause insufficient change and thus lead to no benefit.



Picture 2.4.1

Drilling a sharp nail into a wall allows for increased mechanical stress without having to increase the applied force.

Picture 2.4.1 shows, on side A, a character that is drilling a wall by applying a force of 10 Newtons directly with the hammer and a relatively wide contact surface. On side B, he man applies exactly the same force, but now the object impacting into the wall is sharp and therefore the contact area is much smaller. Both objects impact on the wall with equal force. However, because the force applied is equal on both sides but the object impacting the wall on side B has a much smaller surface, the wall on side B receives a greater mechanical stress.

Any force that contacts a body, does so by inflicting mechanical stress upon that body. For any given amount of force, the mechanical stress that gets generated will depend on the size of the contact area. This is why taking the contact area into consideration is an essential matter when applying a force.

MECHANICAL PRESSURE

Pressure and stress are, mechanically, both measured in the same units, and can therefore be considered to be the same. Mechanical pressure is the amount of force applied per area of surface where it is applied.

The term pressure is commonly used to measure the force of gas or liquid over a given surface in different circumstances. Examples are atmospheric pressure, water pressure at varying depths, etc.

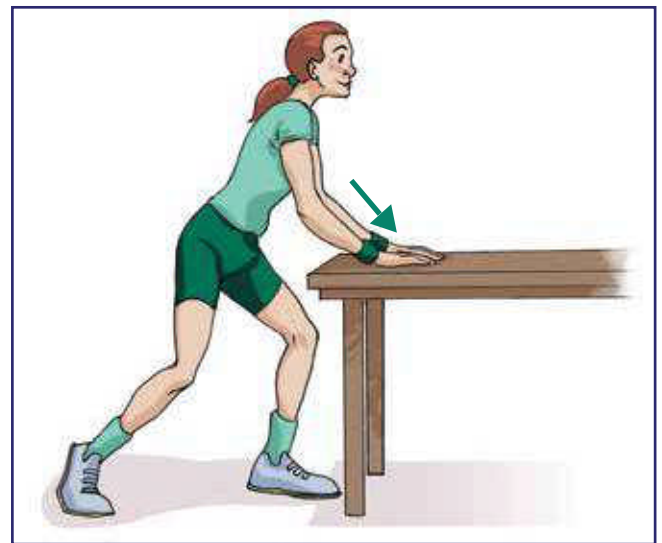
Up until 1971, the most frequently used unit of measurement was N/m^2 , but nowadays we use the Pascal (Pa). A Pascal is equal to 1 N/m^2 . The standard atmosphere (atm) is equal to 1013.25 hectopascals (hPa). Pressure and stress are therefore commonly expressed in different units, but the magnitudes we are referring to (force and area) are the same.

What differentiates pressure from mechanical stress then?

DIFFERENCES BETWEEN PRESSURE AND MECHANICAL STRESS

Even though both terms are virtually equal, we could point out a difference, depending on who/what is the exerting agent and who/what is receiving the force. Pressure is usually associated with whatever is exerting the force. Stress is the preferred term when talking about the structure that is receiving and tolerating the force.

Picture 2.4.2 may help understand what has just been said.



Picture 2.4.2

As the lady exerts pressure on to the table, the table receives stress.

The character on picture 2.4.2 is leaning against a table. She is doing pressure on the table, and the table receives stress, caused by the lady pressing on it. It is the character that is doing the pressure, and the table that is receiving the stress. It is worth mentioning, that the lady's hands are also receiving stress, as a consequence of the reactive forces exerted by the table.

TENSILE FORCES

Two equal forces (action-reaction) that are applied and act on the same line will generate traction (tensile load). They will produce a traction or tensile stress and a proportional traction compliance (tensile strain) on the structure or material that they affect.

$$\text{Tensile stress (N/m}^2\text{)} = \text{tensile load (N)} / \text{cross section area (m}^2\text{)}$$

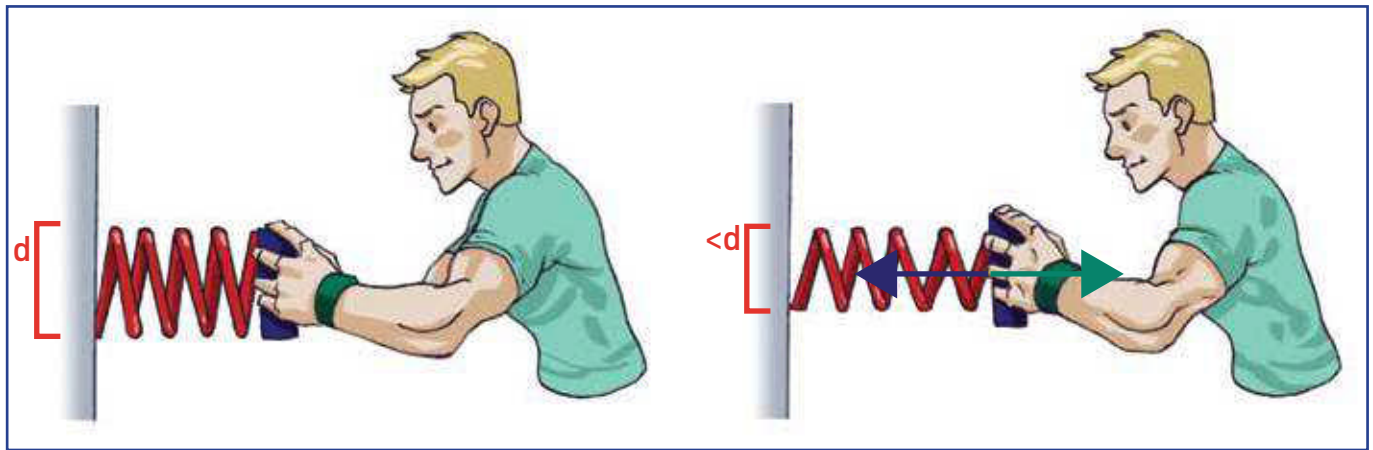
(area is perpendicular to the direction of applied force)

When a traction force is applied, stress is defined as the intensity of the force relative to the cross-section area of the affected structure.

The term strain measures the percent of longitudinal elongation that is experimented by the structure or material acted upon.

The above-named elongation (strain) is expressed by comparing to the affected material's initial size.

A structure's longitudinal elongation –produced by a tensile load– presents a certain amount of proportional transverse shortening, called lateral strain.



Picture 2.4.3

When tensile stress is applied to the spring, the spring lengthens out but the distance -d- shortens (lateral strain).

COMPRESSIVE FORCES

Two equal forces (action-reaction) that are applied and act upon the same line by diverge will cause a compressive load and a compressive stress. This will generate a proportional compressive deflection of the structure, named compressive strain.

$$\text{Compressive stress (N/ m}^2\text{)} = \text{Compressive load (N)} / \text{cross section area (m}^2\text{)}$$

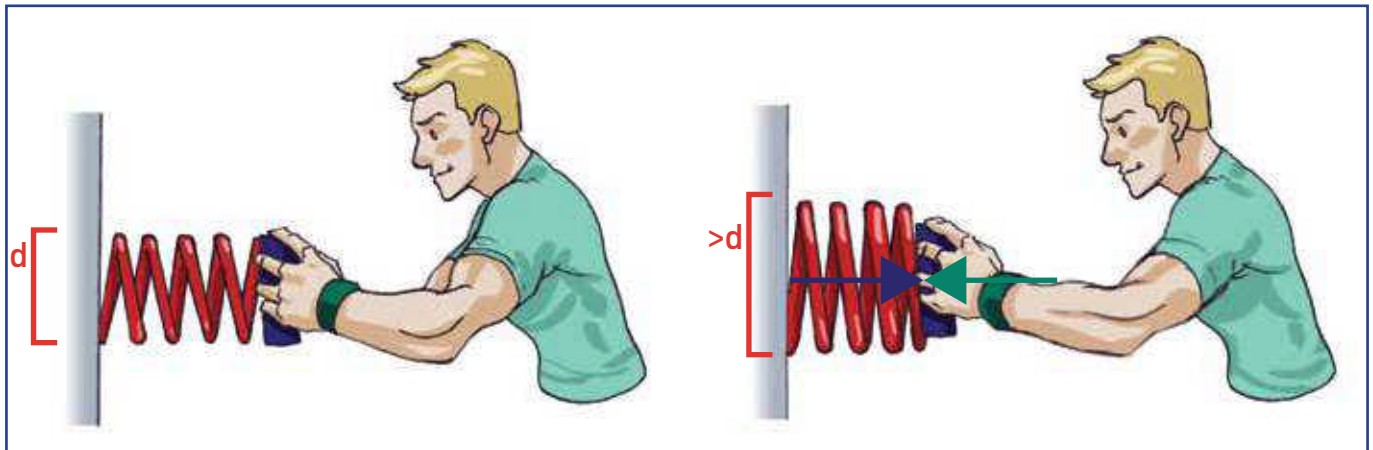
(area is perpendicular to the direction of applied force)

When a compressive load is applied to a structure, the stress can be considered as a way to measure the intensity of the force, relative to the cross section.

The compressive strain measures the percentage of longitudinal shortening that a given structure or material experiments.

As was also the case with traction loads, the deformation (strain) is expressed as a comparison to the affected material's original dimensions.

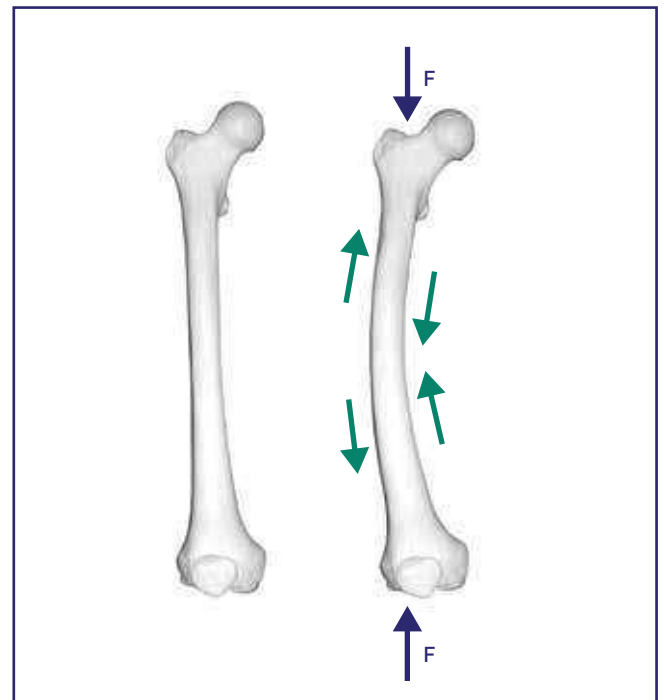
The structure's longitudinal shortening that is caused by the compressive stress, is joined by a certain amount of proportional transverse elongation (lateral strain).



Picture 2.4.4

Applying compressive forces decreases spring length and increases distance, d.

Whenever a structure –i.e. a bone– experiences a bending moment (torsion moment), then it is exposed to both compressive and tensile stresses. Using the longitudinal axis of the bone as a reference, we can say that the tensile stress occurs on the convex side, whereas the concave side experiences a compressive stress.



Picture 2.4.5

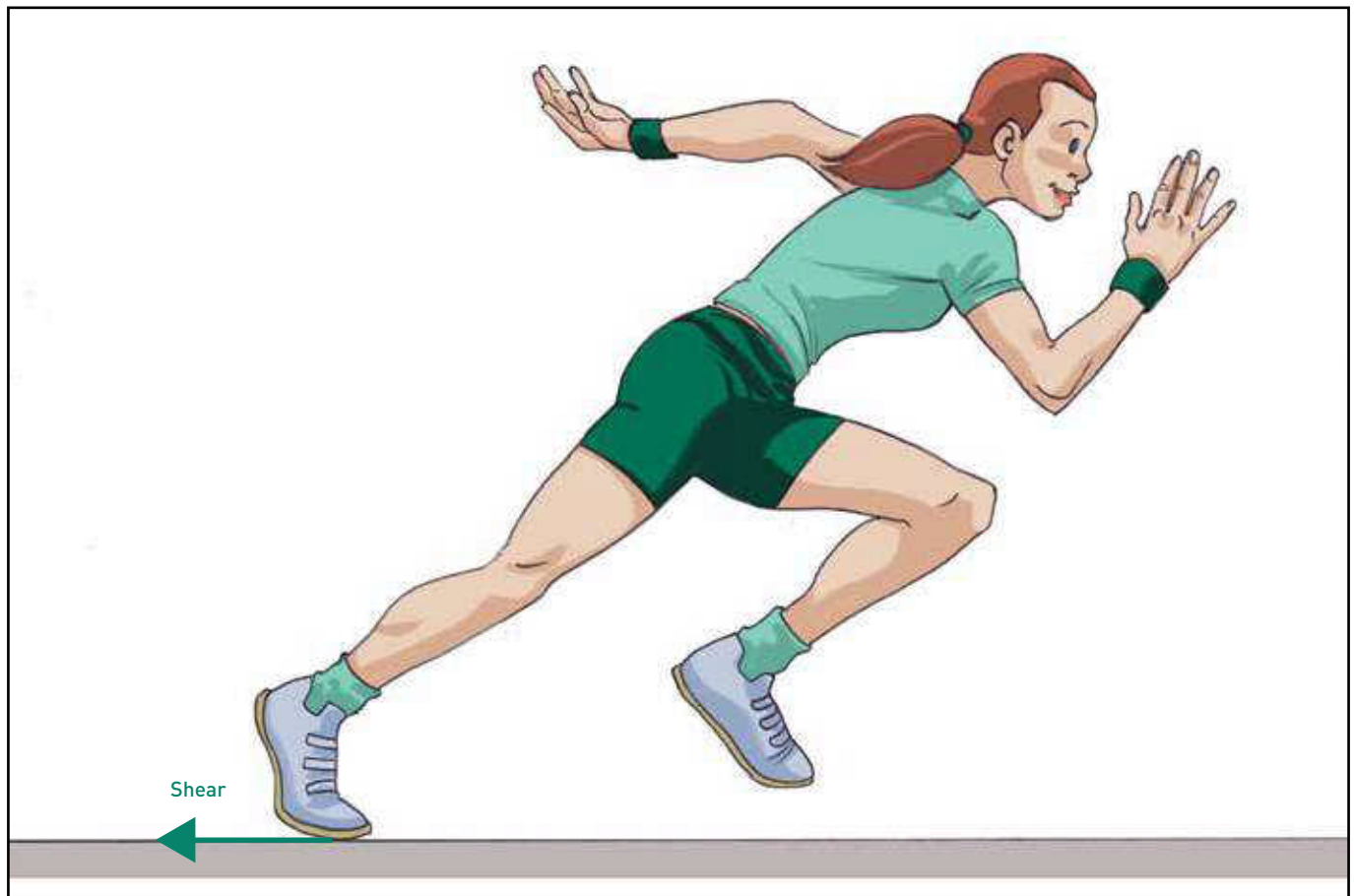
Tensile and compressive stress taking place on a femur, when exposed to bending forces.

SHEAR FORCE

Shear force arises from a force (or a force component) that is applied parallel to the area of contact. The force applied will generate a shear load, which will cause a shear stress and will, consequently, produce a proportional deformation in the structure, known as shear strain. The reactive force to the original force (or a part of the latter) is called friction. This friction will be parallel to the shear force, will run in the opposite direction, does not necessarily have the same magnitude and will not share the line of the shear force.

Shear stress (N/m²) = shear force (N)/cross section area (m²)
(area is parallel to the direction of the force applied).

In the following pages, we will go into further detail about shear forces and their applicability in the exercise field.



Picture 2.4.6

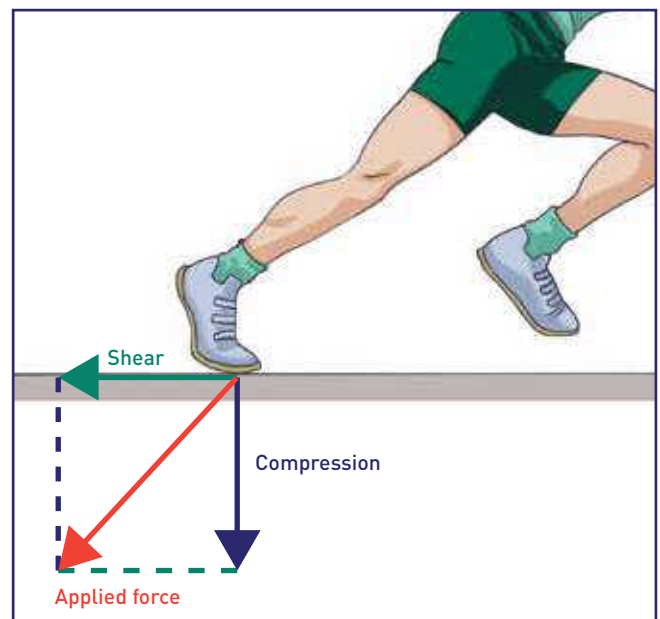
The illustration represents shear forces. The reactive force to shear is friction. In this example, it is the ground that is being affected by a shear force.

COMPONENTS OF AN APPLIED FORCE

As you may have already figured out, applied forces are not necessarily purely compressive, or tensile or shear. Instead, they can be made out of a combination of the various types. In fact, this is commonly the case.

When an applied force is not exclusively of one single type, it can be decomposed, and the compressive, tensile or shear components be figured out.

It is impossible to simultaneously find components of compression and traction (tensile), for these two are directly opposed to each other.



Picture 2.4.7

While sprinting, the lady applies an oblique force that can be resolved into two rectangular components. The shear component and the compressive component.

FRICTION

Potentially, friction forces occur when two objects contact each other. Contact is a result of each object's reaction forces. We will cover Newton's 3rd Law later on in this chapter.

In order for friction to have a magnitude, there needs to be another force acting on at least one of the two contacting objects. That other force is shear, acting on one of the bodies.

Friction forces are therefore related to shear forces. Friction can thus be considered as a kind of reactive force to shear loads.

Friction forces depend on:

- The magnitude of the contact forces between both surfaces. In the exercise field, these will be forces of compression.
- The smoothness/roughness of the contact surfaces (coefficient of friction).
- The magnitude of the shear force.

The coefficient of friction (μ) is constant for two given objects and depends on the type of surfaces that are in contact (i.e. ice on ice is 0.05 μ)

Picture 2.4.8

Depiction of friction and shear forces during a sprint. As shown, both forces are parallel to each other, but do not fall on to the same line: shear force affects the ground and friction affects the foot.

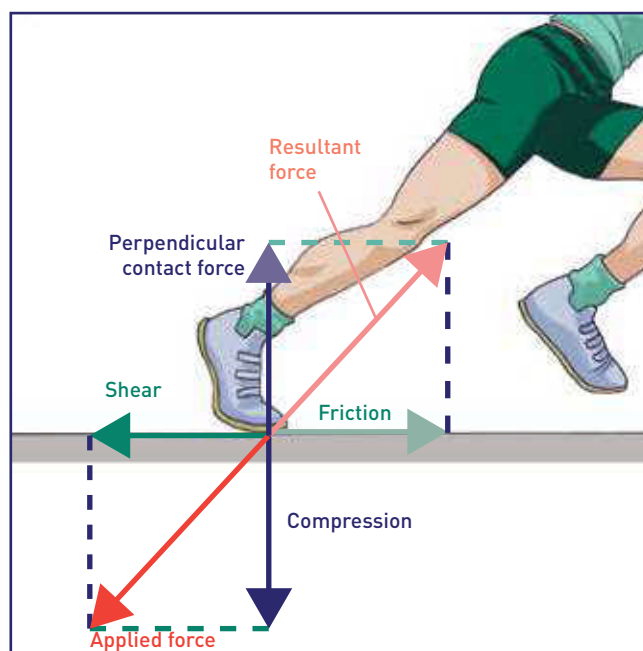
Whenever two contacting objects are not moving respect one another, but there are shear forces involved, that means that there must be other forces present in the opposite direction, to balance out the shear force. This other force is friction. At the most, the magnitude of friction on the given object can equal the product of the coefficient of friction times the contact forces (reactives) exerted between both objects.

$$F_x \leq \mu * F_c$$

F_x = Friction force

μ = Coefficient of friction

F_c = Perpendicular contact force



POWER

Power is defined as force times velocity.

$$P = F * v$$

It is a frequently used term in spoken language, but a complex one to analyze when it comes to physical exercise. Confusion often arises from misunderstanding power and torque (moment of force).

As will be explained later, velocity is constant, so that a force is needed in order to be able to accelerate a given mass. Velocity offers no information on how much force is needed to reach that velocity, nor for it to be maintained.

What is relevant when talking about exercise, is to be able to determine what the forces involved are and how they manage to accelerate a given segment. The force of gravity, or an inertia-free load can cause a body segment to accelerate with little or no muscle force needed.

The most common way to determine muscle power is to assess it relative to an external object that is in contact with the body. That could be a bar, a dumbbell or a force measuring system applied to them. Sadly, this procedure has lead researches to pay more attention to the power exerted on that object (output) than to mechanically analyzing muscle force for each of their involved joint axes.

Correct specific muscle functioning will result in an optimal transmission of the applied forces onto the external object. On top of that, when focus is set upon evaluating force on the object (output), it is often ignored that there could be muscles that may not be generating tension adequately and could, therefore, affect the forces applied to the object. When assessing muscle power in that manner, it cannot be specified the power of which muscles or muscle fibers is being measured. That doesn't mean, of course, that they cannot be useful if well interpreted.

As mentioned before, it is important to take good note that it is forces that are primarily responsible for causing a mass to accelerate. This topic will be further explained later on in

the text. And it being forces that cause acceleration, they too are needed in order to alter the power. Do not be mistaken by thinking that an object's gravitational force (weight) is equal to the total force that given object will exert on the human structure, when accelerated.

We will go into this later again but pay attention to the fact that an object not only exerts the force corresponding to its weight. Apart from that, objects that are accelerated by a body force generate an inertial force that noticeably alters the resulting power. In a similar manner, that given object can gain velocity by being accelerated by the force of gravity, without the need of muscle force.

RESISTANCE OR TORQUE

In exercise mechanics, the force that acts opposite to that of muscle action, and around the same axis, is called resistance. Resistance takes place within the plane of movement, or potential movement. From the above, it can be stated that resistance is a force that produces, or could potentially produce, a rotational movement on a joint lever system.

The word torque is equivalent to the expression moment of

force. It talks about the force that causes rotation around a spinning axis. Because of this, it is torque and not weight that is essential to determine resistance. This is an important factor that often gets miscalculated.

Resistance and torque will be explained in more detail in the coming chapter.

2.5 CREATING RESISTANCES

As explained in the introduction, the starting point for Resistance Institute was this innovative way to interpret and analyze forces and human lever systems. This book comes out after compiling our notes, thoughts, experience and reasoning.

The whole cognitive process revolves around the thought that 'it is all forces'. Because of this, it is of the utmost importance for all exercise professionals to understand forces, and to be able to apply them on to the body we are working with. Maybe we could make an analogy: think of an architect who has been put in charge of restoring a bridge. You would count on that professional's skills to accurately

interpret the structure and all the loads affecting the bridge. It is with these data, that the architect will start making whatever adjustments they consider appropriate in order to restore the functionality of that bridge and for it to become safe again. Exercise and health professionals who apply any kind of force on human bodies should be asked for the same skill. Whenever work is done on living structures that are subject to stress, the applied forces should be correctly interpreted, in a way that will allow for a personalized procedure.

DIFFERENCE BETWEEN LOAD AND RESISTANCE

Knowing the meaning of resistance and clearly being able to differentiate it from the concept of load is of paramount importance. These two terms are often used as if they were synonyms, but they actually aren't.

Load is simply a force that is applied to a structure. When talking about exercise, load can come from a barbell, an elastic, a cable, a medicine ball, the hand of a therapist, etc. Resistance is a force that produces, or could potentially produce, rotational movement on a joint lever system. It could also be described as the forces that opposes to that of the muscles, around a given axis. It takes place on the plane of movement, or potential movement.

Taking this into consideration, it becomes clear that it is possible for a load to not cause any resistance. That would be the case if it produced no movement or potential movement in a joint. Or in other words: if it didn't oppose to a muscle action around the axis of movement.

Picture 2.5.1 shows two examples that aim to clarify the differences between load and resistance. The red dot in the picture stands for the axis of rotation. The drawing on the left side shows how a load is able to cause resistance (it is causing extension, on the mediolateral axis of the elbow. This means it is the flexors of the elbow that will have to come to action in order to gain control over the joint). In contrast, the second drawing shows a load that is not causing resistance on the mediolateral axis of the elbow. In this specific scenario, the load causes no rotation around the axis.

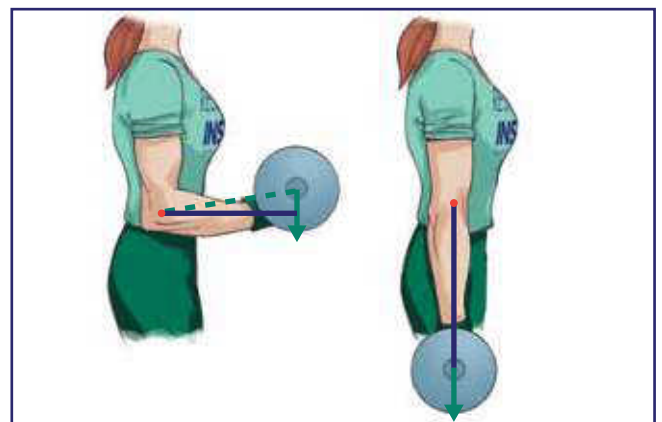
Summing up, there could be load but not be any resistance. You could say that resistance arises from a certain

relationship between the load and the joint axis. It is also true that load and resistance are inter-related, but it is essential to understand the difference between the two.

This difference may offer some hint of why Tom Purvis once mentioned this:

'The weight in your hand is no indication of the forces on your body'

The resistance may as well be provided by an internal force that opposes muscle action around an axis. For instance, the viscoelastic force of the muscle-tendon complex could, when being stretched, cause rotational forces in a direction opposite to that of another muscle that is trying to shorten.



Picture 2.5.1.

The drawing on the left shows how the dumbbell needs to be slowed down by muscle force, thus creating a resistance. There is no resistance in the drawing on the right.

2.6 TYPES OF LOAD

Now that the difference between load and resistance has been explained, let's learn about the characteristics of each of the different loads than can be applied inside the human body. Loads can be strategically used, in order to achieve a greater or lesser resistance. By doing so you will be able to better reach the individual goals of the person you are working with.

Professionals working in gyms or rehabilitation centers will have access to an array of forces (loads). Each one of these loads will provide with unique features, different physical properties and a distinct way to stimulate the neuromuscular system. As a consequence, each load will lead to specific adaptations of the structure, the neurological system, etc. In order to correctly apply these forces, it is vital that professionals know them well.

Just like carpenters know, or should know, the properties of the woods they use for their work, health and exercise specialists should have a deep knowledge of the properties of the loads they use with their clients/patients. These loads may come from their hands, they may be called stretches, maybe manipulation, manual therapy, and others.

TYPES OF LOAD:

- Gravitational load
- Inertial load
- Elastic load
- Load of water or compressed air
- Load in the water
- Friction load
- Manual or external muscle resistance (MARES)
- Viscoelastic load (muscles and connective tissue)

GRAVITATIONAL LOAD

Most of the loads that affect a human body on planet Earth belong to this category. Even the body itself exerts this kind of load over its own joints.

Gravitational load is that which comes from the weight of objects on the Earth. Weight is the force with which a mass is attracted by the Earth, the mass falling within the Earth's gravitational field. All objects that are on Earth and have a mass can potentially generate this type of load. Weight is how gravity affects a given mass (structure), and its equation is well known:

$$W = m * g$$

$$\text{Weight} = \text{mass} * \text{gravity (acceleration)}$$

In fact, we are dealing here with the same equation we saw previously when talking about force, but we now substitute the acceleration for gravity, which is a constant ($G = 9.8 \text{ m/s}^2$ at sea level).

Strange as it may sound, and contrary to what most people think, weight is measured in kiloponds or Newtons, not in kilograms.

Weight, or gravitational load, is a measure of force and, therefore, not a scalar but a vector quantity (with a line, a direction and a magnitude). The point of application for this load will be the center of mass (later explained) and it is

directed towards the center of the Earth. As far as mass goes, it is a scalar quantity and is indeed measured in kilograms.

Taking the above into consideration, we see that weight will depend upon two main factors:

1. Intensity of the gravitational field.
2. Mass of the object.

Because the Earth's gravitational field remains always the same, the only part of this equation that we can alter is that corresponding to mass. Gravity is an acceleration constant that causes masses to be attracted towards the center of the Earth (even though masses, too, attract the Earth towards them).

Where does all this confusion between weight and mass come from? –They are very closely related, since the intensity of the gravitational field does not change much (variability ranges only up to 0.5% in different locations). And because of this being constant, the force (gravity) applied by the Earth on objects, that is their weight, is directly proportional to their mass.

That is the reason why the unit kilopond (kp) was created. The kilopond is a unit of force which makes an equivalence between mass and weight.

The force of gravity corresponds to the weight of objects. Thus, the weight of objects does not significantly change on Earth, unless their mass gets altered.

Putting this all into a real-life example, you could say that a person who's mass equals 70 kg, weighs 686.7 N on Earth. The same person, if placed on the Moon's surface, would still have a mass of 70 kg but would only weigh 114.4 N, due to the fact that the Moon's gravitational field is much lighter than the Earth's. That same person, if on Jupiter (with a noticeably higher gravitational field) would weigh 1,815.1 N.

As a consequence, one single given object (with a given mass, i.e. a dumbbell or a stack) can create a constant gravitational load when the object contacts another body. Do note that gravitational loads are not the only type of

load that an object can cause on another object. The reason for that is that, under certain circumstances, any force originating from a mass can create inertial loads. We will go into this in the following section.

Everything that has been stated up till now can easily be proven by connecting a dynamometer to a dumbbell and causing it to move. The force measured by the dynamometer will keep changing. That wouldn't happen if the only load implied was that of gravity.

INERTIAL LOAD

The term inertial load has been developed by Resistance Institute over the past years and is, therefore, rarely found in existing bibliography. This concept helps explain the effects of inertia while resistance training. Professionals don't usually regard it and very rarely make a strategy to use it. A sound knowledge and the application of this concept is key to correctly applying resistances onto an individual.

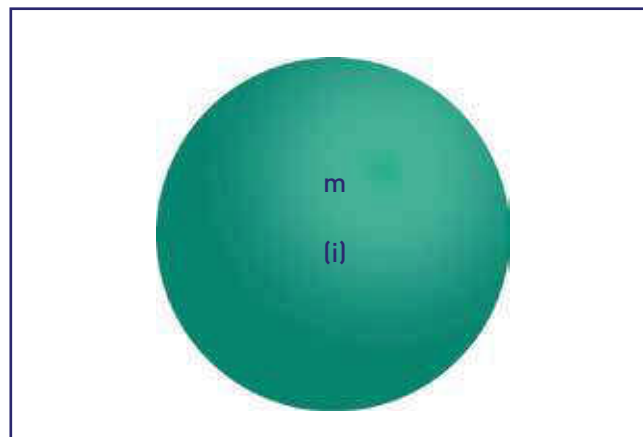
Inertia is the resistance of any mass to linear change. The body must have a mass in order to have inertia. It is necessary to clearly understand the concept of mass, in order to fully grasp the concept of inertia.

Mass is the amount of mass in an object. According to the International System of Units, it is measured in grams, kilograms, pounds, etc.

As already mentioned, mass is not equal to weight, even though they are both intimately related. Weight is dependant on the mass lying within a gravitational field.

SO... IS MASS THE SAME AS INERTIA?

The mass of an object indicates the amount of inertia that given object has (therefore, inertia depends solely on the mass and not on any other factors, such as velocity or acceleration). In fact, both inertia and mass use the same measurement unit. If an object's mass is 5 kg, then its inertia is 5 kg.



Picture 2.6.1

The mass of an object is a measure of the object's inertial property.

CONCEPT OF INERTIA: MOVING FORWARD

The paragraphs above have shown how, for a given object, inertia is directly proportional to mass.

Inertia simply indicates how strongly a mass resists to any change of state. It's the tendency of an object to resist any change that affects its current state of movement or rest (standing still). Whether there is movement or not, that given object will maintain its state, as long as no other unbalancing external force acts on the object and modifies its state.

Therefore, if an object has a mass of 50 kilograms, its inertia is 50 kg, regardless of its state of velocity or acceleration. Inertia is only measured in units of mass.

In order to differentiate between mass and inertia, the following could be stated:

- The word mass is used when referring to an object's amount of matter.
- When talking about how strongly a body resists from changing its state, the word inertia is used.

INERTIA AND NEWTON'S FIRST LAW

It's almost unbelievable that a person could come to such conclusions so many years ago, especially if we consider the kind of resources that were available back then. Newton's first Law is known as Law of Inertia and here is what it states:

'A body at rest remains at rest, and a body in motion remains in motion, with same speed and direction, unless it is acted upon by unbalancing forces. A body's inertia is proportional to that body's mass. An object has greater inertia than another object of the same size, if it is more dense. Consequently, it takes more work to make a denser object start moving, and less work to make less dense body move. And it is therefore harder to stop a heavier object than a lighter object, if they were both moving at the same velocity.'

From analyzing Newton's First Law, it can be noted that neither velocity nor acceleration are related to inertia at all. Inertia is exclusively and solely dependent on an object's mass.

INERTIAL LOAD versus INERTIA

In spite of all of the above, inertia is not able to generate any load (force on a contacting object) on its own. In order to be able to do so, the mass needs to become accelerated by a force. And it is in that precise moment that an inertial load shows up.

When taking a look at the equation for inertial force, it can be seen that it is almost the same as that for Force or Weight, except for some small details, which make it a special one:

$$F_i = I \cdot a$$

Inertial force (F_i) equals inertia times linear acceleration.

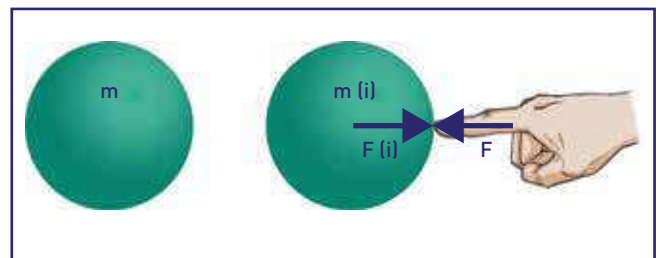
When a force is applied on a given mass, and an acceleration on the object is produced, the reactive force to the

acceleration is what could be considered inertial load.

A close look to the formula shows that the mere fact that an inertial mass exists, does not necessarily mean that an inertial load is present. Just the same way that the presence of a mass, does not necessarily mean there is a weight. In order for weight to exist, the mass needs to be placed in a gravitational field (gravity on Earth equals 9.8 m/s^2). For an inertial load to exist, a force has to be applied onto a mass and, thus, the mass needs to be accelerated.

It is important for exercise professionals to consider inertial loads as reactive forces. We will see why when we get to address Newton's Third Law.

It is also important to focus on the exact objects that are being affected by that inertial force. When a force is applied to a dumbbell, the dumbbell offers a reactive inertial force on the body which initially caused the force.



Picture 2.6.2

Inertial load occurs when a force is applied to an object and creates a linear acceleration. In this picture, the inertial load affects the hand.

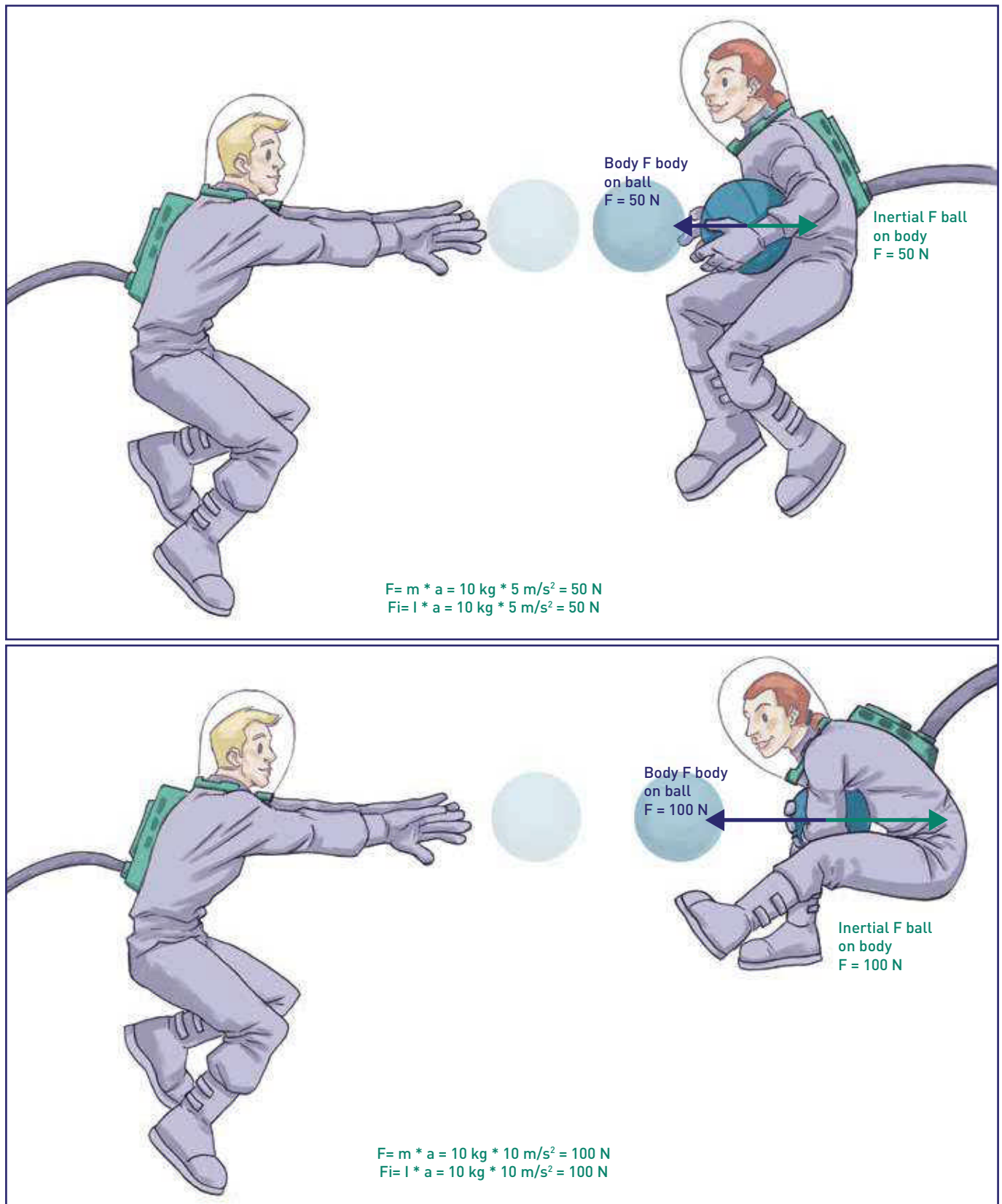
Inertial force (inertial load when applied to a body during exercise) is related to mass and to the amount of linear change. Going back to the formula:

$$\text{Inertial Load (Li)} = \text{Mass (resistance to change)} * \text{Linear Acceleration (amount of change)}$$

According to what was previously explained, in order for inertial forces to exist, two things have to be true:

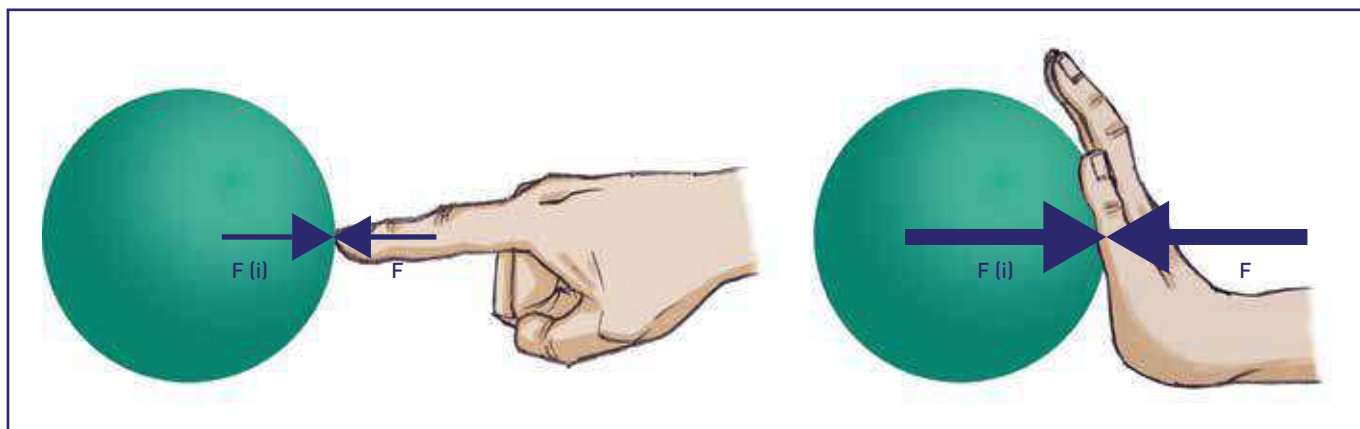
1. There has to be a mass (the greater the mass, the greater the resistance to change)
2. There has to be a force to cause a change (the greater the linear acceleration per time unit, the higher the change).

For objects on planet Earth, inertial forces add themselves to gravitational forces. They increase the forces that cause contact with the body.



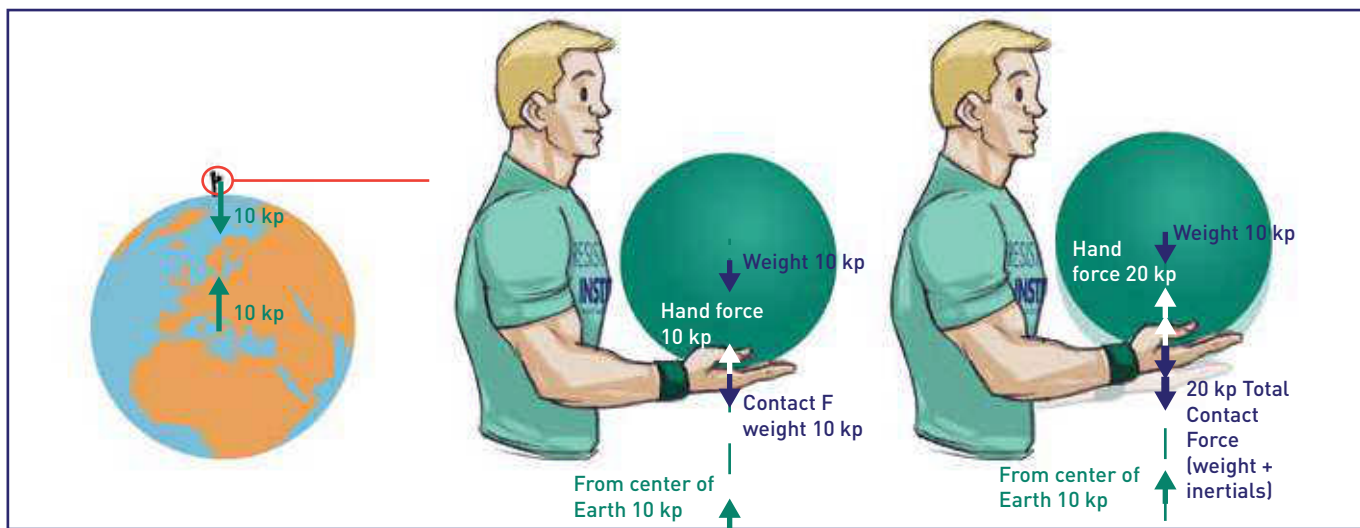
Pictures 2.6.2.a and 2.6.b

The drawings show how, by applying twice as much force on to a given mass, the mass will offer twice as much inertial load to the astronaut on the right.



Picture 2.6.3

The greater the force applied to a mass, the greater the inertial force. The latter is a reactive type of force.



Picture 2.6.4

Within a given mass on Earth, gravitational, as well as inertial forces, can be present.

INERTIAL LOAD ON PLANET EARTH

For any mass located on planet Earth, gravitational loads cause an unbalancing force that brings the mass to accelerate by 9.8 m/s^2 , unless other forces come into play. Masses resist to the force exerted by gravity. They do so by offering a force of equal magnitude and opposite direction to the force of gravity, which affects the Earth (Newton's Third Law). It is logical to think that the mass of the Earth offers a gigantic inertia and that whatever acceleration it may get is virtually inexistent, probably even counterbalanced by orbital forces.

The resistance to change that a given mass offers as a response to the gravitational force is, thus, absorbed by the Earth and does not affect the human body. In fact, gravitational force turns into weight, which is what actually affects the body. That is the reason why body weight should

be regarded as an external force.

From the above, it can be concluded that a object's acceleration, whether positive or negative, with a modulus ranging from 0 up to 9.8 m/s^2 , directed towards the center of the Earth, is caused by the force of gravity and, in general terms, has nothing to do with inertial loads applied to the human body.

Inertial load (force applied to the human body) on Earth comes about whenever a mass that holds contact with the human body becomes accelerated—whether positively or negatively—by an internal force.

INERTIAL LOAD'S LINE AND DIRECTION

Inertial loads are reactive kind of forces (Newton's 3rd Law). As can be seen in pictures 2.6.3 and 2.6.4,

inertial loads fall on the same line but opposite direction to that of the internal force that causes change in that object.

INERTIAL LOAD AND NEWTON'S SECOND LAW

LAW OF ACCELERATION (Newton's second law)

What this law states is that a force (F) that acts on a body, with a given direction is equal to the mass (m) of the body times its acceleration (a) in that specific direction.

$$\text{Force} = \text{mass} * \text{acceleration}$$

If examined, this formula shows that the mass of an object plays a crucial role in understanding the amount of force that will be necessary in order to change the objects state.

Acceleration gives us the data regarding the amount of change.

Two terms are usually mistaken: resistance to change and amount of change. In the equation above, mass represents the inertia or resistance to change, whereas acceleration represents the amount of change.

Mass and acceleration are inversely proportional to each other, when related to a given contact force that is applied to a body.

The contact forces that an object exerts on to the human body are not only related to the weight of that object, but also to the forces that are reactive to the forces causing the object to accelerate.

It is often forgotten that the human body gets affected by reactive forces (inertial loads), that arise due to the internal forces that cause acceleration –be it a positive or negative one. Most commonly we forget to acknowledge negative accelerations against the force of gravity. How an object's weight may be an insignificant contact force –if compared to the forces caused by highly accelerating it with muscle force– can be proven by using a dynamometer. Workout sessions should highly take inertial forces into consideration. Imagine you were working out with moderate weight, for instance. You are trying to deliberately keep away from training your maximal strength. However, high peak maximal forces could be taking place in your joints due to high non-gravitational accelerations.

This is why it is important to consider that weight is merely one of the factors to be considered, together with all other forces that oppose muscle action.

Once again, Tom Purvis's statement comes into play:

'The weight in your hand is no indication of the forces on your body'

INERTIAL LOAD IN SPACE

Resistance Institute often makes their students think about what it would be like to work out in outer space. Professionals often think that, because there is hardly any gravitational pull acting on any given mass, working with an object would cause no weight (force) on the subject. However, when thinking this they forget that, even though there is no weight, the mass can offer a number of other forces when a change in it is caused. As soon as a force causes the object to accelerate, inertial loads come about. Every single time that some force affects a mass, other forces arise that oppose the change of state in that mass. For instance, if a person in outer space applies a force of 10 N to a still-standing 10kg-dumbbell, the barbell will accelerate by 1m/s² y will apply an inertial reactive force of 10 N in the opposite direction, that is, against the person (Newton's 3rd Law).

THE EFFECTS OF INERTIA WHEN RESISTANCE TRAINING

When taking inertia into consideration, the trainee will realize the following. Every time they apply an internal force and cause an acceleration (positive or negative) on the object, throughout the joint ROM, the object will respond by executing a reactive force which will increase the load that rests upon the point of contact.

On Earth, this concept is harder to understand. Because objects over here have a weight, determining how inertial forces act on that object with mass is complicated. Newton is therefore very worthy of merit.

Next, we will take a look at the effects of a given move in space (no gravitational force involved). A dumbbell press has been chosen for this example.

On the initial phase, while the dumbbells become accelerated, it is the anterior musculature that needs to activate (push). When the dumbbells become negatively accelerated, the body needs to activate the posterior musculature (traction), in order to stop the movement. Interestingly enough, it is the anterior muscles who make the acceleration and, following, the posterior muscles who –within the same phase– act as a brake in an eccentric kind of joint motion. This would probably need to be called a pull, instead of a press. The same thing happens on the way back to the starting position. But not only that: the movement can be done in any position (sitting, lying, etc.), because inertial loads only act in a direction opposite to that of the applied force (we will see how this correlates with Newton's

2nd Law later on). In outer space, the trainee will need a point of support. Otherwise, applying any force would cause the body to move.

On planet Earth, as has already been mentioned, the extent to which the effects of inertial loads reach the human body is modified by the gravitational force. Inertial loads could, therefore, be considered as the invisible loads.

INERTIAL LOADS AND NEWTON'S 3rd LAW

Newton's 3rd law states the following:

LAW OF IMPACT AND REACTIVE FORCES

*For any given force that is exerted by a body onto another body, the latter exerts **an equal force but opposite direction**. Therefore, every action is encountered by a reaction, of equal magnitude but opposite direction.*

Even though they cannot be seen, forces on Earth are everywhere, since all masses here are affected by gravity. Whenever a mass is still (not moving), it is a sign that the forces acting on it are balanced (net force equals zero), due to **reactive forces**.

Reactive forces exert a force of equal magnitude to that of the initial force, but in the opposite direction. However, these forces act on different bodies, which allows for the presence of unbalancing forces on either one of the bodies. For instance, take a body that exerts an impact force of 100 kp onto the ground. The ground will, consequently, exert a force of 100 kp in the opposite direction (reactive force). On top of that, the body weighs 70 kp, which means that it is experiencing unbalanced forces for 30 kp, which will cause the body to move relative to the ground.

This thought process is key to understanding the foundations of mechanics when applied to resistance training.

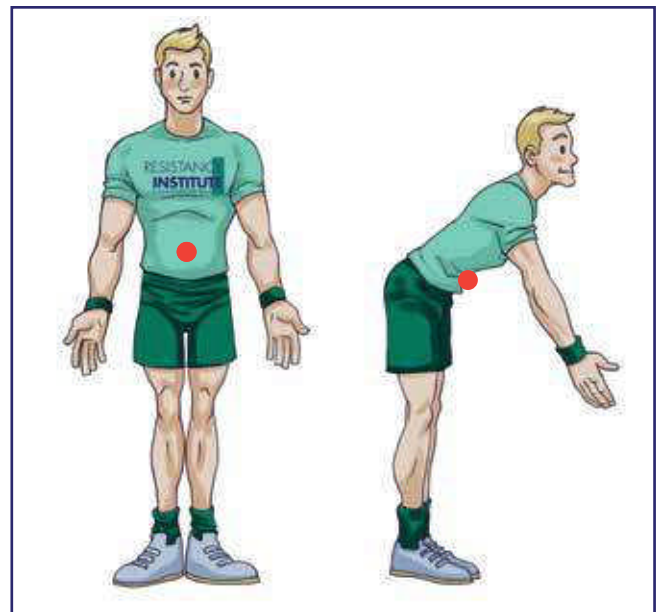
EL CENTRO DE MASAS Y EL CENTRO DE GRAVEDAD DE UNA MASA (C.G.)

An object's center of mass is the spot at which all its mass virtually concentrates, when we want to do static calculations. In other words, it is the spot where its mass finds balance in all planes, and where force should be applied in order to produce pure translation of the object. On planet Earth, the center of mass is also the spot where gravity exerts its action, so it is equal to the C.G.

The C.G. is useful to determine the movement of a mass.

When an object is homogeneous, the C.G. can be located in the object's geometrical middle, but it is more complicated than that when trying to locate it in the human body. To find it then, the spot where the body's mass is balanced out in all its planes should be calculated. However, being it a non-homogeneous object, and because of the fact that its segmentary centers of mass (the centers of mass for each body segment) change, the C.G. can change spots and can, therefore, affect the joints in varying ways. In certain body positions, the C.G. can even be located outside the body's boundaries.

The above-mentioned C.G. could be increased by the weight of an external load, such as a suitcase or a dumbbell, causing the total C.G. to change, as well as the contact forces and the load being applied onto the joints.



Picture 2.6.5

The red dot represents the center of gravity, for different body positions.

Controlling one's own C.G., relative to the forces acting on the body, is a human's first challenge in life. During the first weeks of life, controlling the C.G. of the head is a major challenge. Later on, it's controlling the rest of the C.G. of the whole body that causes a struggle when sitting, standing or walking. **This is what is often referred to as balance.**

For practical calculation purposes, we think of gravity as a force that acts on the C.G., although in reality, gravity affects all the spots within a given object.

Unfortunately, analysing the C.G. in people (and athletes in particular) is hardly of any use. The C.G. only offers information on how strongly the Earth attracts a mass down, but it should not be mistaken for the contact force of that mass on that ground, or whichever other surface.

This is an important factor for those who study Biomechanics. The forces in the body joints depend greatly on the reactive contact forces and on the forces exerted by specific segmentary centers of mass. It is primarily contact forces that affect human joints and the former are not necessarily equal to the gravity. If the forces of gravity that act on a mass were equal to the reactive forces, bodies would not move.

Biomechanics could find an interesting topic here for debate. Usually, more thought is given to the location and displacement of the center of gravity than to contact forces and how internal forces manage to change their amount and direction, which in turn achieves the displacement of the C.G. and human motricity.

Please note that calculating the displacement of the C.G. could be useful, but certainly the internal forces are not yet receiving enough attention as the agents that modify contact forces and cause the C.G. to move.

The limbs, as well as the vertebral spine hold their own segmentary center of gravity. When a load (i.e. dumbbell) is placed on a limb, and if we aimed to be precise, then the integrated limb+dumbbell center of gravity should be calculate, for the different joint axes. In order to make the learning process more simple, this calculation is often left aside when studying lever systems. Nevertheless, the integrated center of gravity should be kept in mind, because the central nervous system is indeed dealing with it.

WHICH KIND OF EQUIPMENT SHOWS INERTIAL LOAD?

First and foremost, one should realize that the potential inertial load that is always present on Earth corresponds to one own's body. The body is a mass, which weighs and has inertia. The neuromuscular system is primarily ready to deal with the resistance that its own mass exerts in the joints. This resistance can either stem from the effect of the weight, or the inertial load caused by accelerating body segments.

Most of the time, joints receive force from the weight (total or partial) and, upon being linearly accelerated, also from the body mass's resistance to change. Forces in the joints will increase with any non-gravitational acceleration,

change in direction, change in segmentary posture, etc. The neuromuscular system will have to do its best in order to control the joints so as to prevent injury and maintain maximal structural integrity.

In addition to the inertial load caused by one own's mass, most of the loads that are usually applied to the body can also cause inertial load. For instance, the so-called free weights –dumbbell, barbell, discs, medicine balls, etc.– and many fitness machines, as well –with a plate stacks, cable systems, etc.–

To summarize, any load that has a mass can cause inertial loads.

INERTIAL LOADS: ADVANTAGES AND LIMITATIONS

Because inertial loads come in different shapes and forms, its characteristics depend both on the type of load itself and on the equipment design (guided machines), the characteristics of the guides, cable systems, on the amount of friction between components, the degree of movement restriction, etc.

The fact that inertial loads are influenced by acceleration makes them resemble daily-life loads. That in itself constitutes an advantage, and they should be the preferred type of load in any training program.

Accelerating, as well as slowing down movements using internal forces implies placing great force requirements in the joints. This happens during many daily-life activities – especially those involving one own's body– and it happens when using inertial loads, as well. Muscle and joint structural integrity could be threatened as the accelerations and decelerations caused by internal forces decrease the degree of control. This constitutes a disadvantage.

What has just been explained does not mean that accelerations are bad for you because they may increase the force

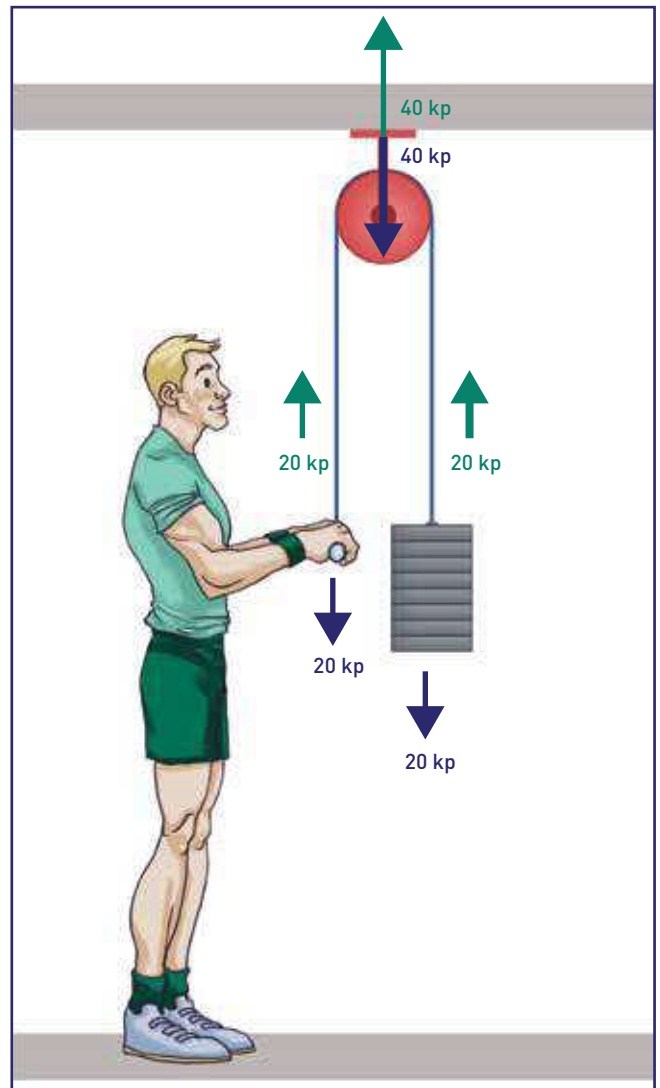
requirements in the musculoskeletal system. It is all about understanding the involved factors and enabling the professional to appropriately use each of the available tools, as they understand how varying stimuli can lead to different effects. When talking about inertial loads in particular, their internal acceleration will strongly determine the path to progression. It should be borne in mind that most of our daily-life force requirements fall within the reactive inertial category. Inertial loads, in the specific context of the individual, could be key to the achievement of many of the desired adaptations.

PULLEY SYSTEMS

Pulley systems are primarily designed to alter the direction of the load. In other words, a pulley's main function is to re-direct the line of force.

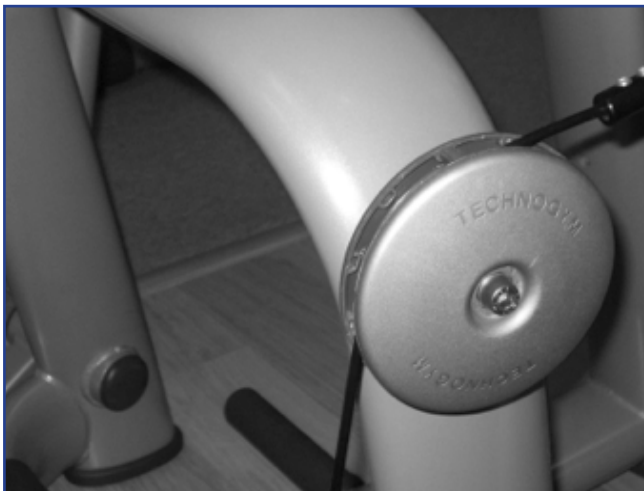
When talking about pulley systems, people tend to think that they are aimed at causing the load to de-multiply. The truth is that a pulley itself does not necessarily cause the load to de-multiply. That would only happen under specific circumstances.

Pulley systems make it possible for us to adapt the direction of the force to the different planes of work. This enables us to alter resistances and adapt them to our different goals. An example of this can be seen in picture 2.6.6.



Picture 2.6.6.

In this example, the direction of the force exerted by the plates runs downwards. If the character were to lift the plates without a pulley, he would have to exert force in the opposite direction (pulling up). But because he has a pulley, he can simply cause a downward pull (traction).



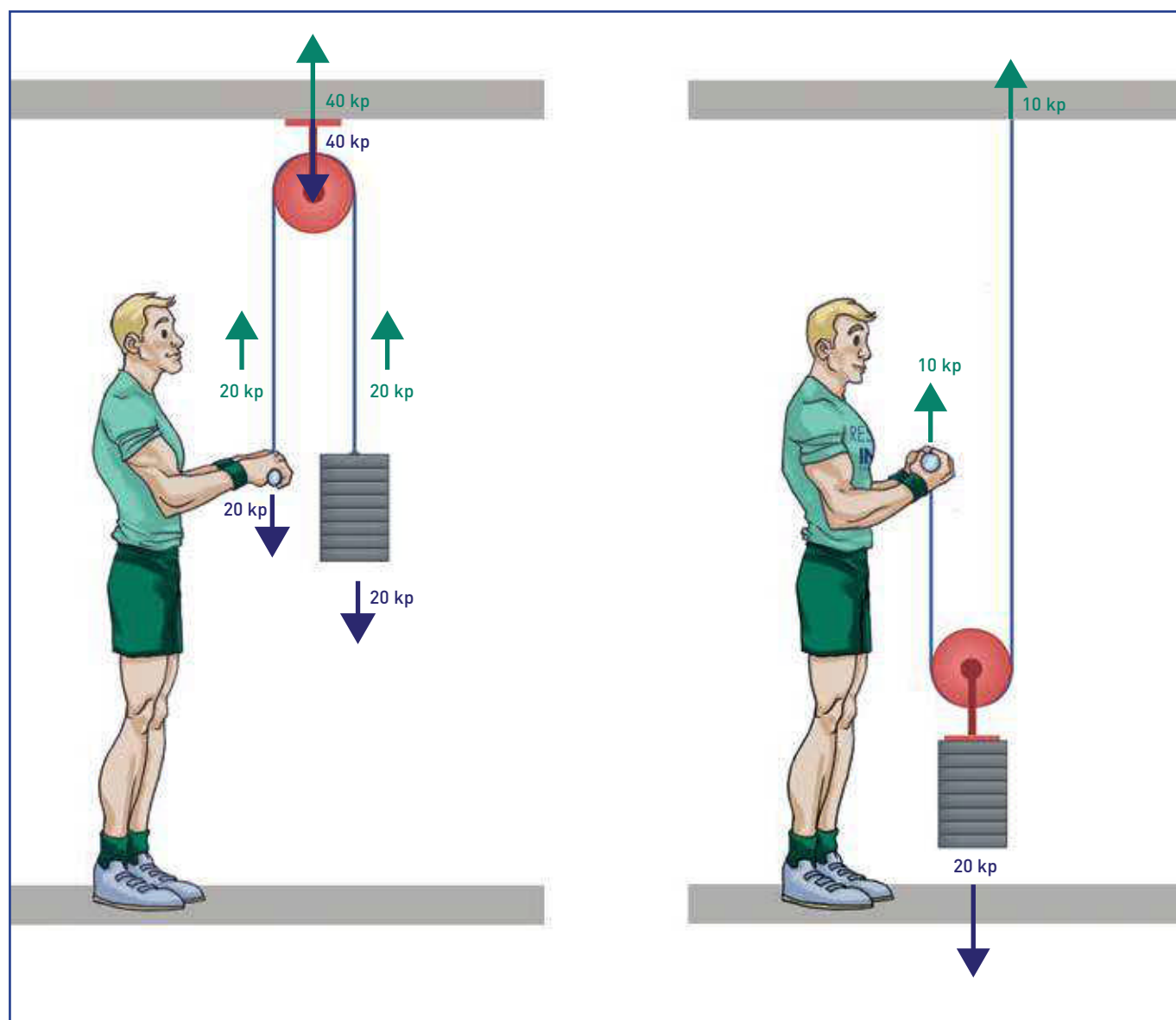
Source: Resistance Institute, 2005

IN WHICH CASES CAN A PULLEY SYSTEM CREATE MECHANICAL ADVANTAGE OR DISADVANTAGE?

A mechanical advantage or disadvantage can occur when the pulley within a pulley system is not fixed and can move linearly, together with the load. This happens with the cable around the pulley is fixed on one end and movable on the other.

When the pulley forms a unit with the load, force is divided into both sides of the pulley. Being that only one side will be moving and the other end is fixed, the load that reaches each of the ends becomes de-multiplied.

Because one of the ends is connected to a fixed surface, only one end can move. The load is split between both sides of the cable, so that the force that will be required to move the load will be half. This is known as de-multiplication of the load and creates a mechanical advantage. Mechanical advantage refers to the fact that twice as much load can be moved for any given effort, or a given load can be moved with half the effort. In Physics however, nothing is for free. In return for this 'ease', the load will be moved half of what it will move on the other side of the pulley system. Thus, mechanical work (W) will be equal on both sides of the system ($W = \text{force} \times \text{distance}$). Picture 2.6.10a shows an example of this. Mechanical work will be explained in chapter 3.



Picture 2.6.7

The picture on the right shows a mechanical advantage pulley system. For a 20-kp force, the cable receives a load of 10 kp. Picture on the left shows a simple pulley system, in which 20-kp plates transmit 20 kp on to the cable.



These pictures show a pulley system that de-multiplies by two. When the exercise is done unilaterally, one of the ends gets fixed, and de-multiplacion takes place.
Source:
Resistance Institute, 2005

When the load is united to more than one movable pulley, the force is divided by, at least, four (depending on the number of pulleys).

The following is then possible:

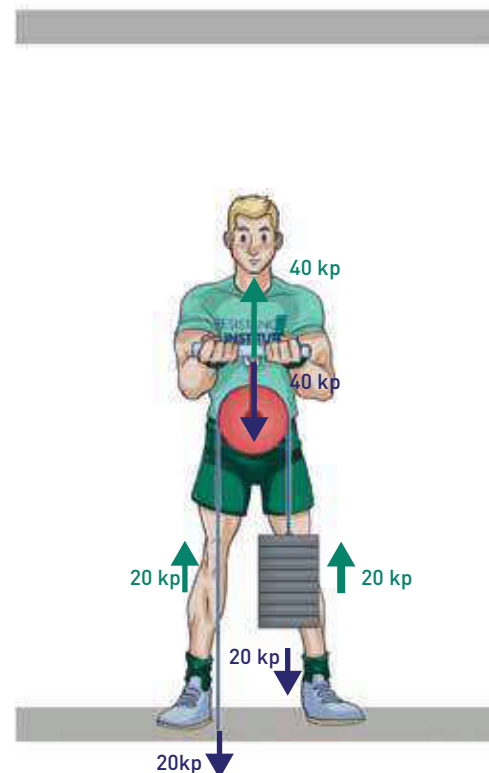
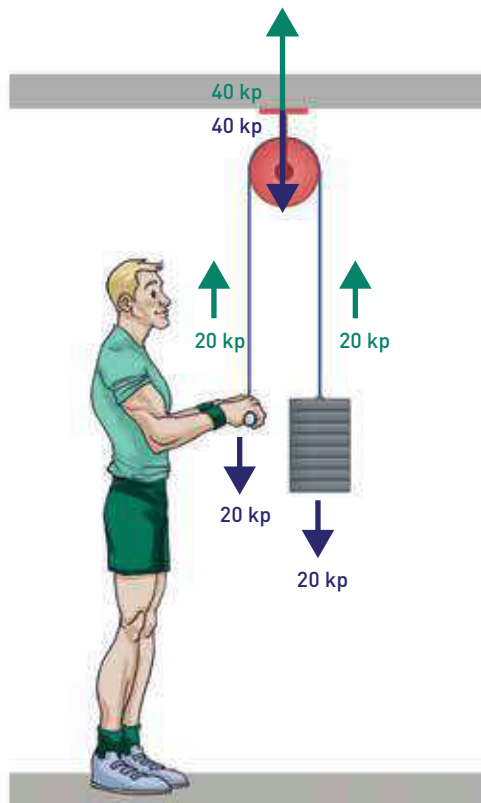
- With one pulley: de-multiplication by two (load: 50%; displacement: 50%).
- With two pulleys: de-multiplication by four (load: 25%; displacement: 25%).
- With three pulleys: de-multiplication by 6 or 8 (load 16.6%/12.5%, displacement 16.6%/12.5%).
- And so on

It is also possible to obtain a mechanical disadvantage by using a pulley system.

By placing a movable pulley on the side of the power, a multiplying effect is created, and tension on the cable is doubled. In return, twice as much displacement is obtained (picture 2.6.10b).

Picture 2.6.8

The drawing on the right depicts a mechanical disadvantage pulley system. With it, the cable receives 40 kp, for a 20-kp force. The example on the left shows a simple pulley system in which, for 20-kp plates, 20 kp are transmitted onto the cable.



CLOSED-LOOP PULLEY SYSTEMS

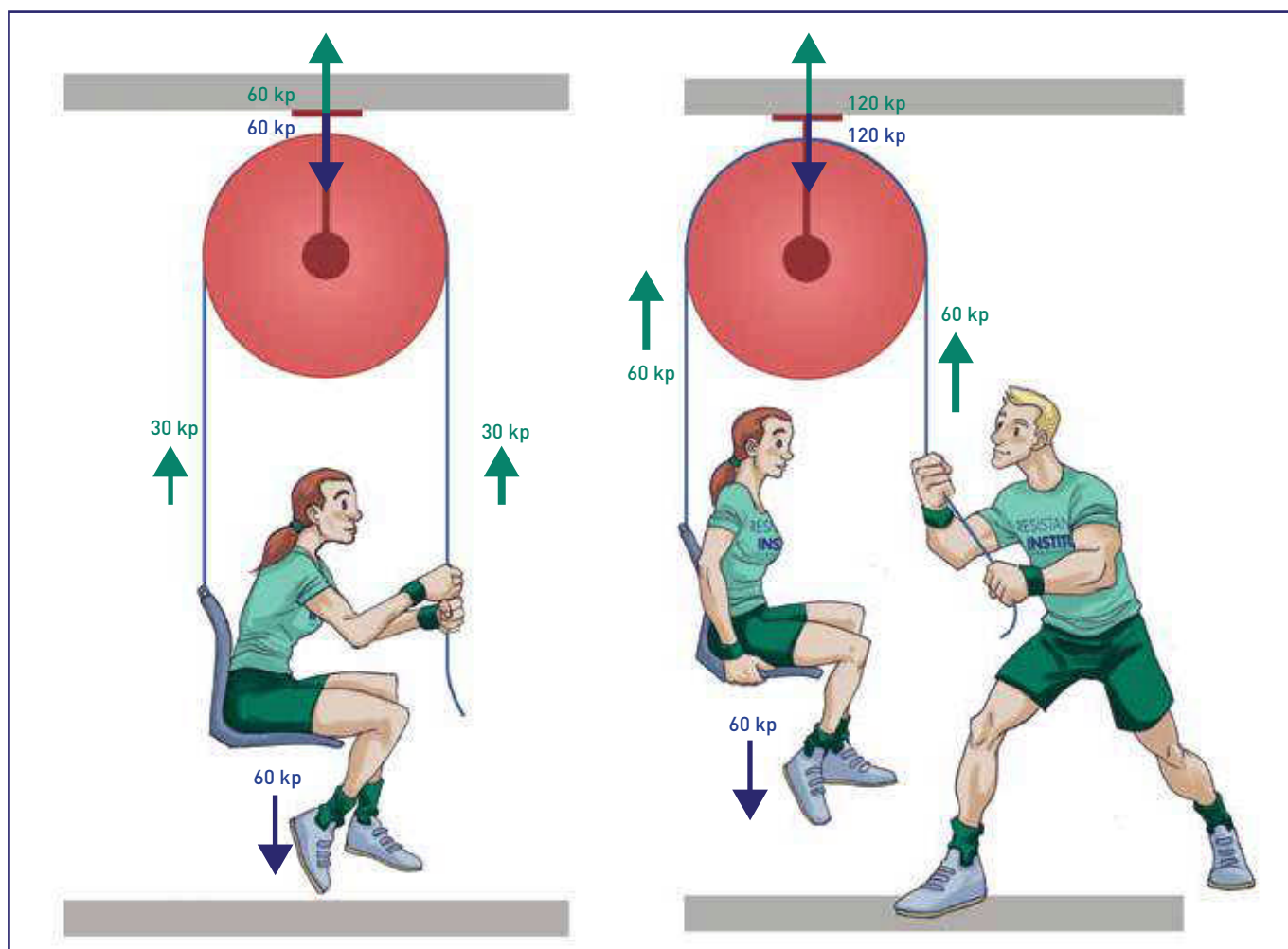
An interesting phenomenon takes place in a pulley system whenever the pulley closes the system between the load and the power.

Even if the pulley does not linearly move, a de-multiplying effect takes place whenever a closed-loop is formed between the power and the load.

The above described phenomenon can be observed in fitness equipment such as the Pilates reformer, Gravity, etc. In order for this phenomenon to take place, the person has to be a part of the system, for it is strictly necessary that the

power be integrated with the load.

The underlying physical grounds for this are exactly the same we have seen in previous sections. The load gets divided into the different cables (the more cables in direct contact with the load, the greater the de-multiplying effect in the system). Watching a graphical representation of a simple system will probably help to understand this phenomenon. Picture 2.6.9 shows, on the left, how two cables are in direct contact with the load, and cause it to de-multiply by two.



Picture 2.6.9

Closed-loop pulley system on the left. Simple pulley system on the right.

The left-hand side of picture 2.6.9 shows how load and power are located inside one single system, thus causing the load to de-multiply into two cables. For that reason, the load is divided by two and the lady pulls from just one of them.

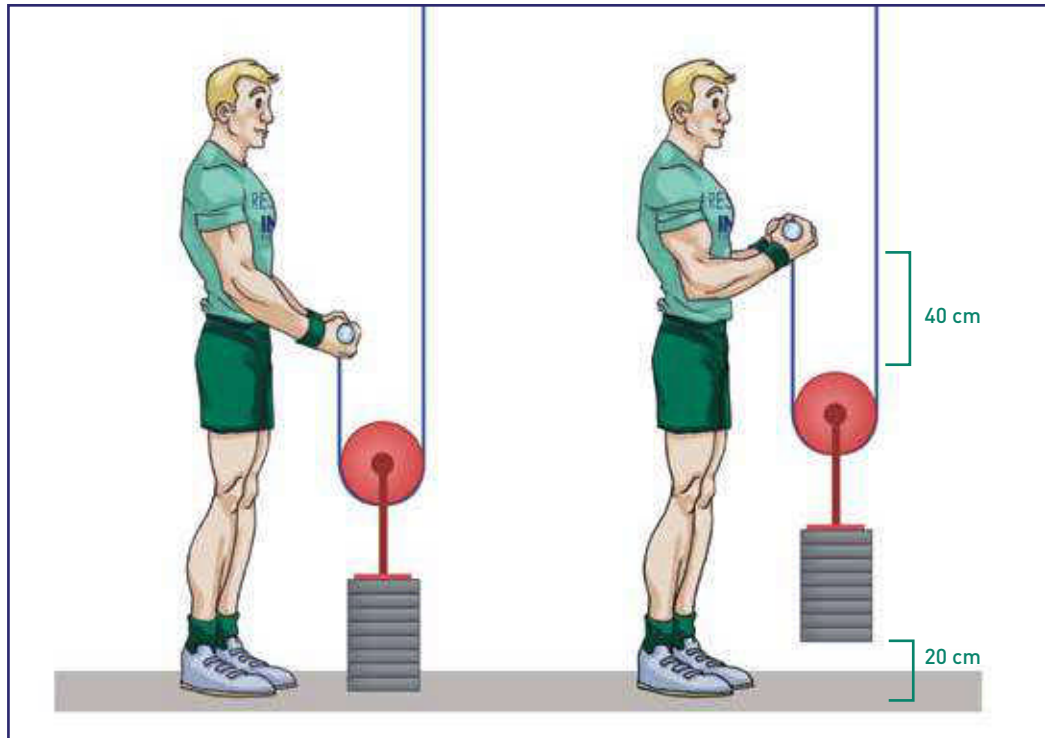
However, when load and power do not belong to one single system, the latter must exert a force that is equivalent to the load in order to keep balance. In this case, the pulley is simply changing the direction of the force.

CABLE SYSTEMS AND INERTIAL LOAD MULTIPLICATION OR DE-MULTIPLICATION

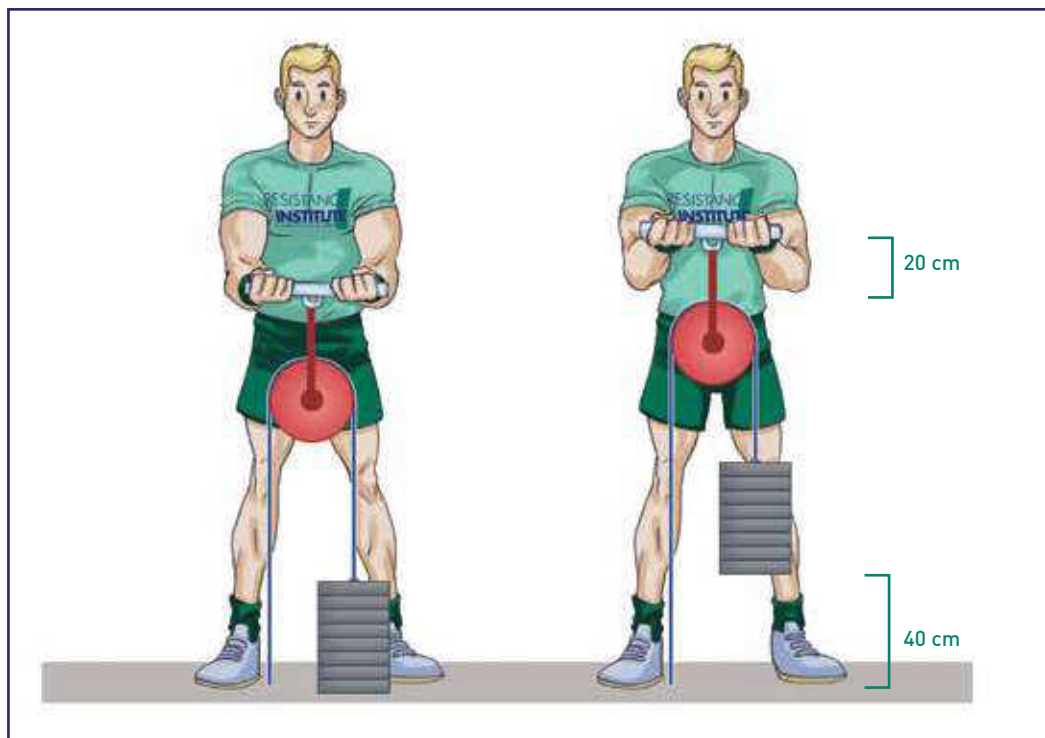
As mentioned in previous sections, pulley systems can alter how much the load is displaced, as compared to how much the power moves.

When the load is de-multiplied by two, the load will move only half of what the segment moves, and inertial forces are cut in half, for any given acceleration-deceleration ratio.

That is to say, the load only experiences half the change, which means that inertial forces are also reduced by two, for a given acceleration-deceleration ratio.



Picture 2.6.10a
A mechanical advantage pulley system. Displacement of the load is half that of the power's.



Picture 2.6.10b
A mechanical disadvantage pulley system. Displacement of the load is double than the power's.

Let's take a look at it this way. If we place twice as much load in a de-multiplying pulley system and we compare it to placing half as much load in a non-de-multiplying system, the resistance obtained would be the same in both cases, as long as the system stayed static. However, in an acceleration were to be applied, the load on the de-multiplying system would only be affected half as much, for a given acceleration-deceleration ratio.

When a system multiplies by two, the load moves twice as much, if compared to how much the segment moves. Also, inertial forces become double, for any given acceleration-deceleration ratio.

Examples of these can be seen in pictures 2.6.10a and 2.6.10b.

DE-MULTIPLYING AND INERTIAL-FREE PULLEY SYSTEMS

Many different types of de-multiplying pulley systems exist. Most of the available pieces of equipment for pulling exercises do not de-multiply, though a few of them do. Interestingly, all of these different machines are used interchangeably and little or no care is taken when numbering the stack, or specifying how much tension actually reaches the point of contact with the body.

Most pulley systems de-multiply by two, which reduces the load to just half. Other systems de-multiply by four, and there are even pulley systems that allow for the user to choose between a by 2 or by 4 de-multiplication.

This is why many gym enthusiasts state that cables offer 'a more constant type of resistance'. The truth is gravity is always constant, but the loading curve in de-multiplying pulley systems is more constant than that in non-de-multiplying systems. The reason for this is that, for a given positive or negative acceleration, a de-multiplying system reduces the inertial forces.

There are de-multiplying pulley systems in the market that allow the user to change the direction of the force, in many different positions. This way, it is possible to work in various planes of movement and to create a number of stabilizing possibilities.

Some guided machines that work on a pulley system basis, cause a de-multiplying effect when used unilaterally. By doing so, the user does not need to change the load when they want to work with just one limb. In this case, the effects of inertia will also change, as a unilateral exercise will only cause half as much displacement of the stack, if compared with what would happen when working bilaterally.

You can also find pulley systems that can totally eliminate the inertial load from the plates. This can be achieved by simultaneously doing bilateral, asymmetric moves, as as to not cause any movement in the stack. By doing so, the only load that has to be dealt with would be the weight, and the effects of inertia will be inexistent.

The so-called closed loop machines, such as the commercial brand Kinesis, are variations of pulley systems that exponentially increase the load, depending on the angle formed between the cables and the grip. In this case, the resultant line of force between both cables has to be calculated. The inertial load is also exponential and depends on the angle, even if the segmentary velocity remained constant.

ADVANTAGES AND LIMITATIONS OF LOAD DE-MULTIPLICATION

Being able to de-multiply a pulley system and, consequently, the effects of inertia, can be handy when the user still lacks the ability to control accelerations and, most especially, the decelerations. As seen earlier, the inertial load caused by accelerating body segments decreases, the more de-multiplied the system might be. Therefore, this is a convenient tool when learning a given resistance move, and it is specially useful for health specialists.

De-multiplying pulley systems can constitute an interesting alternative whenever the user shows proprioceptive difficulty. The reason for this is that accelerations here do not alter the force curve as much, which makes this system a bit more similar to an inertia-free type of load. Nevertheless, we should bear in mind that de-multiplying pulley systems still cause inertial effects, due to the resistance to change offered by body segments. de-multiplying pulley systems would probably be a good middle-step when progressing from an inertial-free type of training into full inertial loads, in specific planes of force.

Because for any given acceleration rate the inertial load decreases, exercise will show softer peaks of force. This fact may be a good reason to use this type of equipment for post-injury strengthening, to better help adapt the structures towards resistance training.

Clearly enough, all available tools have both advantages and disadvantages. And choosing to move in one direction, you potentially lose the advantages of the other.

For the above stated reasons, working with de-multiplying pulley systems can lead to the user acquiring motor sequences that do not resemble those required for daily activities, which often involve higher inertial loads. Abusing the use of de-multiplying pulley systems may not be beneficial. Exercise

should aim for an adequate transferral into actions that require enough muscle force to be able to control positive or negative accelerations. Failing to do so may keep the person away from really adapting to pure inertial loads (weights, suitcases, bags, balls, etc.) and eventually limit their neurological skills.

But not only that, acceleration should also be one of the variables included when progressing into motor patterns that resemble sporting moves.

This is why it is important to highlight that there is no superior pulley system. The key to achieving the desired beneficial

effects is to use this or other types of systems in a strategic way, taking the individual's skills and goals into consideration.

TENSION AND ELASTIC LOADS

Elasticity is defined as the characteristic by which a material recovers after being deformed by a force. Elastic materials are capable of returning to their original shape, once the forces that produced their deformation cease to act on them.

Elasticity is a qualitative (uncountable) characteristic, and Resistance Institute has insisted on this matter during its many educational programs. It is complicated to explain in a book, but it becomes easily understandable when debated amidst a highly-qualified group or professionals.

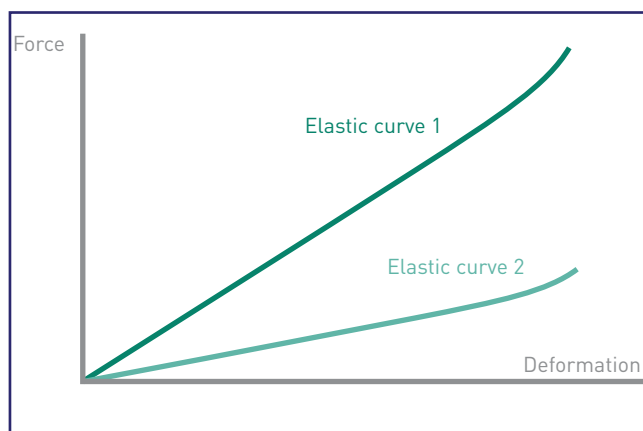
Any given material either has or does not have elastic traits. That is why it is a qualitative characteristic, and cannot be measured. As a consequence, it is incorrect to say to describe a person as more or less elastic. A person is simply elastic. Or to be more precise, when talking about people we should consider their viscoelasticity, instead of their elasticity. Materials with elastic characteristics also show other inherent traits, that come together with elasticity and can be measured. Examples of these would be, for instance, rigidity, hysteresis, deformation within the elastic area, etc. All of them can be measured along the load-deformation curve.

The way elastic materials resist elongation is not dependent on their mass. Because of this, they are neither affected by gravity nor by the effects of inertia. Therefore, elastic-type resistances do not show inertia-related forces.

An object's ability to return to their original shape after having been deformed depends on its structural properties, its length, its diameter, the deformation percentage, etc. All of this data will help determine the so-called modulus of elasticity.

Any given material can reach its elastic limit. When this

happens, any change in length will affect it structurally, it will lose its elastic properties and enter the plastic zone. The plastic zone is the area where a given material experiments structural damage, which prevents it from recuperating its original state (the state shown when no forces had been applied). Young's modulus—also called modulus of elasticity—is used to calculate all of this. **This modulus quantifies (measures) rigidity.** Rigidity refers to how much a given material resists deformation. It is the quotient of applied force by the deformation experimented by a material.



Picture 2.6.11

The graphic shows how two different elastic materials behave and how much force is required in order to obtain a certain deformation.

Elastic materials are often used in fitness rooms. Its characteristics and properties greatly differ from those of inertial loads, and their force profiles are also different.

Following is a description of elastic load properties, which differentiate them from other types of load:

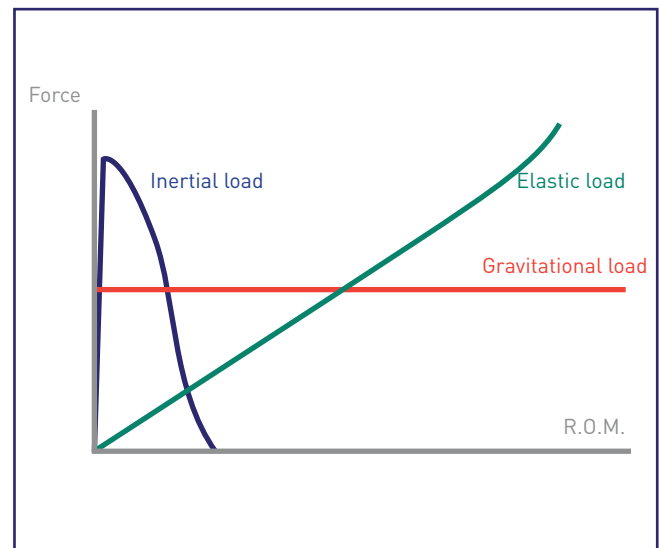
- **they are virtually free of inertia.**
- **the load varies depending on the modulus of elasticity and the deformation percentage.** It is of utmost importance to know the characteristics of each type of elastic material and what area of percentual deformation is being worked in throughout the ROM.

In spite of there being extensive literature stating that dumbbells offer a constant load, whereas elastics offer a load that varies along the ROM, a simple home experiment can demonstrate that this is not always so.

Proof of this can be seen in Resistance Institute's Academy videos, in our Youtube channel, or you may want to test it yourself. For this 'home experiment' you will need to use a dynamometer, a long stretch band with low-rigidity and a dumbbell. Try to do exactly the same articular motion with each of these tools. You will see that the dynamometer shows a much more uneven amount of force when the move is done with the dumbbell, as compared with the stretch band. The main reason for this is that the stretch band offers a low-rigidity with a low percentual deformation (strain) because of its length. These factors combined make it possible to accelerate the band with a low input force. The dumbbell, however, offers reactive forces that resist change. These are the reason why the dynamometer shows considerable changes in the load offered by the dumbbell.

As mentioned above, the amount of force offered by an elastic material as it becomes deformed relates to its percentual deformation or strain, which makes the elastic's length a very relevant factor. For instance, if the work out is done with a 1 m-long stretch band, and it experiments a 1 m deformation, the percentual deformation equals 100%, of its original state. However, if the grip placed one hand on the end and another hand at half-length (or we wrapped it around our hands to make it 50 cm long), the percentual deformation would now equal 200%, even for a given unchanged rigidity level. This would noticeably change how much load reaches the body system.

The elastic load curve provided by a given elastic material can also noticeably change depending on the materials initial degree of percentual deformation. The curve (amount of elastic load, relative to the articular ROM for a given move) would not be the same if, for instance, we started at rest and deformed up to 100%, or if we started with a 100% deformation, and progressed up to 200%. Both of the examples mentioned would imply an equal percentual deformation, but the elasticity curve offered by the material would differ, which in turn will cause different loads onto



Picture 2.6.12

Theoretical representation of different types of load.

the point of application.

ADVANTAGES AND LIMITATIONS OF ELASTIC LOADS

Once again, loads offering certain properties often lack other properties and it falls within reason that no load can offer all properties.

Elastic loads offer inertial free forces. This means that, for a given segmentary acceleration, they will require less force than would working with an inertial load. Apart from that, we have to remember that the elastic curve changes as the material deflects. This fact makes it advisable to adjust the curve so as to match the body's ability to generate force around the involved joints. It is important to be aware of the materials elastic coefficient and to keep in mind that the original length determines the percentual deformation.

Later on in the book, we will explain the concepts of levers and torque. We will then see the factors that need to be taken into account in order to calculate the resistance offered by an elastic force, throughout the articular ROM. These factors include not only the variation of the load, but also the variation of the moment arm, throughout the articular ROM. Only if we do this, will we correctly adjust the resistance profile. Unfortunately, many fitness classes use elastic loads without these considerations.

Elastic materials are inexpensive, show no inertial effects, and reduce load as the eccentric phases progresses. For those reasons, using them can be convenient when pursuing health-related goals, as prophylaxis and after injury.

Elastic loads make it impossible for the user to progress the

profile of force by altering the forces related to acceleration.

This is similar to what we saw happens with de-multiplying cable systems and, most especially, with inertia-free systems. Because of this, they are not the most appropriate loads to use if we want to achieve motor patrons that adapt

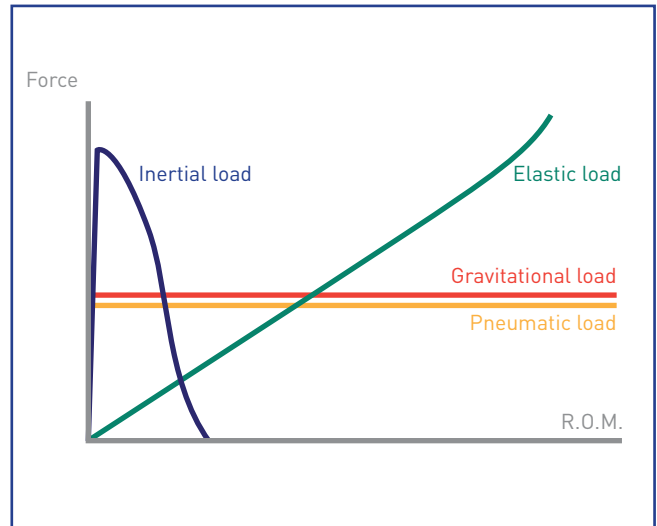
to inertial loads. This fact should not be a problem, as long as transferrals into inertial loading are correctly planned for each individual/athlete.

PNEUMATIC LOADS

Compressing air into a cylinder is yet another way to create virtually inertia-free resistances. At times, their characteristics resemble those of elastic loads.

Some commercial brands have designed cable systems where the load comes from air being compressed, which virtually eliminates the inertial-loads. Unlike with pulley systems, when using pneumatic loads, the move dos not need to be bilateral and asymmetric in order to be freed from the effects of inertia, because it is the force itself that is inertia-free.

Pneumatic loads present the same advantages and limitations as do inertia-free loads. And they also present the advantages and limitations related to that specific piece of equipment (guided, unguided, design, etc.)



Picture 2.6.13

Theoretical representation of different types of load. Pneumatic loads do not change through the movement (as long as the acceleration of body segments is disregarded). Pneumatic loads are free of inertia.

WATER LOAD

Water load is often referred to as an ‘accommodating load’, meaning that it provides a load that is directly proportional to the effort exerted and its direction. Water does not, on its own, offer resistance during the isometric phase. However, this could be achieved by adding an additional type of load, like that of a buoy trying to reach the surface. Because water loads accommodate to the direction of the force exerted by the individual, and it lacks a direction of its own, water load doesn’t provide resistance in the eccentric phase, either. In a way, water load is a reactive force, which could lead to thinking that its characteristics are similar to those of inertial loads. Nevertheless, water load behaves in a completely different way: it doesn’t always resist with the same amount of force, and its direction is always opposite to the direction of movement. This last factor means that there will be no eccentric articular kind of work (force and movement in the same direction) unless, as explained, some device –maybe a buoy– altered the mechanics by exerting a force to try to reach the surface. Water load can exist in all spatial planes of movement.

It now becomes clear that this type of load is virtually inertia-free. Apart from that, if the body were completely submerged, the context would be virtually weightless. Whenever water is not fully covering the body, gravitational forces (weight) coexist with water load and buoyancy aspects.

The load profile (load throughout an articular ROM during an exercise) provided by water loads adapts to the individual’s ability to generate tension. This fact eliminates the peaks of force that usually take place during eccentric phases and avoid going through those ROM areas where there is no control over the load. Needless to say, this significantly reduces the structural damage risk, even though applying any type of resistance holds come potential risk. Water load doesn’t generally offer an articular eccentric phase. To come back to the starting position in these situations, it is actually an articular concentric contraction in the opposite direction that takes place. And this being so, it is easier to work the musculature evenly in this way. It is important,

however to first make sure that there is adequate specific muscle function and that the work-out is balanced. This same advantage, turns into a problem when it comes to transferring the motor patterns into life-on-Earth activities.

Water can be a convenient resource within the resistance application process, and specially useful when our client suffers from certain pathologies, such as degenerative

osteoarthritis, or during bone or ligament post-surgical periods.

Both water and other types of loads should be applied, if we want to make an adequate transition into daily life-on-Earth activities, which will always be subject to gravitational and inertial effects.

PISTON ENGINED PNEUMATIC LOADS

This type of load is similar to the load of water. Because they only offer load when a force is applied, they too can be referred to as 'accommodating' loads. These loads can be considered of a reactive kind. The piston's reactive load can be modified by changing the load that it being applied by the client. The higher the force being applied per unit of time, the more the piston resists the force, leaving little room for acceleration. The hydraulic load can be modified, as well, which will change the amount of applied force that is required to get to move the piston.

From the above, this type of load helps to avoid significant accelerations in the body segment and exerts a force with a direction that is always opposite to the movement (push of the hydraulic piston). This way, the individual does not need to exert any force in order to stop the movement and,

contrary to what happens with inertial loads, there is no resistance to the stopping action, in the direction opposite to the move. This means that this type of load virtually has no eccentric phase. But unlike what happens with water, loads from hydraulic pistons only exert force in the direction of the piston, and they are usually installed within guided pieces of equipment, adding to the list of mechanical handicaps.

The advantages and limitations related to the loads from hydraulic pistons are similar to those of water load. Some differences exist, though, since using loads from hydraulic pistons on Earth implies being subject to inertial and gravitational loads, as well as to the restrictions derived from the guiding system, as well.

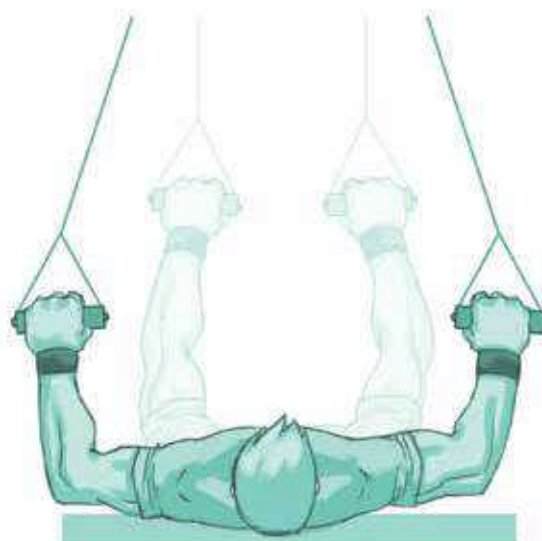
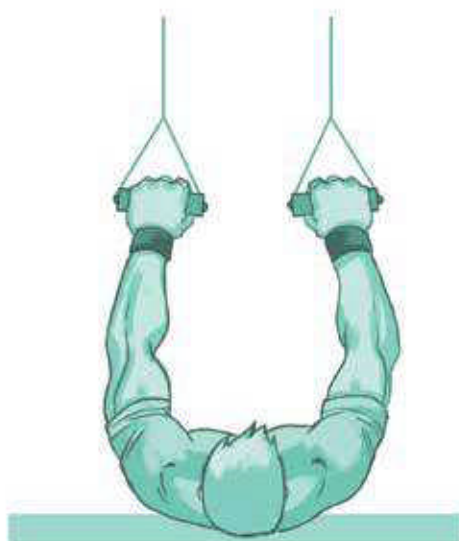
2.7. THE HUMAN BODY RESPONDS TO EXTERNAL FORCES... BUT IT ALSO CREATES REACTIVE INERTIAL RESPONSES THROUGH INTENTION

The joint-muscle system is responsible for handling with the forces taking place in the joints. These forces may be due to external loading, but could also derive from mechanical responses to internal intention that generates within the body system.

Some structures, like the ground or walls, do not accelerate, even when applied forces. When the human body contacts one of this structures, the resistance to change that takes places is due to the body's own inertia. That itself modifies the force being applied to the structure as it becomes accelerated by internal muscular forces. It is essential to bear this in mind, and to distinguish between weight and the direction of the reactive contact force, arising from the supporting areas and into the body joints. Lets take a look at it with the following example. If the body receives a contact force derived from its own weight, the neuromuscular system will respond to it in order to maintain control. However, if that body intended to move, it will have to generate internal forces and the body mass will resist change, which will in turn increase the contact forces, and lead to increased reactive contact forces reaching the joints. From this we can conclude that segmentary acceleration leads to a mechanical response that creates inertial loads, which then transform into resistance at the joint axis.

Reality is such a complex thing that we can sometimes be overwhelming. However, most of the time it is a combination of both scenarios that take place in an alternating pattern, generating motricity and achieving articular stability.

3 LEVER SYSTEMS



In this chapter, you will learn how important it is for an exercise and health specialist to understand lever systems. Examples will be given of how relevant it is to assess the amount of force that access the body via its lever systems. In the following sections, you will be presented lever systems in a new way, specifically adapted for its application to the exercise world.

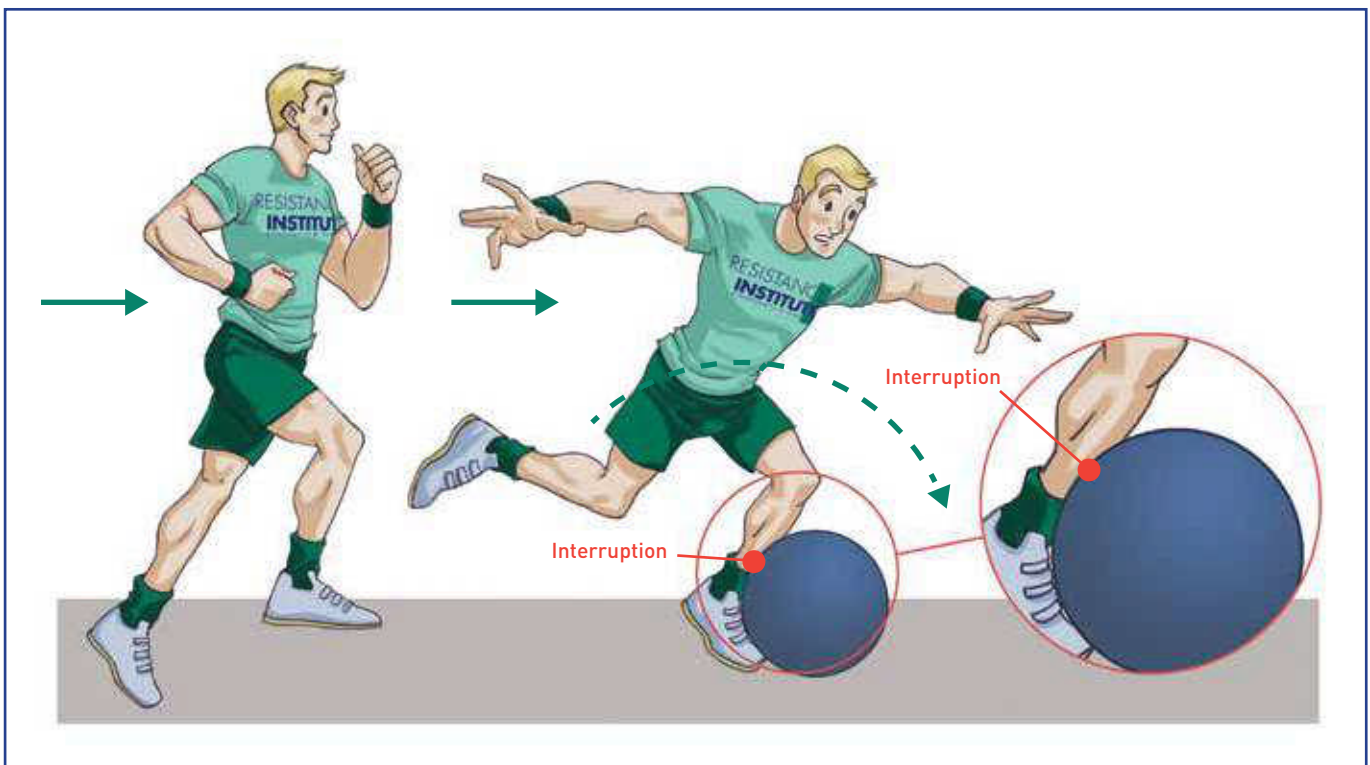
We will cover kinematics –which studies movement, while ignoring forces–, and kinetics –which studies forces relating to movement– and we will integrate both disciplines in order to apply them to real exercise situations.

3.1 CREATING A LEVER SYSTEM. LINKAGES

¿What is actually a lever system?

A lever system is simply the linkage of two or more structures, segments or bodies. These types of linkages have some special characteristics, which we'll go on to cover.

Lever systems occur whenever two structures collide and, in a way, the translational movement for one of them becomes interrupted. When this is the case, a **rotational movement** or potential movement takes place around that point of interruption.



Picture 3.1.1

Lever systems arise when a translational movement becomes interrupted and transforms into a rotational movement. The example above shows how an interruption comes about when the leg contacts a fixed blue object, causing a rotational movement around the interruption.

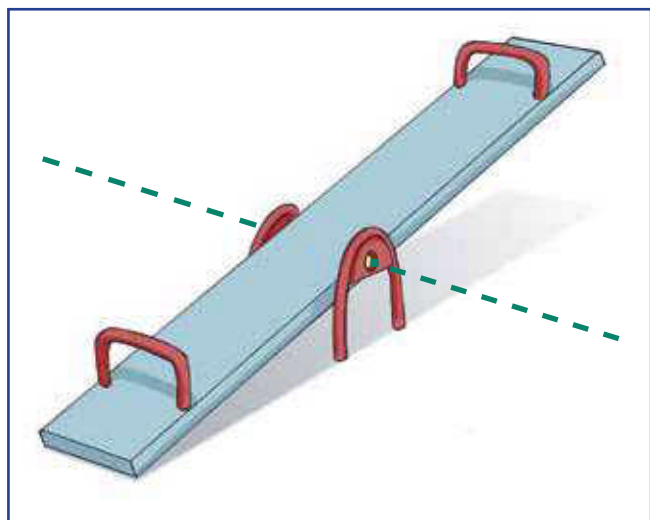
LEVER SYSTEMS

LINKAGES

Linkages are created by forces of interruption. There are many different ways an interruption can take place but, in the human body, they often result from a combination of several forces.

Lever systems in traditional machines usually show interruptions that are defined by geometry and by welding of its components. These pieces, which sometimes share screws or metal plates, form linkages that cannot be deformed and allow rotational movement to happen solely around its axes.

The above described case could be referred to as a **form linkage**.



Picture 3.1.2

The great majority of machines use “shape-derived” unions. In this seesaw, the board is united with its support by a transversal bar.

This way of looking at linkages constitutes a simplification which should facilitate learning and help determine the differences between engineering models and the human body.

Human structures do not generally have such bonded geometrical linkages, and the structures within the linkage are not welded. The bones in the skull resemble that kind of linkages, but most other interruptions in the human body show more complex mechanisms.

Wherever body segments come together, we can usually find some kind of uniting-tissue, known as articular connective tissue. This type of tissue has some viscoelastic properties, which make it different from most materials used in engineering. One of the things it can do, for instance, is allow for the interruption to not only occur at one fixed spot.

Many of the bone linkages in the locomotion system are controlled by the shape of articular surfaces. The way bones are structured in joints varies from joint to joint. But there is one other factor controlling the linkages, and that is the viscoelastic force, which is the force exerted by the articular connective tissue (ligaments, capsules, etc.)

Linkages in the human body, however, are not always capable of fully preserving integrity when acting on their own. The linkage—or ‘rigid linkage’, which is what engineers would probably call it—between segments could be put at stake (do not be misled by the term; the word ‘rigid’ here has nothing to do with tissue stiffness. This fact makes it necessary for the locomotion system to use some other type of tissue, that should fulfil two different and, apparently, contrary functions: to guarantee stability in the interruption and to allow for rotational movement around it. This tissue is, of course, the musculature.

The musculature in the human body serves two functions, then:

1. **Generating movement around the interruption.**
2. **Providing adequate force to maintain the interruption.**

Muscles exert force in two ways: one of them is the contractile ability of its sarcomeres and the other way is related to the viscoelastic properties of the muscle tissue itself.

Good stability (control) around the interruption and adequate mobility in the system are simultaneously achieved by those mechanisms.

Seesaws and machines, which are often used to study levers in the exercise industry are not exactly the same as lever systems inside the human body. These seesaws, when slightly modified can be useful in teaching and help understand several aspects of these systems, as will be shown later on in the text.

LINKAGE KINEMATICS

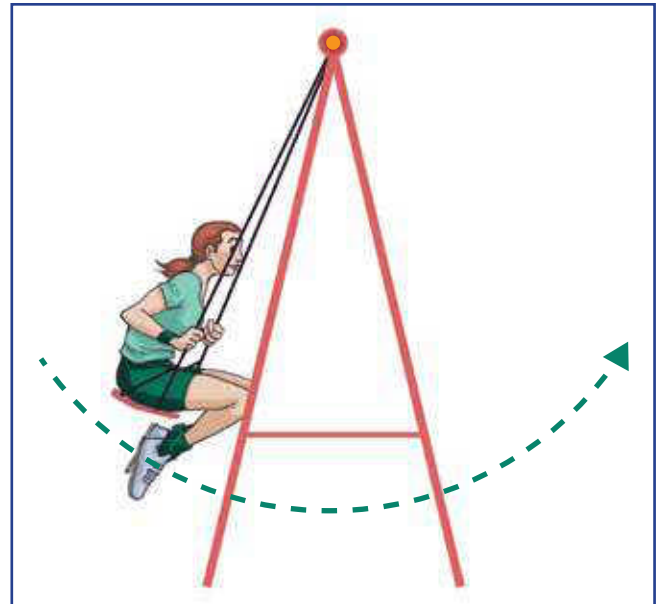
Kinematics studies movement, without taking into consideration the causes for it. The kinematics of a linkage, therefore, deals about the foundations of how movement takes place around a interruptions.

AXIS OF ROTATION (fulcrum)

The imaginary line where the interruption takes place is known as axis of rotation, fulcrum or rotational force axis. It is around this axis that rotational movement occurs. The axis is the only place where there is no movement, when rotation around the interruption. When observing the plan of movement, the axis is seen as a spot, as seen in picture 3.1.3.

Picture 3.1.3

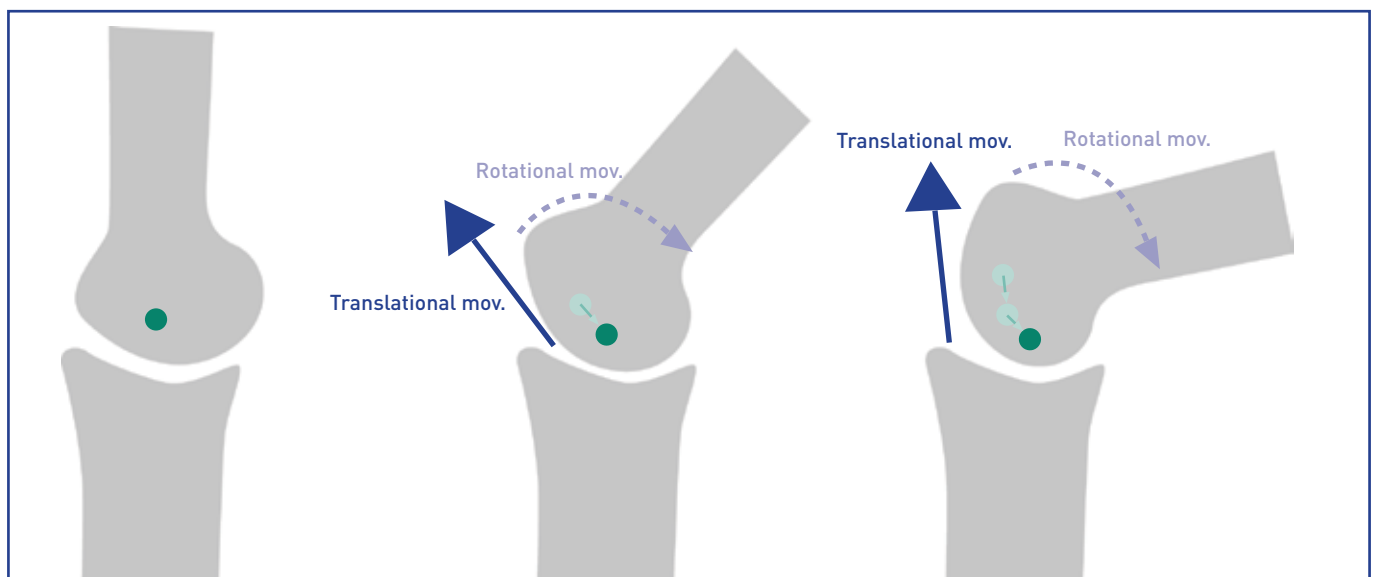
The dot shows the spot where the interruption is generating. Rotational movement takes place around it. In this picture, it is shown on the uppermost part of the swing.



INSTANT AXIS OF ROTATION/DISPLACEMENT

It is the axis that comes out during rotation/displacement at the interruption. Sometimes, the axis moves linearly during the movement, causing the axis to change places as the lever moves. Because of this, in order to analyze movement and forces on a given lever system, the axis of rotation/displacement has to be determined for a specific moment in time (as if a movie was stopped at a specific picture frame).

Whenever linear displacement is present, the resulting movement is not purely rotational. Instead, a curved movement is generated, which varies depending on the ratio between translational displacement of the interruption and rotation around it. Please refer to picture 3.1.4.



Picture 3.1.4

The picture shows how the movement in this structure is not purely rotational. If the axis of rotation did not displace, the structures would collide and not as much rotational movement could be achieved.

LEVER SYSTEMS

ROTATION ACTION. FIXED SEGMENT - MOBILE SEGMENT

A movement happening around an axis of rotation (interruption), within a lever system, can take several forms:

Movement (action) around an interruption can take place in two different ways, depending on the direction of rotation of one segment, in relation to the other.

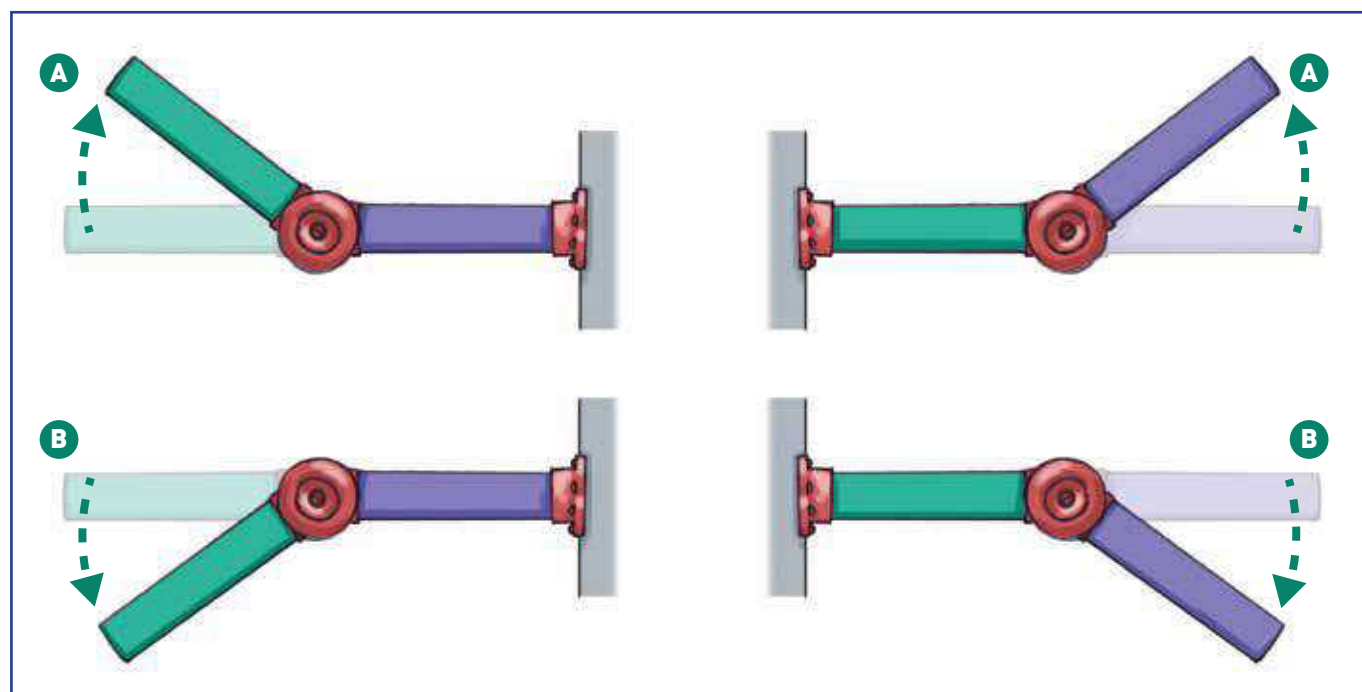
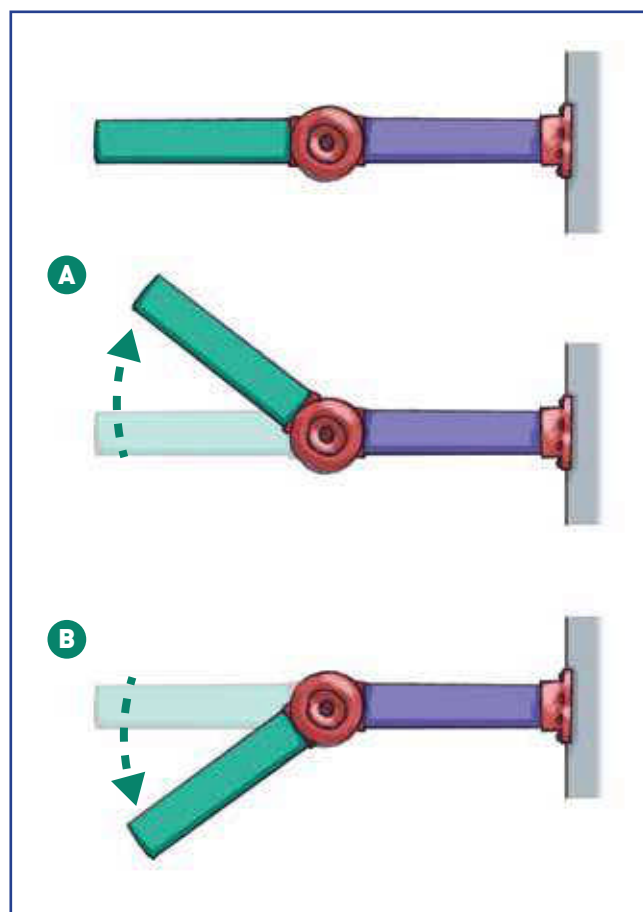
In Kinematics, the segment that stays fixed or still is called proximal segment, and the segment moving is known as distal segment.

Picture 3.1.5

This picture shows how the mobile segment (green) can move in two different directions of rotation, thus generating two different actions: A and B.

REVERSING THE DIRECTION OF ROTATION, FOR A GIVEN ACTION

Reversion takes place when the fixed point turns into the moving one, and the moving point becomes the fixed point thus, reversing the direction in which rotation occurs. Despite the change in direction, the action obtained remains the same. See picture 3.1.6..



Picture 3.1.6

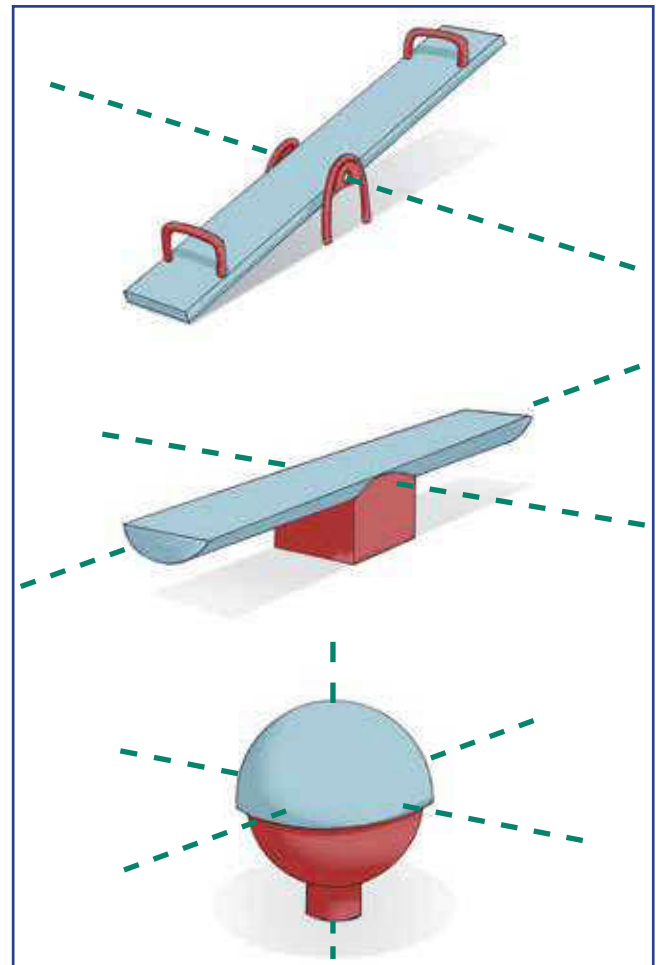
Action "A" has two directions of rotation, depending on which segment is fixed and which one is mobile (green or blue, each segment moves in a different rotational direction, although both situations lead to the same action, "A"). The same thing happens for action "B".

DEGREES OF FREEDOM IN A LEVER SYSTEM

Lever systems may have one, two or three degrees of freedom, depending on whether they allow moment in one, two or three dimensions in space. Picture 3.1.7 shows the degrees of freedom different structures can have, depending on the morphology of their interruptions.

Picture 3.1.7

Different structures, with different degrees of freedom can be seen. On top, there is a one-degree machine. The one in the middle is a two-degreed, and the picture onn the bottom has three degrees. The number of degrees of freedom a structure con move in is determined by the structure itself.



3.2 LINKAGES IN THE HUMAN BODY (ARTHROKINEMATICS)

Interruptions in the human body occur at its joints, and the field of study that deals with their movement is called arthrokinematics of Kinesiology. Kinesiology is the science that studies how the human body moves. There's no need to say that movement in the body works through lever systems (interruptions).

The terms Kinesiology and Biomechanics are commonly confused. The difference between them is that Kinesiology does not take forces into consideration and merely analyzes the movement within the joints, it being monoarticular or multiarticular.

Evidently, this science is closely related to Anatomy, and sometimes included in it, since a thorough knowledge of body structures will help determine a joint's functionality.

The type of movement that can take place within a joint depend particularly on:

- the shape of its contact surfaces.
- the control exerted by musculature.
- the passive stabilization provided by the different structures comprising the articular connective tissue, and most specially, by ligaments.

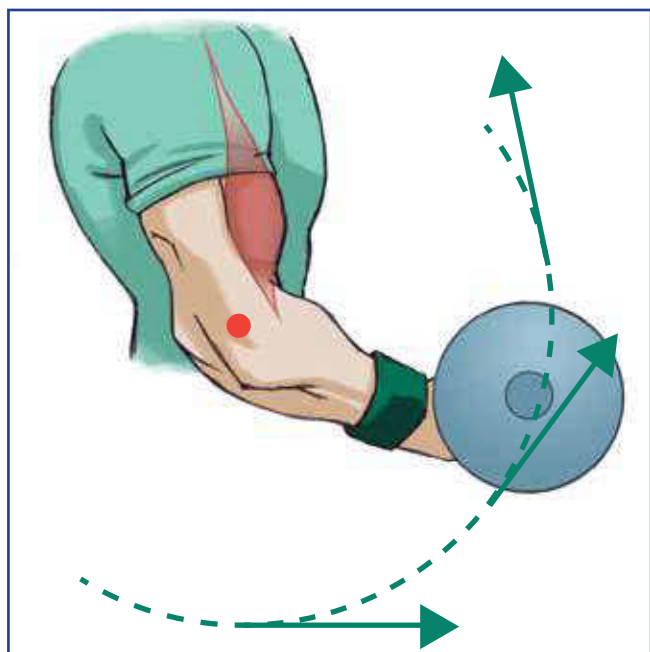
Some joint structures allow for a greater mobility than others. Each joint has its own structural (bony) geometry and a different amount, degree and type of connective tissue. For this reason, different joints provide with different degrees of freedom and articular ROM, but they all essentially depend on musculature to achieve control and maintain integrity in the joint (linkage).

LEVER SYSTEMS

ROTATIONAL MOVEMENT IN THE LINKAGES (lever systems)

Rotational movement talks about how an object moves around a fixed interruption, thus creating a fixed axis of rotation, and describing a circular motion.

The path that is created by the segment while rotating around the axis is called arc. This arc is described within the joint's plane of movement.



Picture 3.2.1

Rotation around the medio-lateral axis in the elbow joint causes the lever to generate an arch. On each of the points that make up the arch, the direction of the lever is tangent to it.

AXIS OF INTERRUPTION AND PLANE OF MOVEMENT

It is important to eventually learn to visualize the axis. The latter is always perpendicular to the plane of movement—or potential plane of movement, similarly to what happens in car tyres.

There is always an axis of force, even when movement doesn't occur. It is a spot, around which rotational movement would take place, if the amount of force were enough to cause movement.

The spin axis created by the interruption is the only spot which doesn't move, it is actually the interruption itself. Potential or actual rotational movement is created by unbalancing forces acting upon it.

LOCATION OF THE AXIS OF ROTATION

The axis of location is not necessarily located within the area of contact or interruption. The axis represents the imaginary line from which rotation is generated. If looked at from a two-dimensional point of view, and from the perpendicular plane, the axis should be seen as a dot. And the dot is located on the center of rotation or potential rotation of the lever.

It is the geometry of each linkage's structure—as well as all its tissues, and the forces keeping the linkage together—that will determine the exact location of the axis, at a given time. Whenever a linkage is created between a concave and a convex structure, the axis is always located on the convex side.

We should also keep in mind, that axes are virtual lines, and as such they do not need to be physically related to the structure. That is to say, if the lever actually moves, but not in a purely rotational manner (this happens frequently within the human joint system), the axis will change from one location of the structure to another, along the way. That same axis could even run along an empty area, without ever touching the actual physical structure of the lever.

PLEASE NOTE

It is important to understand the fact that an axis of rotation does not belong, nor is it linked to, a specific bony structure. The axis is just a virtual line which determines the center of rotation of a joint. It not being linked to a bone makes it complicated to determine an exact position for the axis, especially when evaluating articular motion in traditional anatomic models. When working with models, the axis needs to be virtually located upon a structure. It is then impossible to make the axis of rotation change places.

This limitation has led many professionals to believe that axes need necessarily be linked to a specific bony location.

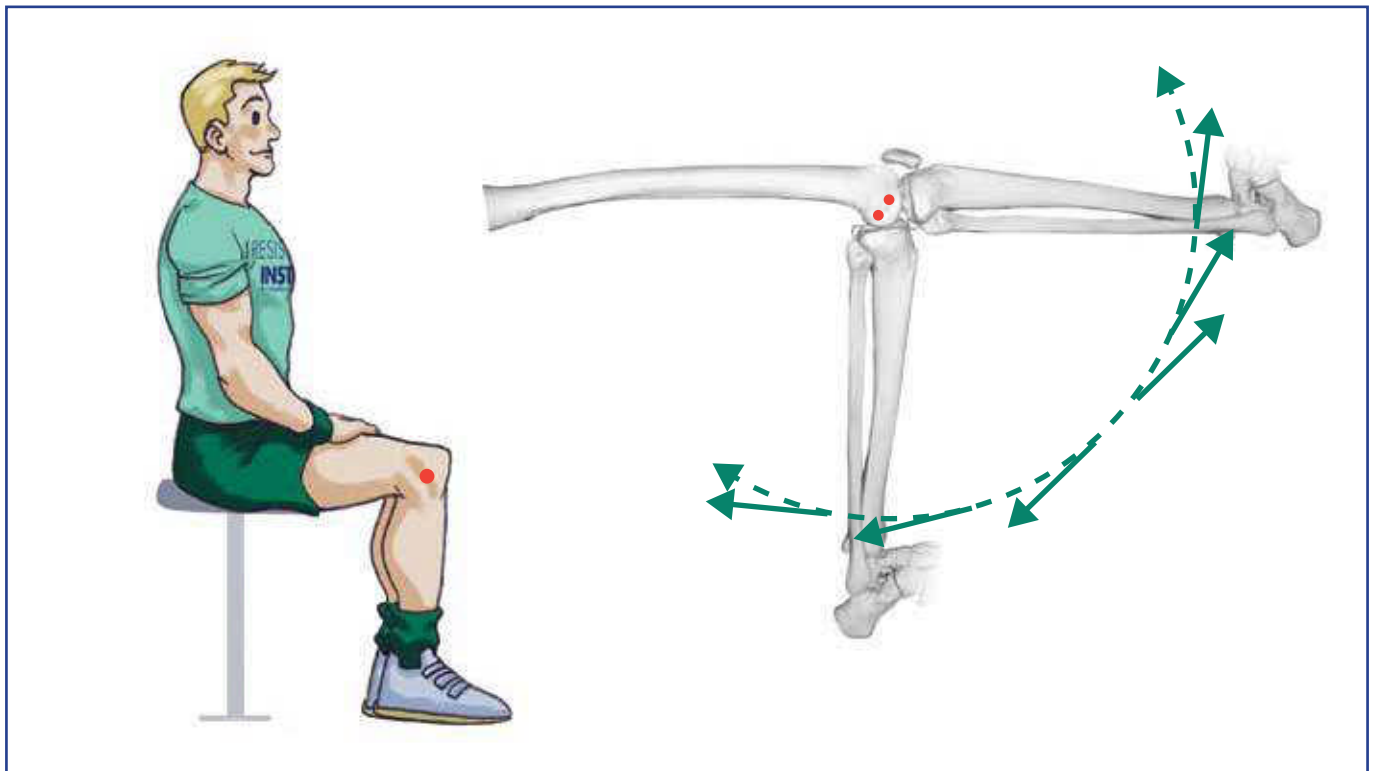
ARTHROKINEMATICS AND ROTATIONAL MOVEMENT (Movement of the linkages within human joints).

Human articular motion is essentially of a rotatory type. These movements occur as a consequence of there being linkages within bony structures (bones). Axes in the human body –unlike those in machinery– change location as the joint moves into different positions.

As mentioned before, even when there is no movement, there still is a so-called axis of force within the joint. We will go deeper into this concept later on, when we get to the kinetics of lever systems. Right now, we will stay focused on

plain movement analysis or kinematics.

The actual movement direction of a lever is tangent to the arc (perpendicular to the lever), at any given point. This arc is, in fact, comprised of an infinite number of tangents.



Picture 3.2.2

The lever (bones) creates an arch that is made up of an infinite number of tangents.

ROTATIONAL ACTIONS IN THE HUMAN JOINTS. FIXED AND MOVING SEGMENTS

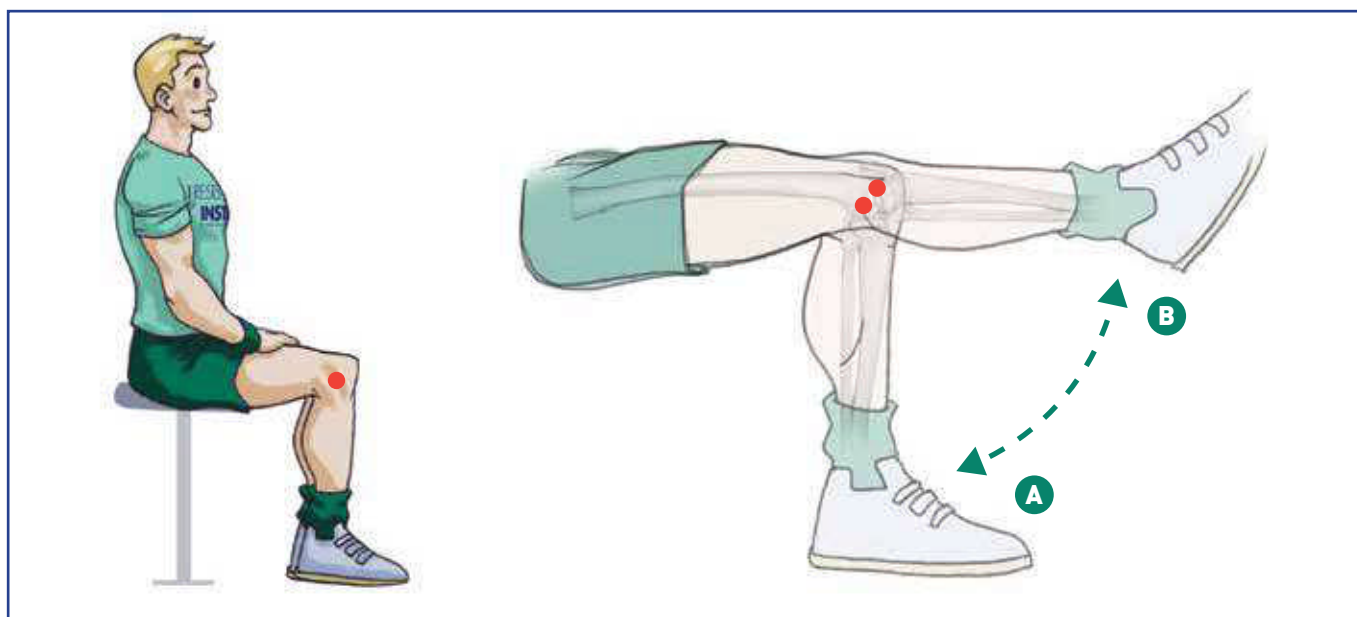
When talking about the human body, and similarly to what has been discussed previously, movement around a rotational axis can have many forms. When talking about kinematic analysis, the first thing to look into is the rotational action.

Two possible movements or actions can take place around a

given articular axis, depending on the direction of movement of the latter when one body segment moves upon another.

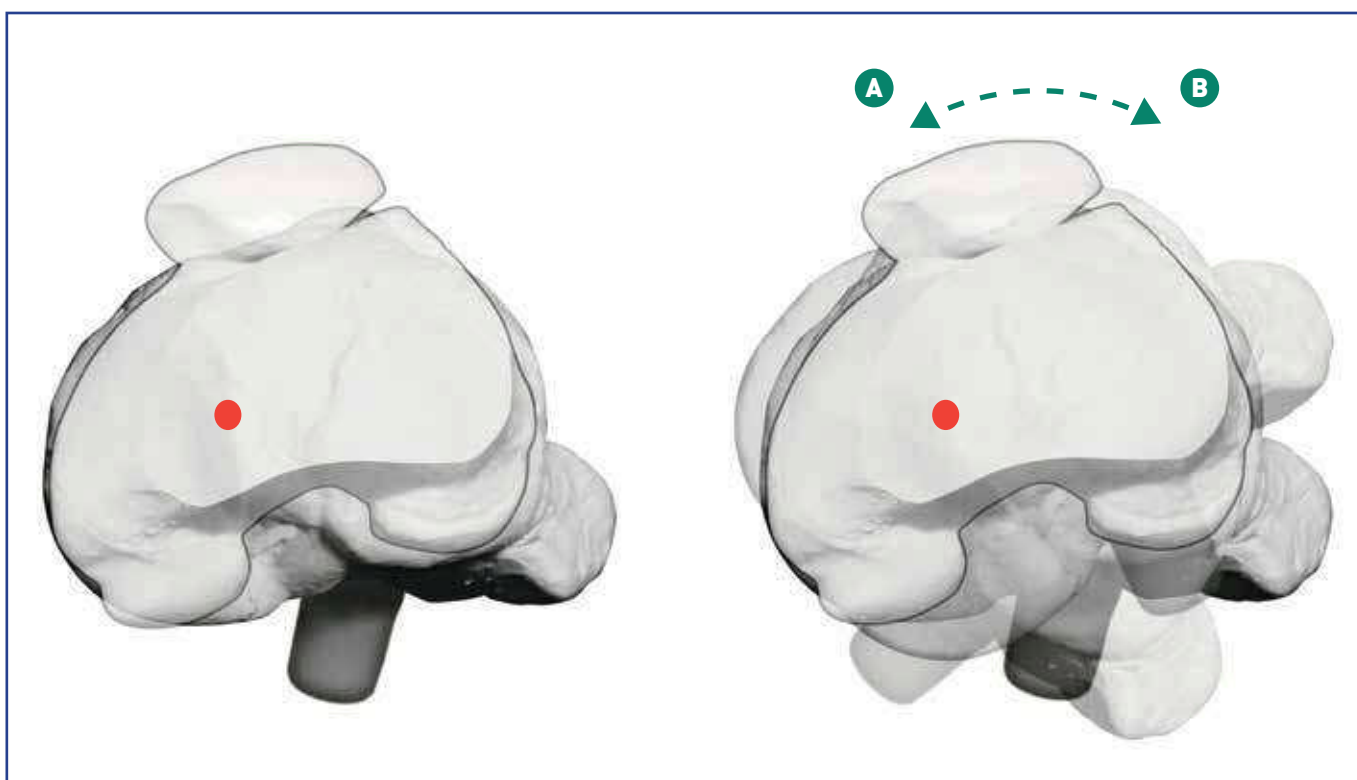
In arthrokinematics, we call proximal segment to the one staying still and we call distal segment to the one that changes position.

LEVER SYSTEMS



Picture 3.2.3

On the medio-lateral axis, the human knee is capable of doing two main actions: flexion (A) and extension (B). In this example, the femur plays the role of the fixed segment. This causes the direction of the rotation to go upwards (counter clockwise). Flexion would occur in the opposite direction (clockwise).



Picture 3.2.4

Eagle view of a femur (see-through) and tibia-fibula, as to the S-I axis (superoinferior) of the knee. If the femur was considered the fixed segment (proximal mechanical segment), two different directions of rotation could take place, and thus, there are two possible actions. Action "A" depicts a medial (internal) rotation of the knee. Action "B" shows a lateral rotation of the knee.

PLEASE NOTE

The placement and mutual relationship of articular surfaces needs necessarily be taken into consideration, since it is them that determine a given joint's motion. Some common misunderstandings should, therefore, be avoided, like those happening when a body segment moves and it gets mistaken with a specific articular action. For instance, the tibia moving medially does not imply there being a medial rotation of the knee. If this was to occur, the movement of the tibia would have to be specified in respect to that of the femur. For instance, take a femur moving in the same direction as the tibia (in our example, that would be medially), without there being any change between their articular surfaces. In the former example, the knee joint would have experienced no articular motion around its S-I axis. Not only that, but the knee could even be laterally (or externally) rotating, if the femur was to move further to medial.

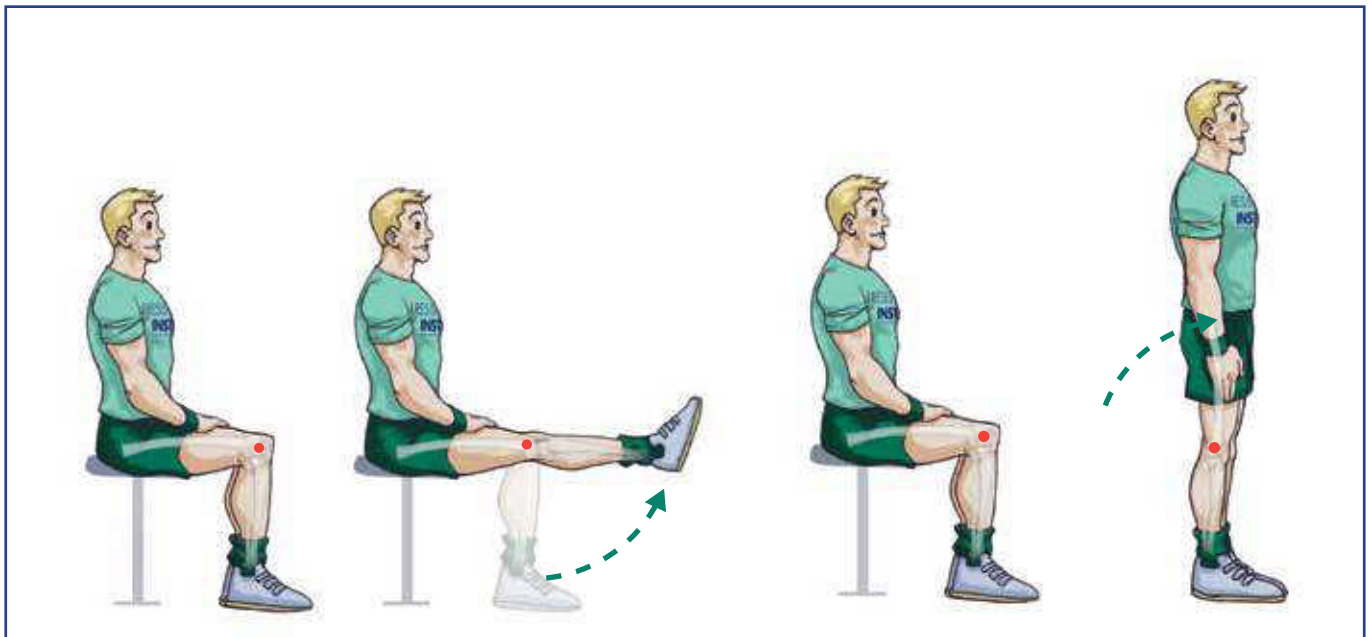
From the above, it becomes clear that it is not just segmentary movement that should be looked at in order to analyze articular motion. Instead, it is a must to gaze deeper into the action and determine how each articular surface exactly relates to its neighbouring one.

REVERSING ROTATIONAL DIRECTION FOR A GIVEN ACTION

For any given action, whenever the relationship between the fixed and the mobile part changes, the direction of rotation inverts.

As you can see from picture 3.2.5, the two directions of rotation depicted imply one single articular motion, which is known as knee extension. The actual difference lies in which

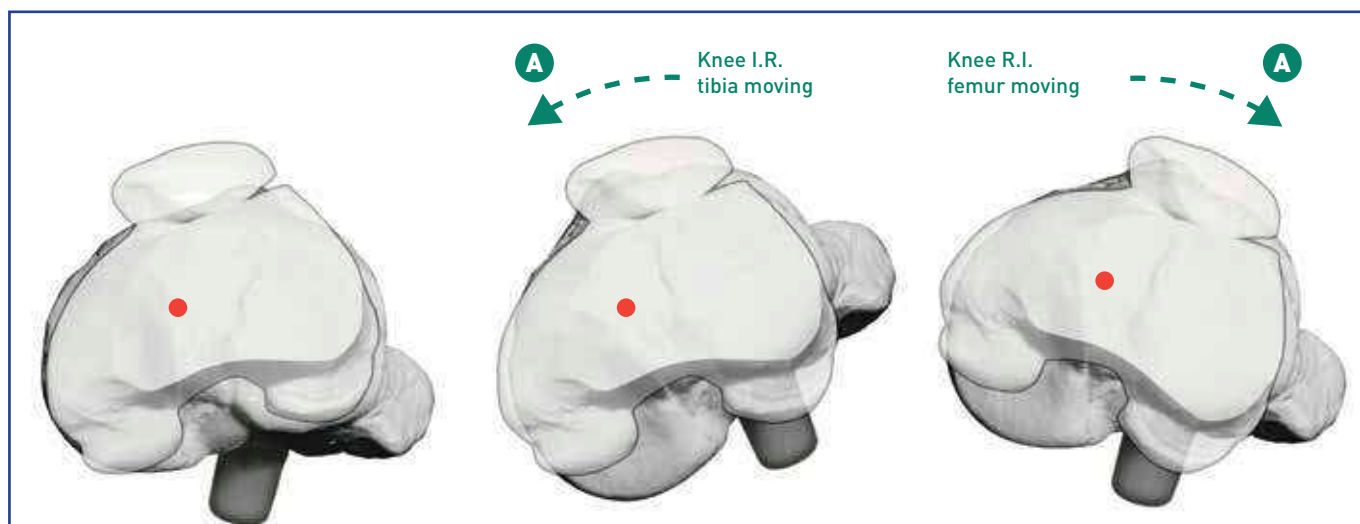
is the body segment acting as a fixed or a moving part. And from that, the direction of rotation also changes. If we wanted to be specific, we could call one of these actions 'knee extension on an inverted direction of rotation'. The term inverted is most commonly used when the anatomically proximal segment is the one moving, which in our example, would correspond to the second illustration.



Picture 3.2.5

The illustration shows how two different directions of rotation can lead to the same action: knee extension. When the character lifts the foot up, rotation around the mediolateral (L-M) axis of the knee takes place, in a counterclockwise direction. Opposite to that, when the character stands up from the chair, rotation around the mediolateral axis in the knee takes a clockwise direction (the femur moves in that direction). Summing up, the picture shows how two different directions of rotation, around one single axis, lead to the one single action (knee extension).

LEVER SYSTEMS



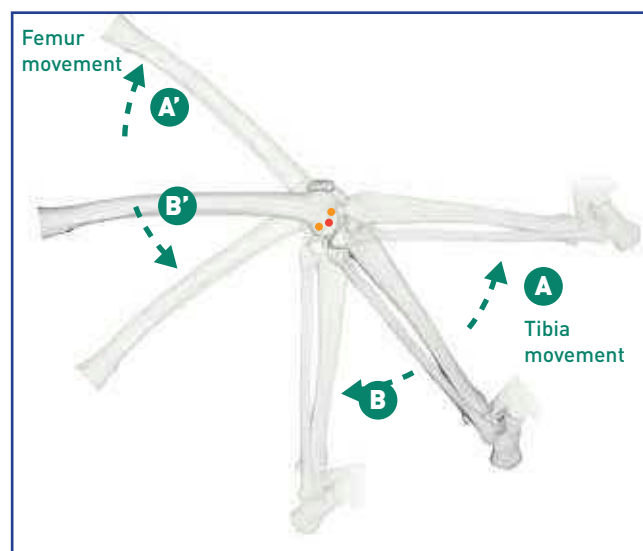
Picture 3.2.6

Eagle view of the femur (see through) and tibia-fibula on a superoinferior (SI) axis of the knee. When comparing the illustration in the middle with that on the left, it's the femur that acts as a fixed segment, being the tibia the moving segment. The knee's direction of rotation takes place on the transverse plane, in a counter clock-wise manner. When comparing the internal rotation depicted on the right side with that on the left side, the fixed segment would be the tibia, and the moving segment would be the femur. The knee's direction of rotation takes place on a transverse plane, in a clock-wise manner.

The picture shown above shows two movements that generate one single articular action: the medial or internal rotation of the knee. Please note that in reality, due to the geometry of this particular joint, the knee would not show this amount of ROM. However, the examples used are meant to help the reader better understand the ground principles of movement and do not aim to accurately describe any joint's real motion).

Picture 3.2.7 shows the four moves that could possibly take place around the depicted axis of rotation:

- Knee extension (A).
- Knee extension upon an inverted direction of rotation (this is usually referred to as reverse knee extension) (A').
- Knee flexion (B).
- Knee flexion on an inverted direction of rotation (commonly called reverse knee flexion) (B').



Picture 3.2.7

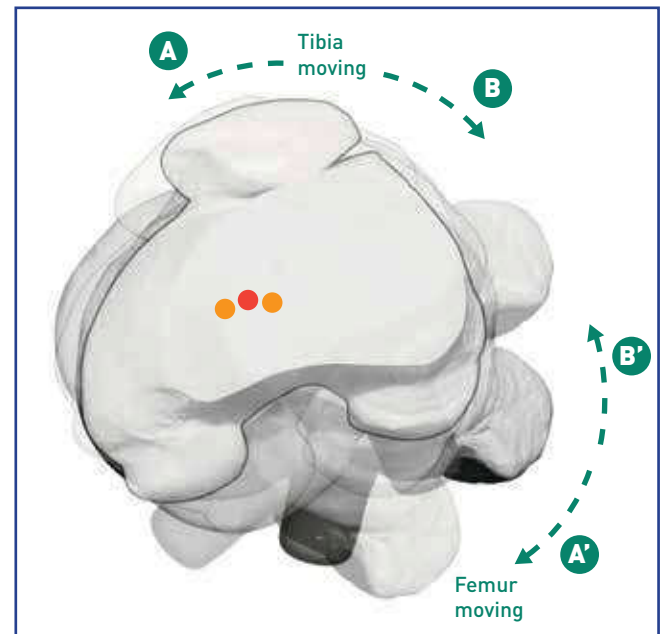
Two possible articular movements around the L-M axis of the knee are depicted. Two actions (flexion and extension) can be seen. Each of the two actions has two possible directions of rotation.

Following are the actions that could take place around the axis of rotation depicted on picture 3.2.8:

- Internal Rotation of the knee (A).
- Internal Rotation of the knee on an inverted direction of rotation (commonly known as reverse Internal Rotation of the knee) (A').
- External Rotation of the Knee (B).
- External Rotation of the knee on an inverted direction of rotation (B').

Picture 3.2.8.

All articular movement options for the knee, around its S-I axis are shown. Two actions can be seen (internal and external rotation), and each action has two possible directions of rotation.



AN EASY-TO-GRASP COMPARISON

In order to better understand the idea of inverted directions of rotation causing one sole action, we could use a water bottle as an example.

If you wanted to open the water bottle, you would have two possible options: to twist the cap in one direction (while keeping the bottle still or fixed); or to twist the bottle in the opposite direction (while keeping the cap still or fixed). Two different directions of rotation that would cause one sole action: opening the bottle.

Talking about the bottle still, there would yet be another possible action of rotation around that axis: that of closing the bottle. Closing the bottle would follow exactly the same rules, but the directions of rotation would be exactly the opposite to those used in order to open the bottle.

DEGREES OF MOVEMENT WITHIN A GIVEN JOINT

Human joints, and machines alike, form lever systems and thus use a three-dimensional space to move in and hold what is known as degrees of movement or degrees of freedom. The latter refer to the anatomical three-dimensional axes, a joint can move upon.

- **One degree of movement:** the joint can only move around one single axis and, consequently, on only one plane of movement. The talocrural joint would be an example of this.
- **Two degrees of movement:** movement can occur around two anatomical axes and, therefore, on two anatomic planes of movement. As a result, movement could also take place on all possible planes of movement derived from a combination of the former two. The joint made up by the humerus and the radius is an example of this.

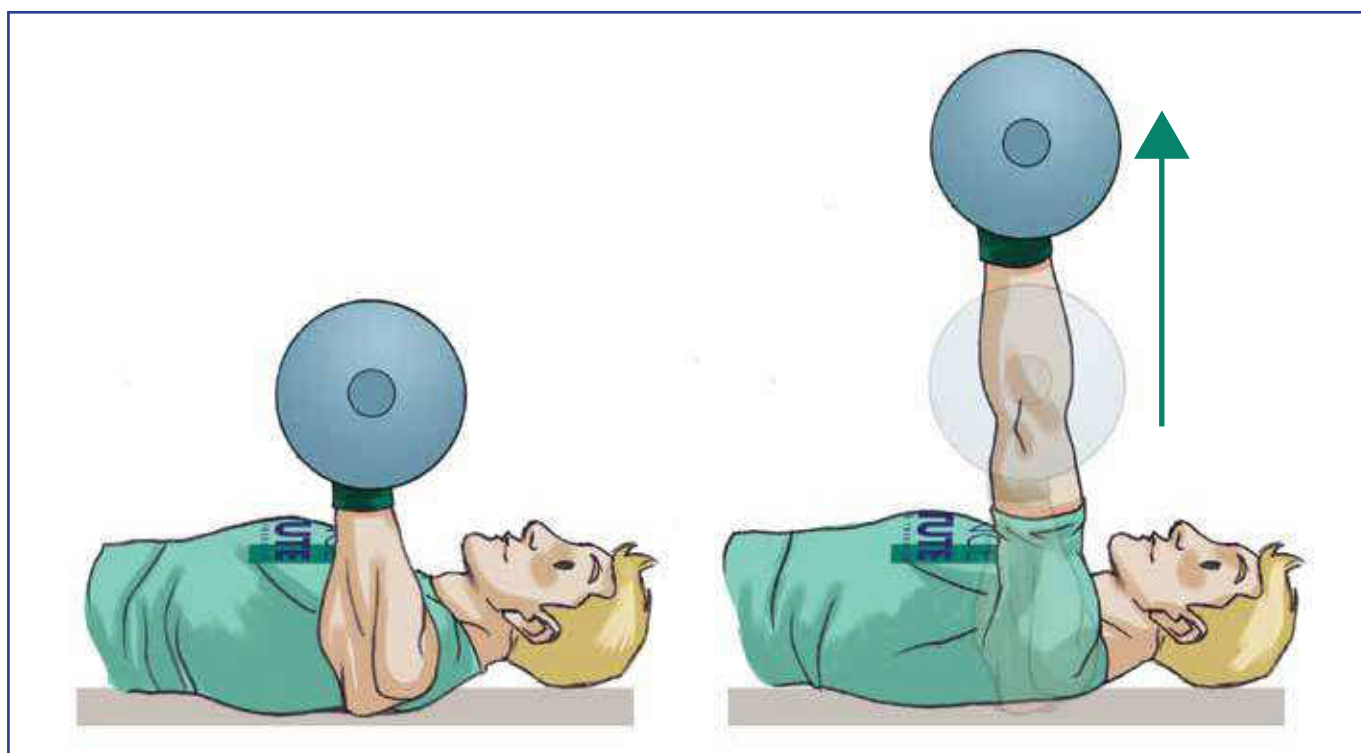
- **Three degrees of movement:** movement occurs throughout all three anatomical axes and thus, on all three anatomical planes. That is to say, it moves within the three dimensions of space. Ball and socket-type of joints (also called enarthroses), such as the glenohumeral or the coxofemoral joint allow for three degrees of movement (three-dimensional movement). This type of joint allows for very large movements, both for each of the anatomical planes, as for multi-planar moves.

LEVER SYSTEMS

TRANSLATIONAL MOVEMENT AND EXERCISE

Translational or linear movement is that in which the object travels along a straight line. Translational movement rarely takes place in a joint's physiological movement. However, since joints don't usually have a single, fixed axis of movement, linear movement could happen at some point throughout a full articular motion.

When looking at human movement as a whole, or at movement of its individual segments, translational movement can be perceived. However, this is actually the result of the combination of different rotational movements around several joints. We often see translational movement in some 'compound moves' with resistance, such as loading in a press or pull action.



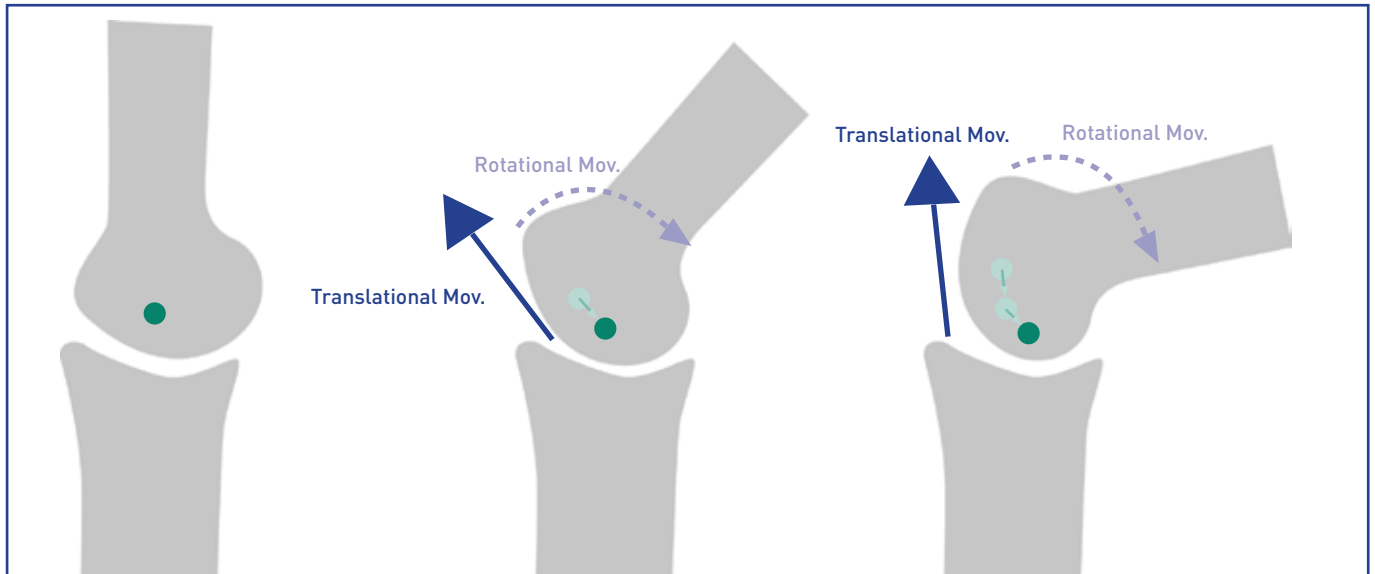
Picture 3.2.9

When looking at a press move, the viewer perceives a linear (translational) movement. This translational movement results from a combination of various rotational moves in the elbow and shoulder girdle.

INSTANT ROTATIONAL/GLIDING AXIS OR ARTICULAR CURVED MOVEMENT

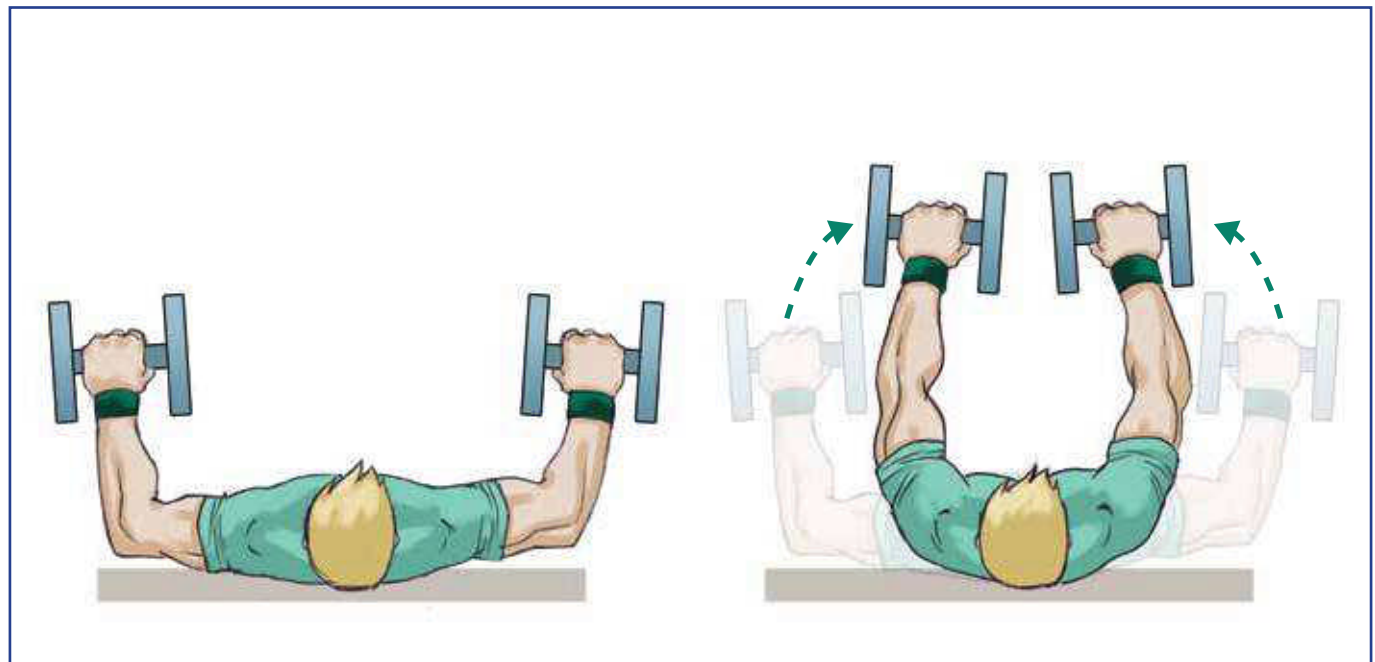
The combination of translational and rotational movement is called curved movement.

This type of movement takes place around a rotational axis that changes in place, causing the length of the radius to change, as well. This type of axis is known as instant axis of rotation/gliding and is kept under control by musculature, as well as passive structures.



Picture 3.2.10

Curved movement can be easily identified in this union, which could resemble a knee joint, on its L-M axis.



Picture 3.2.11

Curved movement in a benchpress. In the example above, the curved movement is due to the proportionally different movements within the elbow and the shoulder joint. It is that difference that generates an instant center of rotational and, thus, a kinematic gliding. These concepts are covered later on in the book.

LEVER SYSTEMS

JOINT PLAY WITHIN AN ARTICULAR LEVER SYSTEM

In order for a joint to be able to move normally, it needs to have some degree of joint play. Joint play is made of translational movements that take place within the joint. These movements cause deformation in the connective and muscular tissues surrounding the joint and helps in keeping the linkage rigid. The joint needs of this viscoelastic deformation in order to be able to produce physiological movements. The only way to isolate joint play is by manually generating passive translational movements of rotation and traction. Joint play movements cannot be demonstrating by active motion, even though normal physiologic movement is a proof of preserved joint play.

Joint play within normal articular motion is usually considered pretty much irrelevant. However, a lack (or an excess) of this movement could lead us to thinking that there might be some neuromuscular or articular issues (they are both interrelated) going on.

Should the structures surrounding the joint offer less stiffness than necessary, the joint could present excessive joint play and become unstable. On the other hand, excessive stiffness in the surrounding structures could restrict normal articular motion and structures could be subject to inappropriate forces.

Connective tissue (i.e.: ligaments, joint capsule, articular surfaces, meniscus, labra) is responsible for the passive stabilization of a joint. That is to say, it is responsible for the maintenance of an adequate relationship between the articular surfaces. Neuromuscular structures also play a very important role in the maintenance of articular integrity and stability, for they are able to alter the amount of stiffness within linkages.

It is equally important to note that the different factors influencing joint play highly vary between one individual and another.

AMOUNT OF MOVEMENT AND ARTICULAR ROM

The amount of rotational movement is expressed in degrees. Range of motion (ROM) refers to the extent of articular motion that is possible within a single plane and the anatomical limits of that articular structure.

ROM is usually considered pathological when it is greater or less than what would physiologically be expected. Whenever a joint's motion exceeds the physiological limits, it is called hypermobile. Similarly, whenever movement does not reach the anatomical physiological limits of that joint, it is called hypomobile.

Unfortunately, determining what is exactly the physiological range for each joint is not an easy task. Different individuals show different structures, which have been determined by genetics of phenotype (adaptations to forces), making it difficult to compare specific cases with normal ranges of motion.

In real life situations, comparing a joints range of motion with expected normal values does not proof very useful. Instead, you can compare a joint with its homonymous on the other side. In the next chapter, you will be able to further learn on articular asymmetries and potential muscular imbalances.

3.3 ARTHROKINEMATICS IN PHYSICAL EXERCISE AND HEALTH

When studying about mechanics, lever systems and exercise, kinesiology is evidently an important field of knowledge to take into consideration. Because of this, it is advisable to use precise terminology, so that all professionals who deal with human bodies can communicate with as much accuracy as possible.

Following is a list of terms that have been extracted from the mechanical field of articular linkages. The use of these terms may prove beneficial for all professionals of human health:

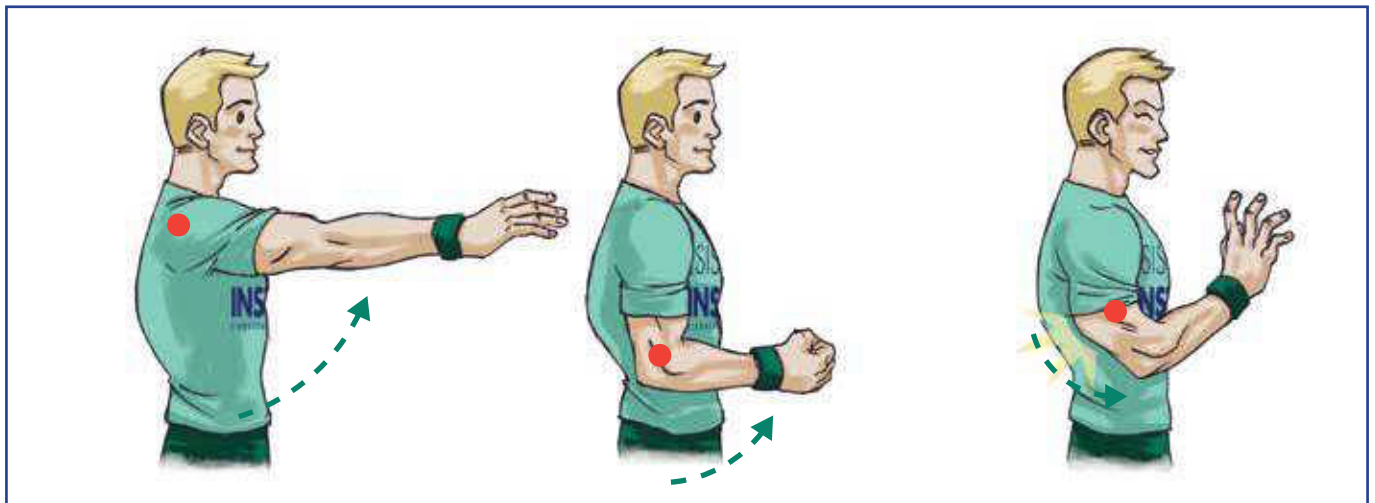
A- ARTICULAR MOVEMENT versus SEGMENTARY MOTION

Many professionals, and some literature as well, describe actions by mentioning how body segments move in space instead of talking about **how the joint is moving**. There are situations in which segmentary motion is relevant but it should be noted that it is not equal to articular motion. Following an example previously seen in this book, the tibia turning medially (internally) on the transverse plane, does not necessarily mean that the knee joint is internally rotating.

Bodily movements are generated in the joints. Anatomy and kinesiology dictate that, in order to speak about movement, one should focus on what is going on around the axes of articular movement.

Commonly used expressions, such as 'arm flexion' or 'leg extension' are often used instead of 'shoulder flexion' or 'hip extension'. This is already confusing in itself, but it becomes even more of a blur when other references use 'arm flexion' to talk about what goes on in the elbow joint, or 'leg extension' when referring to an action going on in the knee.

Being able to identify the plan and axis of movement for each one of the joints is important. That is why it is advisable to thoroughly comprehend how lever systems and articular structures work, for this is the only way to truly being able to understand human movement.



Picture 3.3.1

Movement of the musculoskeletal system is described within the joints. The drawing on the left depicts a shoulder flexion action. The man in the middle is doing an elbow flexion. Talking about 'arm flexion' should be avoided, as it would imply having to bend the humeral bone, in order to place an axis in that body segment (image on the right).

LEVER SYSTEMS

TO BE CONSIDERED:

In order to analyze body movements and muscle mechanics, it is articular movement and the relationship between linkage segments (joints) that should be considered. However, the ultimate purpose of articular movement remains always segmentary motion.

Being able to analyze human body movement is key. Behind any act of segmentary motion, there is always a specific articular movement that professionals should be able to understand and visualize.

There is a commonly used expression, namely 'rotation of the sacrum'. This usually refers to a situation where the lumbar spine goes on rotation (when compared to a neutral spine) while lying prone. What really happens is that the movement taking place in the spine, because of the prone position, causes the pelvis to move around a fixed thorax, which in turn makes it look as if the sacrum were rotated.

For a correct mechanical analysis, and thus –later on– for a better application of muscle work, the joints that are not moving as expected should be identified, since they could be the ones causing the imbalances in segmentary motion (for instance, the perceived lateral rotation of the sacrum).

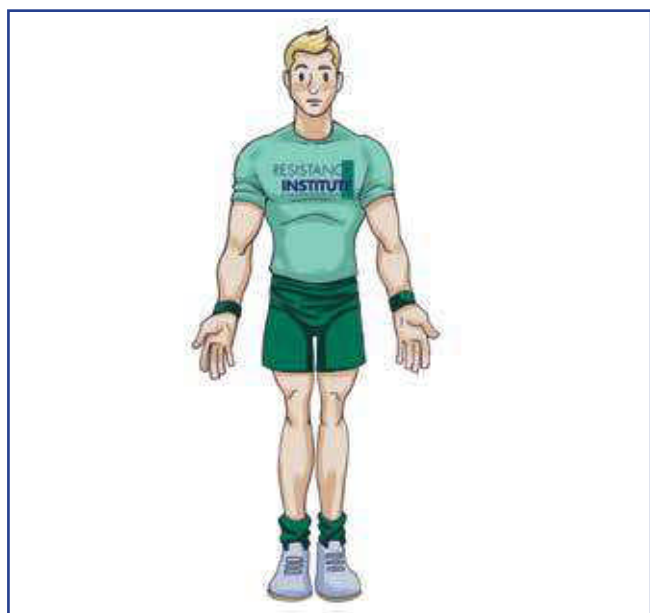
B- REFERENCE SYSTEMS- ANATOMIC AND FUNDAMENTAL POSITIONS

For several centuries now, the anatomic position has been the reference point. Generally speaking, it consists of the individual standing up, with their radioulnar joints (proximal and distal) in a supinated position.

Because the purpose of a reference point is to offer a stand point or starting point, the anatomic position (AP) is not quite an ideal one, since the radioulnar joints there do not start from 'cero'. The reason why it was agreed to use this position as a reference, was actually that it makes it possible

to associate the movement of the thumb with those of the torso.

The fundamental position simply reorients the (proximal and distal) radioulnar joint, bringing it to a neutral position, in which the palms of the hands face the thighs. By doing so, all joints are virtually brought to 'cero'. This is a more reasonable stand or reference point when talking about articular positions.



Picture 3.3.2

Anatomic position.



Picture 3.3.3

Fundamental position.

C- AXIS OR BODY PLANES-REFERENCED MOVEMENT

In this book, it is very important to make clear that all positions and movements will be egocentrically referenced.

EGOCENTRIC REFERENCES versus GEOCENTRIC REFERENCES

Geocentric references are those which we talk about a body, focusing on the geographical space that surrounds it. This means that, even if the body changed positions or moved, any representative spot (axis, plane, mark, etc) is referenced as a function of the space around it.

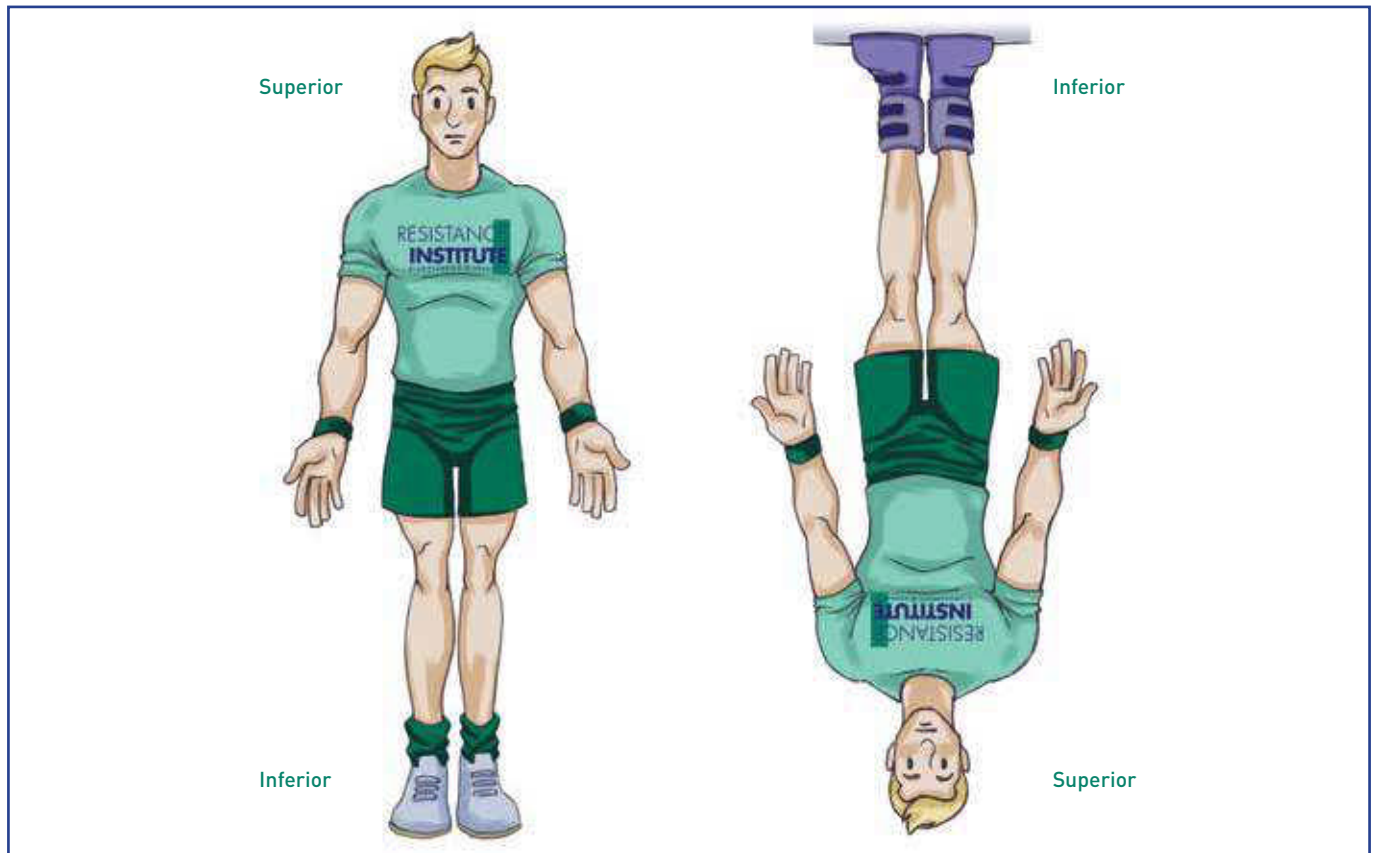
Egocentric references are those which are expressed in relationship with the human body. This means that, in spite of the body moving or changing positions, any axis, plane or bony landmark will be referenced as a function of the body itself. That is to say, joint movements are referenced against the body itself, independently from the physical space around it. By doing this, the existing relationship between body segments becomes highlighted, and their relationship with the ground, the wall or the bench not so.

ANATOMIC POSITION: EGOCENTRIC REFERENCED DIRECTIONS

References about whether something is going to be anterior, posterior, medial, superior, etc are taken from the anatomic position. Even when the body moves out of this position, the

references are still expressed from the anatomic position.

For example: if the head is superior to the feet, the head does not become inferior while on a handstand.



Picture 3.3.4

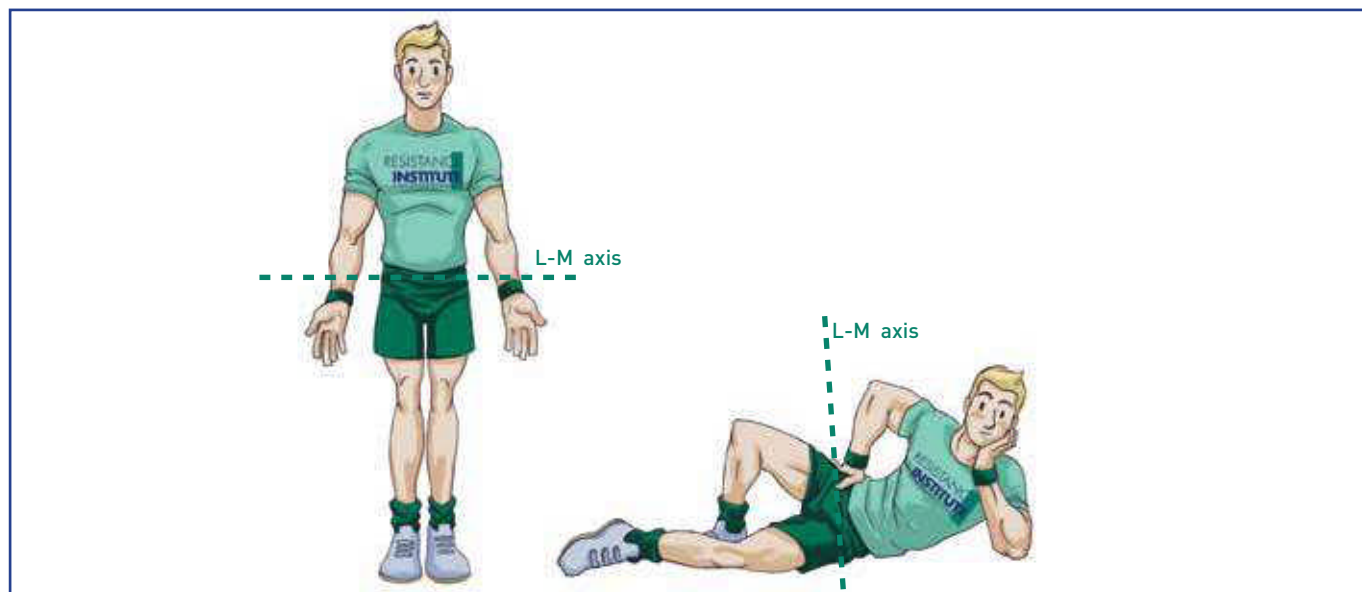
Anatomic references remain the same, no matter what the position of the body is in space.

LEVER SYSTEMS

ANATOMIC POSITION-REFERENCED ARTICULAR AXES

Information on articular axes is expressed according to the segmentary relationships of the anatomic position.

For example: Let's consider the anatomic position as the reference. If we then pick the medio-lateral axis of the hip joint, that imaginary line would still be the medio-lateral axis, no matter what the person is doing.

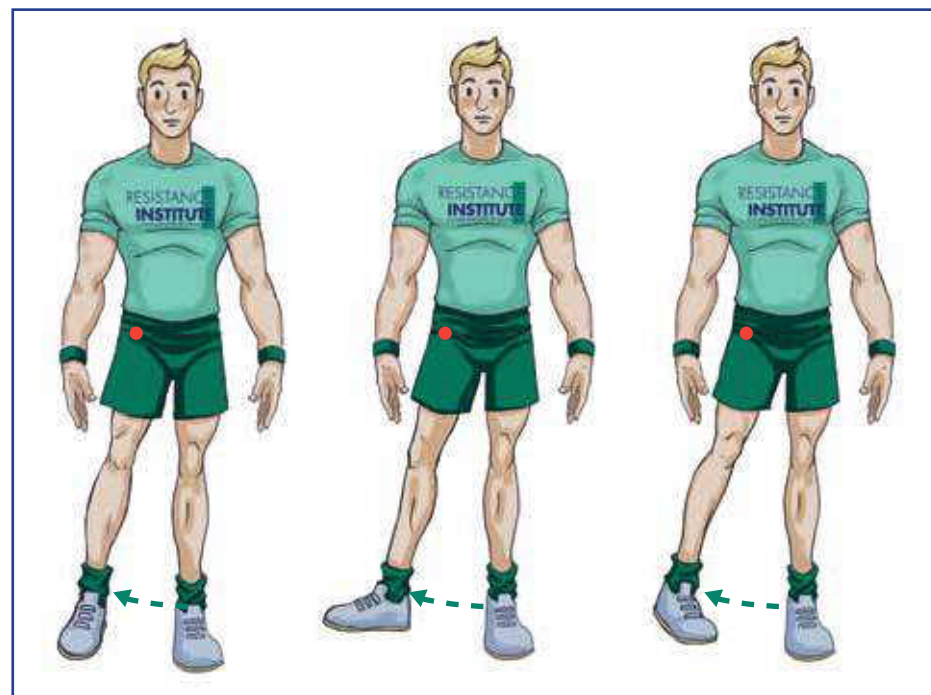


Picture 3.3.5

The picture above shows how the M-L axis does not change, in spite of the body's position in space.

PLEASE NOTE:

Consider for instance, the anterior-posterior (A-P) axis of the hip joint, with a reference on the pelvis (this is the body segment that is used as a reference when talking about the A-P axis from an anatomical position). Every time there is to be movement around the A-P axis, the action will be called abduction-adduction. It won't matter if the hip were positioned in different degrees of rotation, on a superior-inferior (S-I) axis. See picture 3.3.6.



Picture 3.3.6

The drawing in the middle depicts a man who is doing an abduction of the hip, with the hip in external rotation, but the A-P axis remains in the same place as that on the man on the left, because the pelvis remains the reference.

The same thing happens to the man on the right, who is doing an abduction with the hip internally rotated.

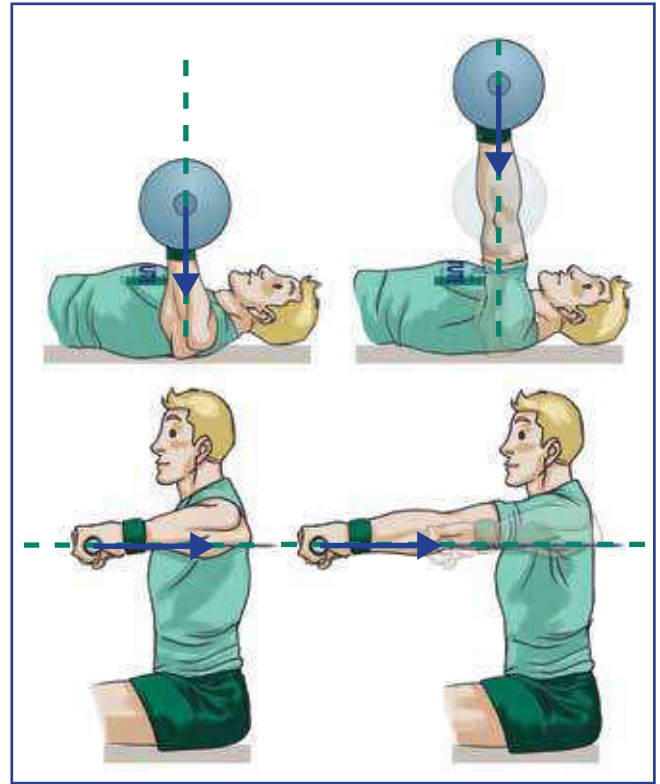
ARTICULAR POSITIONING AND MOVEMENT. USING THE FUNDAMENTAL POSITION AS A REFERENCE

The way joints are positioned, and the movements arising from that, should be egocentrically referenced. Additionally, what should be highlighted is its relationship to the fundamental position –which equals neutral– and not the anatomic position.

These references are certainly not a must, and referential systems can be modified although, by doing so, two issues arise. The first issue is that only one single referential system should be used throughout. The referential system should not be arbitrarily changed, for that lack of consistency would cause the information to be misleading. The second issue is that different institutions and professionals using different referential systems make it difficult for them to efficiently communicate.

The suggested referential system is not necessarily the best one, and there can certainly be different opinions about it. However, the Resistance Institute believes this one to be the simplest, and the one that makes the most sense, for professionals to use.

One obvious example of how to apply the egocentric referential system would be to look at an upper extremity pressing move, on the transverse plane. As far as the joints go, the movement will remain the same, whether the subject is lying on a bench, or sitting on a weight machine. Being clear about the referential systems is key to understanding the mechanics of any given move.



Picture 3.3.7

When doing a transverse press, joints move the same way even if the position of the body in space varies.

POSITIONS AND HUMAN MOVEMENT RELATE TO THE BODY ITSELF AND ARE INDEPENDENT FROM THE POSITION IN SPACE OR THE BODY'S RELATIONSHIP TO ITS SURROUNDINGS.

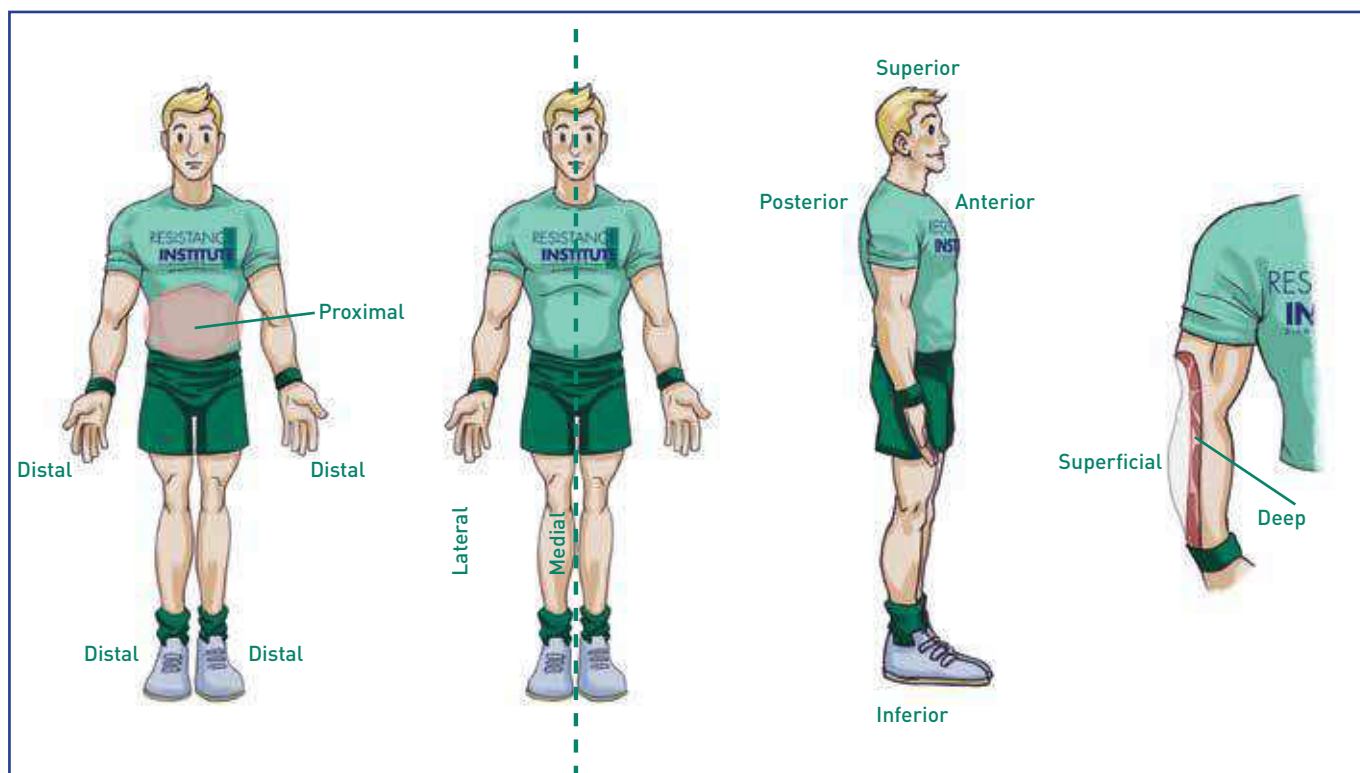
D- KINESIOLOGY: POSITIONS AND MOVEMENTS, NOT MUSCLES.

As was previously stated in this text, forces are not taken into consideration when using kinesiology. This implies that, when speaking only on kinesiological terms, articular positions or movements cannot be associated to muscle work. If this were done, important mistakes could arise, like for instance associating a specific articular movement with specific musculature being challenged. Because of this, it is essential to analyze all forces in the first place, since it is them that will determine the neuromuscular response.

LEVER SYSTEMS

3.4 ANATOMICAL DIRECTIONS

- ANTERIOR OR VENTRAL: It refers to the frontal part of the body. When taking about the relationship of one segment to another, it indicates that one of them is more to the front than the other.
- POSTERIOR OR DORSAL: It refers to the back part of the body. When speaking about two different parts, it indicates that one is more to the back than the other.
- SUPERIOR-CRANEAL-CEFALIC: It is a position that, relative to another, is closer to the head.
- INFERIOR-CAUDAL: It is a position that, relative to another one, is closer to the feet.
- MEDIAL: It indicates proximity to the mid-line (it can be referenced to the body or to a limb itself).
- LATERAL: It indicates that something is far from the mid-line (it can be referenced to the body or to a limb itself).
- PROXIMAL: It indicates proximity to the trunk. It is often used to express the relationship between two spots within a single joint.
- DISTAL: It indicates being distant to the trunk.
- BILATERAL: It refers to both sides.
- UNILATERAL: It refers to only one side.
- SUPERFICIAL: It indicates proximity to the surface.
- DEEP: Underneath what is superficial.



Picture 3.4.1

3.5 AXES AND PLANES OF MOVEMENT

Before we go on to define axes and planes, one should bear in mind:

What is a line?

A line is simply a sequence of points. A straight line is a sequence of points within one single dimension.

And what is an axis?

An axis is described as a line that travels through the center of movement or potential movement around an interruption. axes are, by definition, perpendicular to the plane of movement.

And finally, what is a plane?

A plane is formed by more than two unaligned points. Planes of movement are always perpendicular to the rotational axis. Body segments move within the plane that gets created from spinning around the axis.

Planes are often tangible objects, like for instance a think sheet of paper. However, when talking about the human body, they are hard to visualize, especially when unaccustomed.

LEVER SYSTEMS

ANATOMICAL AXES

There are infinite axes for movement within the human body. However, similarly to what happens with other references such as North/South/East/West, three axes have been chosen to serve as three-dimensional references in space. They are known as anatomical axes and include the following:

- **Superior-inferior axis (S-I).** Also known as **longitudinal or axial.**

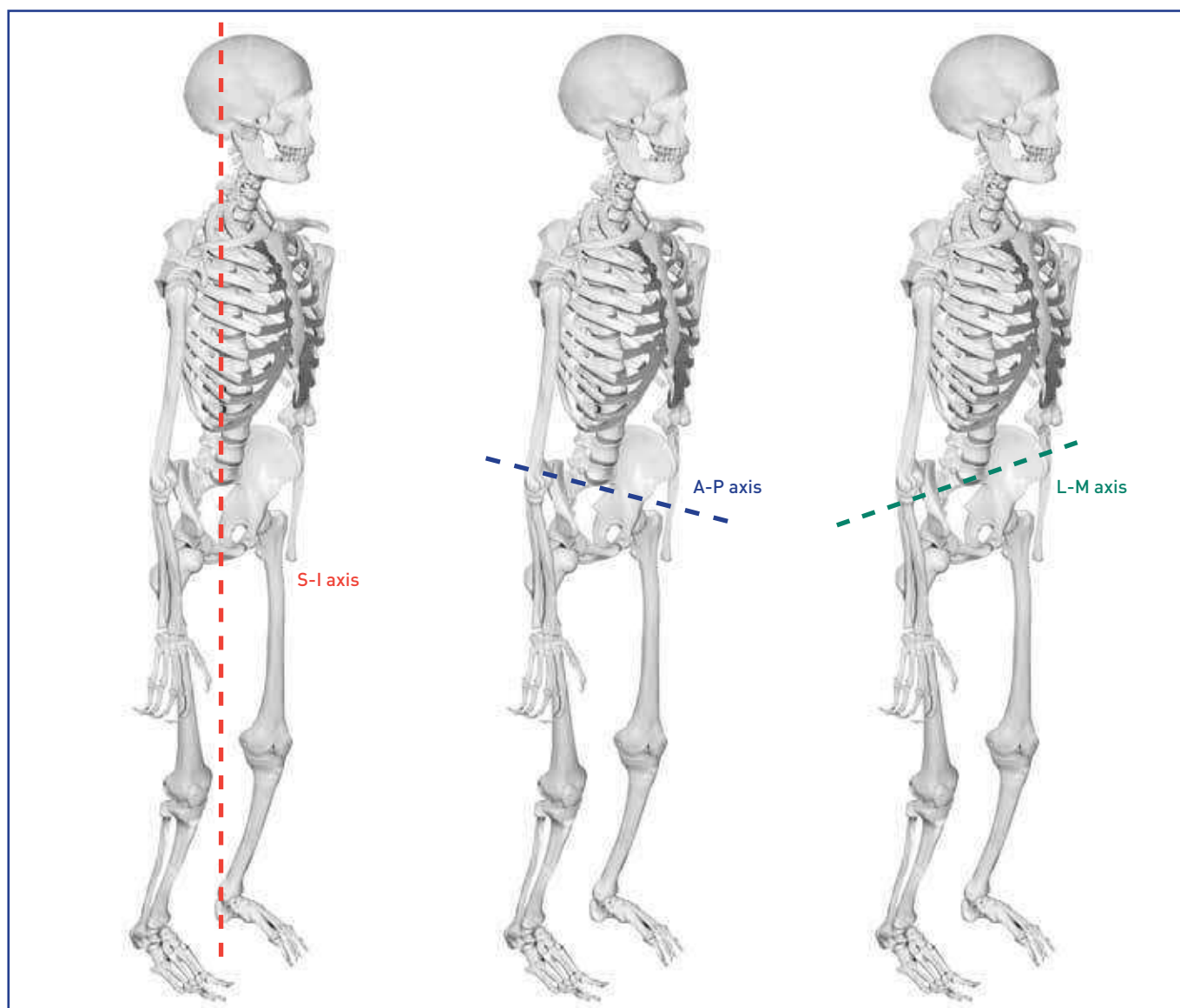
It is the axis that runs through the center of rotation, with a superior-inferior direction, with reference to the body. This axis is also known as the vertical axis. However, calling it that way could lead to misunderstandings, especially when the body is not in the anatomical position (i.e. when lying on the floor).

- **Anterior-posterior axis (A-P).** Also known as **sagittal axis.**

It is the axis that runs through the center of rotation, along an anterior-posterior direction, with reference to the body.

- **Medial-Lateral axis (M-L).** Also known as **transverse axis.**

It is the axis that runs through the center of rotation with a medial-lateral direction, with reference to the body. It is also frequently called horizontal, although this could be misleading when the body is not in the anatomic position (for instance, when lying on the floor)

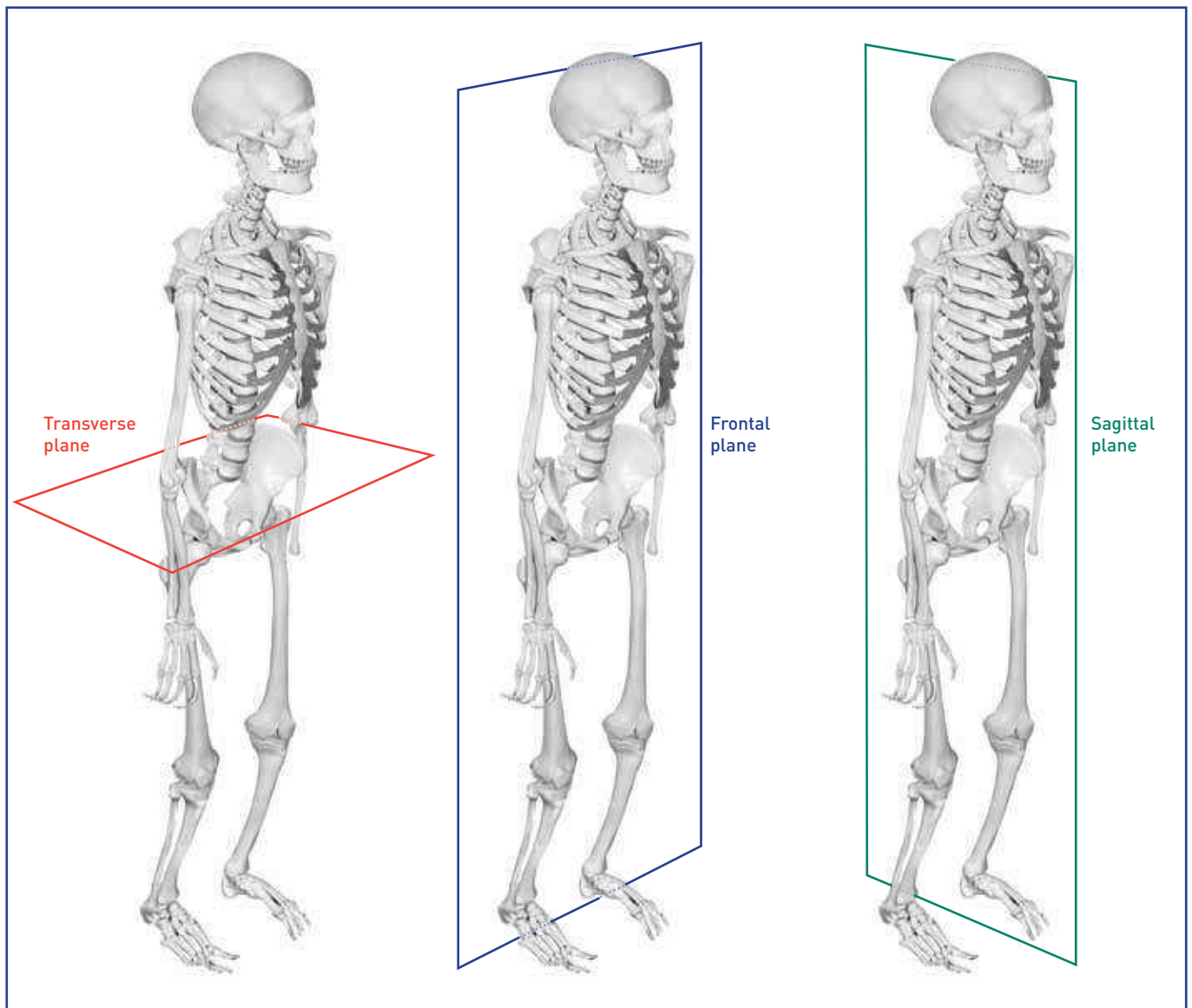


Picture 3.5.1

ANATOMICAL PLANES

Similarly to what happens with axes, the academic community has agreed on three anatomical planes, which relate to the three dimensions of space and human motion.

- **Transverse or horizontal plane:** it is usually described as that which divides the body in a superior and an inferior portion. The transverse plane is described when movement takes place around the S-I axis.
- **Frontal or coronal plane:** it is usually described as that which divides the body in an anterior and a posterior portion. The frontal plane is described when movement takes place around an A-P axis.
- **Sagittal plane:** it is commonly described as that which divides the body in a right and a left portion. The sagittal plane is described when movement takes place around a M-L axis.



Picture 3.5.2

LEVER SYSTEMS

In the medical field, as far as magnetic resonance images (MRI) and computerized axial tomographies (CT-scan) are concerned, the transverse plane is also called axial plane. Back in the days, these imaging procedures could only be done along the body's axial (S-I) axis, and therefore the name. Therefore, when reading the results of an MRI or CT-scan, what is written as axial plane should be understood as the transverse plane that was described previously in this text.

Even though every-day motion occurs around an infinite number of planes, or even with no specific plane (like what happens with circumduction), the three above-described

planes are known as the anatomical planes. They relate to the three dimensions of space, and their combination allows for all kinds of possibilities.

Planes are described through the joint and just as we saw earlier for the anatomic position, they are a mere reference to help describe motion.

Sometimes, when a given plane does not exactly match with the three established ones, it gets called oblique plane. Oblique planes are described by specifying how far (in degrees) they find themselves, in regard to the anatomical planes.

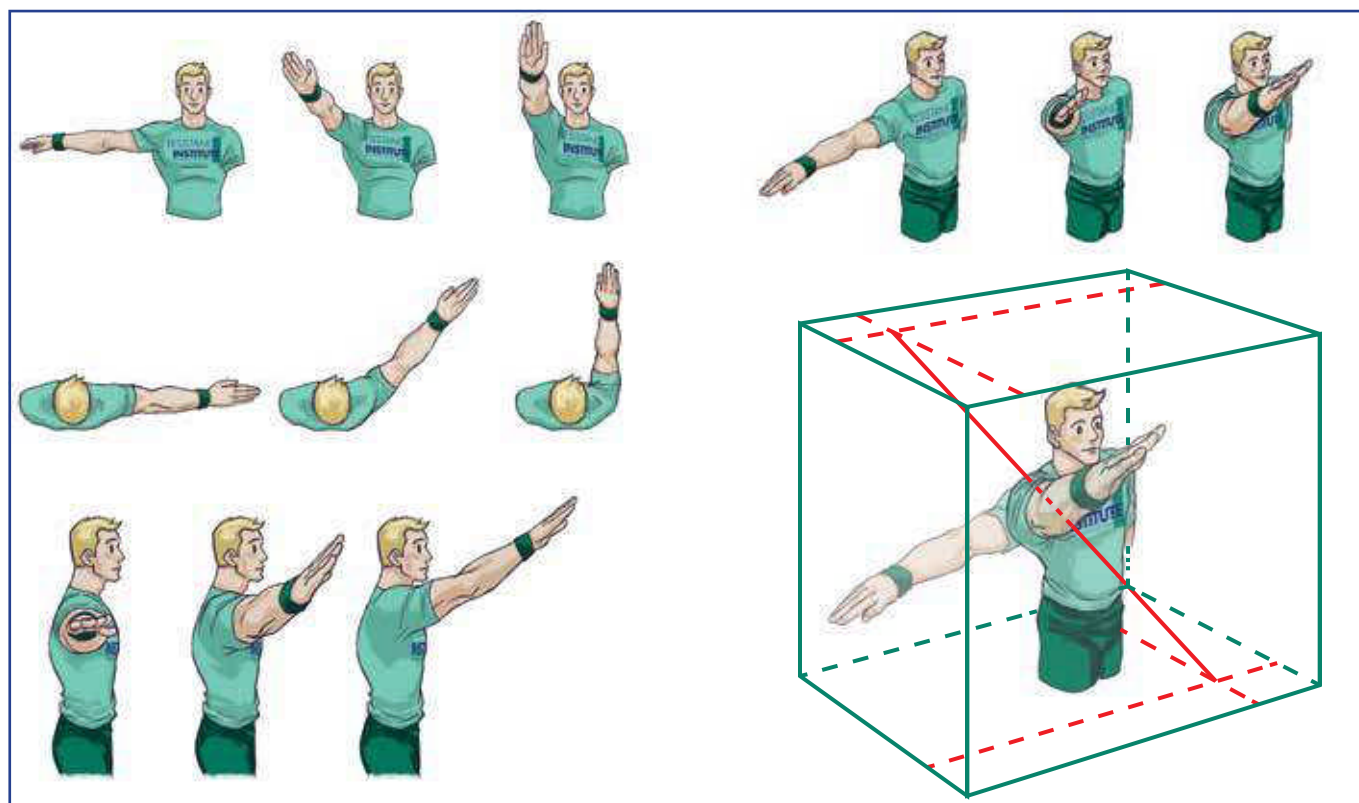
IT IS IMPORTANT TO BEAR IN MIND THAT PLANES ARE ALWAYS RELATIVE TO THE BODY, AND NOT TO THE SPACE THAT SURROUNDS IT.

MOVEMENT THROUGH AN OBLIQUE PLANE

People often mistake movements that happen on an oblique plane multiple plane movements. The fact that there may be an axis on an oblique plane, crossing two or more anatomical axes, does not mean there have to be two or more degrees of movement. See picture 3.5.3.

Whenever there is movement relative to an oblique axis, it is possible to have it decomposed, according to the extent of movement taking place on each of the anatomical planes that it crosses.

Kinematic decomposition of oblique movements can be carried on, by proportionally relating movement to each of the planes of movement.



Picture 3.5.3

The illustration shows one single movement, around an oblique axis, from different perspectives.

PLANE OF MOVEMENT IN THE BODY AND IN THE JOINT

In spite of it seeming simple to interpret the planes and axes of movement, the situation becomes a bit more confusing when trying to relate the movement of a limb to the body axes of movement.

As has already been mentioned, a joint's position and movement should be referenced and described according to the anatomic position.

The text above implies that a limb's plane and axis will match that of the body's as long as the limb is in the anatomic position (since that is the position from which joints' axes got referenced). However

WHAT HAPPENS WHEN THE LIMB CHANGES ITS POSITION RELATIVE TO THE TRUNK?

Following is a practical example, that will help to understand what goes on in such situations.

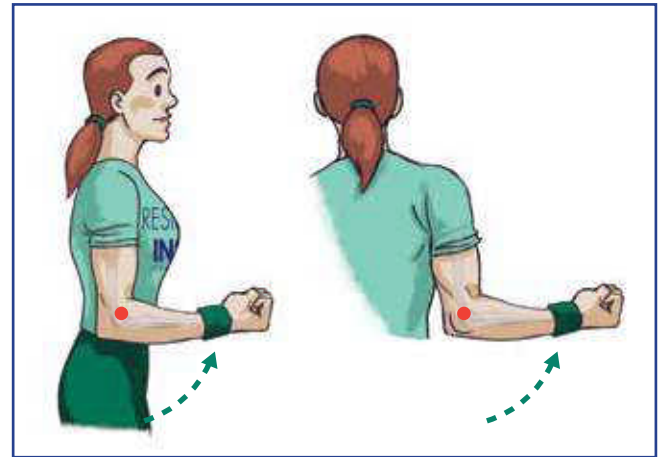
When the arm is held in the anatomic position, the elbow's flexion and extension movements occur in the sagittal plane. If the arm rotates by 90° (that is 90° of shoulder external rotation), many professionals would say the flexion and extension of the elbow now take place in the frontal plane. Many texts, however, describe flexion and extension as movements that solely happen in the sagittal plane.

When the shoulder externally rotates, references as to what is anterior or posterior, medial or lateral in the arm do not change. What has actually changed is the position of the arm relative to the trunk. The relationship of the different components of the joint remain the same, and so does the rotation axis of the joint, which remains M-L, relative to the arm. Taking this into consideration, it is advisable to think of the movement as happening in the arm's sagittal plane, mostly because its M-L axis has not changed.

The movement of the joints is described by they use from the anatomic position. And we already mentioned that movement takes place inside the joint itself and is independent from the space surrounding the body.

Looking at things in this way is important for several reasons. On the one hand, it helps clarify communication among professionals and identify what is really going on during a move. But most importantly, it is absolutely essential when

attempting to figure out movement that happens from a non-anatomic-position start-point. For these reasons, it is advisable to note that articular movements are basically a description of how joint surfaces change positions, in respect to one another.



Picture 3.5.4

The elbow always flexes in the sagittal plane (medio-lateral axis), even if it would be a different plane to the body (frontal plane; second drawing).

JOINT/ARTICULAR POSITIONS

Joint/articular positions describe the relationship between the joint surfaces in static.

One should take into account that a certain joint position does not necessarily throw information about the kind of movement that brought the joint into that positioning.

Here is an example: a person is holding their elbow in a 20° flexion position. This doesn't mean that they just flexed their elbow. They may as well have extended from a prior more-flexed position.

Bearing this in mind, we can conclude that in order to determine an articular/joint position, only two factors should be considered:

- The angle (how many degrees) relative to the axis or plane in the neutral position.
- The kinesiologic side—relative to that joint's anatomically neutral position— the joint has moved to (flexion vs extension, internal vs external rotation, etc.).

LEVER SYSTEMS

WHAT IS NEUTRAL POSITION?

The anatomical neutral position is the line that divides positions of different names. It is much more precise to describe positions by stating the distance (in degrees) which separates it from the neutral position, being neutral equal to 0°.

Negative degrees are used to indicate that the position goes further to the other kinesiological side, beyond neutral position.

Here is an example: One may say that the shoulder has 30° of extension. Or it could look at from another perspective: by stating that it has -30° of flexion.

The glenohumeral joint is said to hold a functional ROM from -30° to 110° of flexion. But one could also say its functional ROM ranges from -110° to 30°. Or it could even be put this way: its ROM goes from 30° of extension to

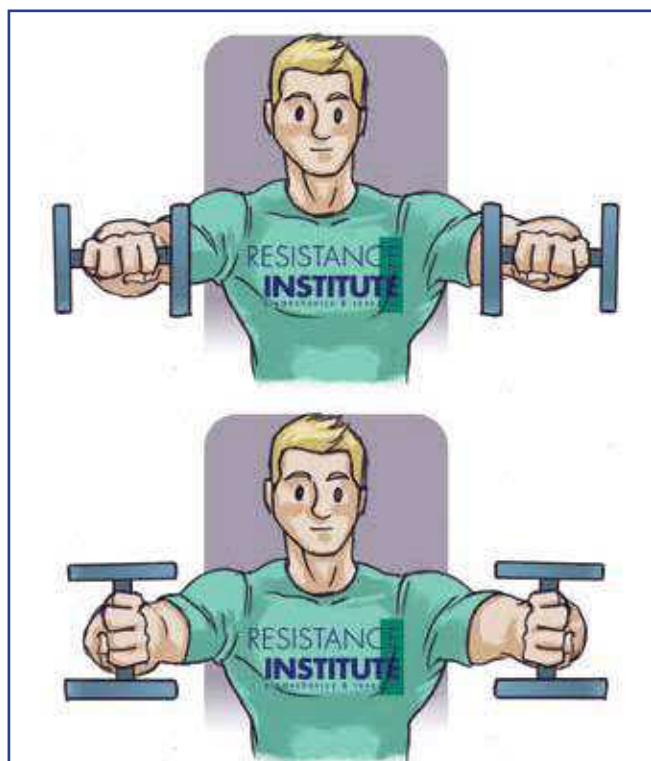
110° of flexion. The most common way to make this kind of statements is to refer it to the position which holds the most positive degrees of ROM. From our example, most people then would use the first alternative.

Do bear in mind that hearing about a '**neutral**' position, indicates a line which divides positions of different names and it does not take into consideration what moves were undertaken in order to achieve that position. Once again, positions should not be mistaken for movements, and viceversa.

In order to correctly describe a movement, one should speak about the direction within its plane of movement. This would make clear how the movement is taking place, whether it is happening to one or the other side of neutral position. If you take this into consideration, a flexing action can take place in a flexed or extended position, and it could as well go from an extended and into a flexed position.

JOINT POSITIONS AND MOVEMENTS: COMMON MYTHS

IS THE ATHLETE'S HAND PRONATED AS HE BENCH PRESSES?



Picture 3.5.5

Top drawing: dumbbell press with the radioulnar joint in a neutral position.

Bottom drawing: dumbbell press with pronation in the radioulnar joint.

Even though the previous section on pronation may sound obvious, it is frequent to make mistakes when trying to apply knowledge into real life situations. One can often read misleading statements talking about the positioning of the radioulnar joint, that is to say talking about pronation and supination (internal-external rotation) when the body changes position in space. This misunderstanding arises from the (wrong) believe that the above-mentioned joint is either pronated or supinated, when it is not. The joint has, as a matter of fact, maintained its neutral positioning, its articular surfaces of the radioulnar joint have not moved, and stay just the same they used to be in the fundamental position.

Bench pressing is one of the clear examples where one is often mistaken by considering that the radioulnar joint is in pronation.

Another very frequent mistake, also related to the example above, is thinking that the position of the radioulnar joint somehow depends on the rotational or flexing movements of the shoulder. And the same occurs the other way around. At this point, it may sound like understanding this is a trivial matter. However, it is essential to correctly lay the foundations. This will, later on, allow for a deep analysis of muscle participation and mechanics, as well as for the comprehension of how forces affect human connective tissue and joints.

AN EFFORT SHOULD BE MADE IN ORDER TO DETERMINE THE POSITION OF A JOINT BY HOW ITS SURFACES STAND AND NOT BY ITS RELATIONSHIP TO THE ENVIRONMENT OR OTHER BODY SEGMENTS

HIPEREXTENSION OF THE SPINE

Hyperextension of the spine is typically described as an extension movement that goes beyond the anatomic position (beyond neutral position), in the sagittal plane. Nevertheless, most people interpret the prefix **'hyper-'** as meaning 'excessive'. Therefore, it would seem more reasonable to speak about extension beyond neutral, at least when the ROM does respect the individual's functional abilities for that specific joint (we are talking spine here, but the same would apply for the shoulder, the hip or the knee joint).

From the above, it would seem more reasonable to only use the prefix **'hyper-'** when referring to actions which exceed an individual's limits.

Imagine you were doing an extension movement with your

spine. If you passed beyond the anatomic position and stopped, you would have created a position of extension, not hyperextension. Then, moving from the extended position towards the anatomic position is considered a **flexion** movement.

TO BE CONSIDERED:

If it were decided to call extension movements that went beyond neutral 'hyperextension', flexion movements that passed beyond neutral would have to be called 'hyperflexion'. Referential systems need to be used consistently throughout all possible situations.

For all of the above, caution is advised when using the term 'hyper-'. It should only be used for movements that go beyond the individual's physiological boundaries.

ROTATION AND ARTICULAR MOVEMENTS

All motion originates from a rotational movements around articular axes (we have covered that, already). That is a characteristic of the human musculoskeletal system.

Confusion may arise from using kinesiologic terms to only refer to rotation when it is happening around the S-I axis. However, all movements around an axis should be considered rotational movements, whether they take place around the S-I, A-P, or M-L axis.

Following is the correct nomenclature for movements. In the following pages, each term will be looked at in further detail:

- Flexion and extension are rotational movements around the M-L axis of the joint.
- Abduction and adduction are rotational movements around the A-P axis of the joint.
- Medial and lateral (internal and external) rotations are rotations around the S-I of the joint.

TO BE CONSIDERED:

The kinesiologic term rotation –which refers to movement around the S-I axis– should not be mistaken for articular rotation, which takes place during almost all human moves.

LEVER SYSTEMS

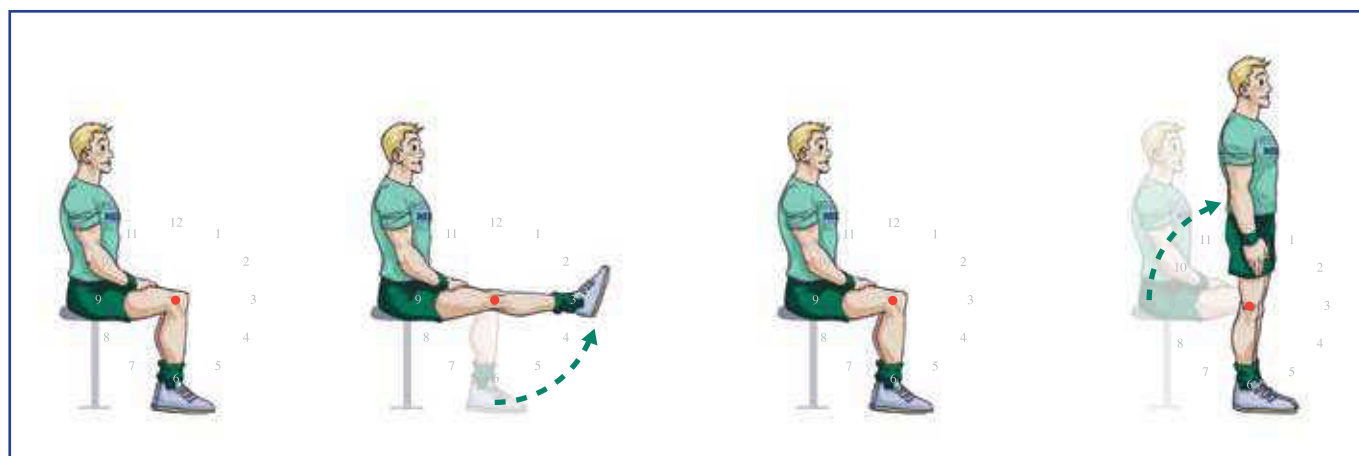
ARTICULAR MOVEMENTS IN THE SAGITTAL PLANE

FLEXION AND EXTENSION:

Flexion and extension are movements that take place around the M-L axis of joints.

Following are the necessary steps, one would need in order to define which specific action is taking place. First, the specific

joint to be evaluated needs to be determined. The joint should be visualized from the plane that is perpendicular to the axis, and relative to the side we are watching (example: sagittal view). Then, a direction of rotation –clockwise or counterclockwise– should be determined, relative to a point that stays still and a point that moves. As seen in previous chapters, two different directions of rotation are possible, for one same given action.



Picture 3.5.6

About knee extension: the direction of rotation in a knee extension differs, depending on which segment stays still and which other segment moves. In the first sequence, a counterclockwise direction can be seen. The second sequence shows a clockwise direction of movement.

ANOTHER CONFUSING ISSUE: ANKLE FLEXION AND EXTENSION

Ankle flexion and extension is yet another subject over which the specialised bibliography seems to have found no consensus, yet.

One can often read about professionals talking about ankle flexion, where other professionals would talk about ankle extension.

One possibility is this misunderstanding arises from trying to associate how a foot is made up, to the way a hand is made up, in the anatomic position, that is. In order to avoid mistakes, you can follow this simple hint: follow the movement of the joints in the toes and in the metatarsophalangeal joints, just as you do with the hand. Flexion of the joints in the toes and metatarsals follow the same direction as flexion in the ankle. This brings us to the conclusion that the direction of rotation that would imply flexion of the interphalangeal joints in the foot, is the same as the direction for flexion of the ankle.

WHAT EXACTLY ARE DORSAL FLEXION AND PLANTAR FLEXION?

The medicine and physiotherapy community often use the term dorsiflexion (bringing the dorsal or upper side of the foot toward the leg) and plantar flexion (bringing the sole or plantar side of the foot toward the leg).

The former are very commonly spread terms, especially in medical and rehabilitation bibliography, and thus have been accepted as scientific terminology. But in fact:

- **DORSAL FLEXION** equals ankle extension.
- **PLANTAR FLEXION** equals ankle flexion.

ARTICULAR MOVEMENTS IN THE FRONTAL PLANE

ABDUCTION AND ADDUCTION:

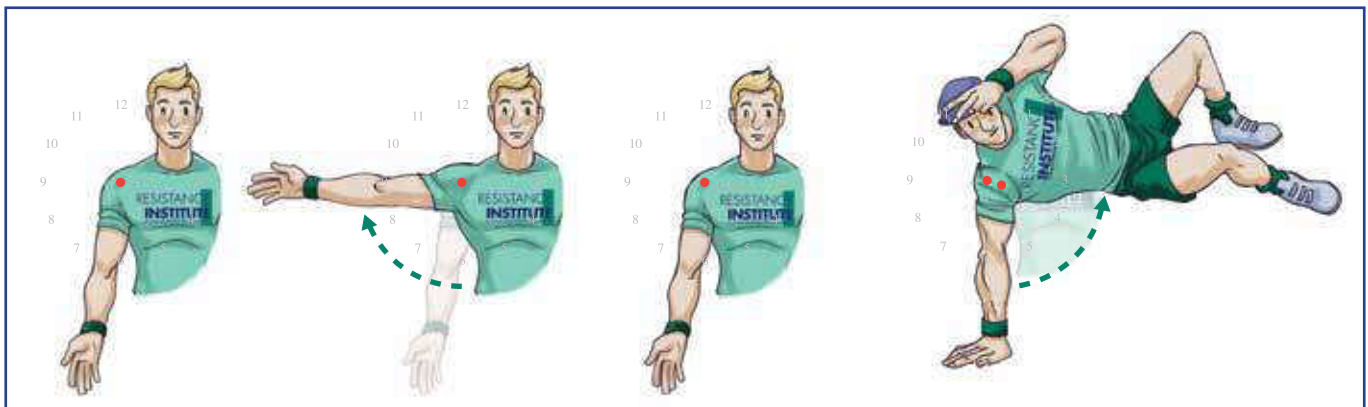
Abduction and adduction is how rotational movements around the A-P axis of joints are called.

In bibliography, it is common to read how abduction and adduction are defined as movements that separate a body segment (ABD) or brings it closer (ADD) to the mid-line. But let's take a look at how the upper extremity moves. One can easily observe that, when moving beyond 90° of shoulder abduction, the limb begins to get closer to the midline until the arm reaches the head. And this voids the initial definition.

LATERAL FLEXION:

Lateral flexion is the movement that the vertebral spine describes around the axis. It is described as lateral flexion to the right or left, depending on the direction the body is moving in.

In order to determine actions such as abduction or adduction, right or left lateral flexion, one should: be specific on the joint being evaluated, watch it from a plane that is perpendicular to the axis (frontal view) and determine in which direction movement occurs (clockwise or not), relative to a point that stays still and one that moves.



Picture 3.5.7

Both sequences represent one single action (shoulder abduction), but they show different directions of rotation.

ARTICULAR MOVEMENTS IN THE TRANSVERSE PLANE

MEDIAL (INTERNAL) ROTATION AND LATERAL (EXTERNAL) ROTATION

These are the rotational movements that take place around the S-I axis in the joints.

It was pin-pointed earlier, that because kinesiology only considers as rotational movements those happening around this axis, people are often misled. In reality, all movements are rotational movements.

There is one other important aspect that should be carefully considered. It is easy to make mistakes if, when the hip or the shoulder do not start out from the anatomic position, one looks at the movement being described at the distal end of the limb. For instance, take an internal rotation of the hip, with the knee flexed. If we pay attention to the foot, we will see it moving outwardly, and we could easily be misled and think we are watching an external rotation of the hip,

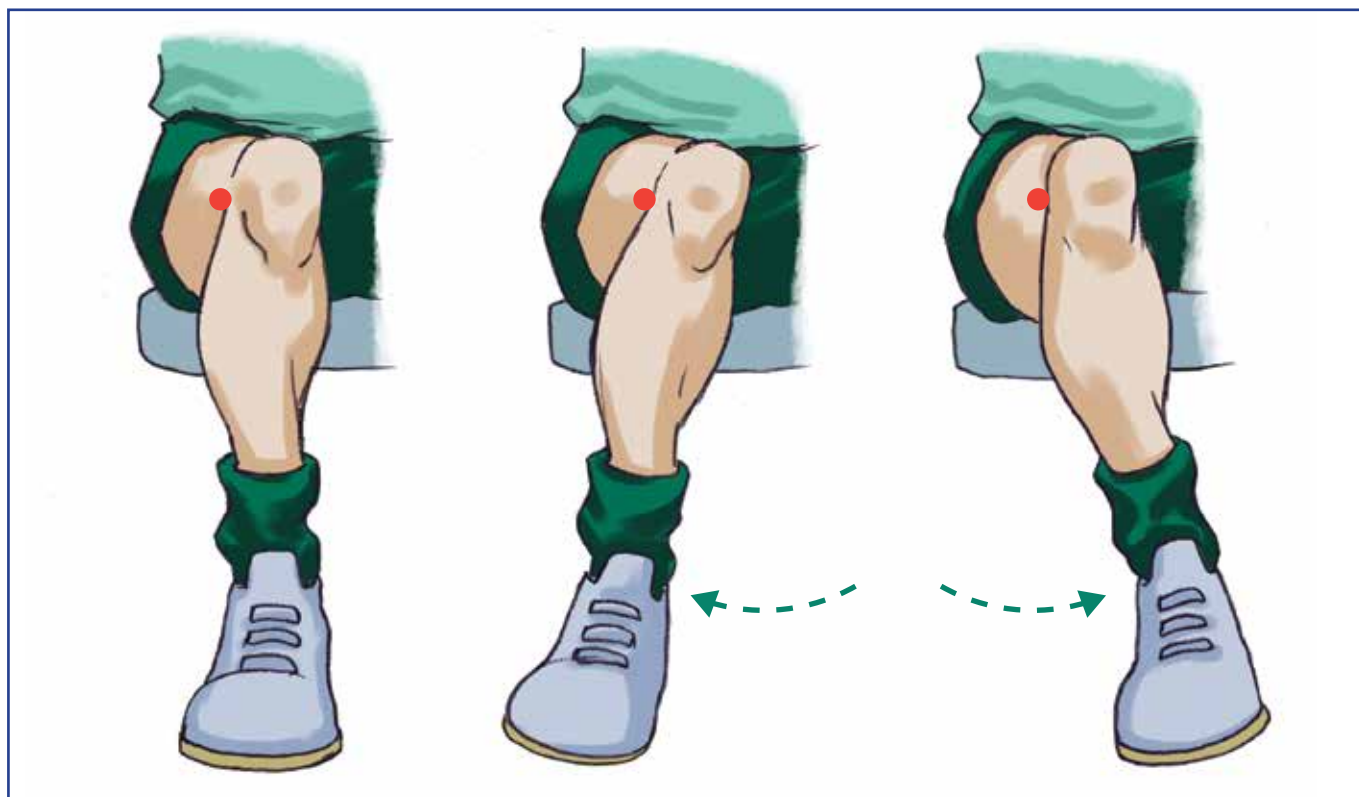
when it is actually an internal rotation (see picture 3.5.8).

Rotational movements around the S-I axis of the elbow are usually called pronation or supination.

- **Pronation** is the medial rotation of the radioulnar joint.
- **Supination** is the lateral rotation of the radioulnar joint.

In order to determine a medial (internal) or lateral (external) rotation, the specific joint should be picked out and watched from a plane that is perpendicular to the axis and relative to the visualization plane (superior-inferior inferior view). After that, a direction of rotation needs to be determined –clockwise or counterclockwise–, relative to a point that stays fixed and one that moves.

LEVER SYSTEMS



Picture 3.5.8

The looks might be deceiving but, in this position, should the hip joint move around its superoinferior axis, the foot would move out in internal rotation and in during external rotation.

MOVEMENTS THAT TAKE PLACE IN MULTIPLE PLANS

Throughout the anatomical ROM, joints which hold more than one axis can –and usually do– describe movements using more than one plane. There is no kinesiology name assigned for most of these movements, although certain multiplanar movements do have one.

MOVEMENTS OF CIRCUMDUCTION:

Circumduction is defined as a non-coplanar, conical movement. The cone's vertex lies within the joint, and its base corresponds to the most distal part of the limb. Circumduction is a combination of movements taking place around the M-L and the A-P axes, and it requires no rotation around the S-I axis.

INTEGRATED MOVEMENTS IN THE HUMAN SYSTEM

So far, we have been describing articular movements as isolated subjects. In real human movement, however, they do not usually take place in an analytical manner. Most movements are single components of a more complex and integrated system of articular movements.

Global motion is generated by the addition of multiple analytical movements taking place in the joints, which then form a chain of movement. It is the synchrony of these movements that allows for efficient and integrated global motion.

Following are the most common integrated moves: pressing actions, pulling actions and integrated rotations.

PRESS AND PULL ACTIONS, AND THEIR KINETIC RELATION

Professionals in our field often do not come clear as to whether press-pull actions constitute kinetic or kinematic actions.

When considering press-pull actions from a kinematic point of view, forces should be left aside and only the combined articular movement should be looked at.

Lets use the weight room as an example and go on to analyze a dumbbell press in the transverse plane. For this move, we'd be better off calling only one of the phases press, since the way back to the starting position would actually cause the opposite movement, that is a pull. By solely considering the kinematics, dumbbell presses would have to be called dumbbell press-pulls. Similarly, pulls would have to be called pull-presses, because it requires both those kinematic movements to make for a complete move.

The above stated nomenclature is rarely used, and not very logical. Therefore, press-pull actions lie more within the kinetic than the kinematic classification model. This means that, in order to determine whether an action is called press or pull, it is not the articular movement that matters. What actually matters is the relationship between the direction of the force being applied to the human body and the resulting movement (see picture 3.5.9).

We will analyze this type of interpretation in a section to follow, which is dedicated to the kinetics of lever systems.

'THE HUMAN SYSTEM DOES NOT THINK ABOUT MUSCLES, IT ONLY THINKS ABOUT MOVEMENTS'

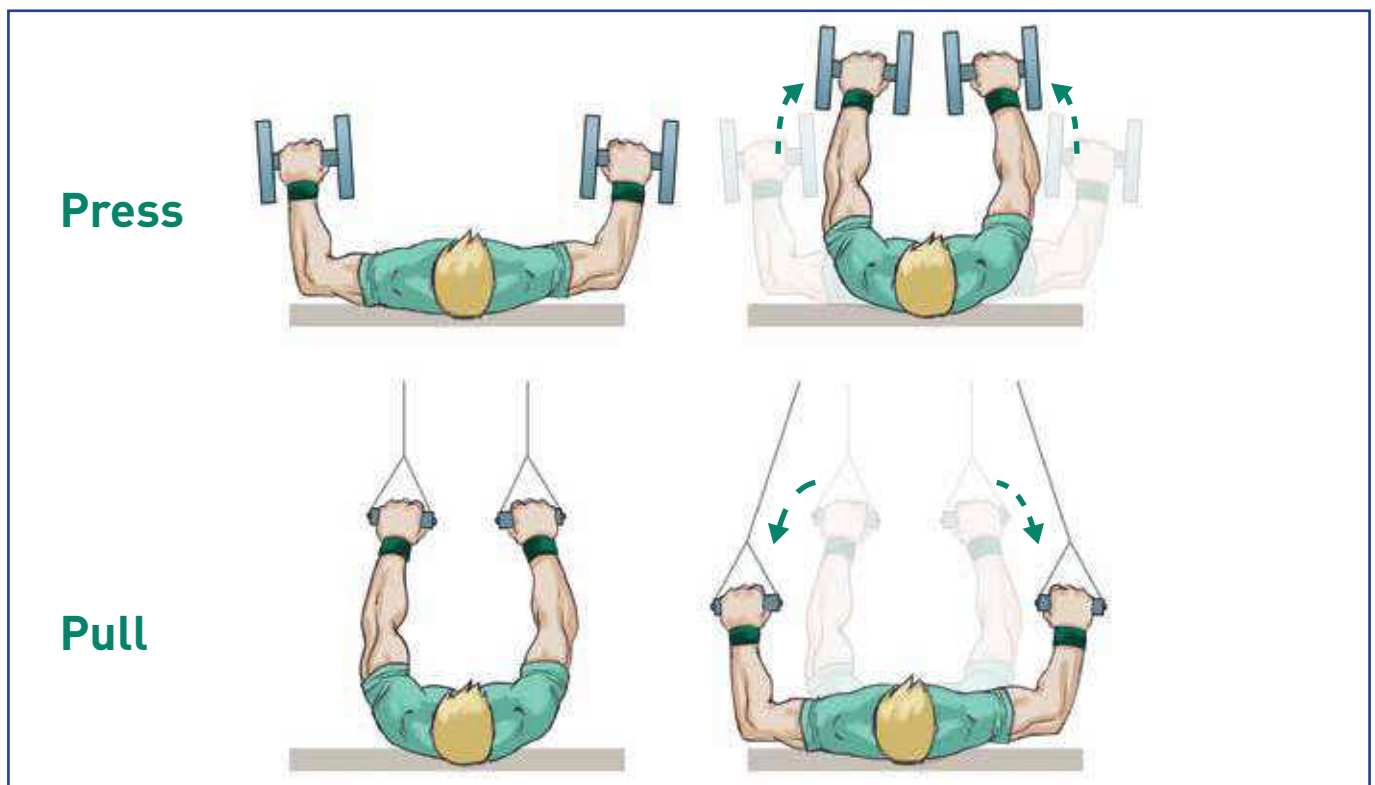
Many fitness professionals use this expression 'the human system does not think about muscles, it only thinks about movements', when trying to explain that motion is mainly

composed of integrated actions. Oftentimes, the example set are press-pull actions, and it is argued that the body thinks merely of executing the move. However, one should take into consideration, that those actions are not only determined by movement itself, but by the relationship between forces affecting the system and articular movement.

Things like this are proof that it is advisable to acquire a deeper understanding of exercise mechanics, which would avoid getting confused between movements and forces.

When an individual is executing an action of press or pull, their neuromuscular system focuses only in dealing with the forces which try to bring it out of balance. It is more about this than about the move being done. This also is the reason why actions are either called press or pull –both in their positive as in their negative phases. Because it is not the movement, but the direction of the force that matters.

At Resistance Institute, it is believed that the body thinks about forces, since both musculature and movements are both dependent on them.



Picture 3.5.9

LEVER SYSTEMS

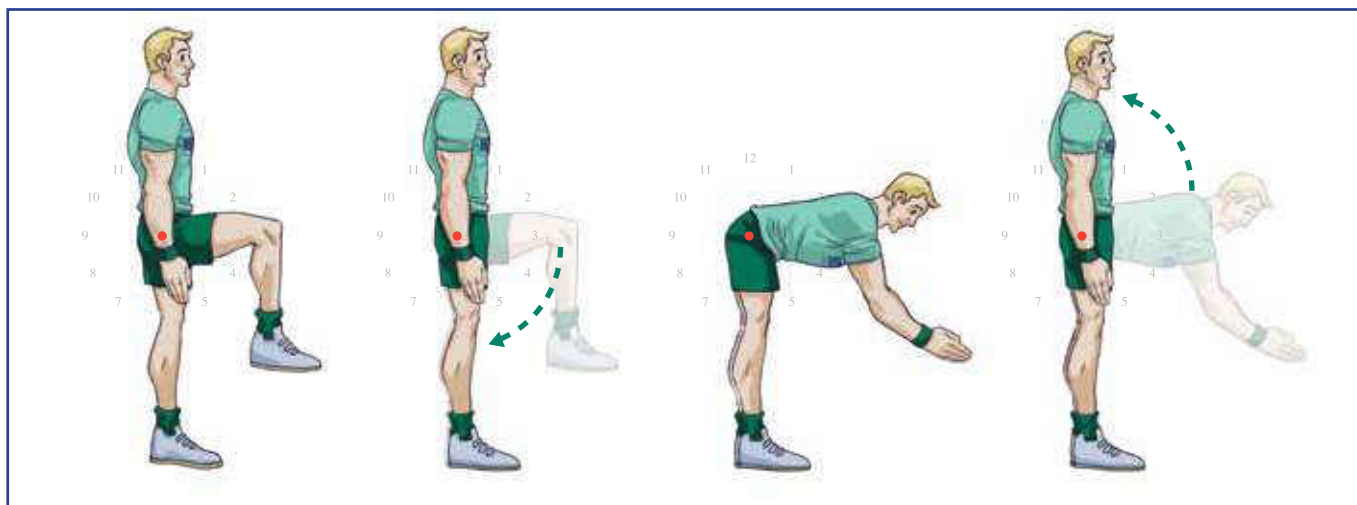
INVERTING THE ARTICULAR DIRECTION OF ROTATION

Most commonly, movements described in kinesiology assume that there is a point that stays still. This point that doesn't move is usually considered to be the most proximal bone or body segment. An example would be the arm moving over the fixed trunk.

It is also feasible that the body's proximal segment turns

into the moving point, which in turn causes the segment to become the distal point within the lever system (of movement).

Do bear in mind that, as seen before, in the above example, the rotational action remains the same, even though the direction of rotation has been inverted.



Picture 3.5.10

Both sequences show a hip extension action, taking place from 90° of hip flexion and into a neutral hip position. However, when the fixed and moving points change, one could easily be misled to thinking that the articular movement has changed, when it hasn't.

3.6 THE KINEMATICS OF LEVER SYSTEMS (LINKAGES)

In this chapter, we will look closely at linkages and the kinematics in them. We will then go ahead to analyze human articular linkages (athrokinematics). At the end of the chapter, you will be introduced to the kinetics of human lever systems.

LINKAGES

As far as human articular movement and control are concerned, it is joints (linkages) that make up the actual 'battle field'. They are key to human motion.

It is inside the joints that external forces and those forces generated from within the human system (muscle and connective tissue) make contact.

The final goal is to be able to guarantee a controlled articular system and thereby movement and locomotion.

In order to be able to cater for these two functions, the human system employs some truly astonishing solutions. The human body needs to combine articular movement with as much articular control as possible. That is to say, the human body needs stability.

The way joints tolerate force and maintain adequate mobility, while preserving the integrity of the system's individual components is a masterpiece of bio-engineering, resulting from millions of years of evolution. Never mind that, the extent of tolerance depends greatly from one individual to another. What an individual can easily bear might be excessive for a different individual and cause them to lose the physical integrity of their joints.

It has been observed that several factors affect the ability of a given articular system to tolerate forces. However, the neuromuscular system is one of the key aspects to work on if correct functioning and joint integrity is our goal.

THE STRUCTURE OF LEVER SYSTEMS (LINKAGES)

The design of the skeletal system is based on the principles of lever systems. By negotiating with the forces that affect it, the skeletal system is designed to allow for movement, usually of a rotational kind.

A lever is a rigid beam with a fixed point (linkage), around which rotation can occur if an external force is applied onto the lever.

The fixed point is known as fulcrum (f) or axis. The lever rotates around an imaginary line called axis of rotation. This imaginary line is perpendicular to the plane of movement. In the human system, joints act as fulcrums, to allow for angular movement of the bones.

In order for rotation to be able to start, there has to be an imbalance between the forces causing rotation in the

system.

Two different types of forces can be present in muscle-joint lever systems. We call one of them moving force or effort (E), and the other one is the resistant force or resistance (R). The direction of rotation taking place around the axis will depend on these two forces.

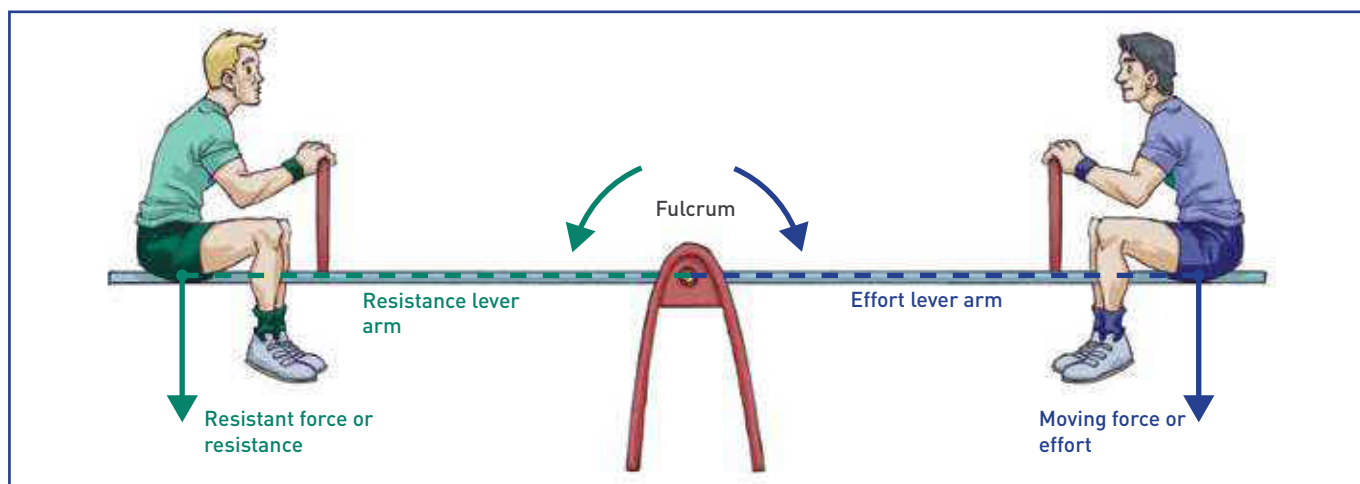
The shortest distance from the fulcrum (f) to the effort's (E) point of application is known as effort lever arm or effort arm (Ea). The shortest distance between f and R is called resistance lever arm or resistance arm (Ra).

You can look at an example of a simplified lever system in picture 3.6.1.

Summing up, the elements that form a lever are:

f = FULCRUM
E = MOVING FORCE OR EFFORT
R = RESISTENT ARM OR RESISTANCE
Ea = EFFORT LEVER ARM (effort arm)
Ra = RESISTANCE LEVER ARM (resistance arm)

LEVER SYSTEMS



Picture 3.6.1

Representation of a simplified lever system.

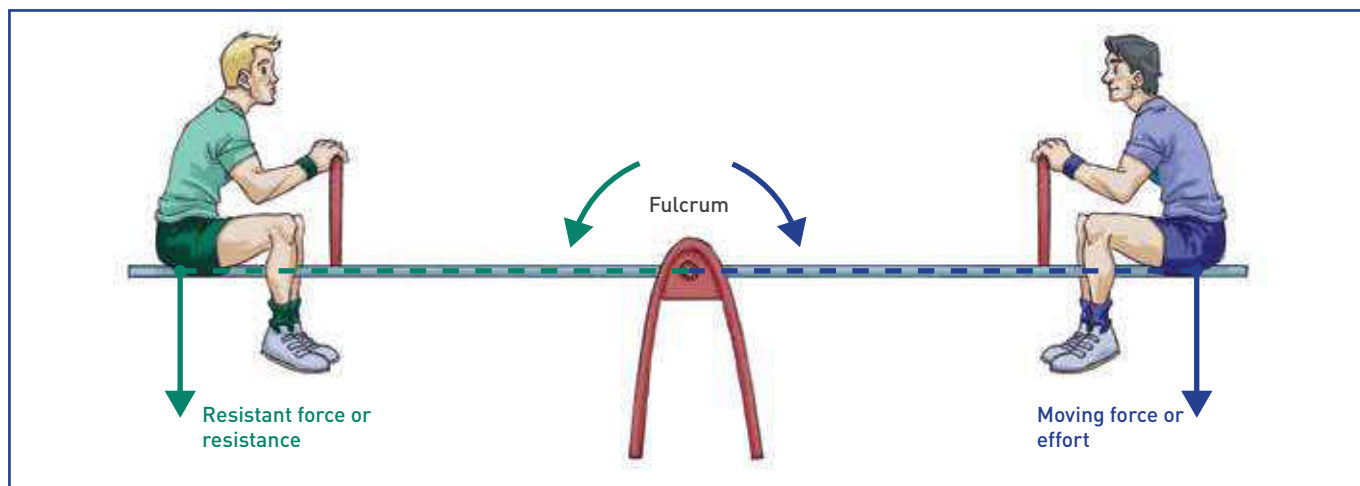
CLASSES OF LEVERS

Levers are typically classified into three categories, depending on how fulcrum, effort and resistance relate to each other.

This book will eventually clarify that this type of explanation has become a *cliché* with little practical applicability. It remains an easy resource to make for exam questions in college, but is far from being a useful explanation.

CLASS 1 LEVERS

In a class 1 lever, the fulcrum lies in between the resistance and the effort. An example of a class 1 lever can be seen in picture 3.6.2.



Picture 3.6.2

CLASS 2 LEVERS

Levers are defined as class 2 levers when the resistance is found in between the effort and the fulcrum. The effort arm in this type of levers is always greater than the resistance arm. Most commonly, it is the effort that holds mechanical advantage here, although this is not always so.

An example of a class 2 lever can be seen in picture 3.6.3.

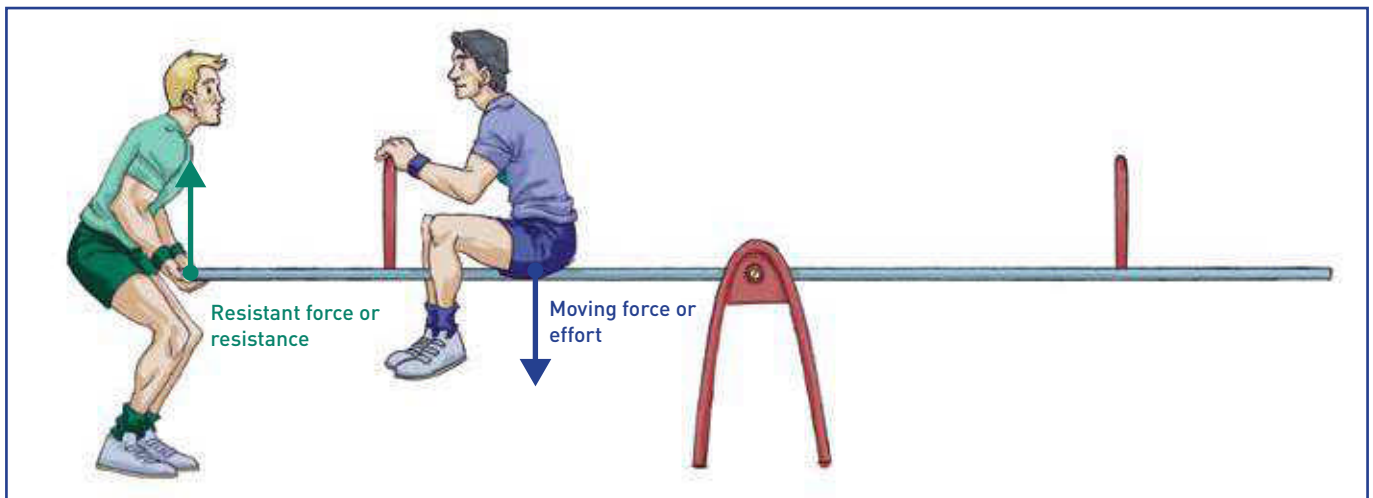


Picture 3.6.3

CLASS 3 LEVERS

Class 3 levers occur when the effort is located between the resistance and the fulcrum. This type of levers are often seen within the human locomotor system.

An example of a class 3 lever can be seen in picture 3.6.4.



Picture 3.6.4

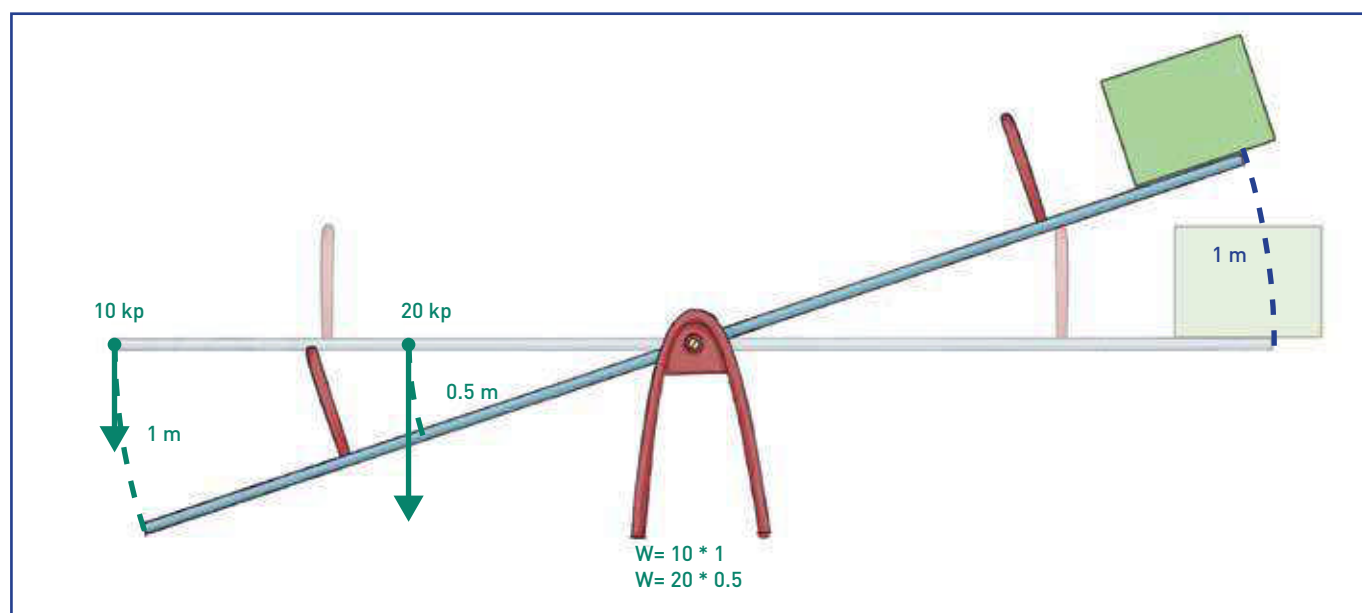
LEVER SYSTEMS

DISTANCE TO AXIS AND DISTANCE MOVED

One way to determine a lever's mechanical advantage is to observe the distance moved by the effort's point of application, and compare it to the distance moved by the resistance's point of application. Mechanical work (W) is determined by the relationship between the force being applied and the amount of movement that it produces.

$$W = F * D$$

Take a look at picture 3.6.5. It shows a lever system where a 10 kp and a 20 kp force are being applied. The 10 kp force is applied at twice the distance-to-axis than the 20 kp one. In such situations, and for any given angular movement, the 20 kp force will move half as much as the 10 kp one. As a result, both forces will make the same amount of mechanical work.



Picture 3.6.5

The following formula is needed in order to calculate the arc's length: $L_a = \frac{2 \cdot \pi \cdot r \cdot \alpha}{360^\circ}$ α represents the angle of rotation, whereas r equals de radius (distance). In order to balance out a lever system, twice as much force would be needed if it were applied at half the distance, in comparison to the force on the other side of the axis. In return, should the lever move, the force at half the distance would move half as much. That is to say, the force on the other side of the lever would move twice as much.

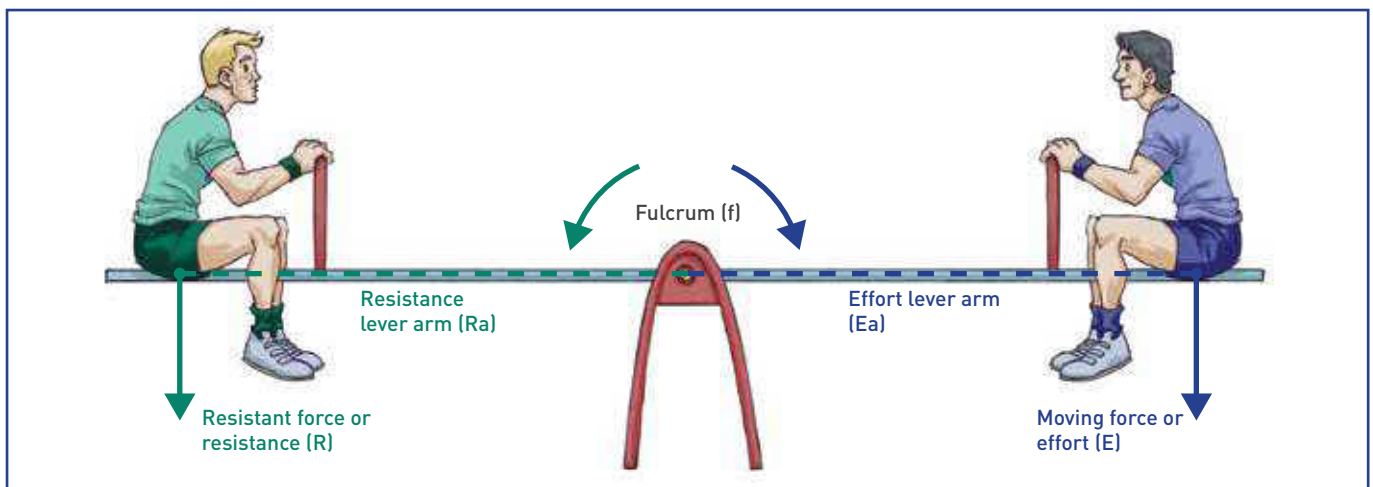
Taking a close look at picture 3.6.5, it can be observed that, as far as mechanical work goes, forces are inversely proportional to the distance moved. In order to move a given load of 10 kp, the most distant force would move 1 m, while the force that is closer would only move half as much. The magnitude of that second force would, in return, need to be double.

3.7 INSTRUCTIONAL PROGRESSION OF LEVER SYSTEMS FOR PHYSICAL ACTIVITY

Lever systems may look tangible and easy to analyze. When applied to human systems, though, their analysis becomes more complex. The importance of lever systems is often underestimated and one should not consider themselves good biomechanists until thorough understanding of the matter is achieved.

In the following pages, you will see a progression that should help you to analyze levers. A simplified analysis (phase 1) will be presented first. Then, variations to the standard seesaw will be explained, that should help to better transfer knowledge to the human body environment.

PHASE 1: SIMPLIFIED LEVER ANALYSIS USING THE STANDARD SEESAW



Picture 3.7.1

The picture represents a simple lever system, where forces apply at a 90° angle to the lever arm.

The picture above shows the elements that make up a lever system. The fulcrum (axis) is the place around which rotation occurs.

Levers are commonly exemplified in a seesaw manner. This constitutes a simplification of reality, and oftentimes two forces are represented. The forces are usually shown on different sides of the axis, and the lever depicted normally belongs to class 1.

In this type of examples, the forces always act on a 90° angle to the lever.

There are two lever arms, one per force. One of the forces is typically called Effort, and the other one gets called Resistance. Because of this, it is reasonable (in the sports

industry, as well) to call one of the arms effort lever arm or effort arm (Ea) and the other one resistance lever arm or resistance arm (Ra).

Lever arms are the shortest lines from the point of application of the force to the axis. And the shortest line is always a straight line. Simple.

This type of lever systems usually show a fine, straight line that coincides with the lever arm (both the resistance's and the effort's). This makes figuring out the lever arms in these structures easy.

Before we go on to phase 2, let us proceed to analysing one of the key concepts for the understanding of lever: the concept of torque.

LEVER SYSTEMS

TORQUE (MOMENT OF FORCE)

Torque is the equivalent to the fraction of force that generates rotation around an axis. It is also known as moment of force or moment, and the concept is gradually entering the fitness industry.

Lever systems are commonly known by their class, but it is torque where the importance actually lies.

Torque could be considered as the force causing (or aiming to cause) movement around an axis of rotation. It must be emphasised that, even if no movement occurs, that doesn't mean that there are no forces trying to generate torque around the axis. It is possible for two forces to generate counteracting torques, which leads to no resulting moment of force, as exemplified in picture 3.7.1.

This causes torque and force to be virtually the same thing. The difference between them lies in the kind of movement produced. Torque, however, depends on the distance to axis.

$$\text{TORQUE (simplified lever analysis)} = \text{FORCE} * \text{LENGTH OF LEVER ARM}$$

Please note: This equation can only be used for the analysis of simplified levers, where force acts on lever at a 90° angle.

TO BE CONSIDERED:

Because joints are basically ruled by rotational movements, torque is the key factor one should take into consideration when referring to muscle force or resistances being applied to body joints.

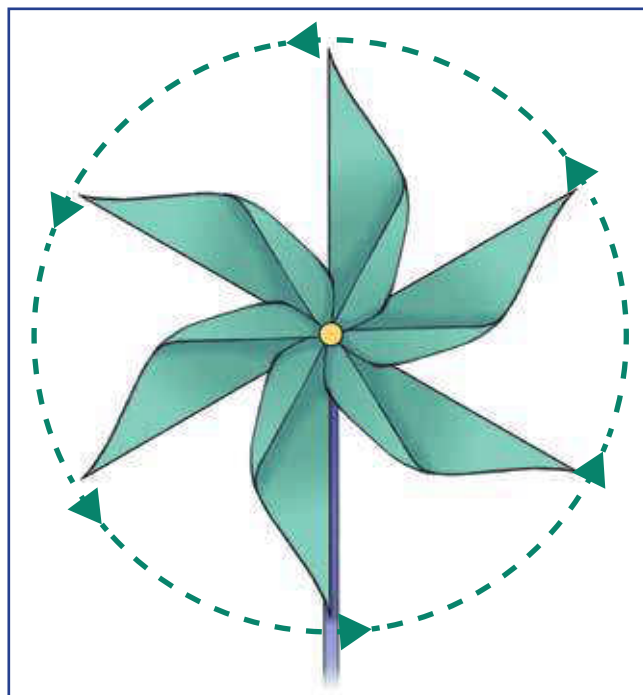
TORQUE'S MODULUS AND DIRECTION. WHAT DO THEY MEAN?

When a linear force contacts and causes rotation on a lever system, the resulting rotational force hold some distinct properties. These properties are different to those of linear forces. Modulus is the rotational force generated on the system, for its multiple positions, around the axis. A given modulus can cause rotation solely in one of the two possible directions. Once the direction of rotation has been established, the direction of the force will always

be perpendicular to the lever. The direction will gradually change, and follow the arc generated by the rotation (see picture 3.7.2).

For simplified lever systems, torque can be calculated by simply multiplying force times that force's lever arm.

This is as much as books usually tell us about levers. But if we want to learn and apply levers to human and exercise reality, it is a must to know about some other variables. Let's take a closer look.

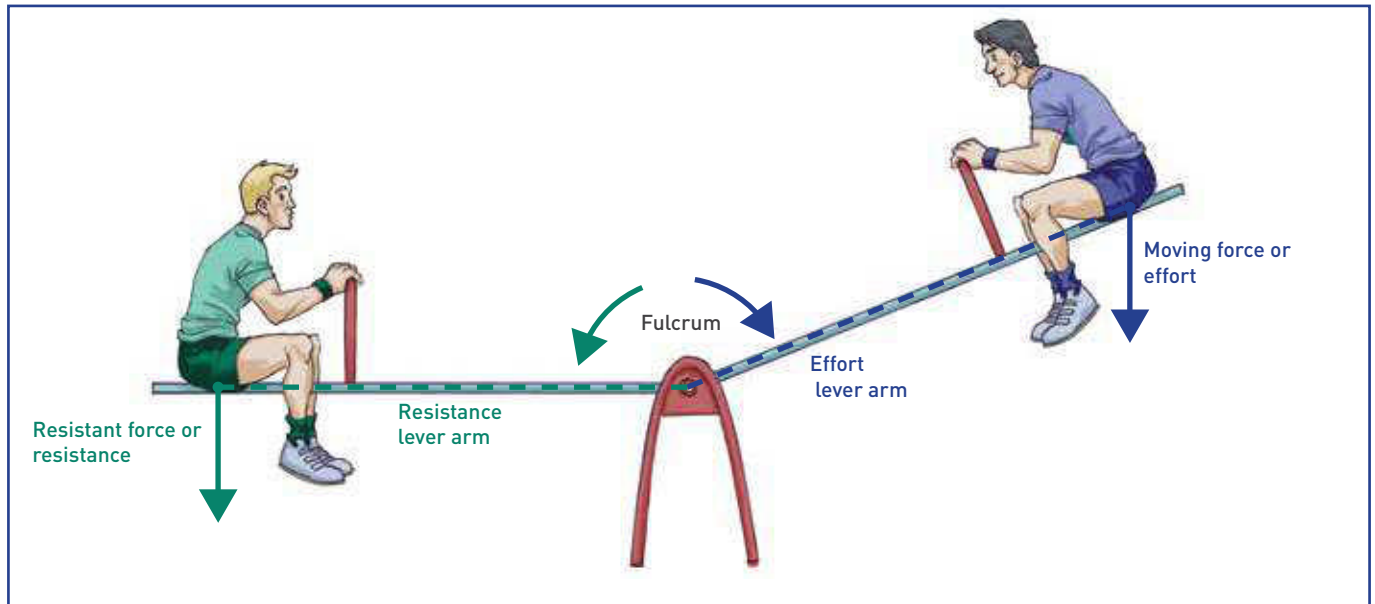


Picture 3.7.2

The direction of a rotational force is tangent to the lever and it changes with movement around the rotational axis.

PHASE 2: DIRECTION OF THE FORCE AND MOMENT ARM

The reality of levers differs from that simplified analysis that was covered in phase 1. Forces do not always act on the lever at a 90° angle. In fact, they rarely do.



Picture 3.7.3

The drawing shows a lever system where the moving force does act upon the lever at a 90° angle.

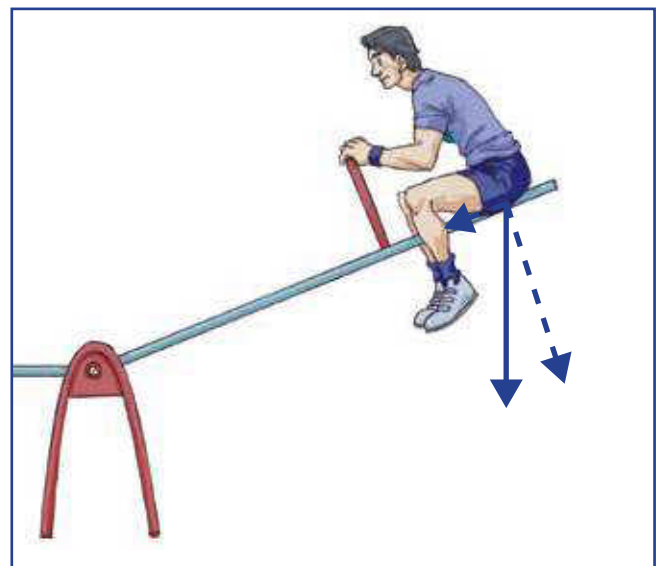
In the picture above, two forces with lever arms of equivalent length can be appreciated. If we were to follow the rule we saw earlier, we would expect both forces to generate the same amount of torque (rotational force around the axis). But, by simple common sense, we can guess that the forces will generate unequal rotational forces around the axis of the seesaw. Important factors are being overlooked...

COMPONENTS OF FORCE, WHEN APPLIED TO AN INTERRUPTION

Forces acting upon a lever, usually consist of two components. There is a rotational component and a translational or axial component. Do note that the latter should not be mistaken for translational movement.

Torque has to do with the rotational component of the force; it is the part of the force that is perpendicular to the lever, and which causes it to rotate. The axial component runs along the lever itself and generates forces within the linkage).

These two components are, by definition, perpendicular to each other.

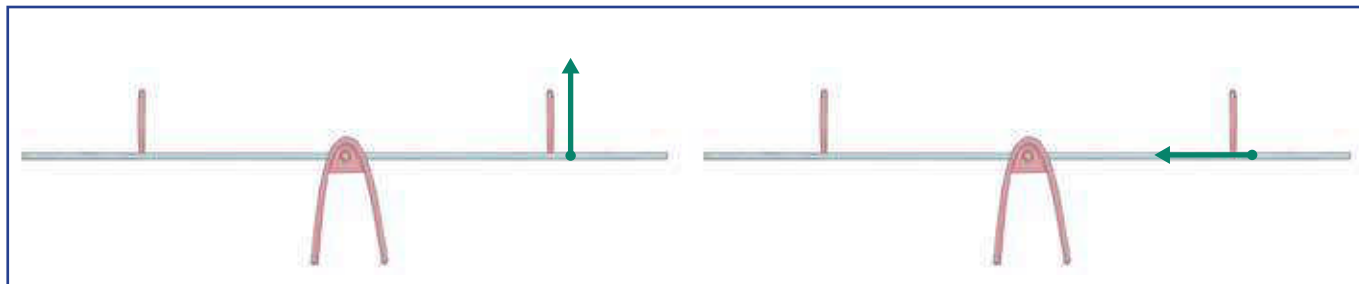


Picture 3.7.4

By resolving the force into its rectangular components, we obtain a rotational component (90° to lever) and a translational component (0° to lever).

LEVER SYSTEMS

At times, forces present solely a rotational component. This happens under one specific scenario, that being the force acting on the lever at a 90° angle.



Picture 3.7.5

The picture on the left shows a force that is entirely rotational, since it acts on the lever arm at a 90° angle. The picture on the right shows a force that is entirely translational, causing no rotational force whatsoever.

Whenever a force has a 90° angle, the whole of it will cause the lever to rotate. It is in these cases that the length of the lever arm can directly be used to calculate torque. This is what most books and study plans show. Nevertheless, this situation is seldom seen in human joint systems.

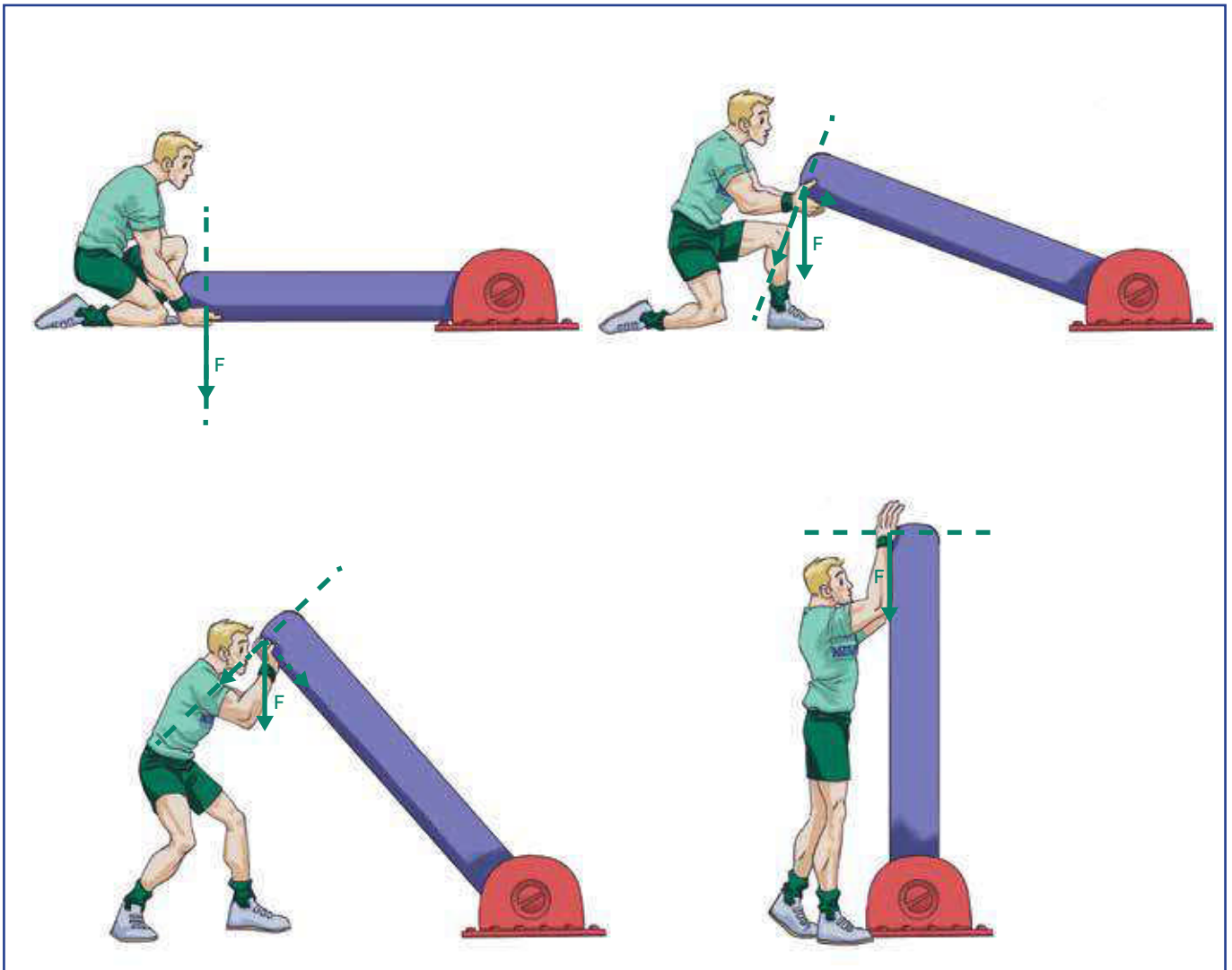
In phase 1 we learned that this equation was sometimes useful:

$$\text{Torque (simplified analysis)} = \text{Force} * \text{Lever arm}$$

However, because of what has just been stated, whenever forces do not act at a 90° angle, the classical equation is of no use.

When a given force does not act upon the lever at a 90° angle, only part of the force will cause rotation around the axis (rotational component). Picture 3.7.3 exemplifies that scenario.

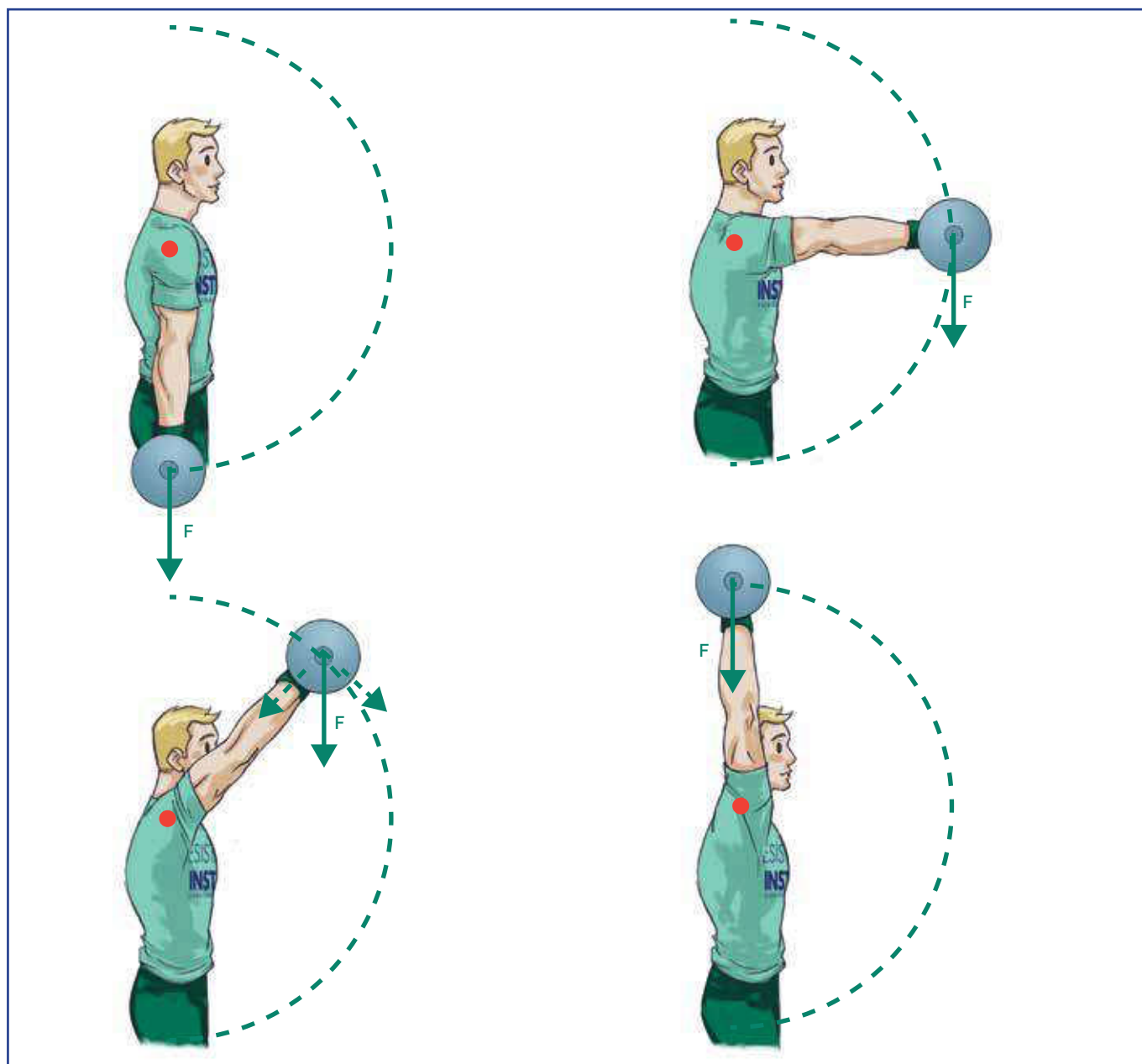
If we go back to what we learned before about directions of rotation and the direction of the torque, it will become clear that: the direction of the lever is tangent (perpendicular) to the lever itself, which means that the direction has a 90° angle. In picture 3.7.4, the force shown in the previous picture has been resolved.



Picture 3.7.6

This sequential image shows a character who is lifting an object around an axis of rotation. The more vertical the object becomes, the less rotational force the object offers. This is due to the direction of the force (gravity) gradually becoming more and more distant to the direction of rotation around the axis (dotted line). When the object reaches the vertical position, the direction of rotation is perpendicular to the direction of gravity, and the angle of the force equals 0° . At 0° , forces cause no rotational effects.

LEVER SYSTEMS



Picture 3.7.7

We see movement around the M-L axis of the glenohumeral joint. Throughout the move, rotational forces generated by the dumbbell vary. If we consider the full ROM, we can see the dumbbell generating no rotation at some positions, or fully causing rotation at others.

Whenever a force affects a lever at a 90° angle, it follows the same lever direction, and allows for all of it to generate rotational force (torque) on the lever.

WHEN ONLY PART OF THE FORCE CAUSES ROTATION AROUND THE AXIS

Whenever a force does not act on the lever at a 90° angle, only part of the total applied force generates rotation. The fraction of force causing rotation is known as rotational

component.

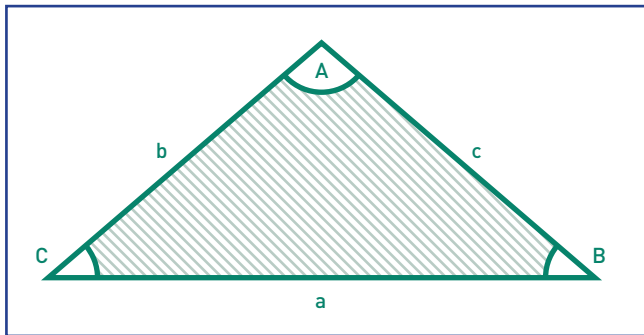
The value for that fraction of force causing rotation can be obtained by using simple trigonometric calculations. We will now take a look at them.

BASIC TRIGONOMETRY

You will need to remember some basic trigonometry concepts if you want to understand what happens when forces act on levers at an angle different to 90° .

DEFINITION OF TRIANGLE:

A triangle is a polygon with three sides (or legs) and three angles. Sides are named with small letters. Angles are assigned the letter of its opposing side, just in block (capital) letters.



Picture 3.7.8

TYPES OF TRIANGLES:

Triangles can be classified in several ways. Following are the two most common forms of classification:

■ Classification according to their angles:

RIGHT TRIANGLE: Right triangles are those which have one 90° -angle, also called a right angle. The two sides that form that right angle are known as catheti (singular: cathetus). These have been labelled a and b in picture 3.7.9. The side opposite to the right angle is known as hypotenuse and has been marked as h in picture 3.7.9. This type of triangles, and the mathematical calculations for them, will be key to understanding how trigonometry can be applied to the study of forces in the human system.

ACUTE TRIANGLE: All of its three angles are acute (measuring less than 90°).

OBTUSE TRIANGLE: Has an obtuse (more than 90°) angle.

■ Clasificación en función de sus lados:

ISOSCELES TRIANGLE: Has two sides of equal length.

EQUILATERAL TRIANGLE: All three sides are equal, and thus all of its three angles are also equal (60°).

SCALENE TRIANGLE: All three sides are different.

COMPUTING TRIANGLES:

Following is how to calculate the length of a side or the measure of an angle.

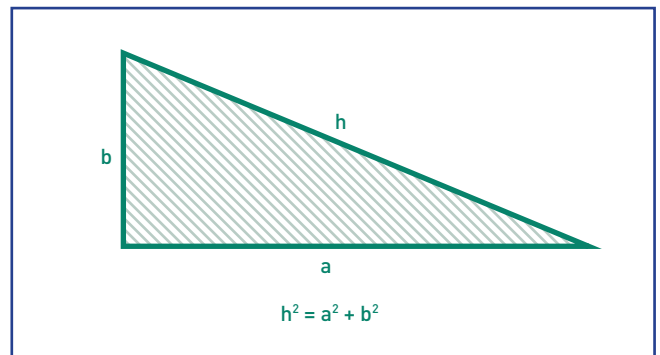
Basic Trigonometry:

No matter what kind of triangle we talk about, the sum of its internal angles will always be equal to 180° .

Two triangles are said to be similar, when two of its angles are equal (and thus, all three of them). And because of this, the sides of both triangles are proportional.

The sides in a right triangle follow a rule known as the **Pythagorean theorem**, which states that:

'The square of the hypotenuse equals the sum of the squares of the catheti'

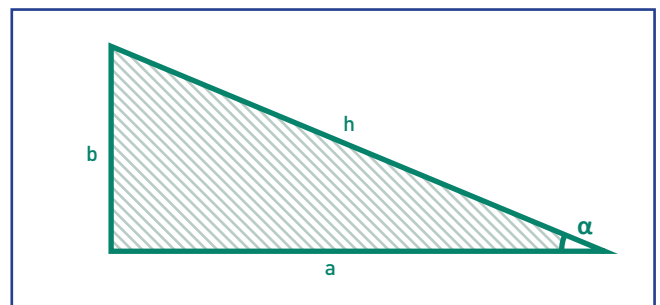


Picture 3.7.9

TRIGONOMETRIC RATIOS IN RIGHT TRIANGLES:

A ratio is the quotient of two values.

$$\begin{aligned}\sin \alpha &= b/h \\ \cos \alpha &= a/h \\ \tan \alpha &= b/a\end{aligned}$$



Picture 3.7.10

LEVER SYSTEMS

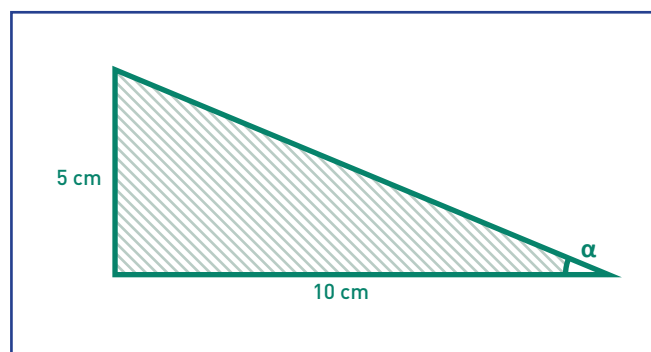
SINE (sin) α : It is the ratio of the opposite cathetus to the hypotenuse. It represents the percentual length the opposite cathetus to the hypotenuse.

COSINE (cos) α : It is ratio of the adjacent cathetus to the hypotenuse. It represents the percentual length of the adjacent cathetus to the hypotenuse.

TANGENT (tan) α : It is the ratio of the opposite cathetus to the adjacent cathetus. And thus the proportion of one to another.

Properties:

Trigonometric ratios do not depend on the sides' values but on the relationship between them. In order for two triangles to be considered similar, both triangles need to have the same trigonometric ratios. This fact implies that trigonometric ratios for any given triangle are **non-dimensional**.



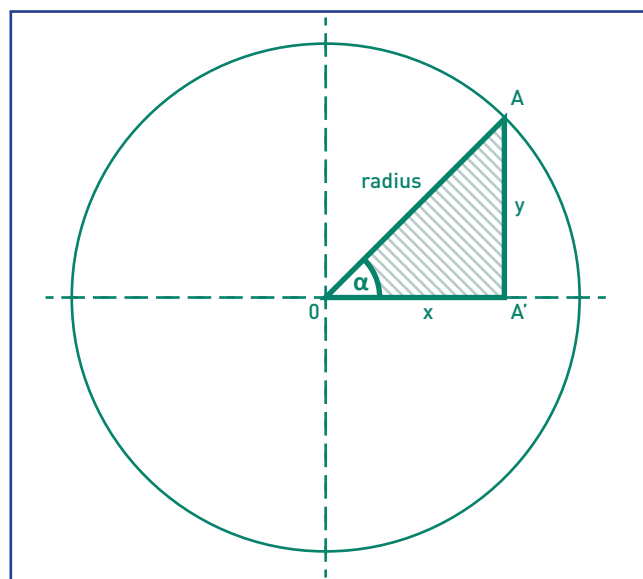
Picture 3.7.11

By taking a close look at the picture above, one can tell that the tangent 'measures' the relationship between the catheti and is non-dimensional.

$$\tan \alpha = 5 \text{ cm} / 10 \text{ cm} = 5/10 = 0,5 \text{ (the first one if half the other one).}$$

The value of sine and cosine of a given angle will lie between -1 and +1, since they represent quotients of catheti to hypotenuse. Tangents can have any value.

Trigonometric ratios can be applied to any angle. This means that one can pick out any spot of a circumference and create a trigonometric value.



Picture 3.7.12

$$\sin \alpha = \text{opposite/radius; } \sin \alpha = y/r.$$

$$\text{For radius} = 1, \sin = y$$

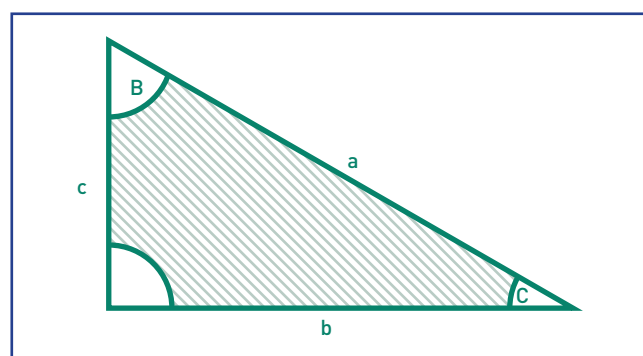
$$\cos \alpha = \text{adjacent/radius; } \cos \alpha = x/r.$$

$$\text{For radius} = 1, \cos \alpha = x$$

$$\tan \alpha = \sin \alpha / \cos \alpha; \tan \alpha = \text{opposite/adjacent}$$

$$\tan \alpha = y/x$$

RELATIONSHIP BETWEEN THE ELEMENTS OF A RIGHT TRIANGLE



Picture 3.7.13

The value of the 'b' cathetus equals the hypotenuse times the cosine of the adjacent angle ($b = a * \cos C$). OR the hypotenuse times sine of the opposite angle ($b = a * \sin B$).

$$b = a * \sin B$$

$$c = a * \cos B$$

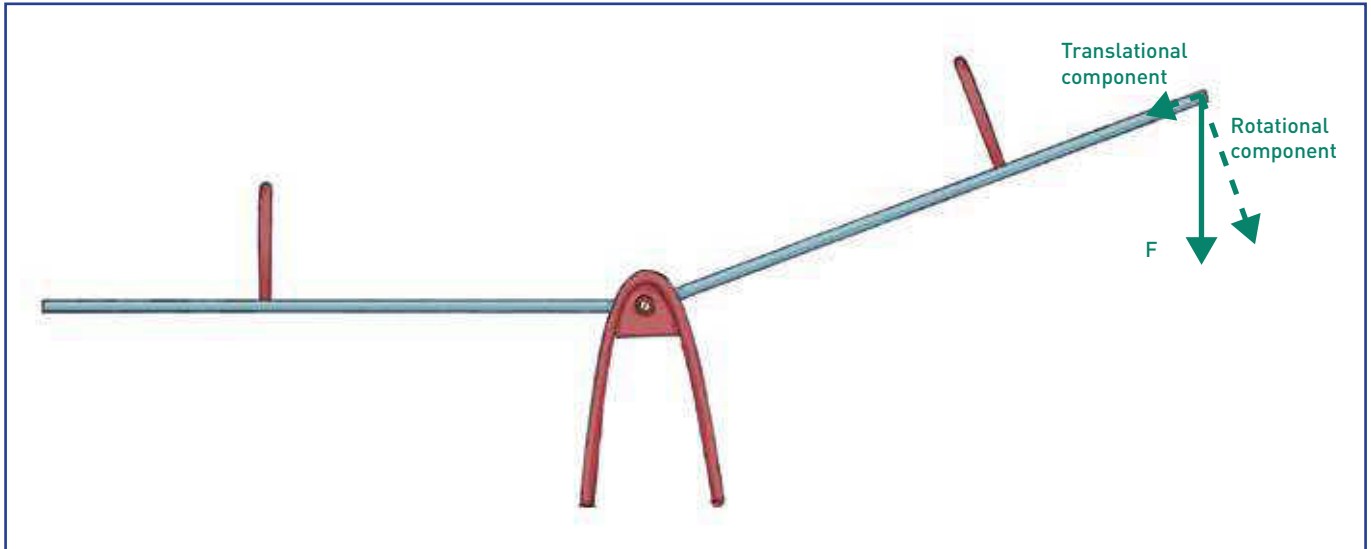
$$b = a * \cos C$$

$$c = a * \sin C$$

In summary, if we know the values for one side and one angle (apart from the right angle), we can figure out the rest of the components of a triangle.

APPLYING TRIGONOMETRY TO LEVER SYSTEMS

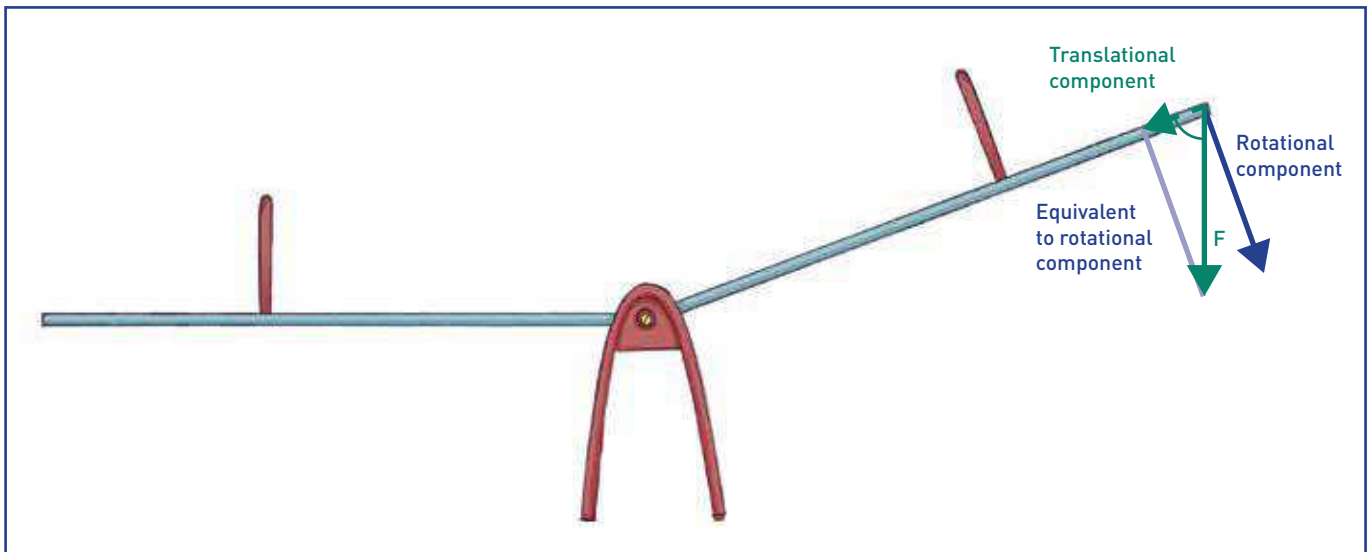
The following trigonometry concepts apply to lever systems, whenever the angle of the force is different to 90°.



Picture 3.7.14

Trigonometry is a key field of study to understanding forces that act upon lever systems.

In the picture above, only the rotational component causes the lever to rotate. Note that this component is tangent to the direction of the lever. The translational component runs along the lever itself and causes other types of forces.



Picture 3.7.15

In the picture above, it is the triangle formed between the resultant and the translational and rotational components that should be looked at in order to make trigonometric calculations. The rotational component represents a portion of the force which relates to the angle of application of the resultant force vector (force applied to the lever):

$$\text{Rotational component (cathetus opposite to } \alpha) = \text{Force applied} * \sin \alpha$$

LEVER SYSTEMS

The sine of α shows the portion of rotational component relative to the resultant force.

The sentence above explains why, for forces acting on an 90° angle, the sine of 90° equals 1. This is what most simplified see-saw examples illustrate, since the force lies in the direction of movement and only has rotational effects upon the lever.

The equation we learned about earlier on to figure out torque needs to be modified if the force does not act on the lever at a 90° -angle. In that case, sine of α has to enter the equation.

$$\text{Torque} = \text{Force} * \text{Lever arm} * \sin \alpha.$$

ANGLE OF THE FORCE (FA)

The angle of the force corresponds to the angle found between the lever arm and the force. This angle holds a direct relationship to the portion of the force generating rotation upon the lever. This relationship has to do with the optimal angle of application for a force to generate rotation. This optimal angle is 90 degrees ($\sin 90^\circ = 1$). In trigonometry, this ratio is demonstrated by the sine of the given angle.

Degrees	Sine	Cosine	Tangent
0	0	1	0
1	0,017452	0,999848	0,017455
2	0,034899	0,999391	0,034921
3	0,052336	0,99863	0,052408
4	0,069756	0,997564	0,069927
5	0,087156	0,996195	0,087489
6	0,104528	0,994522	0,105104
7	0,121869	0,992546	0,122785
8	0,139173	0,990268	0,140541
9	0,156434	0,987688	0,158384
10	0,173648	0,984808	0,176327
11	0,190809	0,981627	0,19438
12	0,207912	0,978148	0,212557
13	0,224951	0,97437	0,230868
14	0,241922	0,970296	0,249328
15	0,258819	0,965926	0,267949
16	0,275637	0,961262	0,286745
17	0,292372	0,956305	0,305731
18	0,309017	0,951057	0,32492
19	0,325568	0,945519	0,344328
20	0,34202	0,939693	0,36397
21	0,358368	0,93358	0,383864
22	0,374607	0,927184	0,404026
23	0,390731	0,920505	0,424475
24	0,406737	0,913545	0,445229
25	0,422618	0,906308	0,466308
26	0,438371	0,898794	0,487733
27	0,45399	0,891007	0,509525
28	0,469472	0,882948	0,531709
29	0,48481	0,87462	0,554309
30	0,5	0,866025	0,57735
31	0,515038	0,857167	0,600861
32	0,529919	0,848048	0,624869
33	0,544639	0,838671	0,649408
34	0,559193	0,829038	0,674509
35	0,573576	0,819152	0,700208
36	0,587785	0,809017	0,726543
37	0,601815	0,798636	0,753554
38	0,615661	0,788011	0,781286
39	0,62932	0,777146	0,809784
40	0,642788	0,766044	0,8391
41	0,656059	0,75471	0,869287
42	0,669131	0,743145	0,900404
43	0,681998	0,731354	0,932515
44	0,694658	0,71934	0,965689
45	0,707107	0,707107	1

Degrees	Sine	Cosine	Tangent
46	0,71934	0,694658	1,03553
47	0,731354	0,681998	1,072369
48	0,743145	0,669131	1,110613
49	0,75471	0,656059	1,150368
50	0,766044	0,642788	1,191754
51	0,777146	0,62932	1,234897
52	0,788011	0,615661	1,279942
53	0,798636	0,601815	1,327045
54	0,809017	0,587785	1,376382
55	0,819152	0,573576	1,428148
56	0,829038	0,559193	1,482561
57	0,838671	0,544639	1,539865
58	0,848048	0,529919	1,600335
59	0,857167	0,515038	1,664279
60	0,866025	0,5	1,732051
61	0,87462	0,48481	1,804048
62	0,882948	0,469472	1,880726
63	0,891007	0,45399	1,962611
64	0,898794	0,438371	2,050304
65	0,906308	0,422618	2,144507
66	0,913545	0,406737	2,246037
67	0,920505	0,390731	2,355852
68	0,927184	0,374607	2,475087
69	0,93358	0,358368	2,605089
70	0,939693	0,34202	2,747477
71	0,945519	0,325568	2,904211
72	0,951057	0,309017	3,077684
73	0,956305	0,292372	3,270853
74	0,961262	0,275637	3,487414
75	0,965926	0,258819	3,732051
76	0,970296	0,241922	4,010781
77	0,97437	0,224951	4,331476
78	0,978148	0,207912	4,70463
79	0,981627	0,190809	5,144554
80	0,984808	0,173648	5,671282
81	0,987688	0,156434	6,313752
82	0,990268	0,139173	7,11537
83	0,992546	0,121869	8,144346
84	0,994522	0,104528	9,514364
85	0,996195	0,087156	11,430052
86	0,997564	0,069756	14,300666
87	0,99863	0,052336	19,081137
88	0,999391	0,034899	28,636253
89	0,999848	0,017452	57,289962
90	1	0	Infinito

Trigonometric table

LEVER SYSTEMS

THE MOMENT ARM

Trigonometric calculations are, understandably, not very handy when it comes to exercising with clients. Not only that but, when humans move, lever arms can change along with the angles, which makes calculations even more difficult.

Because of the handicaps explained above, it is worth mentioning that there is a very useful term in exercise mechanics called moment arm. Moment arm is a popular concept in Mechanical Engineering as well as in Physics. These fields of study commonly include moment arm in their lever analysis but surprisingly, it is not as common to use it in physical activity. In the following sections, you will discover that Moment arm and Lever arm are not synonyms.

What is moment arm a measure for?

Moment arm measures the mechanical characteristics of a given applied force, relative to its rotational effects on a lever system.

It is a way to graphically make trigonometric calculations, and it represents how much of the lever arm actually causes rotation. Using moment arms simplifies the process of determining the mechanics of an applied force on a lever. It is handy for exercise environments and it does not require the use of a calculator. All you need is to multiply this moment arm by the force, in order to obtain torque.

Moment arm also expresses how mechanically advantageous an applied force can be, throughout the rotational movement of the lever (articular ROM)

Following is the equation for the arithmetical calculation of moment arm:

$$\text{Moment Arm} = \text{Lever arm} * \sin \alpha$$

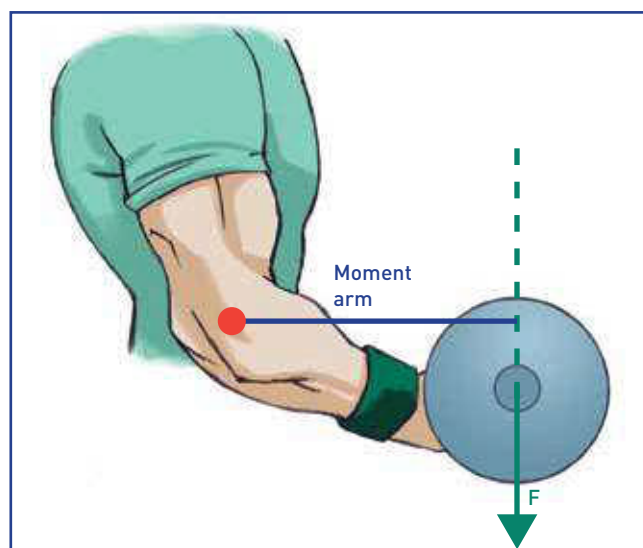
How can it be graphically calculated?

Moment arm can be calculated by drawing a straight line, running through the lever's axis and perpendicularly to the line of force. Moment arm equals the shortest distance from the axis to the line of force, that runs perpendicular to the latter.

From the statements above, it can be deduced that moment arms have to do with the resistance's force angle, and with the length of the lever arm.

Whenever the line of force is at the axis's height, moment arm equals zero, since the angle formed between them is

0° ($\sin 0^\circ = 0$). In this situation, the stated force would cause zero torque.



Picture 3.7.16

Representation of moment arm (blue line).

MOMENT ARM AND TORQUE. WHAT HAPPENS WHEN NOT ALL OF THE FORCE IS CAUSING ROTATION AROUND THE LEVER SYSTEM

Whenever forces acting upon a lever system attack at a 90° angle –as is often the case in book illustrations–, moment and lever arm become equal. However, this is rarely the case in real life situations.

Lever systems are often taught in mechanical scenarios where moment arm and (resistance or power) lever arm coincide. It is, therefore, understandable that the differences between moment and lever arm are not always perceived.

It should be noted that, in real life, power/resistance lever arms coincide with moment arms solely when all of the applied force acting upon the lever does so at the direction of the lever (that is, at a 90° angle). For this given scenario, all of the applied force would cause rotation upon the lever.

Moment arm is a measure that is proportional to the power/resistance arm. Therefore, every time a lever system is contacted by a force, acting at any degree that is not 90° (that accounts for most situations), the moment arm diminishes.

Think about it: whenever a force attacks a lever at an angle different from 90° , not all of the force would be causing the

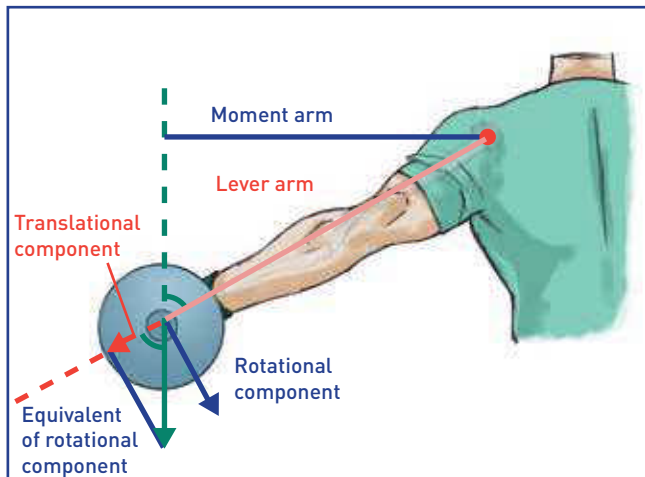
lever to rotate. And only the fraction of a force that causes rotation (that is, torque) is reflected on the moment arm.

Why can moment arm be compared to the rotational component within a lever system?

In picture 3.7.17, you can see two triangles that share two of their lines. This causes them to be considered similar triangles. And they share the same proportions and trigonometric reasons.

As shown in the drawing, the blue cathetus in the small triangle equals the rotational component. The rotational component holds exactly the same proportion to the amount of force, that holds the moment arm in relationship to the length of the lever arm.

One could say that moment arm is a representation of the fraction of rotation applied on the distance of the lever arm. Meanwhile, the rotational component would represent that same fraction applied to the force. Both calculations offer the same result, for both proportions are identical; the only difference is one is applied to the distance, while the other one acts on the force.



Picture 3.7.16

Because of them being similar triangles, the equivalent of the rotational component holds the same proportions as the moment arm, within the triangle.

CALCULATING MOMENT OF A FORCE RELATIVE TO THE MOMENT ARM:

$$\text{Moment of a force} = \text{Force} * \text{Moment arm} \\ (\text{Lever arm} * \sin \alpha)$$

CALCULATING MOMENT OF A FORCE RELATIVE TO THE ROTATIONAL COMPONENT:

$$\text{Moment of a force} = \text{Rotational component of force} \\ * \text{Lever arm} = (\text{Force} * \sin \alpha) * \text{Lever arm}$$

For a given $\sin \alpha$ the moment of force is the same (as long as the force and lever arm considered are the same).

Both of the ways explained for the calculation of force mechanics are equally valid. However, in the world of exercise, using moment arm is often more convenient. Moment arm is a visual method that is used to represent a specific distance. It is therefore more simple to visualize it than it is to calculate the percentage of a force in relationship to the angle of application. And this is even more so in cases where the lever arm also varies, which is true to many of the moves in a gym.

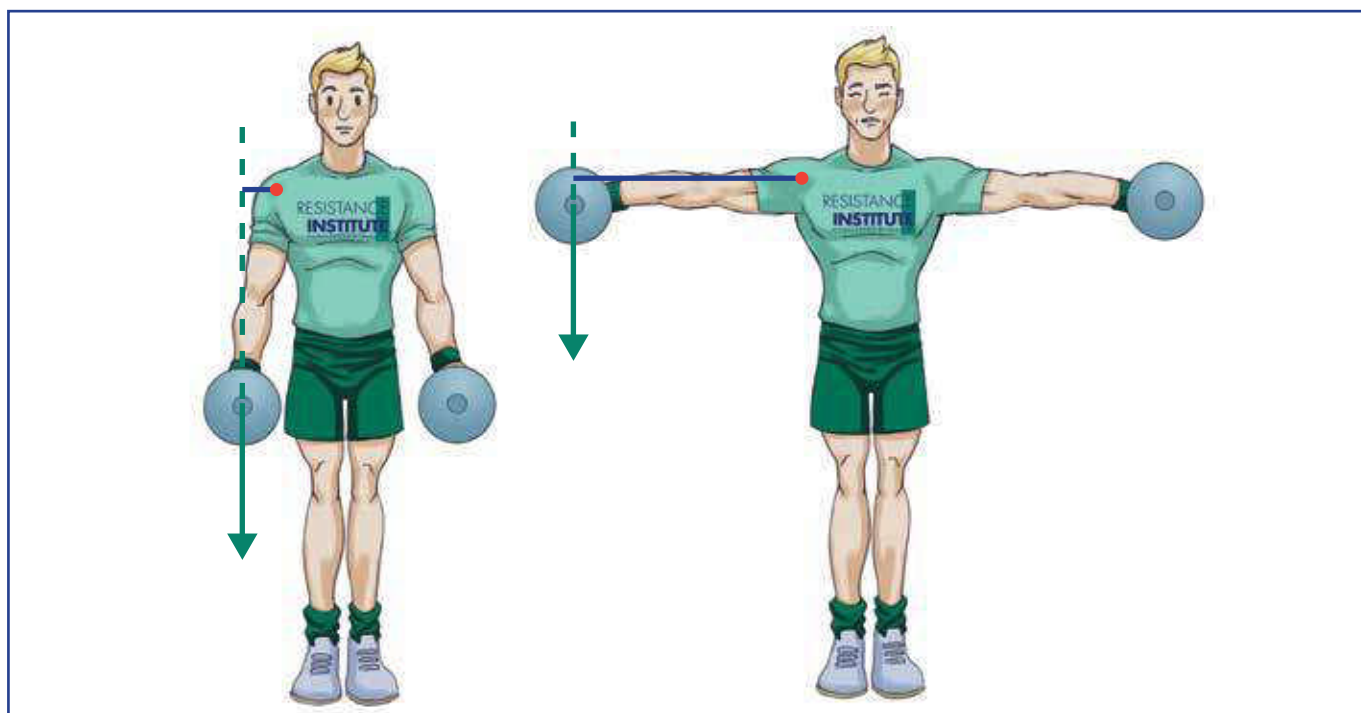
¿HOW CAN I CALCULATE THE MOMENT ARM IN A LEVER SYSTEM? (calculation in static)

An applied example can be seen in pictures 3.7.18 and 3.7.19.

Illustration 3.7.18 shows a shoulder abduction (lateral raises) using dumbbells. In this case, the load remains unchanged, and so does the resistance lever arm and despite all that, the force causing rotation on the lever changes depending on the position. You can see how the moment of the force increases as the shoulder abducts, until it reaches its maximum –see picture on the right.

The evolution of a rotational force throughout the ROM, as shown, only applies to calculations for moment of a force considered in static. Torque fluctuates as a consequence of dynamics and due to the effects of inertial reactive loading. This idea was pointed at in chapter 2 and will be taken a closer look at in the following pages.

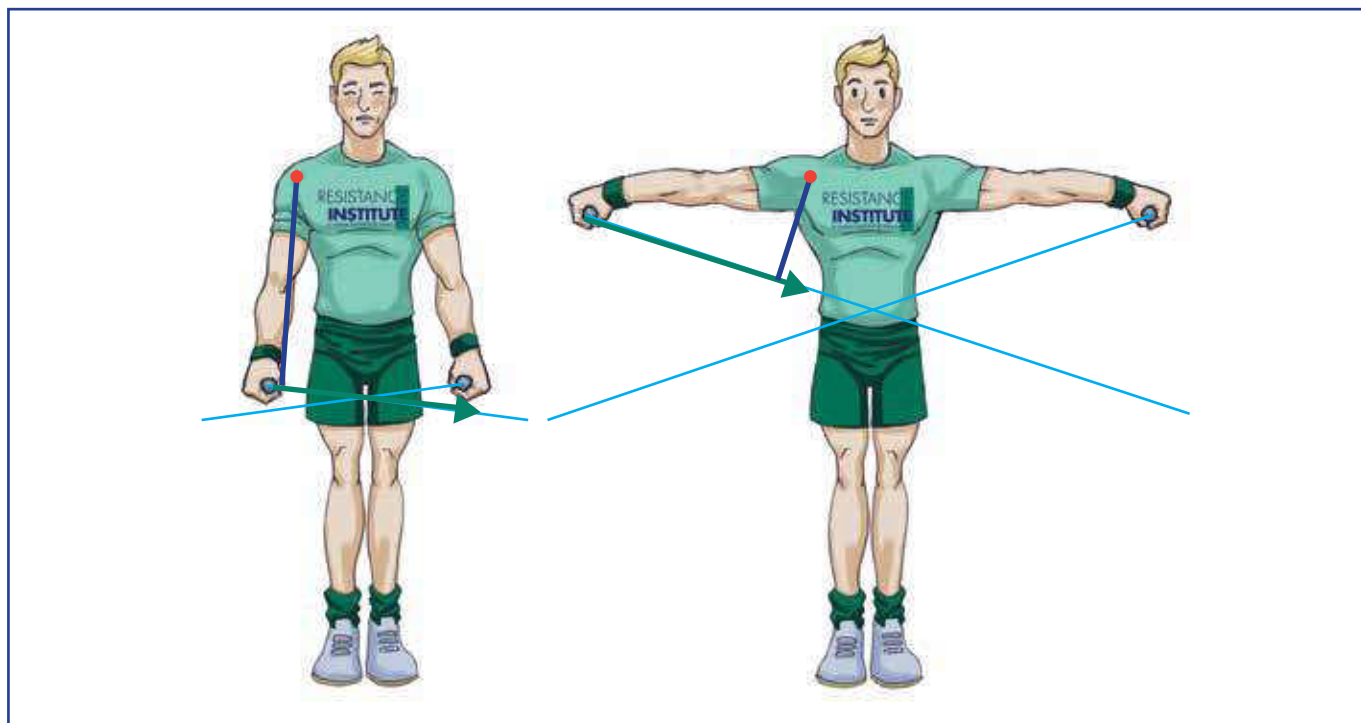
LEVER SYSTEMS



Picture 3.7.18

What happens when the direction of the force changes (for a given move, with a given load and a given ROM)?

Picture 3.7.19 show how the direction of a resistance can be modified by using a cable machine, even if the character still performs the same move.



Picture 3.7.19

For an equivalent load --and differently to what was depicted in picture 3.7.18, the moment arm here is greatest at 0° abduction of the glenohumeral joint, which is when the character receives the highest amount of torque.

HOW MUCH ROTATIONAL FORCE IS TRANSMITTED ONTO THE JOINT DOES NOT ONLY DEPEND ON HOW FAR THE POINT OF APPLICATION IS LOCATED. IT ALSO DEPENDS ON THE DIRECTION OF THE FORCE!

THE MOMENT ARM EXPLAINS WHY:

- It is not only the weight that matters, but the force provoking rotation in the joint; that is torque.
- This force causing rotation, varies throughout the joint's ROM. It does so as the moment arm changes, and even if the load (applied force) remains the same.
- The direction of the force is key to understanding the evolution of the moment arm.

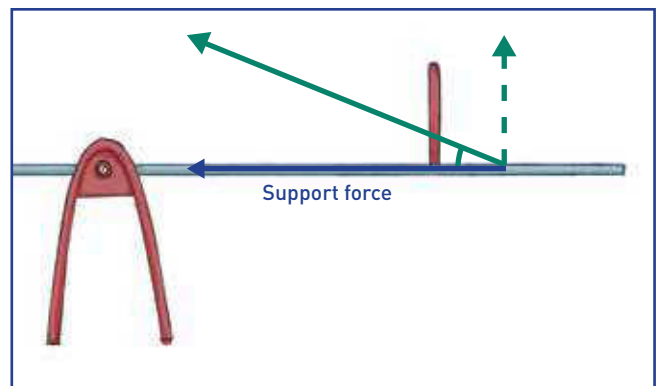
AND IT ALSO GIVES US SOME CLUES REGARDING:

- Which group of muscle fibers will be affected throughout the joint's range of motion.
- The type for motor units and muscle fibers that will be recruited, as well as their activation within a continuum depending on intensity by unit of time (torque determines the necessary intensity of response to be exerted by the neuromuscular system).
- Joint forces that get caused by an external load's willingness to generate force, and its relationship to joint muscle forces that arise in order to counteract the former.
- The effect of moment arm upon contacting articular surfaces, which in turn change position along the way and receive greatly varying forces throughout the ROM (would it be possible that, at some point within the move, force could be excessive?).
- The neurological activation that will be required to respond to the different resistances generated by the changing moment arm and torque, throughout the ROM. Please note that this could be adequate for some scenarios and inadequate or insufficient for others.

SUPPORT AND SUSPENSION FORCES IN LEVER SYSTEMS

Whenever a force contacts a lever at an angle that is different to 90°, a force component of translational nature is created.

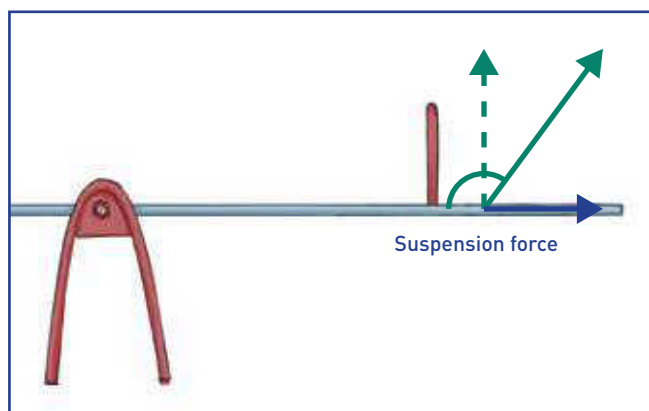
If this component is directed towards the axis, it will be considered a support force. If the component travels away from the axis it is considered a suspension force.



Picture 3.7.20

In this picture, the translational component is directed towards the axis of rotation. This is a support force applied to the lever. As you can see, when the angle of force is smaller than 90°, translational forces are lever supportive.

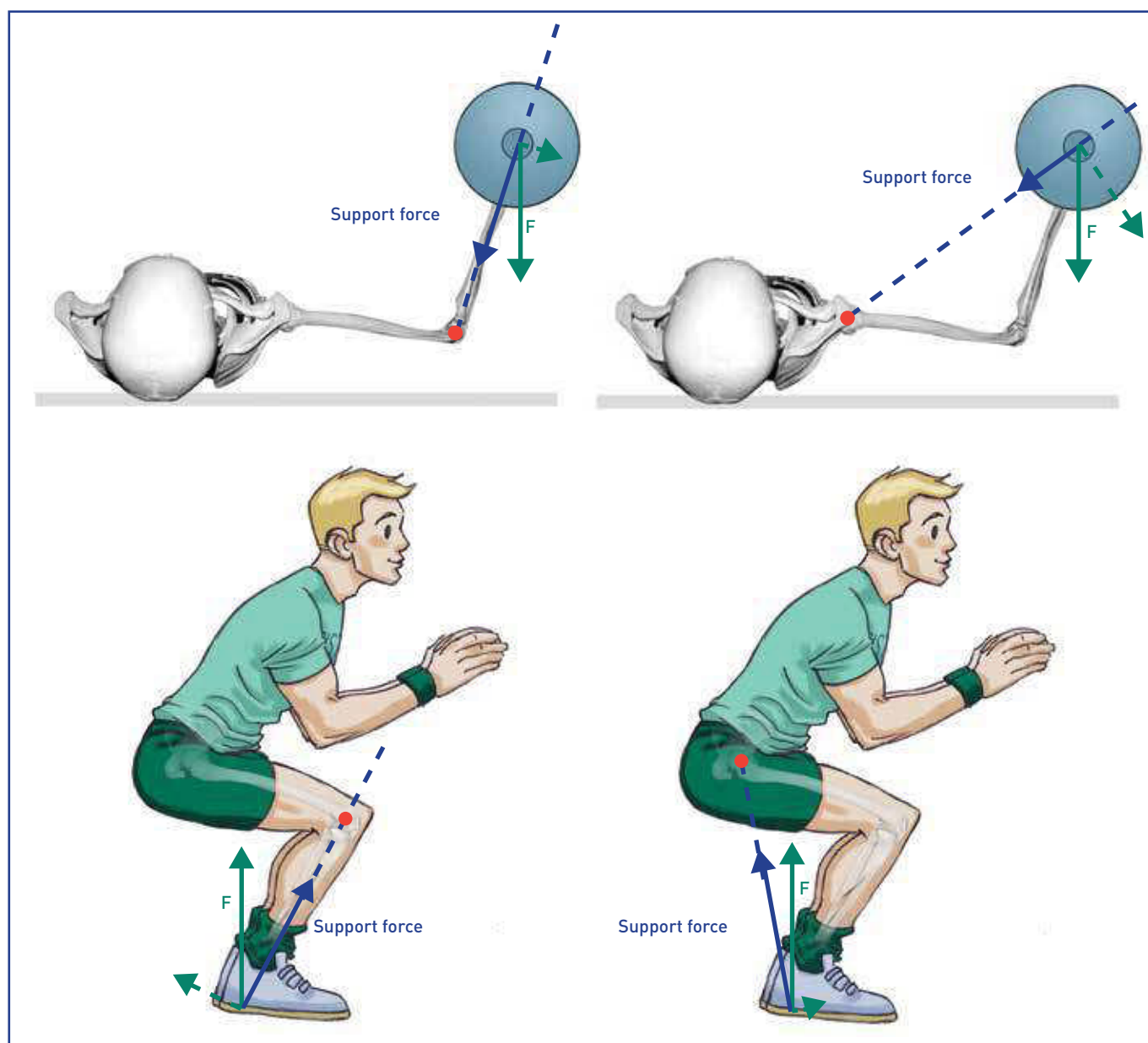
LEVER SYSTEMS



Whenever a force causing a pushing action (press) is applied, it is usually within a support system (because under usual circumstances, forces that reach different joints are of the support type). The opposite is true for pulling actions, that usually lay within suspension systems.

Picture 3.7.21

In this picture, the translational component aims opposite to the direction of rotation of the axis. The force applied to the lever here can therefore be called suspension. Simply put, whenever the angle of force is greater than 90° , forces act as suspension on the lever.



Picture 3.7.22

Examples of support systems, where the translational component aims towards the different joint axes.

PHASE 3: CHANGING THE GEOMETRY OF THE LEVER

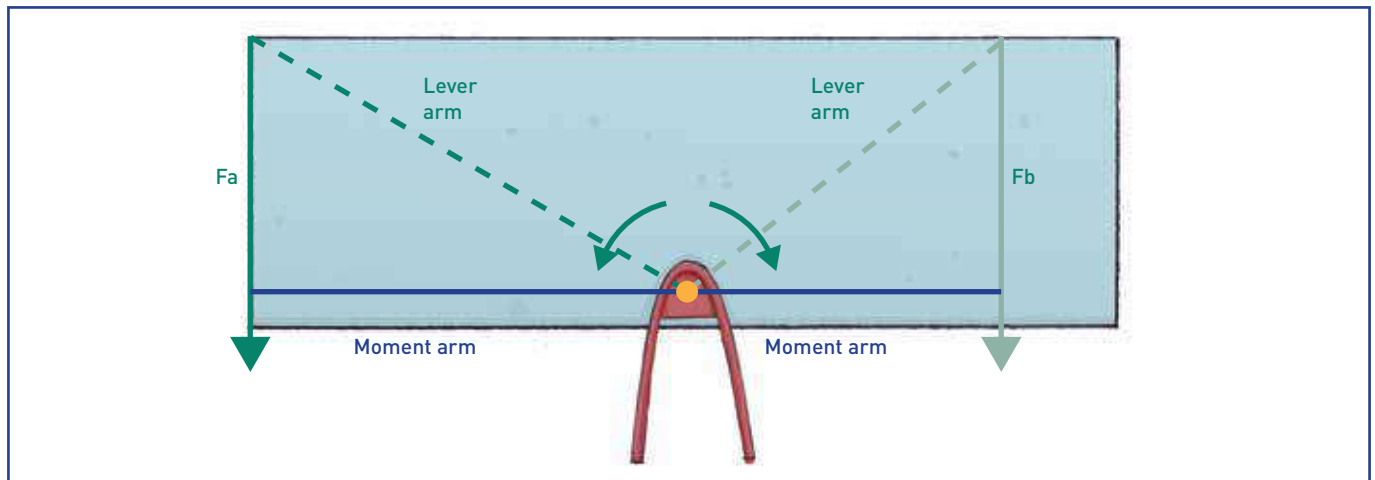
Engineers define levers as lines that run directly from the axis to the point of application of the force.

How is this different to the examples of levers we have been looking at so far?

The difference is that engineers ignore the visual shape of the object. Instead, they merely consider the actual lever that is hidden within.

This is a key concept if we want to understand what really constitutes the lever, upon which the force is exerted.

The simplified model, often depicted by a seesaw, must be surpassed. In order to allow for a deeper comprehension, a series of lever systems have been developed. These should be useful to fully understand the reality of lever mechanics.

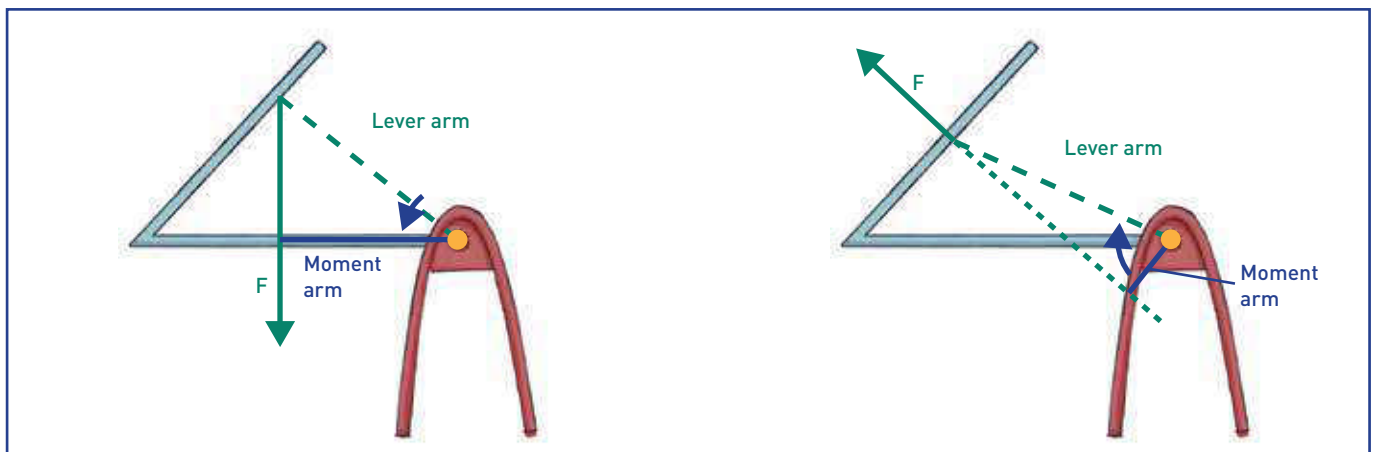


Picture 3.7.23

Lever arms do not follow any specific structure; it is always defined by drawing a straight line from the point of application of the force to the axis of rotation.

The appearance of the lever could be misleading. The real lever in the illustration above actually corresponds to the imaginary, dotted, line. This line will be key when trying to visualize mechanical levers and determining the direction of rotation generated by a force around a given axis.

Lever arms are straight lines running from an axis to the point of contact of a force, and they create mechanical levers.

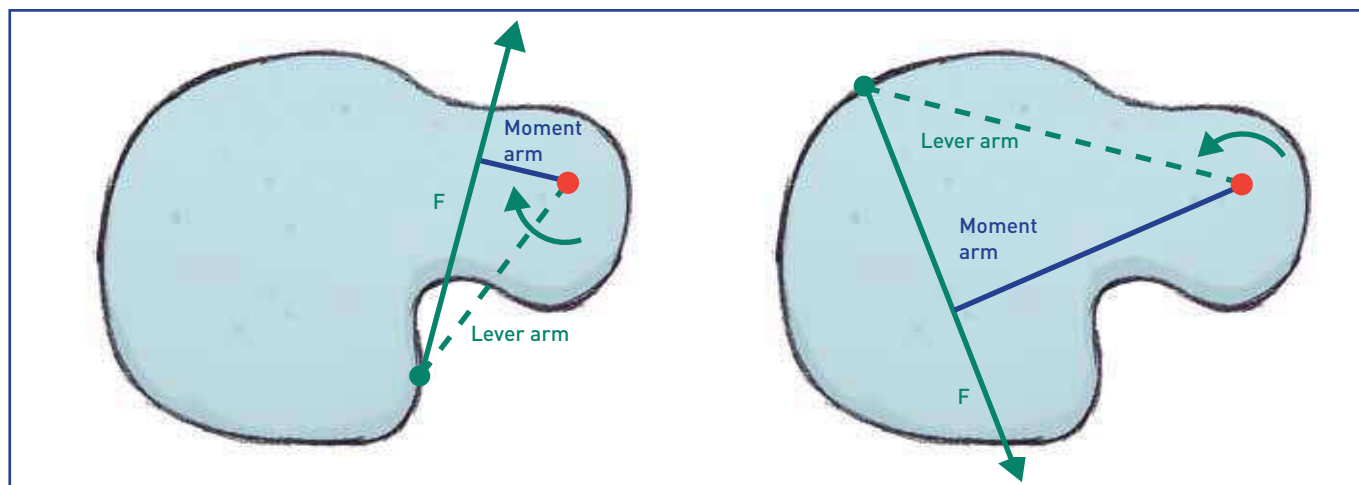


Picture 3.7.24

One can easily be misled when trying to establish what makes up the lever. Lever arm is the line that goes from the point of application of the force to the axis of rotation.

LEVER SYSTEMS

Picture 3.7.24 exemplifies how a lever does not necessarily need to be a straight bar. It also shows how the geometrical shape of a structure not always correlates to the lever arm. Unfortunately, the traditionally simplified lever models tend to confound shape and lever arm.



Picture 3.7.25

Levers may come in various shapes and be different from traditional seesaws. In order to calculate the lever arm, however, only one thing should be considered: the shortest straight line between the axis and the point of application of the force.

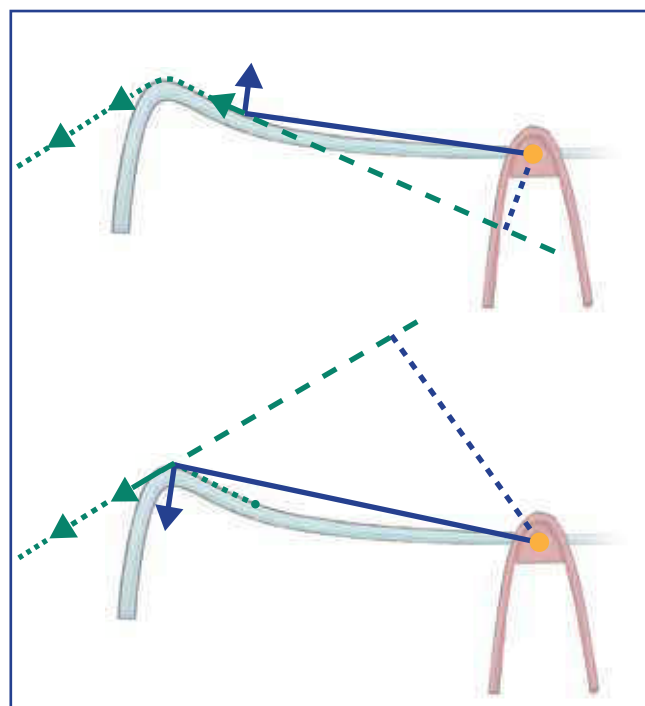
PHASE 4: REDIRECTIONING OF A FORCE AND LEVER SYSTEMS

There is something else which often comes along in human lever systems: forces –and muscle forces in particular– do **not at all times act linearly** upon levers.

Sometimes, forces do not attack structures (levers) in a regular linear manner. Whenever this is true, the direction of the force responsible for the generation of torque (i.e.: rotation around the axis) equals the direction the force had prior to it losing contact with the structure. This small detail greatly changes things when talking about the human body. Many calculations for muscle-generated torques will be affected by this, and this content will be covered later on in the muscle mechanics section.

Arthur Jones had this precise thought in mind when he designed eccentric pulleys for variable resistance machines. In them, the point of contact between the force and the pulley remains constant (it's always at the site where the cable wraps around), but the oval shape of the pulley redirects the force, causing the moment arm to change as the lever spins around the axis.

In the following chapter, we will see redirections of forces being applied to muscle mechanics. These cause significant mechanical alterations to much of the human musculature.

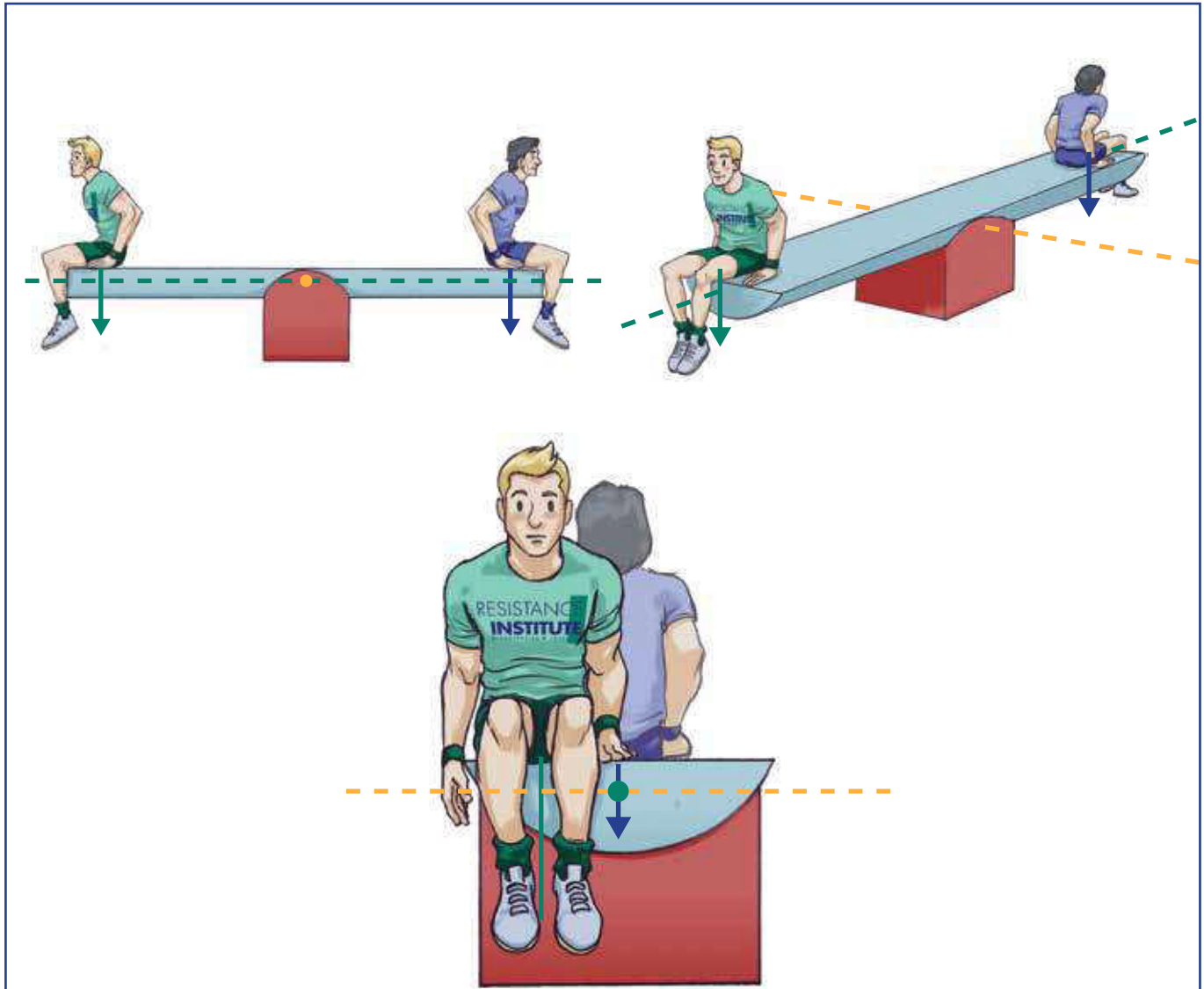


Picture 3.7.26

The picture on top shows how the muscle's insertion site has been chosen as the point of application of the force. THAT IS NOT CORRECT. The bottom picture shows how it should be done, instead. The correct line of force is that which originates from the last point of contact between the force and the structure.

PHASE 5: THREE-DIMENSIONAL LEVER SYSTEMS. NON-COPLANAR FORCES ACTING UPON LEVER SYSTEMS. THREE-DIMENSIONAL TORQUE

WHEN A GIVEN FORCE ACTS ON MORE THAN ONE PLANE



Picture 3.7.27

In the upper left hand picture, it may look as if both acting forces would counteract each other. However, by looking at it from a different perspective (see bottom picture), it becomes obvious how the character in green is additionally generating a rotational force around the longitudinal axis of the seesaw.

Another noticeable handicap of traditional simplified lever system models is that they don't allow for a full analysis of the force. For a full analysis, three-dimensional views and calculations are a must-have, must-do. Traditional seesaws are depicted from a specific view. However, this perspective to be changed, like it has in the picture above. And watching from a different perspective could bring up new information, like there may be other planes in space where that given force may be generating a rotational force.

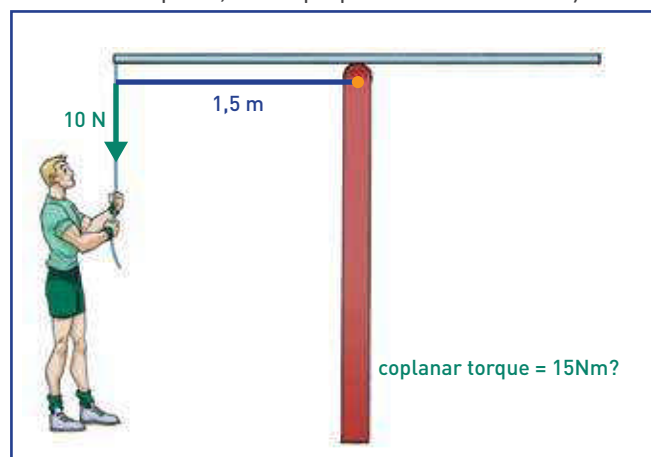
In picture 3.7.27, it can be seen how two forces that cause equal torque in the transverse axis (bi-dimensional perspective), do not in fact cause equal three-dimensional torque.

In order to be able to see all rotational forces affecting a system, one would have to either look at the system from different perspectives (three axis of space), or make calculations with three-dimensional programs.

LEVER SYSTEMS

CALCULATING WITH NON-COPLANAR FORCES

Whenever a force is not fully on the plane that is perpendicular to the axis, only part of that force will generate torque around that given axis. To be more specific, torque will only be caused by the component of that force that is on the plane, and is perpendicular to the axis).



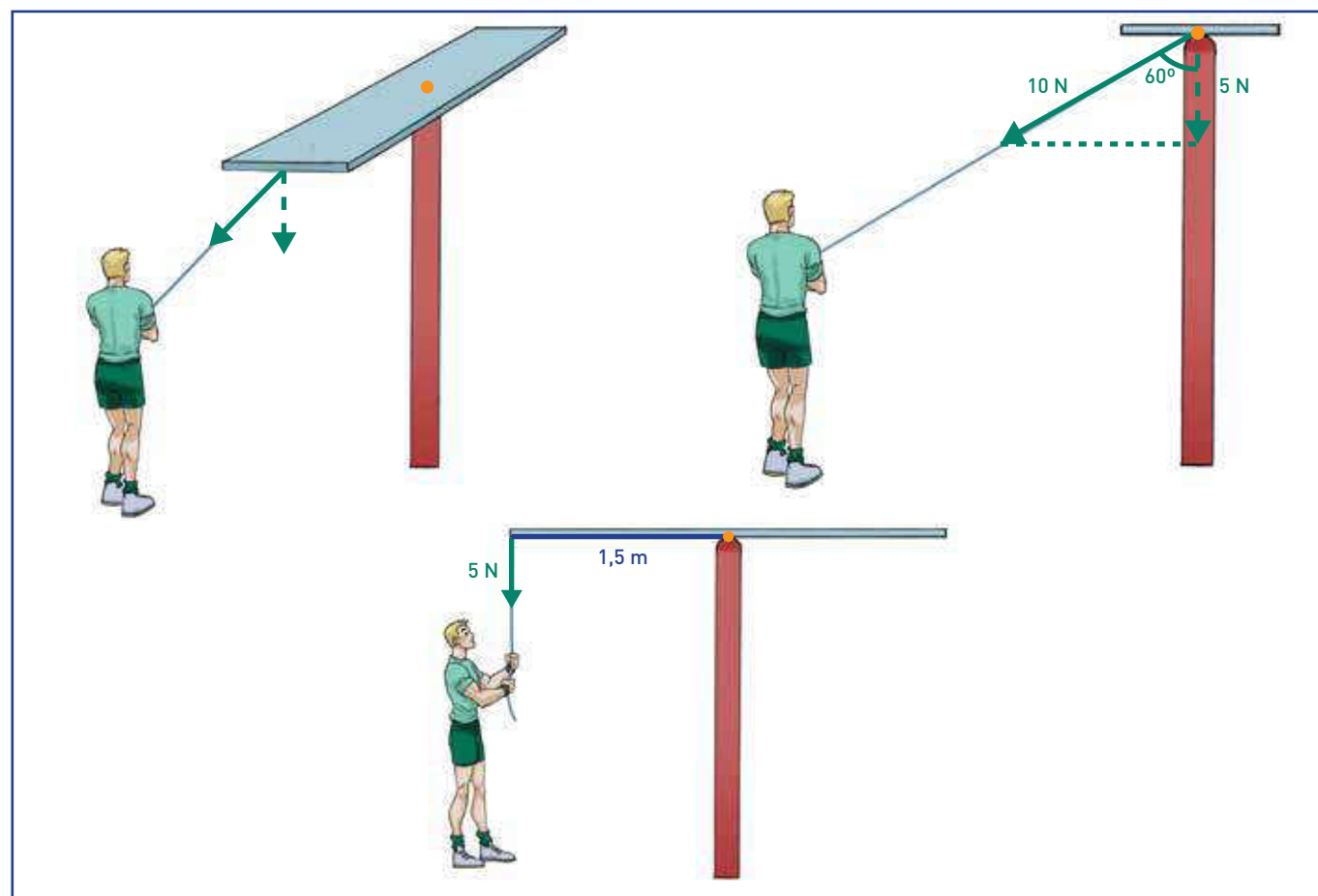
Picture 3.7.28

This means that, in order to calculate how much torque is generated by a force on a specific plane that is not in the plane of the force, only the part of that force that is applied on that plane holds mechanics over it (see picture 3.7.29).

It should also be borne in mind that it is possible for a part of the force to apply in a direction where the joint doesn't allow for rotational movement. Should that be the case, forces in the linkage/joint would be created, which would not result in rotational movement. Rotation will only result from the component of the force that lies on the plane that is perpendicular to the axis causing the rotation. This fact adds on a new variable in the studying of lever systems.

Co-planar torque (in a spatial axis) =

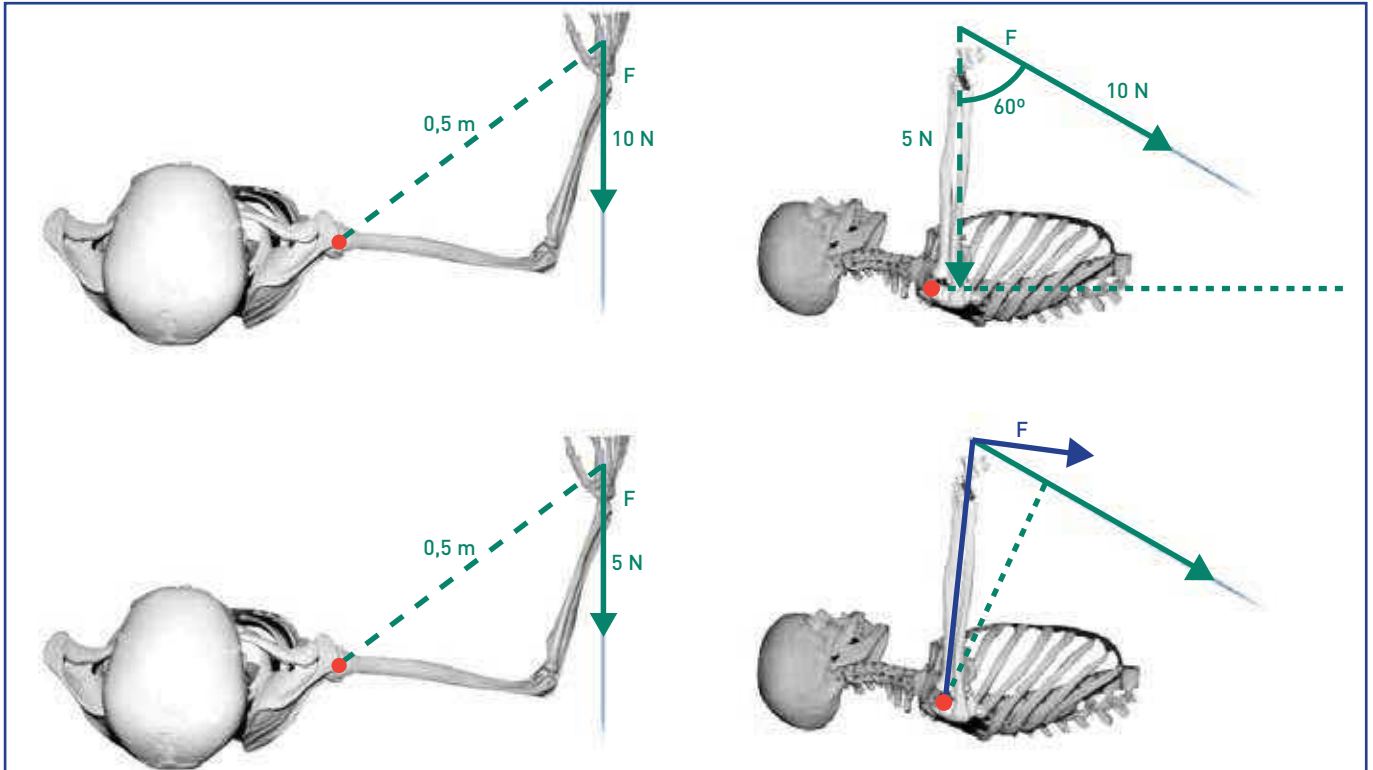
Component of the force (co-planar) that is in the plane * Lever arm * $\sin \alpha$



Picture 3.7.29

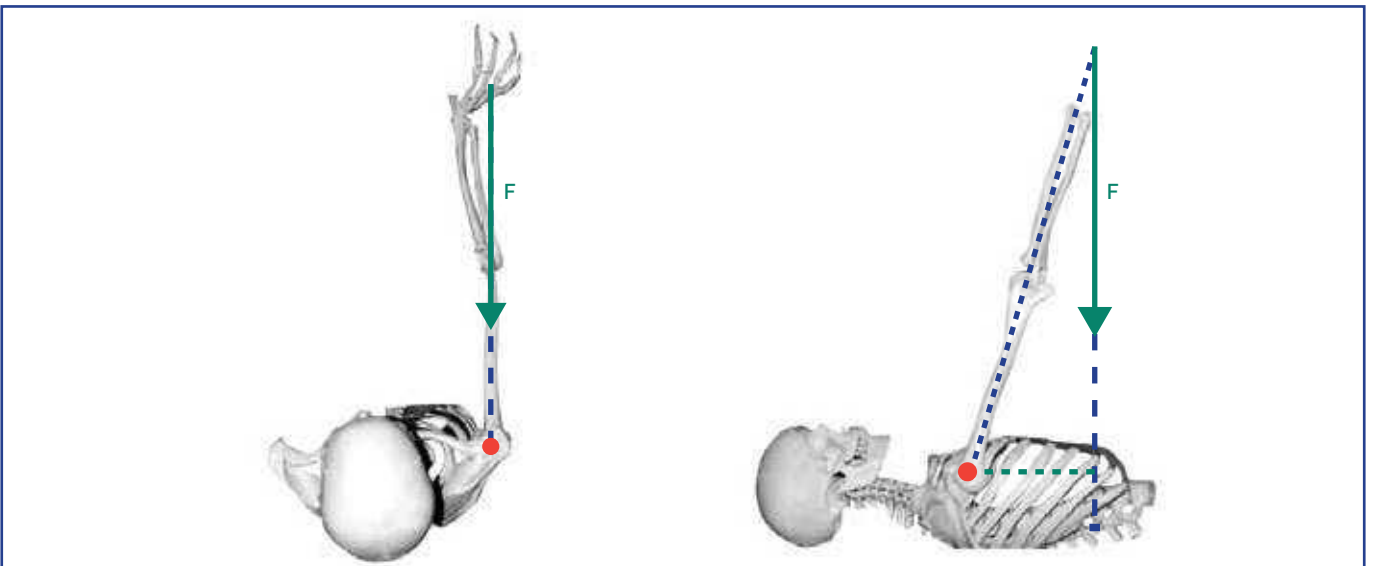
One single mechanical situation, seen from different stand points. When a non-coplanar force is being applied upon a given axis, only the part of it that acts on the plane of the axis will generate torque. In this specific scenario, only 5N of the 10N being applied by the character are, indeed, in the plane of force.

Given the three-dimensional structure of our bodies, it is highly probable that forces acting upon it—specially muscle forces—cause torque around more than one axis within a given joint.



Picture 3.7.30

By looking at the upper left hand picture, we can see that the cable applies a 10N force, and it causes a rotational force around the shoulder S-I axis (horizontal abduction). If we look at it from a different place (right lateral view), it stands out that the same force attacks at a 60° angle, relative to the transverse plane (upper right hand picture). The present angle of application of the force implies that only 5N are being exerted within the transverse plane, and not the 10N available in the cable (bottom left hand picture). By taking another right lateral view in the sagittal plane (bottom right hand picture), it can be observed that this force also generates internal shoulder rotation.



Picture 3.7.31

Whenever a force is only just analysed from a single perspective (see picture on the left), it could be assumed that the force causes no torque whatsoever. However, the same picture being looked at from a different perspective, makes it clear that the given force is causing shoulder extension (picture on the right). This is the reason why, in bidimensional analytics, various perspectives should be analysed, prior to calculating rotational forces.

LEVER SYSTEMS

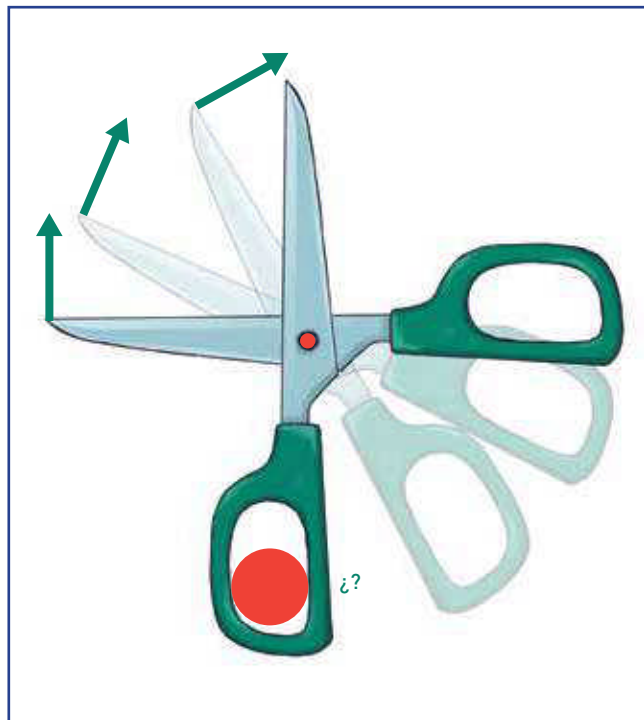
PHASE 6: LEVER SYSTEMS WITH MULTIPLE INTERRUPTIONS (multiarticular). THE HUMAN BODY AND ITS RIGID LINKAGES. EFFECTS OF BIOTENSEGRITY

Studying multiple-lever systems in the human body is a highly complex task because, unlike industrial machinery, human joints usually combine shape closures with closures by biotensegral force.

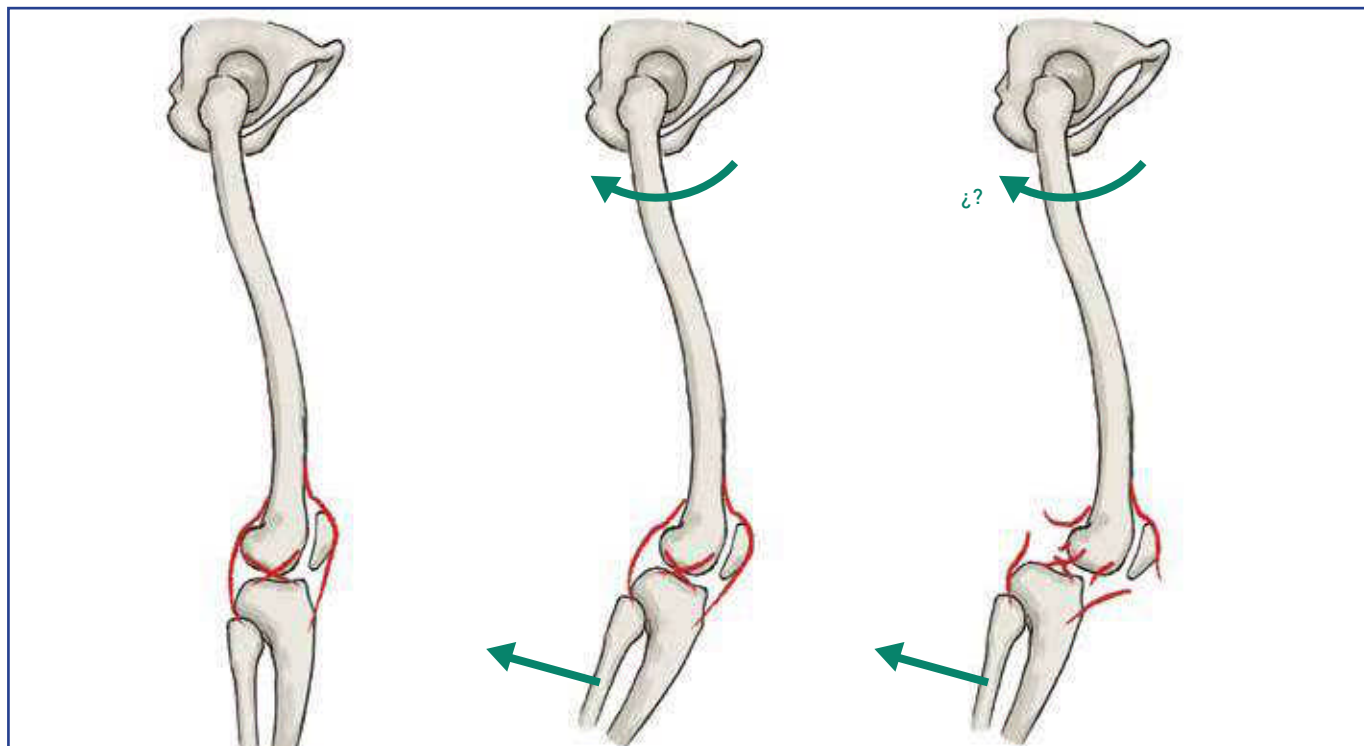
Resistance Institute undertook a simple experiment, in which a pair of scissors got attached to an external linkage site, thus creating a new axis (picture 3.7.32.a). The result differs from what happens in the human system. With those scissors, as long as the force got applied at a 90° angle to the blade (green arrows), no torque was generated on the external axis (large red dot).

What happened with the scissors does not happen in human body systems. Whenever a force is applied to a human body segment, the force is transferred to another segment, regardless to the angle of application.

In the picture below (3.7.32.b), you can see how tissues in one linkage (joint) transfer the force on to another joint. The tissues that have been depicted on this drawing have been very much simplified. However, they are clearly the reason why this is referred to as a rigid linkage, without which forces reaching the tibia would not go on to cause torque at the hip joint.



Picture 3.7.32.a



Picture 3.7.32b

HUMAN CLOSURES AND BIOTENSEGRITY

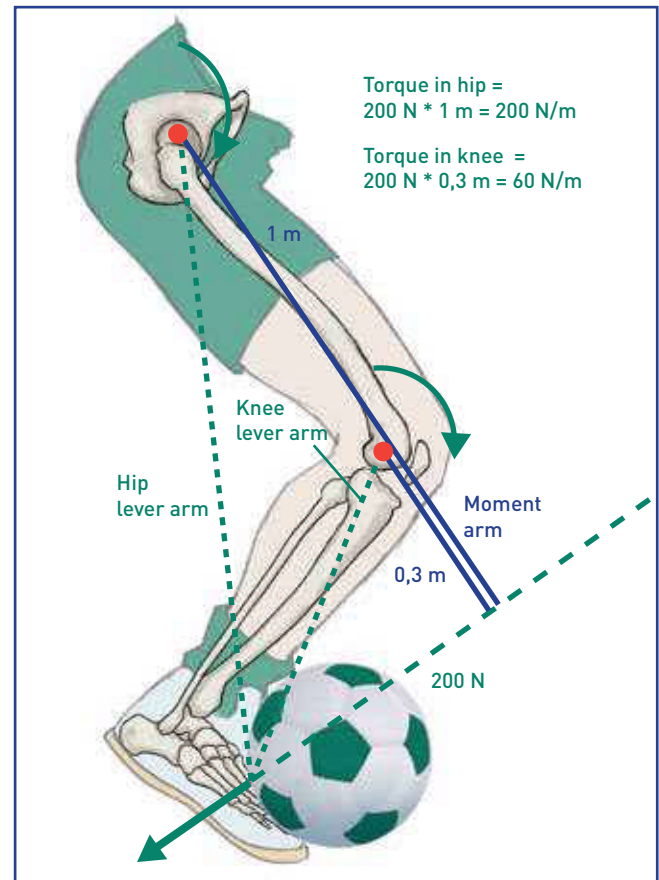
The relationship between lever systems and Dr. Stephen M Levin's research on biotensegrity is still being analyzed. It is possible –though currently just a hypothesis, that the key to understanding the relationship between the various human joints may lie in the combination of a shape closure (the linkage's structural geometry) and a force closure (tension between the linkage's tissues).

Tensegrity may be key to understanding some of the behaviours observed within human levers.

It could also be responsible for the force transmission between linkages and segments. This would explain how a force being applied upon a distal segment is able to cause torque on more proximal joints.

Whenever the structure itself (geometry of the linking segments) is unable to guarantee integrity, the system takes on force closures by means of tensors, which help to bring about an interruption.

These characteristics generate a rigid union (through tensegrity) in between the human joints. As a consequence, forces are transmitted to the different segmentary and joint levels, as if it were a rigid union. That is, as long as the stability created within the different joints is adequate.

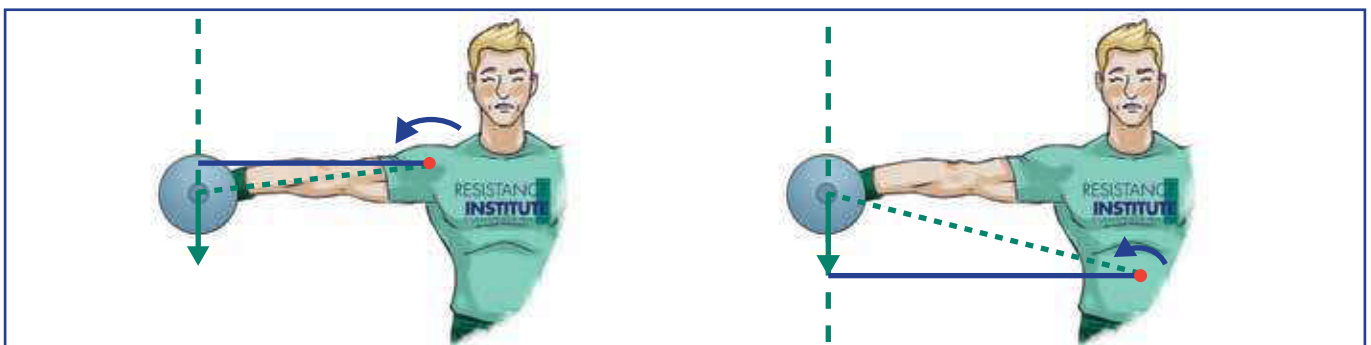


Picture 3.7.33

Because the knee and hip joints are rigidly linked by tensors, the contact force exerted by the ball will affect both joints. The rigid union will take place whether or not there is movement at the knee.

PECULIARITIES:

- Integrity in the linkage depends on the modulation of the different tensors involved in this tensegrity. Most specially, it relies on muscle tension, which is regulated by the nervous system.
- In order to calculate torque in joints distal to the point of application of force, all proximal joints should be contemplated as rigid linkages. Picture 3.7.34 shows how, in order to calculate torque for the A-P axis of the shoulder, or for the A-P axis of a spinal level, the rest of the body needs to be considered as a rigid linkage.
- A lack of adequate stability within a segment can significantly affect the stability at other levels of the structure.
- The nervous system plays a key role in managing the ideal tension required to maintain integrity.



Picture 3.7.34

LEVER SYSTEMS

KINEMATICS IN MULTIPLE-AXED LEVER SYSTEMS

Whenever a system consists of multiple levers, kinematic instant axes of rotation/glide can exist. The latter do not result from an interruption at a linkage. They only take place when a combined movement occurs, in more than one actual axis, within a lever system with multiple axes.

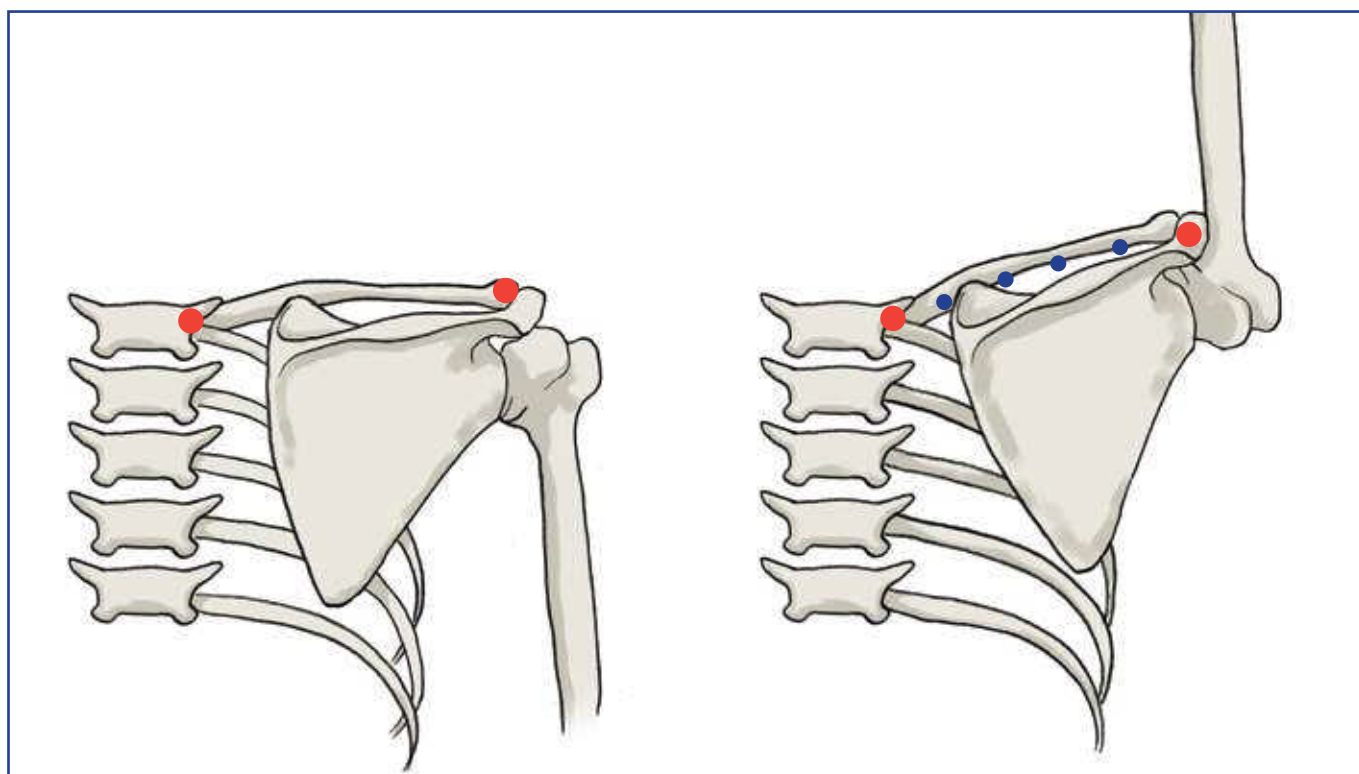
This combined movement is dependent on the relationship between the movement of the axes at each one of the interruptions. Thus, it need not necessarily describe an arch.

Using these axes may be of use when doing kinematic calculations. However, it has so far not proven useful for the calculation of exercise moments, since it does not allow for the calculation of forces nor moments upon the kinetic axes in the linkages.

KINEMATIC VS KINETIC AXES

Kinetic axes, which can be found in linkages, are the real ones. Forces are transmitted on them in order to generate movement and maintain linkage integrity.

Trying to calculate forces around a kinematic axis is inappropriate, for it is the kinematic result of two or more kinetic axes, which could be moving at different velocities and directions, depending on the forces affecting the linkage.



Picture 3.7.35

The picture on the left shows two real kinetic axes, at the sternoclavicular and acromioclavicular joints. In the picture on the right, several additional kinetic axes have been added, which could result from the combination of rotational movements at these two joints.

KINEMATICS OF MULTIPLE LINKAGES, IN ORDER TO ACHIEVE OPTIMAL SEGMENTARY GLIDING

In comparison to that of a simple articular motion, linkages with multiple axes are able to achieve greater segmentary gliding, on any given plane.

The human body presents some clear examples. Human motion results from the sum of various articular axes, that work in a synchronized manner in order to achieve the desired segmentary glide. At times, a given articular axis may 'cover up' for movement. This should not always be considered a problem or a compensation. It could as well be the opposite: motor efficacy.

PHASE 7: ARTICULAR FORCES. MAINTAINING STABILITY AROUND THE INTERRUPTION. FORCE AND STABILITY LINKAGES

Forces that are applied onto a structure within a linkage **do not merely create rotational moment**. It could be stated that moment is created because the linkage holds together and, somehow, the shape of the joint maintains its integrity.

Many of the forces that establish contact with the lever will try to break the interruption.

In shape or structural linkages, since they only allow for movement to take place around the axis, it is the geometry itself that tolerates and counterbalances the attempt to exit the joint. When talking about human joints, the situation is a bit more complex.

For the combined shape and force linkages, tension generated by connective tissue and the musculature itself needs to be adequate, in order to maintain a relative instant axis of rotation/gliding. Not only that, but the generation of an instant axis of rotation/gliding is partly due to the tension provided by the muscles and connective tissue.

In the next chapter, you will discover how one of the keys to maintaining joint stability is having the muscles generate force with the correct timing.

It is at this stage where integration of traditional lever systems with the concept of biotensegrity could make sense. This could create a complex mix of structural supportive forces and tensile forces, which together could help maintain the integrity of human joints.

Moreover, it is frequently stated that ligaments and connective tissue are the sole controllers of the articular axis, and people forget that musculature is one of the keys for the generation of stability around a shape/force linkage. Unfortunately, the following sentence is still too often pronounced: 'muscles generate movement while ligaments stabilize.'

CONTACT FORCES WITHIN LINKAGES

In order to be able to study the forces reaching a linkage, the following differentiation should be made: forces that act upon a structure (as seen in the previous chapter) and the type of forces which reach the linkage.

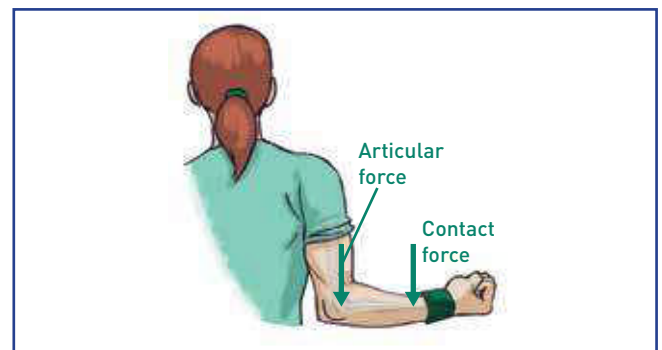
Contact forces within a joint depend upon the following:

- Type of contact force (compression, traction, shear).
- Geometry of the linkage.

The type of force reaching a linkage will depend on the direction and the way of the contact force reaching that structure. And also, on its relationship to the shape of the contact surfaces in the linkage.

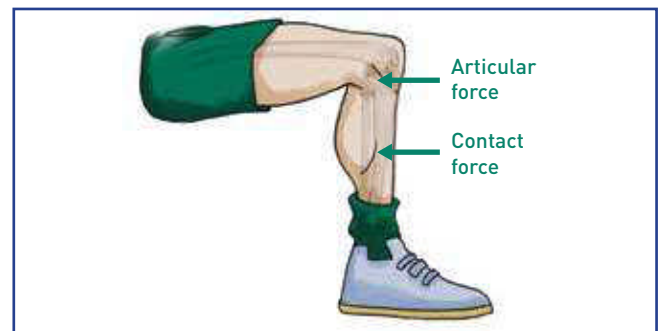
Total forces acting upon a linkage result from the summation of all forces reaching it, whether parallel or perpendicular to the contact surfaces.

In other words, when applying a given force on the body, the fact that its translational component runs in the direction towards the axis does not, by all means, imply that the contact force within that linkage shall be compressive. That will totally depend on the geometry of that given axis.



Picture 3.7.36

Compressive force onto the structure, causing a traction force in the union.



Picture 3.7.37

Compressive force onto the structure, causing shear force in the union.

LEVER SYSTEMS

3.8 RESISTANCE PROFILE AND LEVER SYSTEMS

PROFILE OF A GIVEN RESISTANCE THROUGHOUT THE ROM (STATIC)

Resistance profiles show how much of a resistance affects a given axis, throughout a given articular ROM. It should be noted, that resistance profiles can also be calculated as a function of the application time. Resistance profiles are also known as resistance curves.

Resistance profiles do not necessarily offer exact digits, but rather show **changes in the resistance throughout the ROM**. Because of this, should two different forces hold the same proportions throughout the joint's ROM, both of the resistance profiles would look the same, in spite of both loads being different.

One single resistance profile could also indicate various resistance intensities. When this is so, the curve would show the same shape and proportions, but it would be higher up or lower down, depending on whether the intensity is of a greater or lesser magnitude.

Resistance profiles show how a given resistance behaves throughout the ROM, thus indicating which areas may require of a proportional increase or decrease of the stimulus. This profile indicates how much rotational force exists, at each point of the articular range.

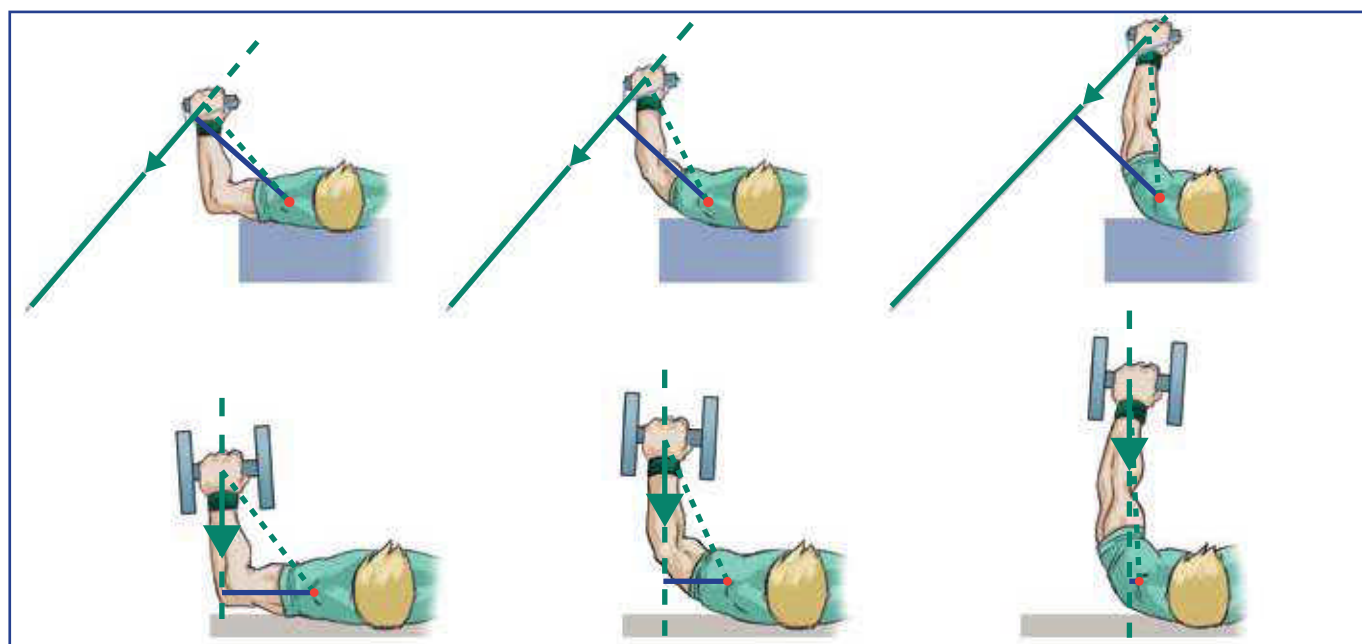
In order to calculate a profile, all you have to do is figure the torque for the various points of ROM, and for each of the articular axes subject to rotational forces.

Following are the variables that are needed, in order to make this calculation:

- **LOAD**
- **MOMENT ARM** (you will need to know the direction of the resistance, as well as its point of application).
- **PERCENTAGE OF COPLANAR FORCE THAT IS PERPENDICULAR TO THE AXIS BEING ANALYZED.**

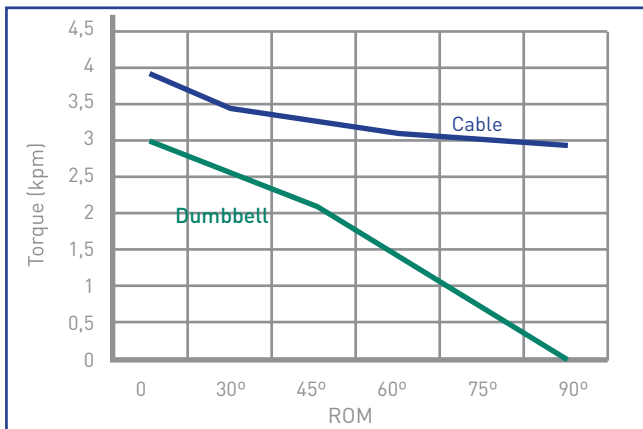
In order to exemplify this, let's take a look at how different the resistance profiles look for one given move (horizontal plane press up, relative to the shoulder joint), depending on whether it is dumbbells or a cable machine that is being used. Picture 3.8.1.

The curves obtained reflect the relative torque received, for the glenohumeral joint, along a 0-90° ROM, considering its S-I axis (relative to the thorax). That is for movements of horizontal abduction/adduction.



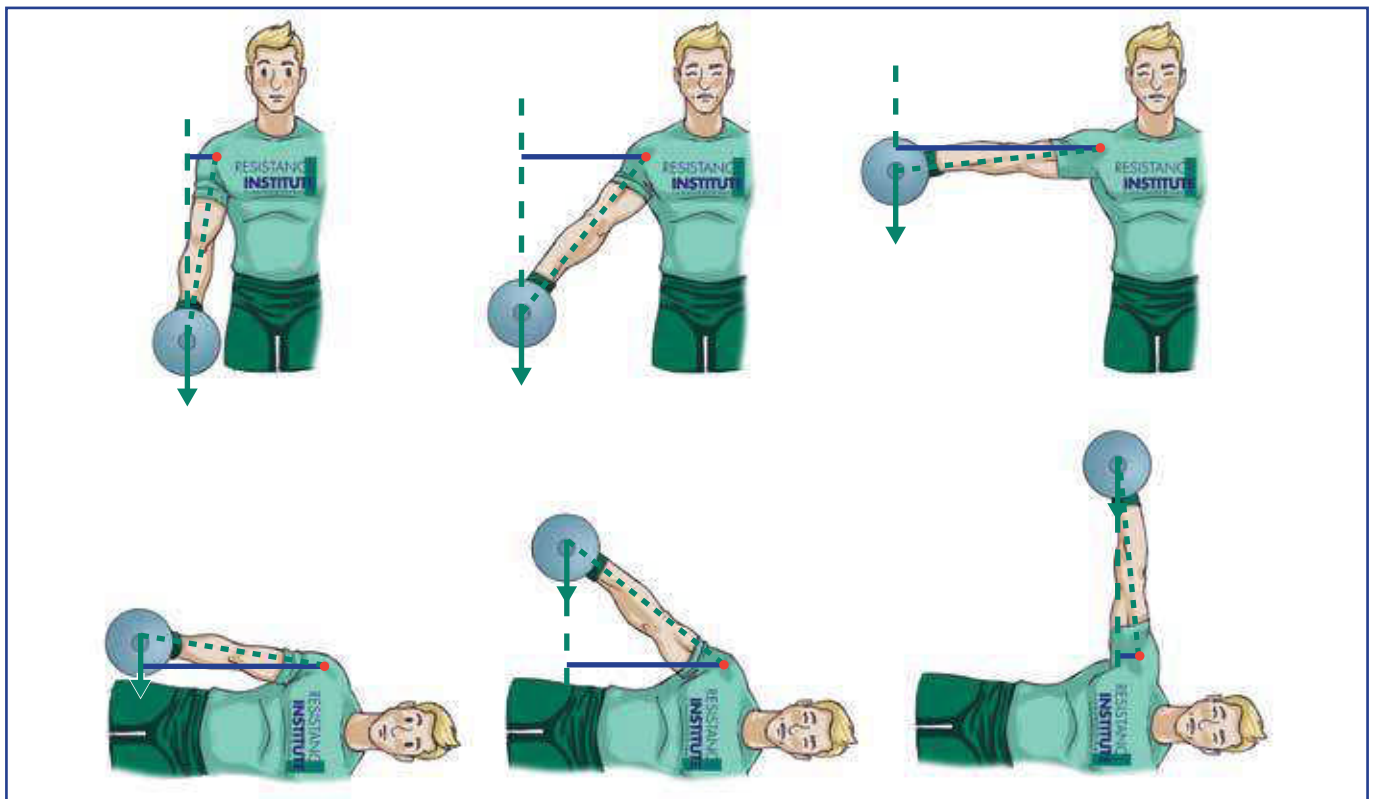
Picture 3.8.1

All of the drawings above consider one given contact force (load) in the transverse plane. The resistance profile will be determined by the varying moment arm (blue line), throughout the ROM. Note how the evolution of the resistance profile is different from one exercise to the other. Even though the load is the same throughout the ROM, the resistance profile in one exercise is different to the other.

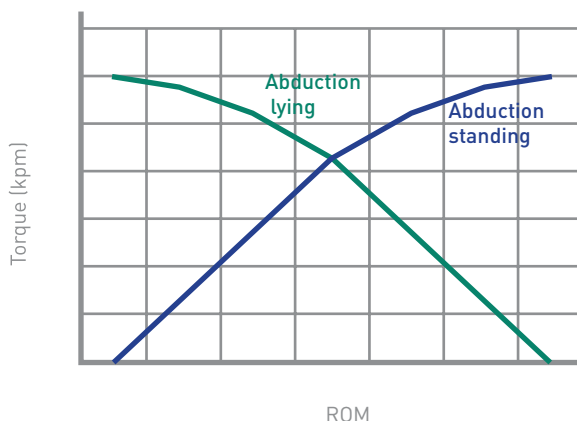


The graph in picture 3.8.2 depicts the resistance profiles that correspond to picture 3.8.1. The blue curve shows the profile for the movement that used a cable. And the green curve is the result of the dumbbell press. Please note that the force (load) has been established at 10 kp for both moves, and inertial loads have been intentionally disregarded. Distance values chosen are those of an average-sized individual. However, these could change from one person to another.

Picture 3.8.2



Picture 3.8.3



The picture above (3.8.3) shows a shoulder abduction move, being executed with a dumbbell, from neutral into 90° of shoulder abduction. The top line shows the man performing this while standing on his feet. Whereas in the bottom row, he is doing it while lying supine. The load in both examples is the same, but the way in which the resistance profile progresses is different. Take a look at the graph in picture 3.8.4. This is the graph for the resistance profile to these two variations. Both are moves in the A-P axis of the glenohumeral joint, for a shoulder abduction movement with dumbbells. And you can tell from the graph that there is a considerable difference between the two.

Picture 3.8.4

LEVER SYSTEMS

The graphs that have been seen in the book so far, showing resistance profiles, are mere indicators of the **rotational force** throughout the ROM of one given joint –the glenohumeral joint.

These curves relate to and are indicative of the examples in the pictures. If we take for instance the picture showing a press with a cable machine, the shape of the curve will also depend on the direction in the cable. This will noticeably affect the resulting torque profile.

For educational purposes, these graphs have been simplified and do not take into account the effects of inertia (inertial loads). The latter would substantially alter dumbbell-induced contact forces and, thus, the end profile for that

resistance.

PUTTING IT ALL TOGETHER

The graph in 3.8.2 indicates the coplanar torque reaching the glenohumeral joint, relative to its S-I axis (reference thorax). This leads us to thinking that, in order to obtain more information about the torque being produced at other axes of that same joint, the same resistance profile calculations would have to be done for the desired axis, should some torque affect it.

Total resistance for one given joint equals the sum of all moments of force, looked at in a three-dimensional manner.

3.9. VELOCITY AND INTRODUCTION TO DYNAMIC MECHANICS

‘IT IS NOT VELOCITY THAT AFFECTS FORCES, BUT FORCES WHICH AFFECT VELOCITY’

Marc Moragues

‘IT IS NOT THE FALL THAT KILLS YOU, IT’S THE LANDING’

Tom Purvis

VELOCITY AND MOVEMENT EXECUTION

Velocity is often regarded as a key factor to understanding the way forces act when movement is involved. However, things are little bit different. What really matters is not velocity, given that velocity remains a constant, as long as no forces enter the equation. What is truly relevant is the force generating acceleration (whether it is positive or negative) and/or a change in the linear direction of a given mass.

It is important to note that accelerations are a consequence of forces and show the relationship between one given force and one given mass. Acceleration can only take place as long as there is a force or force imbalance acting upon a given mass.

What is **MOMENTUM**, then?

Momentum is an expression that is commonly used in the exercise field to describe the factor that affects a badly controlled movement using resistance. Momentum is determined by mass and velocity ($m \cdot v$).

Since velocity remains constant in the absence of forces, momentum is also not as relevant in resistance training as one may think. It will remain constant, as long as no other force interferes.

KINETIC ENERGY (KE)

Kinetic energy is the energy a given object (mass) possesses due to its own movement. Its formula is $EC=1/2mv^2$. In order to allow for a change in KE, acceleration (change in velocity, either positive or negative) needs to take place. And, as stated before, acceleration only happens when a mass receives unbalancing forces.

Question: How much force would be required to stop a given mass, in the same amount of time, at a given distance, but with different speeds?

The lesser the distance, or the lesser the time allowed for the change to happen, the greater the force that will be necessary to accomplish it. Because KE is determined by velocity squared, the change is exponential. This means that a given mass, moving at twice the velocity, has 4 times more KE. By establishing a direct relationship between applied force and KE, it could be stated that: in order to stop that given mass, for the above specified context, the force required would be 4-fold.

Positive or negative accelerations generated due to the force of gravity only affect the Earth, since it is the Earth that creates them. In order for the effects of acceleration to affect the human system, they must be generated by internal body forces.

And thus, practically speaking:

‘A given dumbbell, that moves twice as fast as another one, has 4 times more Kinetic Energy. Therefore, for any given time and distance, and in terms of required force, it will cost 4 times more to stop it.’

INERTIAL LOAD AND THE EFFECTS OF INERTIA ON LEVER SYSTEMS

The effects of inertia on lever systems –better known as inertial moment– constitute an important factor for the understanding of the concept of resistance to change.

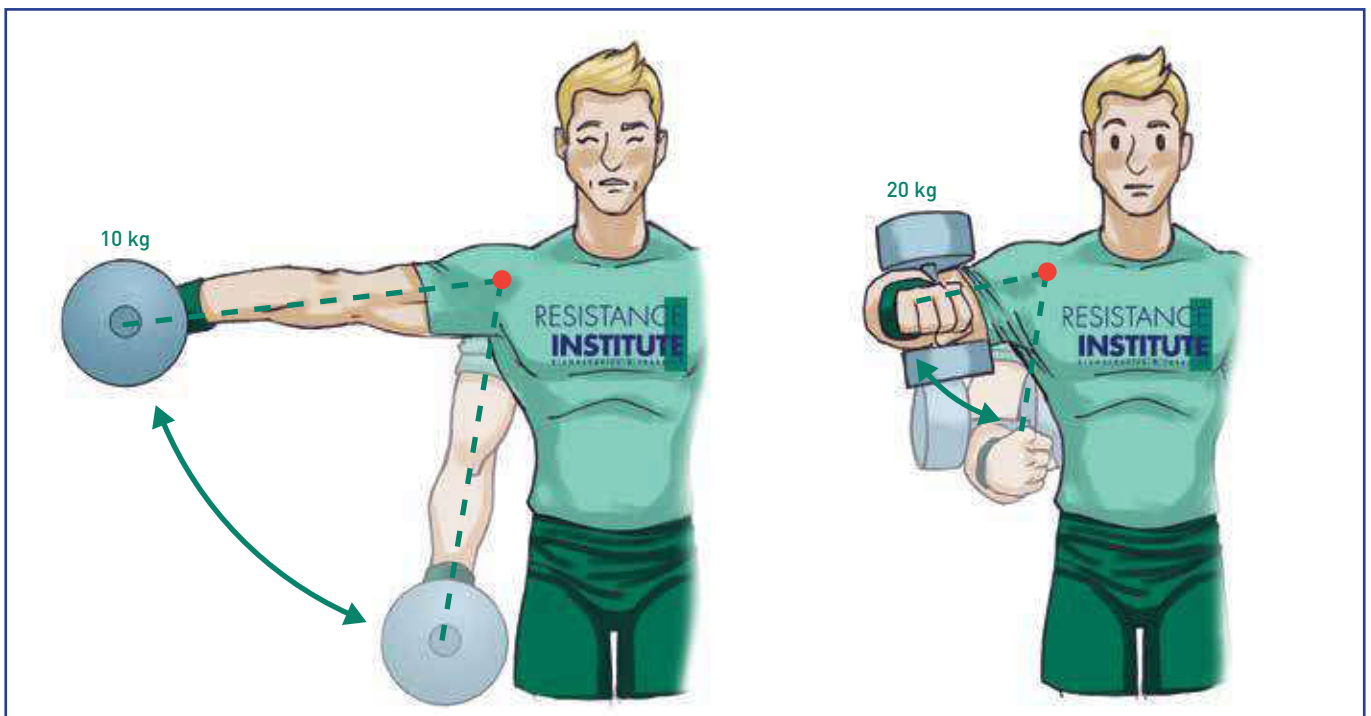
This is the formula for inertial moment:

$$\text{inertial moment} = m * r^2$$

Inertial loads on the body are affected by non-gravitational, acceleration-generating forces, at every point of the ROM. Consequently, the curve changes with every change of state. For any given angular movement, the force causing acceleration increases exponentially, relative to its distance-to-axis.

That means that, in order to achieve an equal amount of angular acceleration, an object that is twice as far to the axis, needs four times more force. Because it covers twice as much distance, with twice the torque.

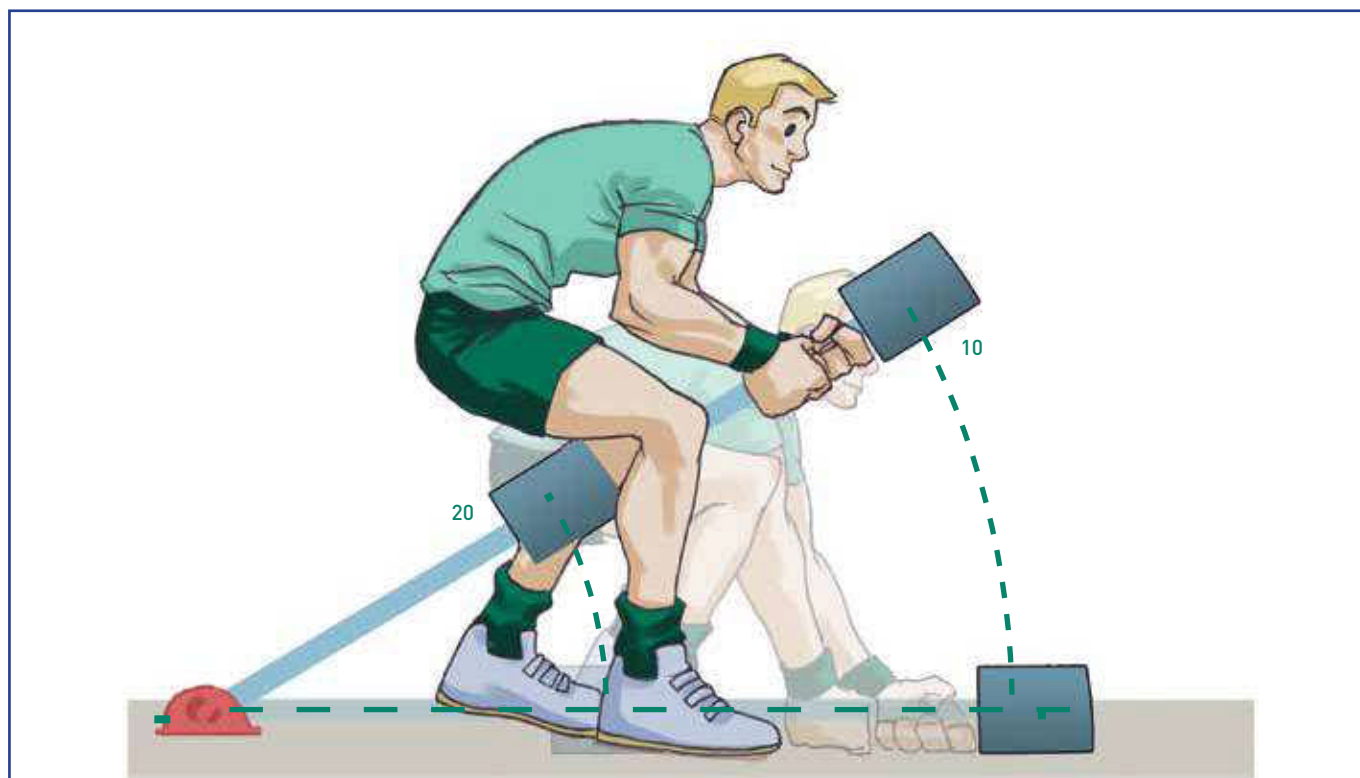
Even if only half as much load was placed at twice the distance, the necessary force to obtain angular acceleration (changes in kinetic energy) would be double. In this scenario, even if the force moment is the same, because that half load is placed at twice the distance, it will take twice as much force to either accelerate or decelerate the movement by means of internal forces. Now summing up on this previous example: the inertial force –reactive to the force generating internal acceleration, will be double for one given acceleration or deceleration ratio; or either the same, for half the acceleration or deceleration ratio.



Picture 3.9.1

The character on the left is abducting his shoulder, with a 10kp dumbbell and an extended elbow. The character on the right is using a 20kp dumbbell, placed at his elbow. The weight on both of the exercises offer equal amount of torque. However, the move on the left causes twice as much inertial torque than the move on the right. This means that, for any given speed of execution, twice as much force will be required by the character on the left in order to be able to accelerate and decelerate movement against gravity.

LEVER SYSTEMS



Picture 3.9.2

The character in this picture is doing a move called 'tip row or T-bar'. Two possibilities can be seen: in one of them, a 20kp load is placed at half distance; in the other, a 10kp load is placed at full distance. As was the case in picture 3.9.1, the static rotational force will be equal in both cases -here, we choose to ignore the bar's mass-. However, when the half the load is placed at twice the distance, the inertial torque also doubles, making it twice as difficult to accelerate and decelerate movement by internal forces.

FORCE PROFILES AND EFFECTS OF INERTIA

The effects of inertia on lever systems cause changes showing up in the resistance profile.

The inertial force derived from an angular acceleration acts on the lever at a specific angle. Needless to say, this angular acceleration is the result of an internal force—such as muscle force, acting upon an object. Whenever a gravitational load becomes angularly accelerated by an internal force, the direction of the inertial force is perpendicular to the lever. The reason for this is that inertial loads are of a reactive nature, thus always directing themselves opposite to the forces causing the acceleration of the lever, at any point of the ROM.

It is important to bear in mind that the lever's direction of movement is tangent (perpendicular) to the lever, throughout the arc of motion. Also remember that, in spite of the inner arthrokinematic motions, uniarticular movement is mostly rotational.

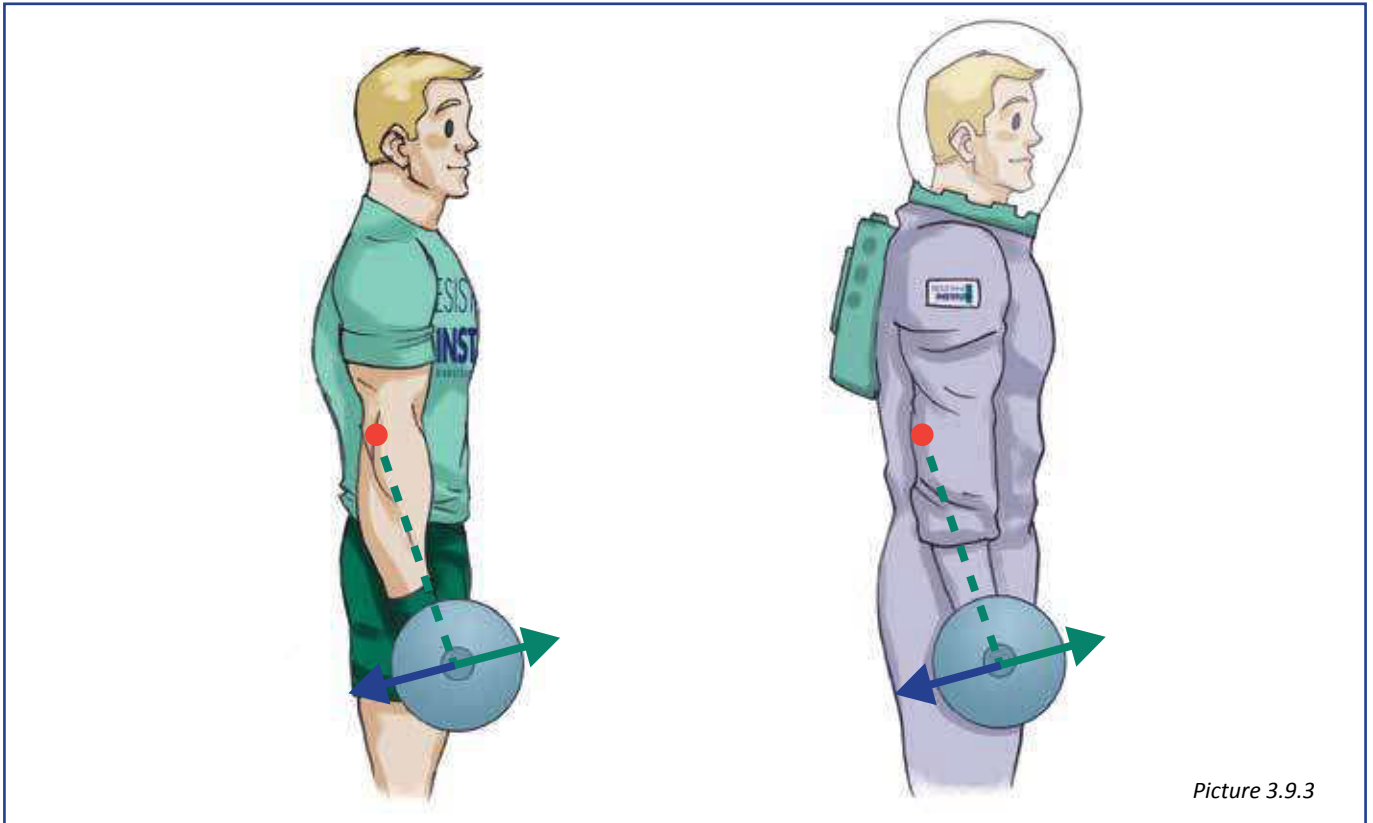
As explained in the previous chapter, reactive forces modify

the contact force upon the body (load), with a precise direction of the vector, and generating changes in the resistance.

THE EFFECTS OF INERTIA BECOME ALTERED WHEN GRAVITY-INDUCED ACCELERATION IS PRESENT, WHETHER THE LATTER IS POSITIVE OR NEGATIVE

As long as a body becomes either positively or negatively accelerated by gravity (not by the body's internal forces), the changes in kinetic energy incurred as a result will not create inertial forces in the joints.

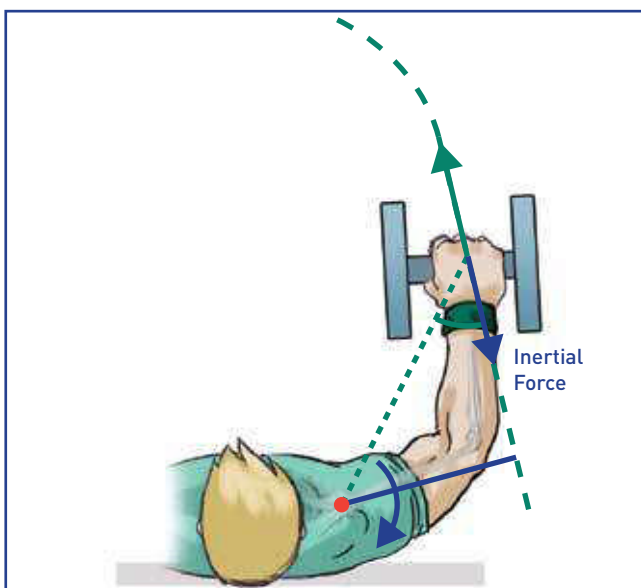
Taking this into account: negative accelerations (breaking) within the articular concentric phase and positive accelerations within the articular eccentric phase are due to the force of gravity (at least the majority of times). Should that be the case, the body would not be receiving inertial forces due to changes in the KE, since it would not have been the body that changed it, but planet Earth, instead.



Picture 3.9.3

The blue arrows represent the inertial forces that are reactive to the accelerating forces generated by the character (green arrows). Inertial forces follow the same line but opposite direction to acceleration forces. Acceleration forces affect the dumbbell, while inertial forces affect the body. Both in outer space as on planet Earth, these forces are present. The exception is that, on Earth, inertial forces will hardly ever show up when acceleration takes place in the direction of gravity, since it is gravity itself which produces them.

In a uniarticular motion, as long as the resistance's lever arm remains constant, the inertial load acts at a 90° angle. On the other hand, for multiarticular actions, the lever arm changes and the object does not necessarily move in an arc-like direction, all of which causes the angle to be different.



Picture 3.9.4

In a press move, reactive inertial forces (in blue) on a dumbbell, generated by internal forces are commonly off the perpendicular line to the lever, since movement is not purely rotational. The line for the inertial load of the dumbbell will be dependent on the line of the angular acceleration created by internal forces. The direction will depend on whether acceleration against gravity is taking place positively or negatively (that is, decelerating the dumbbell on its way down). This load generates both moment arm and torque. Both of them combine with other forces, and modify the forces of contact upon the hand (gravitational force is not depicted on this drawing).

LEVER SYSTEMS

This good 'attack' angle of inertial forces allows them to have significant torque, since most of the times they will almost entirely cause rotation on the lever. Understandably, this will depend on the amount of inertial force, which depends on the mass and the acceleration per time unit.

As mentioned, it works somehow differently for multiarticular systems, where the effects differ slightly for some joints, and the angle is not equal to 90°.

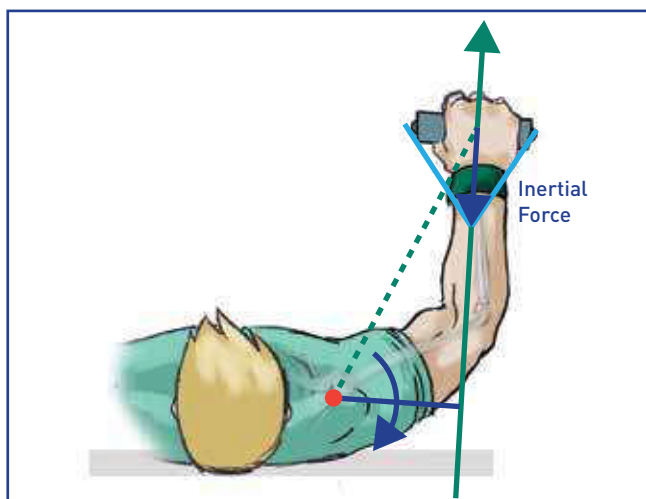
The use of cable systems represents an interesting scenario. What happens to inertial effects then? Keep in mind that in cable systems, inertial forces from the cable do not follow the direction of the segmentary acceleration, but that of the cable itself.

Inertial forces can then be calculated in just the same way. That is by multiplying the inertial load (mass receiving the force * instant acceleration) by the distance of the moment arm.

That is the way to obtain an inertial resistance profile, throughout the articular ROM.

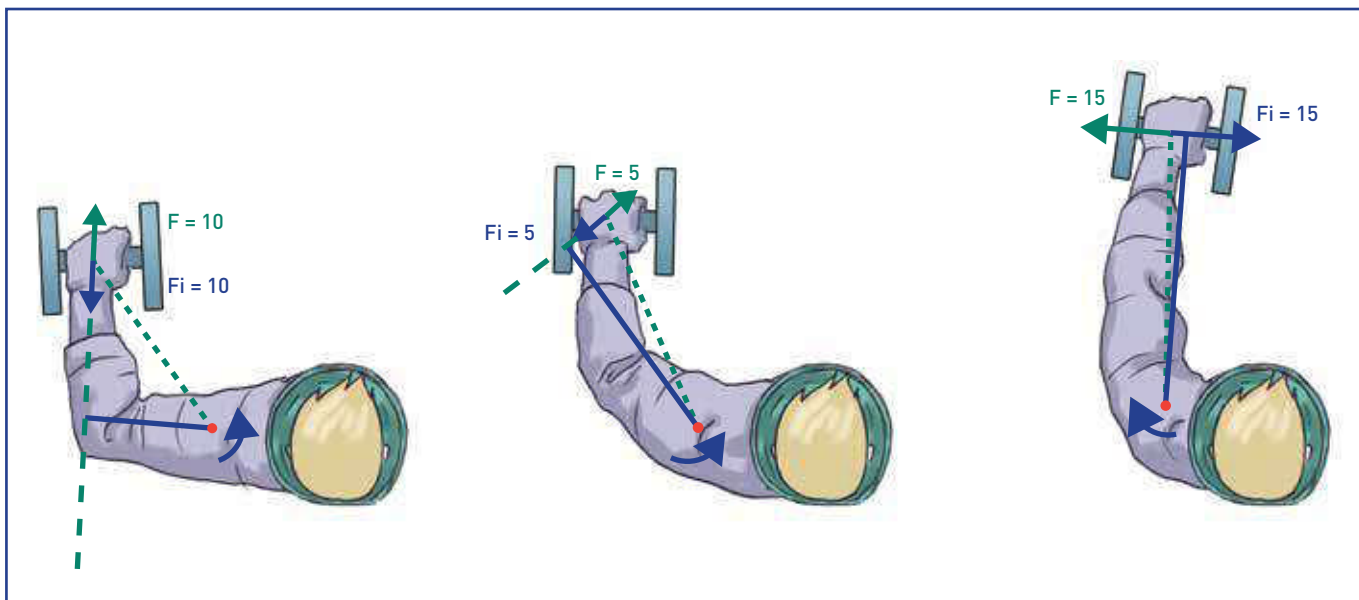
The various inertial resistance profiles resemble pretty much what would happen in outer space, with no gravity (picture 3.9.6).

Total resistance on planet Earth is more complex than that in outer space because there are other forces –such as gravity or the friction of the air– affecting the object. Inertial forces on the Earth only affect body load as long as acceleration and/or a change in direction are present, created by an internal force.



Picture 3.9.5

Doing a press move on a cable machine is different from doing it with a dumbbell, as in picture 3.9.4. Even though the angular acceleration force is the same as in the previous example, the moment arm is reduced. As a consequence of this, the inertial torque diminishes, for the direction of the reactive inertial force follows the cable's line and not the segment's direction of acceleration (as long as the limb's mass gets accelerated, the segment's inertial forces still exist, but they are small in comparison to those from the cable).



Picture 3.9.6

Accelerations on an object are a result of an imbalance of forces upon it. A dumbbell on Earth receives 10 kp of gravitational force. Say a person was holding the dumbbell in their hand and applied 20 kp against gravity. This would cause a 10-kp imbalance in the forces received by the dumbbell. This imbalance creates an acceleration to the dumbbell. The dumbbell will offer a 20-kp reactive force that will affect the person's hand and will become accelerated depending on its resistance to change (inertia). In picture 3.9.6, the arrow in Green represents the force acting upon the dumbbell, while the blue arrow shows the inertial force that the hand is receiving.

The force generated by the individual (which causes an imbalance and a resulting acceleration) affects the dumbbell, whereas the reactive inertial load acts on the person's hand.

CENTRIPETAL FORCE AND ITS INERTIAL REACTIVE IN A ROTATIONAL MOVEMENT AROUND AN INTERRUPTION

Every time a rotation movement occurs, there is a force that runs towards the interruption or axis of rotation. The force would run directly and exactly towards the pure center when the angular velocity were constant. In other words, when there is no angular acceleration. That force is what's called centripetal force or normal force.

Under these conditions, the mass creating the rotation tends to follow the straight linear trajectory. But there is this one

force holding it towards the interruption and which causes its linear direction to change. Should this force vanish, the rotational action would no longer be.

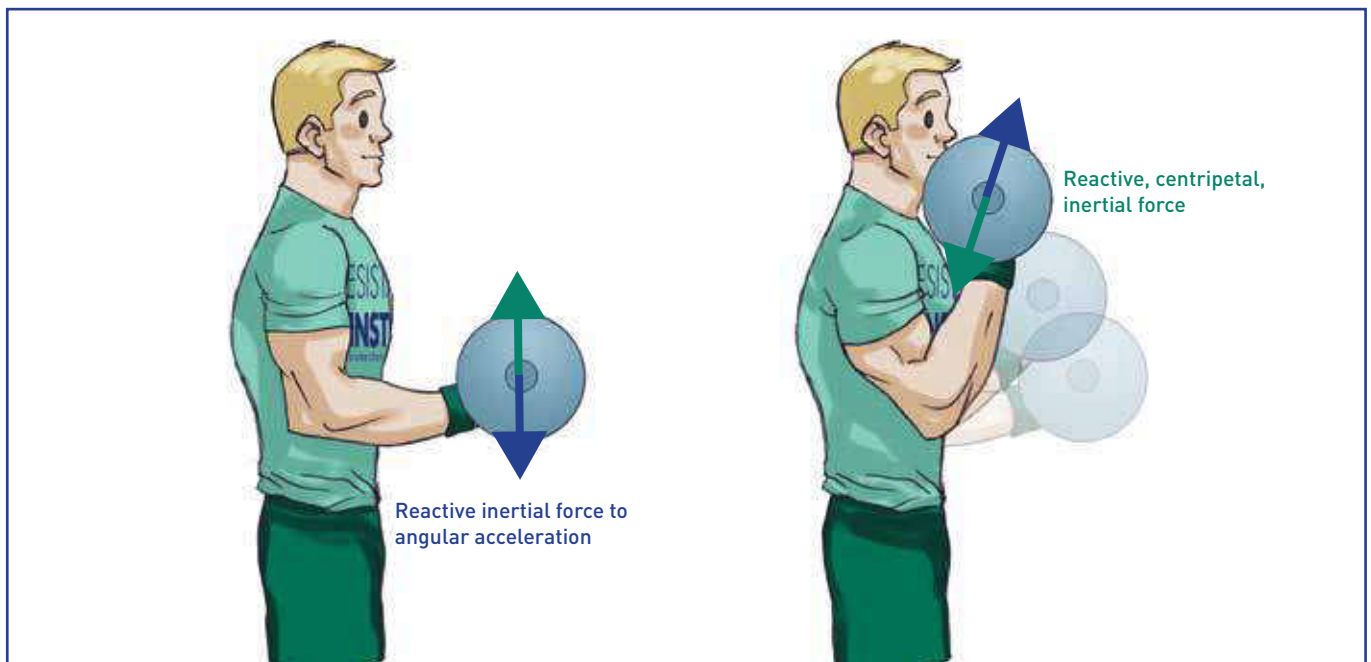
That means that it is the centripetal force that maintains the interruption and the rotation, associated to the change in direction.

The force reactive to that mass's change of direction is an inertial load that responds to that centripetal force.

The term centrifugal force is also very commonly used. However, physicists consider it a fake force, since there actually isn't any force pulling from the object outwards. The actual force is the centripetal one and it is the reactive inertial load that offers resistance against change.

Every single time there is angular velocity, even if it is a constant, there are centripetal forces created by the tissue making up the linkage (bones, connective and muscular tissue), directed towards the axis, and that have to do with the load's kinetic energy.

In spite of the above, and as far as exercise is concerned, neither the centripetal nor the associated inertial force are too relevant. The reason behind this is that they generally have little impact on articular rotational forces, if we compare them to other forces being applied onto the human body.



Picture 3.9.7

Inertial forces acting on the body can come as a result of: A resistance to angular acceleration (in this example, tangent), or a change of direction (reactives to the centripetal force). Inertial forces affect the hand and have been represented by a blue arrow. Forces that alter the dumbbell's state have been represented by a green arrow.

LEVER SYSTEMS

WHAT COULD HAPPEN IF THE EFFECTS OF INERTIAL MOMENT WERE NOT BEING CONTROLLED?

An increase in the distance-to-axis affects the non-gravitation generated segmentary acceleration forces in an exponential manner. This implies that some exercises will be more prone to be influenced by the consequences of inertial resistance.

Moves such as flies, pullover, lateral and frontal elevations can be affected by the inertial moment, due to the existing distance between the point of load support and the joint's axis of rotation. Both the articular forces as well as the resulting rotational forces can be noticeably increased as a result of a greater moment arm for the weight at the time of acceleration. Especially so, for negative accelerations at specific points of the ROM.

When doing these moves, people usually accommodate to the new and greater force requirements by accelerating the load in a slower way, compared to how they would do it for a load that was placed closer. If they didn't do so, it would be much more difficult to move the weight to the same acceleration.

The following, however, are reasonable questions when we think about it:

HOW FAST IS THE LOAD MOVING WHEN IT NEEDS TO BE DECELERATED?

ARE THE MUSCLES READY TO GENERATE SUCH TENSION WHEN NEEDED?

Should they not be ready, that tension would need to be generated by other tissues (passive).

Executing moves such as the above-mentioned, is not bad in itself. However, it should be taken into consideration that the effects of inertial moment could, at specific stages of the ROM, cause the requirements of necessary force to exceed the structures' capacity. If that was the case, and acceleration forces were not kept under control, there would be an increased risk (especially so in the deceleration phase).

When the load being used comes from a cable system which is, moreover, de-multiplied, the forces necessary in order to (positively or negatively) accelerate an object experience a significant reduction. The required forces needed to achieve acceleration are even further reduced when the load is free of inertia, since acceleration then will only affect –inertial load wise– to the mass of the body segment itself (which, indeed, cannot be eliminated).

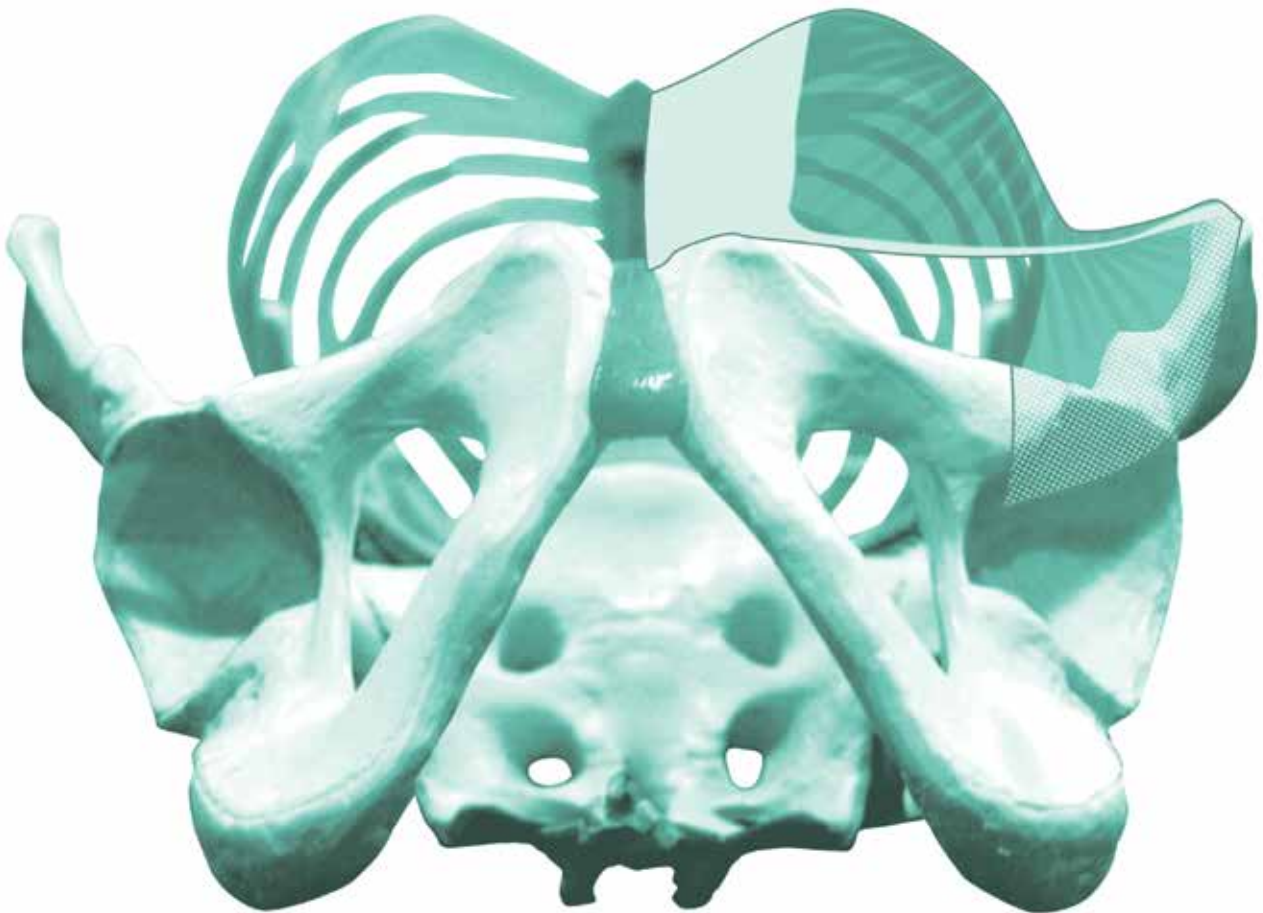
In situations like this, the resistance profile will be different. Its force peaks will not reach as high up, for one given (positive or negative) acceleration of the load.

Logically, for some training goals, making use of the effects of inertial moment would be advised, within the strategic progression, and most specially for those sports requiring of high accelerations.

For other individuals and goals, the risk-benefit factor, as well as the level of progression, should be highly regarded, prior to making use of exercises where the distance load-to-axis might be relatively large. This consideration becomes quite relevant for older individuals, people with low neuromuscular control, the sedentary, people of accentuated ectomorphic morphology (long and thin limbs), etcetera.

4

MUSCLE FUNCTION AND JOINT CONTROL



MUSCLE FUNCTION AND JOINT CONTROL

4.1 CREATING INTERNAL FORCES

In the human articular system, internal forces must be generated in order to be able to control and move the joints. Joints are always (to some extent) subject to gravitational forces (gravity induces resistant torques upon joints at nearly all times). And they are also subject to inertial reactive forces caused by segmentary accelerations.

HUMAN SKELETAL MUSCULATURE IS IN CHARGE OF NEGOTIATING WITH FORCES REACHING THE JOINTS!

The mission of skeletal musculature in the human body is to generate tension. That is to generate force, throughout the human articular system, in order to control and move the joints. It is through this that human motricity is achieved and the survival of the species assured.

The amount of force a muscle is able to produce is dependant upon several factors. Its analysis is somewhat more complex than what has up till now been covered in this book for other types of forces.

4.2 SKELETAL MUSCLE: STRUCTURE AND FUNCTION

As mentioned earlier, it is the skeletal musculature's responsibility to generate and/or to control the movement of the bones (levers) around the articular axes which it runs across (seen in the previous chapter). It should also be noted that musculature is responsible for the force distribution, not only in joints, but all around, via all the tissues it connects (fascia, bone, capsule, etc.). It is also muscles which are responsible for maintaining joint integrity, and they should not merely be regarded as movers of bony levers.

The amount of force a muscle can make in order to control/move joints depends on:

- **Its structure and contractile ability.**
- **Its neurological control.**
- **Its mechanical characteristics.**

4.3 STRUCTURE AND CONTRACTILE ABILITY

There is a wide range of materials and high-quality bibliography on this topic, all of which are of easy access. Because this book aims to offer new lines of thought, only the basic most relevant aspects of this topic will be covered here. For a thorough understanding, the reader is encouraged to study these topics through specialized courses, as well as from some of the numerous human physiology books that are available. You may want to look up 'muscle physiology' and 'exercise physiology'. There are many materials available out there.

MUSCLE STRUCTURE

Under a title like this, it is common to read information on the muscles' characteristics. Especially on those that have to do with its size, arrangement and number of fibers.

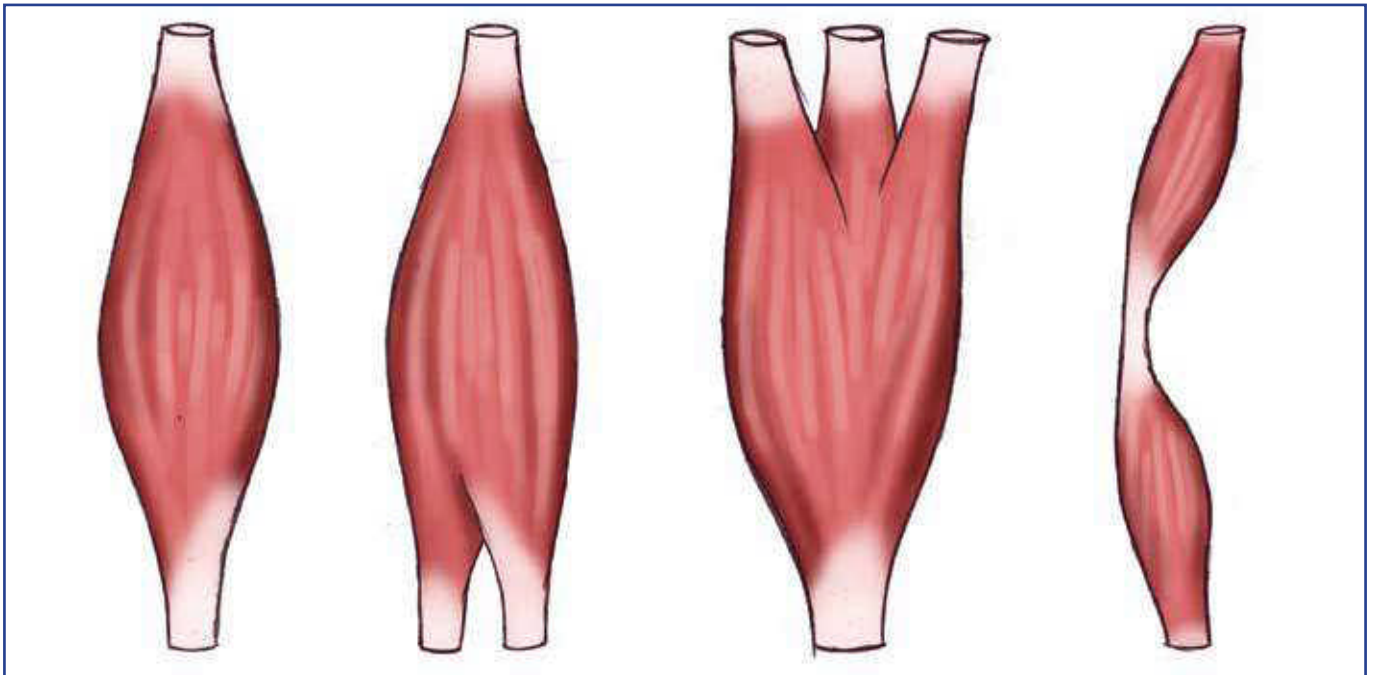
The length of a muscle fiber, the arrangement of its fibers and the number of muscle fibers per muscle are different for every specific muscle in the human body. This variability between muscles affect, not only their shape and size, but also their function. But they all have one thing in common: each muscle fibers is capable of shortening to approximately half its length.

In spite of the existing differences between muscles and between individuals, one thing should be kept in mind: the skeletal musculature pursues one same main and most prioritary goal, to control and move the joints. And it does so by actively contracting muscles, which itself is the main

difference to all other tissues, such as articular connective tissue.

How a muscle's fibers are arranged in space affects the tension-elongation relationship, as we'll discover next.

Fusiform muscles are arranged in such a way that the muscle fibers run parallel to the muscle's longitudinal axis, and to each other. Many fusiform muscles are made of long fascicles that run along the whole length of the muscle. That is the case of the biceps brachi muscle, for instance. In other fusiform muscles, such as in the rectus abdominus, fibrous intersections cause the fascicles to conform into small segments. These divisions work up to create what's called a polygastric muscle. Most fusiform muscles are made of fibers running all the way along, from origin to insertion.



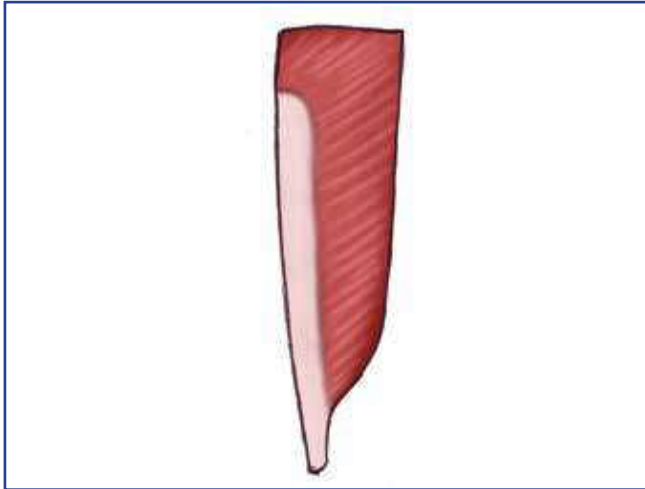
Picture 4.3.1

This picture shows different types of fusiform muscles. Some have one belly, some have two, three, or more (polygastric muscles, such as the rectus abdominis).

Muscle fibers in the pennate musculature run oblique to the muscle's longitudinal axis.

For pennate muscles, fibers making up their fascicles tend to be shorter and more numerous than that of the majority of fusiform muscles.

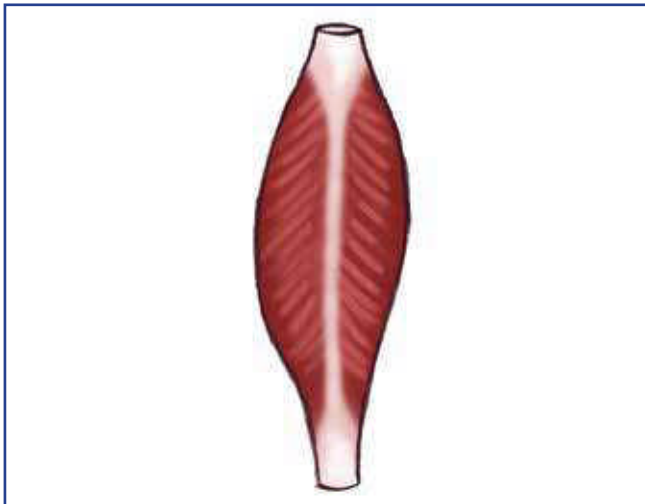
Unipennate muscles display muscle fibers on just one of the sides of the tendon.



Picture 4.3.2

This picture shows the representation of a unipennate muscle. The tibialis posterior is an example of a unipennate muscle in the human body.

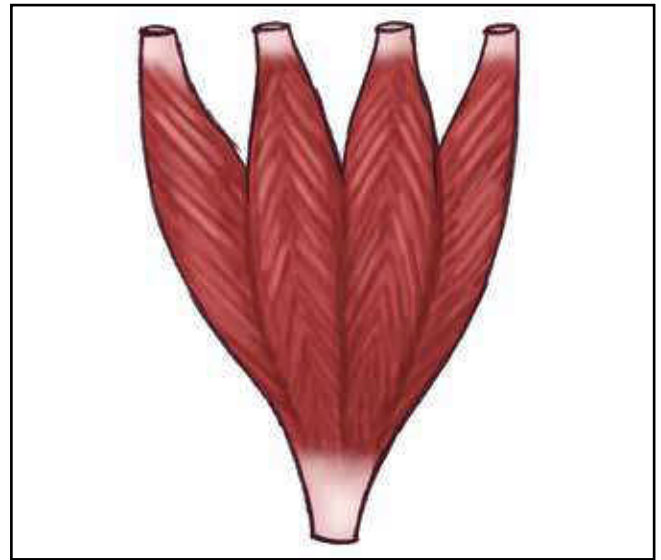
Bipennate muscles have muscle fibers on both sides of the central tendon.



Picture 4.3.3

This picture shows the representation of a bipennate muscle. One such example in the human body would be the gastrocnemius.

In **multipennate** muscles, fibers run in an oblique direction and converge into several tendons.

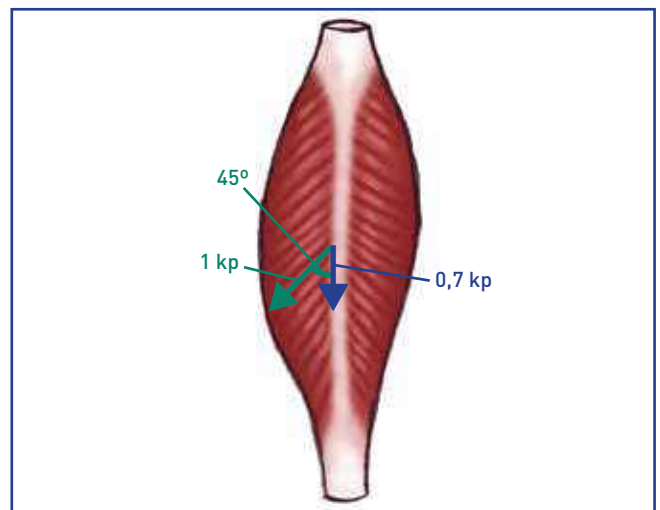


Picture 4.3.4

This picture shows a representation of a multipennate muscle. The human deltoid muscle would be an example of this.

In fusiform muscles, there is a direct relationship between the muscle fiber's linear deformation and that of the central tendon or connective tissue the fibers pull onto. This direct relationship is breached in the case of pennate muscles.

As far as they are concerned, only a part of the longitudinal deformation of their muscle fibers will induce deformation of the tendon, which determines the line of muscle force. The greater the pennation angle is, the smaller the direct relationship will be between the deformation created by the muscle fibers and the deformation induced by them onto the central tendon.



Picture 4.3.5

The pennation angle does not just affect the relationship fiber-to-central tendon as far as the deformation. It also affects the relationship between force exerted by the fiber and force transmitted to the central tendon. Only part of the force generated by the muscle fibers creates tension along the tendon's muscle line of force. As a consequence, only that part contributes to the creation of muscle torque. Based on trigonometry, and as seen in the previous chapter, a pennation angle of 45° transmits approximately 70% of force onto the central tendon. However, because there is a higher number of fibers for one given area, the total amount of force that gets transmitted onto the tendon is usually higher than that in fusiform muscles.

It should be noted that pennation angle and muscle force angle are two different things. The latter will be explained later on in this book. A pennation angle is formed between the muscle fiber and the tendon which indicates the muscle's line of force. A force angle is created between the muscle's line of force and the muscle's lever arm.

CONTRACTILE ABILITY

Skeletal muscle contains, basically speaking, two types of tissue. One of them is contractile tissue (myofibrils) and the other one is non-contractile tissue (muscle connective tissue that will be described later).

The characteristics of these tissues, and how they interact are key elements in the unique traits of muscles.

Muscle tissue –and other biological tissues, alike– is viscoelastic, due basically to the large amount of fluid it contains. This causes it to behave as a semi-solid.

CONTRACTILE TISSUE:

Skeletal muscles are made of thousands of muscle fibers. Each fiber consists of one single muscle cell, that is surrounded by a membrane known as sarcolemma. Muscle cells, that is to say muscle fibers, come together in bundles called fascicles. One single muscle contains several fascicles.

Muscle fibers are filled with sarcoplasm. Sarcoplasm is made of structures called myofibrils, which are themselves the contractile component of the muscle fiber. And it also contains other structures, that are necessary for the cell's metabolism.

Each myofibril is made of small filaments called myofilaments (that is where sarcomeres are found). Myofilaments consist of two types: thin and thick.

Thick filaments are made of a protein called myosin. Thin filaments are made of a protein called actin.

Tension is generated when cross bridges between the actin and the myosin filaments of the myofibrils are established. As long as there is no superposition between the actin and the myosin filaments, crossed bridges cannot take place and, thus, muscle tension will be impossible. Similarly, whenever a muscle is in its shortened most position, and

superposition of actin and myosin filaments is maximal, the body is unable to create any more cross bridges and, understandably, the generation of any more muscle tension becomes impossible.

In vivo, these two extremes of shortening and stretching, where there is absolutely no chance of generating any cross bridges, hardly ever take place. It is nevertheless possible to assess a noticeable decrease of cross bridging generation ability at the extremes of the contractile range.

For shortening muscle contractions, known as concentric contractions (even though some authors, such as Hubbard & Stetson, 1938 prefer to call them myometric), actin filaments are pushed towards myosin filaments, cross bridges are generated, broken and once again regenerated.

For lengthening muscle contractions, known as eccentric or plyometric (Hubbard & Stetson, 1938), actin becomes pushed away from myosin and the existing cross bridges break apart, are rebuilt, and broken once again. This process is known under the name of sliding filament theory.

TYPES OF MUSCLE FIBERS (histology of muscle fibers):

Fibers are commonly defined as being of three different types, according to their histological traits. However, a continuum of different types exists, which can include up to 50 variations of muscle fibers, from the fastest-twitch muscle fibers (FTX) to the low contraction ones (ST).

How many motor units are assigned to each of these muscle fiber types, for one given muscle, actually varies greatly from individual to individual. These variations are considered to be genetically determined.

Even though different individuals may present different characteristics, many muscles do tend to have a greater proportion of one type of fibers over the other. Muscles

that hold a relatively high proportion of slow-twitch fibers, for instance the soleus muscle, are often referred to as postural, stabilizing or tonic musculature. These terms could be misleading, though. From an exercise mechanics perspective, muscles should neither be named tonic, nor stabilizer or postural. While it is true that their differing traits –such as muscle fiber typology– cause muscles to exert tension in different amounts and ways, they all share one same goal: to control and to move body joints. It is how they manage to perform muscle tension, and to what degree they can do so that depends on the percentage of one type of fibers over the other.

Generally speaking, muscles that consist mainly of fast twitch fibers tend to be those involved with the production of large joint ROMs. It is often referred to as non-postural, movement or phasic musculature. Once again, it would be preferable not to differentiate muscles according to

this criteria, since all skeletal musculature is designed to generate movement to some extent, and all of it helps maintain stability, for whatever force scenario that may be taking place. In fact, unless the individual is under general anaesthesia, it is impossible for the musculature not to have some degree of residual tension, and not to be tonic to some extent.

4.4. NEUROLOGICAL CONTROL

El sistema nervioso juega un papel vital en la motricidad humana, aunque tradicionalmente ha sido menospreciado en el sector del ejercicio y actividad física. Una relevancia marcada por la intención, organización y orquestación de la respuesta motora, pasando por la entrada propioceptiva y las respuestas reflejas medulares.

Al igual que se ha descrito anteriormente en referencia a la estructura y capacidad contráctil, se van a describir solamente los aspectos más relevantes en lo concerniente al control neurológico del movimiento, por lo que, para una comprensión más completa en este área, se recomienda estudiar estas materias en cursos especializados y en los numerosos libros de neurofisiología humana y neuroanatomía disponibles en el mercado.

‘MUSCLE FORCE IS MAINLY A FUNCTION OF THE NERVOUS SYSTEM-CREATED TENSION, NOT A MATTER OF MUSCLE MASS’

RTSm Tom Purvis / Mitch Simon

The functional unit in musculature is known as motor unit. Motor units consist of one motoneuron (also known as motor neuron) and all muscle fibers innervated by it. A motor end plate is the place where the motoneuron meets the muscle fiber.

Skeletal muscle contraction comes as a result of various motor units becoming activated repetitively, both in synchronic and asynchronic manners. The magnitude of the muscle contraction can be altered by manipulating the number of motor units being activated and/or the frequency of the impulse (spatial and temporal summations, respectively). The amount of motor units present depends on the specific muscle.

Motor units vary depending on the nervous cell body's dimensions, the axon's diameter, the number and type of muscle fibers. Those motor units with small-sized cell bodies have smaller diameter axons, causing the nervous impulses to travel more slowly.

The size of a motor unit is determined by the number of muscle fibers it contains. The number of fibers within a motor unit ranges from two or three to several thousands. Muscles that participate in fine or high precision motricity (eye musculature, for instance), have smaller-sized motor units and usually present with small cell bodies, as well as small diameter axons. Muscles that have to do with more gross motricity and with the production of large amounts of force have bigger motor units and, generally, bigger cell bodies and axons of a larger diameter.

NEUROMUSCULAR ORCHESTRATION

When forces contact a human body, they affect its neurological orchestration. Controlling their centre of gravity is a human being's very first life challenge, right after they are born. Little by little, babies learn recruiting sequences which improve their control over the segmentary centres of gravity, as well as over the reactive forces to the body's points of support. Meanwhile, the person's musculature involved in that same control task gains strength. In other words, human beings learn to recruit muscles as a key element in order to create control and articular movement, from the very first forces that they receive.

Strength professionals have a complex responsibility. They must be able to detect the variables capable of causing changes in the applied forces. The neuromuscular system will or should at least respond by recruiting motoneuron and controlling or moving the joints, depending on its goals.

If, for instance, we did a seated, two handed transversal press movement with cables and we let go of one of the two cables, a new force causing rotation on the superior-inferior axis of the spine would automatically come about. This new force scenario will lead the neuromuscular system to activate those muscles which avoid the rotation of the spine, as long as the goal is to stay put and to carry on executing the press move.

Can the forces being applied be understood and strategically modified, according to the individual goals, in order to optimize the resistant torques that are being created?

Yes, they can. It is not a simple task to do, though. This, however, should be the goal of all professionals who apply forces for exercise's sake. And this should be so, no matter what the specific method or technique may be called.

THE ACTIVATION OF MUSCLES IS RESPONSIVE TO FORCES, WHETHER EXTERNAL OR THOSE CREATED BY BODY WEIGHT, AS WELL AS TO THE ATTEMPT TO ACCELERATE BODY SEGMENTS (inertial forces).

For any exercise professional, one of the goals that should be priority is to know whether a client/athlete is capable of controlling or moving their joints with an appropriate neuromuscular response to the applied forces.

'THERE ARE TWO ESSENTIAL MATTERS: ONE IS TO KNOW WHETHER MUSCLES ARE BECOMING ADEQUATELY ACTIVATED. THE OTHER IS TO KNOW WHETHER THERE IS SUFFICIENT CONTROL TO MAINTAIN JOINT INTEGRITY'

NEUROLOGICAL AND ARTICULAR INTEGRITY ARE KEY ASPECTS

THE AMOUNT OF MUSCLE TENSION

The amount of muscle tension that musculature generates depends on several variables, such as: the types of motor units being activated, the percentage of fibers of one type over the other for each muscle and the differences between

individuals, the muscle fibers' cross section, fatigue, the relationship between tension and elongation, the corticospinal deficits for certain muscles, the inadequate spinal reflex regulation, etc.

AFFERENT INPUT AND EFFERENT RESPONSE

Correct orchestration heavily depends on the sensory afferent input coming from the rest of sensory receptors. This information is key in order for the nervous system to decide which muscles should be activated and, in that manner, to be able to control the new demands, such as: centre of gravity maintenance, joint control, segmental acceleration, etc. for each of the different scenarios.

Unfortunately, there are many factors that can cause a significant decrease in this sensory input. Some of them

may be trauma, sedentary life styles, Immobilization, or even social or job-related factors, such as sitting all day long and wearing shoes, among others.

There is this old adagio that has to do with IT which we often use at Resistance Institute. It goes like this: garbage in, garbage out. This basically talks about the relevance of this afferent-efferent loop and how a non-adequate sensory input will affect the resulting motor orchestration.

NEUROMUSCULAR COMPENSATION: TRYING TO FIND A SOLUTION

Unfortunately, it is likely that everyday stimuli will gradually lead to a deterioration of the ideal movement orchestration and joint control. Once the system lacks the correct options, it finds itself in a position where it needs to look for new solutions, and that is when compensation and substitution come into play.

This new solution will turn into the new orchestration and will become common or 'normal', as long as that specific force scenario keeps getting reinforced.

It may be commonly believed that integrated moves (a.k.a. functional exercise) help get rid of weak links and neurological compensations. Nevertheless, both rehabilitation specialists and researchers in neurophysiology and motor learning have proven that, unless there is analytic work on the specific area of the weak link, the latter will not regain its correct orchestration.

This does not mean that practicing integrated actions is inadequate or 'non functional'. It rather means that the specific context and the individual's traits (remember that phrase from 'Siff' in chapter 1) should determine the scenario and the dose of the stimulus, which in turn will determine the functionality for each case. And it should not be based on the name of the exercise, nor on how it should supposedly be executed.

*UNLESS WE IDENTIFY AND ELIMINATE THE
WEAK LINKS FIRST, 'FUNCTIONAL TRAINING'
(INTEGRATED EXERCISES) WILL JUST REINFORCE
THE COMPENSATORY PATTERNS'
(GREG ROSKOPF)*

ACTIVE TENSION GENERATION CAPACITY (summary)

The term 'active tension' talks about the tension generated by the contractile structures in the musculature.

- The higher the recruitment frequency of motor units (temporal summation), the higher the tension.
- The higher the number of recruited motor units, the higher the tension (spatial summation).
- Tension increases when the motor units being recruited have high diameter axoned motoneurons, which innervate fibers of faster twitch contractions.
- Tension may be increased by recruiting motor units with a higher number of muscle fibers.
- For one given scenario, a muscle with a larger cross section is able to generate more tension than a muscle with a smaller cross section. But what is really difficult is that the same scenario actually takes place. Consequently, a larger cross section will not necessarily guarantee a greater ability to generate active tension.
- Fatigue alters a muscle's ability to generate tension. It does so throughout the shortening-lengthening cycle in an unproportionate manner. Commonly, within the cycle of contraction, the effects of fatigue are first perceived in the most shortened areas.

TENSION – ELONGATION RELATIONSHIP

There is a direct relationship between the tension being generated in a muscle and its length. There is an optimal length, where the muscle is capable of generating its maximal tension.

According to several experts in muscle neurophysiology, that optimal length equals, approximately, 1.2 times the muscle's length at rest. This optimal length is the position where actin and myosin filaments are so located that the highest possible number of cross bridges can be established. It is almost impossible to determine this optimal length in vivo. Because of this, and to make it feasible in training, what is important is to be clear on how this capacity of generating cross bridges evolves. This is in fact what influences a muscle's ability to generate tension throughout its varying elongation states.

The amount of tension a muscle is able to generate

gradually decreases for each micrometer that it deviates from its optimal length, whether that is in a stretching or shortening mode. Whenever a muscle experiences elongation beyond its optimal length, the actin and myosin in it become progressively less superpositioned. Chances to generate cross bridges then lower. Nonetheless, passive tension from the elastic components running in parallel may become increased as the muscle lengthens.

During a concentric or myometric contraction, muscles are usually capable of generating moderate tension in elongated positions, maximal tension in midrange positions and minimal tension in the shortened most positions of the contractile range. This relationship can, however, be altered due to the effects of fatigue, or mechanics. For multiarticular muscles, the relationship would also depend on how the other joint being crossed over by the muscle is positioned.

PROPRIOCEPTIVE INPUT AND LINEAR DEFORMATION OF MUSCLES. THEIR RELATIONSHIP

According to neurophysiologists, evidence suggests that a muscle being stretched offers the greatest amount of proprioceptive inputs. While this happens, there is also a decrease in active tension and an increase in passive tension. Muscles express their peak active tension capacities at their contractile midrange areas.

Whenever a muscle is in a shortened position, its proprioceptive input is diminished, and so are its active and passive tensions. This is exactly the reason why intrafusal gamma fibers are so relevant when talking about this range of contraction. A correct alpha-gamma co-activation will allow the muscle spindle to continue to send proprioceptive signals towards the spinal cord in those shortened positions.

At midrange and, most specially, stretched positions, muscle spindles are able to send proprioceptive signals in a direct manner, due to the mechanical deformation of the spindle itself. In these situations, there is a lesser need for the gamma fibers to generate an elongation of the mechanoreceptors in the muscle spindle.

In order to better comprehend these processes, you are highly encouraged to search in the bibliography or in specialized texts on nervous system organization and spinal reflexes.

PASSIVE TENSION

We call passive tension to that developed by passive components in the musculature, that is those that are non-contractile.

All of the connective tissue in the muscle is interconnected and makes up its passive elastic component. All of these tissues run parallel to the muscle fibers. These connective tissues (fascia, endomysium, perimysium, epimysium), the sarcolemma and the intracellular elastic filaments make up the parallel elastic component of the muscle.

The tendon follows to this connective tissue, which then

turns into a continuum with the periosteum.

This is why the tendon is considered to run in series with the contractile components and why it is included as just another element within the elastic components in series of muscles.

The collagen fibers in the tendons are usually arranged in parallel. This makes it possible for them to bear the high, unidirectional traction forces that are generated by active muscle tension and/or by the bone. This arrangement makes tendons more resistant to traction stress forces,

rather than to compressive or shear stresses. Please bear in mind that the main role of tendons is to transfer muscle force, as precisely as possible, onto the bone.

Because that is exactly the tendon's main purpose, and unlike common belief, they are not designed to be stretched. In fact, and according to expert muscle physiologists, tendons rarely present with any more than a 4% strain (proportional deformation).

Moreover, it would be impossible for a traction force to place a stretch locally on a tendon, for it holds an intimate, in series, relationship with the muscle.

With this in mind, it is now easier to understand why it is so complex to determine, in vivo, what percentage of the force applied to a bone by a muscle actually arises from the active components (sarcomeric) and how much of that tension is due to the passive components (in series and parallel connective tissue).

MUSCULATURE'S PROTECTIVE RESPONSE TO STRETCHING (is it passive?)

Whenever a muscle is stretched, the most usual response is for the nervous system to command a tension increase, meant for joint protection.

This response is often commonly regarded as passive tension. However, one should not forget that this tension is usually a neurological protective response and it does not only come from the resistance of passive tissues to deformation. Opposite to that, to a large extent, that response is due to reflex active tension (nervous system regulated), which aims to protect the area against the force that is creating the stretch.

OTHER FACTORS INFLUENCING THE MUSCLE'S ABILITY TO GENERATE TENSION

There are other factors that have an effect on how much tension is exerted on the bone lever by the muscle. For instance, the speed of contraction. In this case, it is virtually impossible to be able to carry on in vivo experiments, without the outcomes being altered by the nervous system.

The variables that affect generated tension have been reviewed. The next step should be to analyze how much of that force generates torque. In order to do so, we must learn the mechanic variables within the human articular lever systems.

4.5 MUSCLE FORCE AND MECHANICAL CHARACTERISTICS

Let's now review the concept of torque. It is the force that is applied around an interruption (joint's axis) and with attempts to generate rotational movement. Previously in this book, it has been explained that applied forces require of a mechanic factor that directly connects them to the specific lever. It is these mechanical characteristics that significantly determine the given force's ability to generate rotation and stability around a given joint axis. If we transfer this to human lever systems in particular, it implies that muscle tension (force) is just one more factor within the ability to generate muscle torque.

By integrating the information covered up to this point, and considering those muscle-related-factors with an effect on muscle torque, the following formula would apply:

$$\text{Muscle torque} = \text{Muscle strength (active tension + passive tension)} * \text{mechanical traits (moment arm)}$$

The factors that are related to muscle strength are:

- **Structure and contractile ability**
- **Motor control**

The factor that is related to mechanical characteristics is:

- **Moment arm**

The present book deals with muscle and exercise mechanics. It is precisely in this field that Resistance Institute is able to cater the readers with new concepts and a more advanced vision. This does not mean that those aspects related with strength aren't crucial. Again, there are many publications and specialists in strength, giving out education on this. For this reason, the topic will be thoroughly covered later on.

MUSCLE LINE OF ACTION

The action lines for a muscle fiber are determined by the direction of its fibers, or the connective tissue that binds tendon-to-tendon. As will be discussed later, identifying the direction of the line of force that generates muscle torque around a joint is not an easy matter.

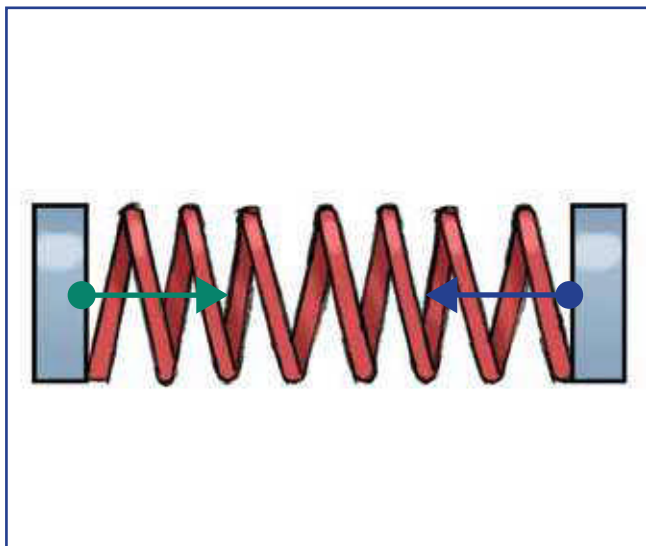
Tension generated by the sarcomeres, as they are trying to cross bridge, generates a convergent type force, towards the musculature's central area. This force creates traction at the attachment sites on both sides. This can take place via the tendon, or via the fascial tissue, which oftentimes also ends up inserting itself into the bone.

This convergent force could be compared to the force exerted by elastic materials, such as a spring or even ligaments, while being deformed into a stretch. However, we should not forget there are important differences between active

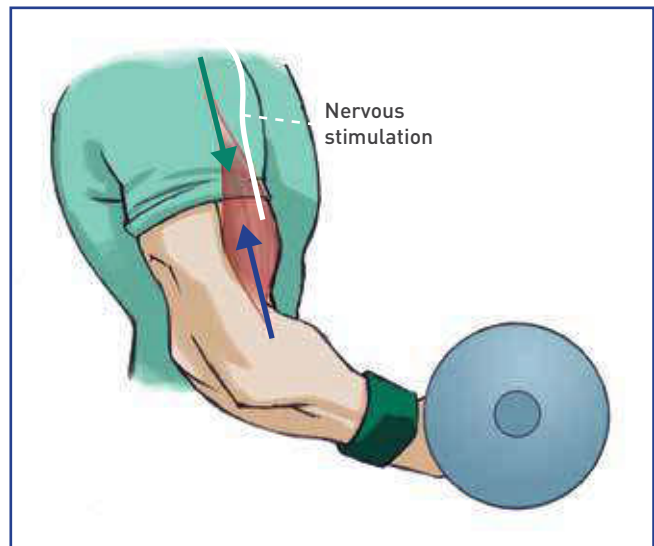
and passive elastic and viscoelastic tissues.

The latter respond with a convergent force whenever an external force deforms them beyond their normal length. Musculature, however, has the ability of generating that force by itself. It does so by overlapping actin and myosin filaments in the sarcomeres, as a consequence of the motor units having been excited. We could say that musculature exerts this convergent force thanks to neural excitability and the ability of muscle fibers to contract. On top of that would be viscoelastic forces, should the tissue be subject to deformation.

Muscles exert a pulling force onto all segments they insert in, no matter which one will eventually experiment movement.



Picture 4.5.1a



Picture 4.5.1b

As you can see from the pictures, a muscle's line of force resembles the elastic line of force a spring, in that they both pull from their extremes towards the center. In spite of this resemblance, musculature can generate this type of force without having to deform, thanks to the excitability and contractility of its muscle fibres.

Moreover, unlike springs, musculature is a semi-solid material and thus confers viscoelastic forces.

It is essential that this type of force generation is understood, since it greatly differs from the forces that have been previously covered and which had to do with lever systems.

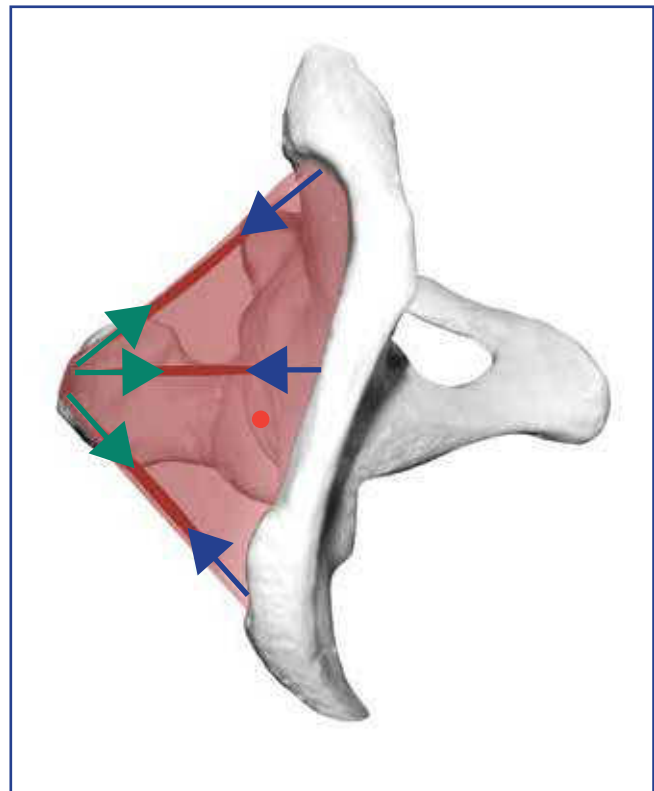
Muscles create tension throughout their full length. They do so either by the length of those fibers running from the origin to the insertion sites, or by shorter fibers transmitting their force onto connective tissue (which constitutes a continuum with the fiber), following the muscle line of action (pennate musculature).

Therefore, muscles exert tension from their attachment sites to the centre.

Picture 4.5.2

The picture shows an illustration of a superior view, in a transverse plane, of the gluteus medius. By looking at it in this way, the S-I axis (red dot) can be observed, and thus the mechanics of this muscle's fibres over medial or lateral rotation.

The different muscle fibres in the gluteus medius, hold different (and convergent) lines of force, depending on their insertion sites. As a result of this, they have different mechanics around this axis. A precise description of muscle function should analyse the specific mechanics of the different fibres in this muscle, rather than establish a generic muscle function.



FIXED SEGMENT, MOVING SEGMENT

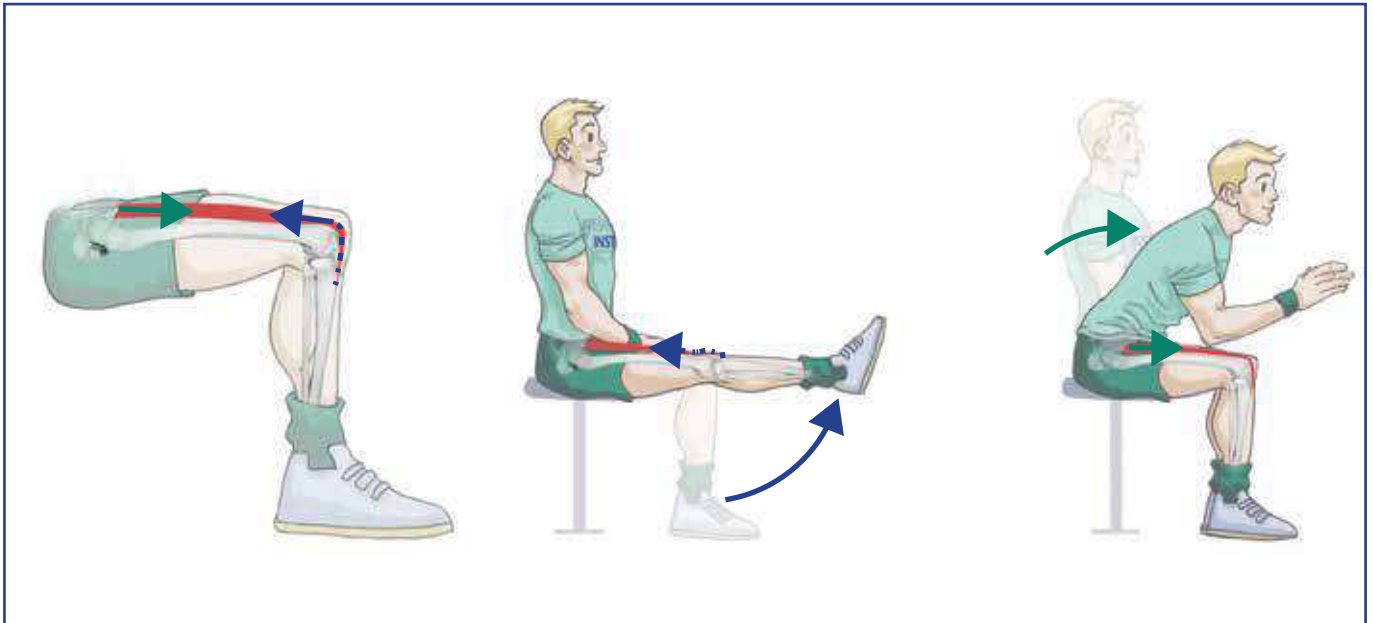
Every time a muscle fiber generates tension towards the centre, it attempts to move both of the attachment sites. Whether those segments actually end up moving, or not moving, and whether they stay stable in spite of movement depends on other musculature regulating tension and governed by the nervous system. Either that or there is an external force (point of support) preventing movement from happening. In the end, it will always be the segment which offers the least resistance that will move.

Consequently, it is not muscles which 'decide' what segment is going to move when they generate tension. Muscles try to generate force towards the centre, from both of its ends. It is either the nervous system, through the

orchestrated activation of several muscle fibers, and/or the force contacting the segmentary points of support which determine the segments that will eventually move.

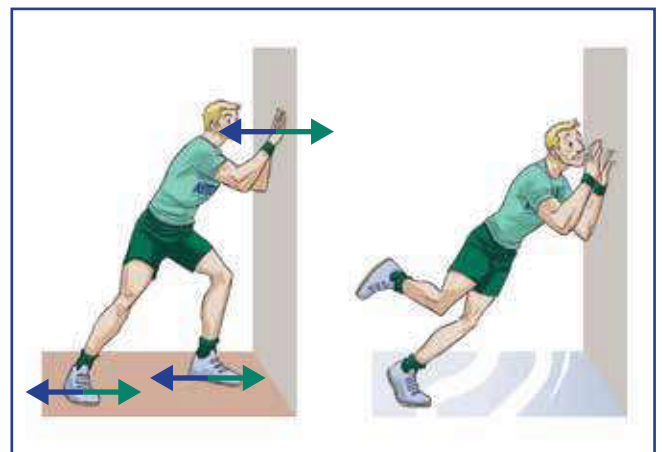
From the example in picture 4.5.3, as the rectus anterior of the quadriceps generates tension, two possible scenarios arise: if the pelvis stays fixed, it will be the tibia moving. This is depicted in the middle drawing. Should it be the foot that stays fixed, the pelvis will move, as seen on the right side of the picture.

By integrating this, it can be stated that the mere fact that a muscle or fiber might be generating force does not necessarily imply that a segment will be moving.



Picture 4.5.3

We need to understand that, in order for the body to be able to maintain certain segments stable (allowing from whatever amount of movement), some other muscles will have to act and guarantee that control. It is very likely that, if the nervous system considered it was impossible to guarantee a given segment's stability, it would probably limit the amount of tension a certain group of muscles was able to exert. It is somewhat comparable to an individual trying to press against a wall, while on a frozen ground. This scenario is depicted in picture 4.5.4. When the ground does not offer an appropriate base of support to the legs, due to insufficient friction-, it is unlikely that the body will be able to exert as much force on the wall.

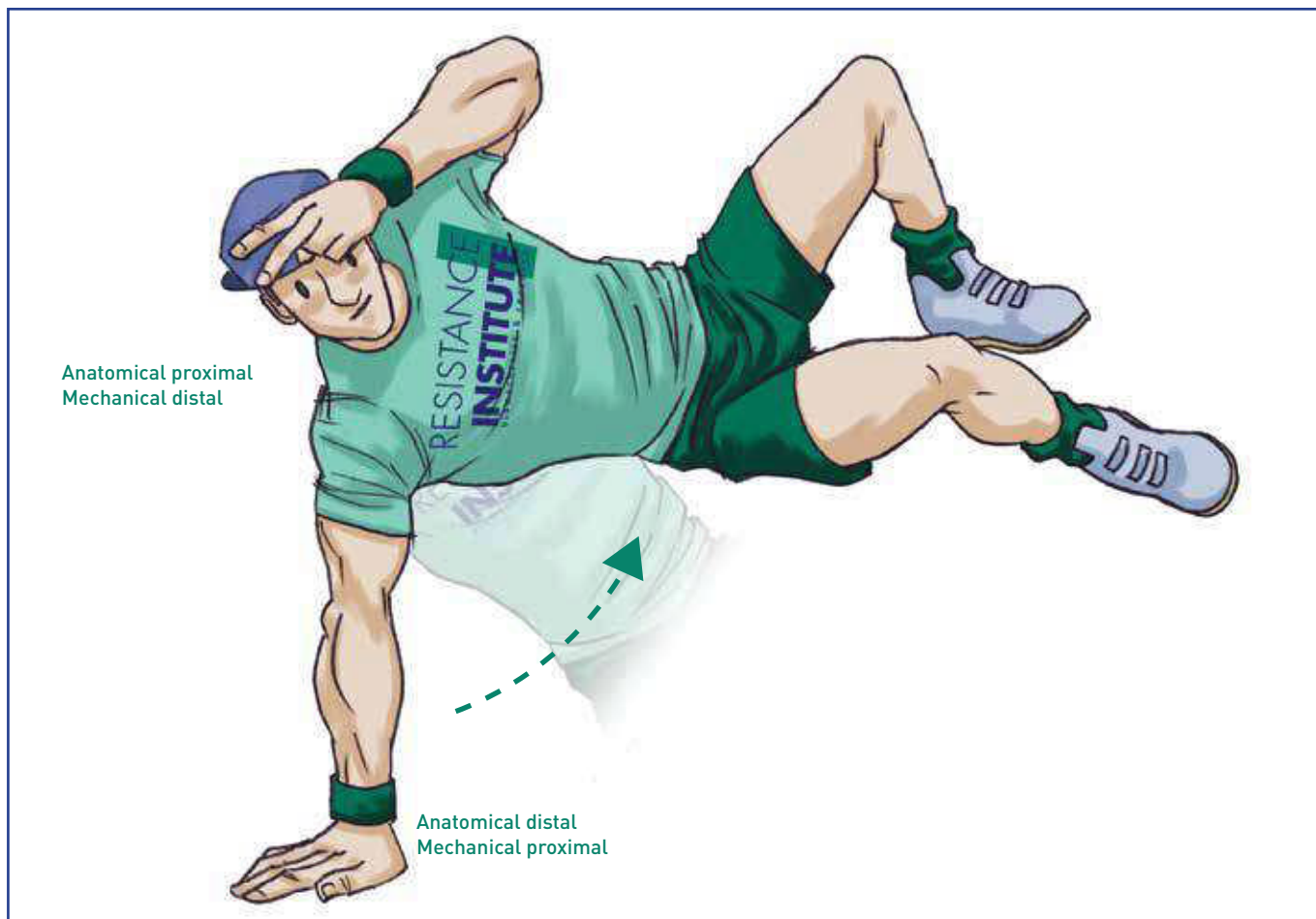


Picture 4.5.4

PROXIMAL SEGMENT AND MECHANICAL DISTAL vs. PROXIMAL SEGMENT AND ANATOMICAL DISTAL

Within a lever system, the distal segment is considered to be that which offers the least resistance. And the proximal segment the one which offers the most. In other words, the first one to move is considered distal.

This fact means that whenever the anatomical distal segment (which never changes, since it is referenced from the anatomical position and its goal is merely descriptive) turns into the fixed segment within a force scenario, it automatically changes to be the mechanical proximal segment.



Picture 4.5.5

This picture recreates a shoulder abduction action, where the torso moves in relationship to the hand. The hand remains fixed to the ground, which causes the anatomical distal segment (hand) to become the mechanical proximal segment. And it causes the anatomical proximal segment (torso) to become the mechanical distal segment.

This might seem like a logical and irrelevant statement. However, the usual confusion between mechanics and anatomical nomenclature has actually led to one of the most well-established myths in the field of exercise.

We are talking about the myth of open and closed kinetic chains. Luckily this has been magnificently explained and dismissed by Tom Purvis, in his RTS programme.

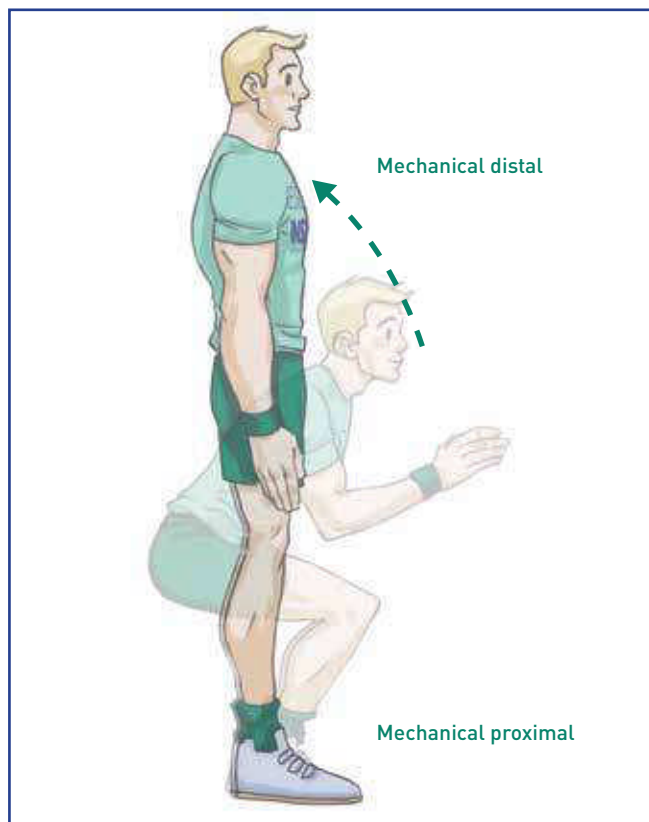
We thank him for his insight and wish to present the following thought:

Squats are considered closed chain because the distal end

is fixed (exercises are very commonly classified as closed chain when their anatomical distal segment is fixed). Nevertheless, this open/closed chain considerations should be based on mechanics, not anatomy. Could it be that the mechanical distal end in this move is not fixed?

Resistance Institute recommends that the reader accesses their analyses and recommendations about kinetic chains.

More information is available at:
www.resistancetrainingsspecialist.com



Picture 4.5.6

This picture examines a squatting action, where the torso moves while the foot remains on the ground. This fact causes the anatomical distal segment (foot) to become the mechanical proximal segment; and the anatomical proximal segment (torso) to become the mechanical distal segment. But then... is the distal segment in a squat fixed?

Anatomical terms are merely descriptive, and movement analysis is supposed to be carried on from a mechanical point of view. In this way, the mechanical distal segment here is not fixed, and the answer to the previous question is NO.

LEVER ARM IN A MUSCLE FORCE

Muscle lever arm is the shortest distance between the axis of rotation and the point of application of that muscle force.

This point of application has always been regarded as the bony site of attachment. However, this book will put forward a variation. The reason for this being that many of the existing muscles present with bony redirections, which radically changes their line of force. This will be looked at in greater detail later on.

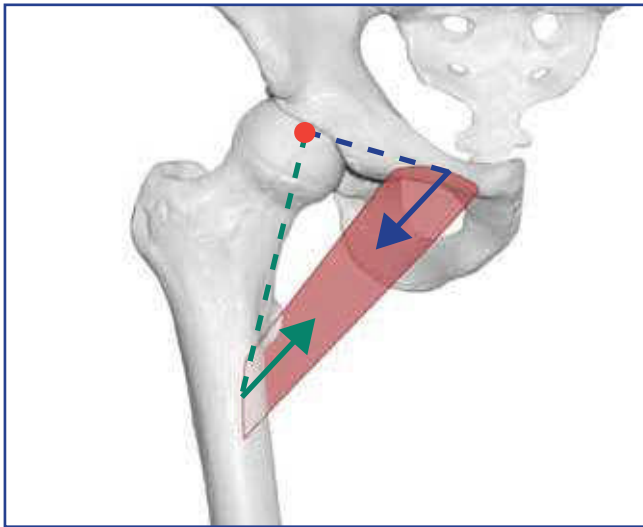
By introducing the length of the lever arm, we can calculate the rotational force or torque that a given muscle force generates on a specific articular axis. This will determine the classical formula for torque:

$$\text{TORQUE} = \text{FORCE} * \text{LENGTH OF LEVER ARM}$$

Generally speaking, and according to what we have already covered in the previous chapter for lever systems, this is how rotational forces are calculated. It is a proportional relation between the force factor and the distance factor.

The above stated formula establishes a proportional relationship between the force and the distance where it is applied. However, it is only valid as long as the angle of force to the lever arm equals 90 degrees, in which case all of the force will be generating a rotation. Basically, this is all that is taught in the majority of courses and college classes of Biomechanics.

In real human bodies, this is not always the scenario we encounter. Consequently, the angle of the muscle force will have to be taken into consideration, in order to calculate what percentage of the applied force actually generates rotation at the axis. Unless this is looked at in this manner, our view of muscle mechanics would be seriously hindered.



Picture 4.5.7

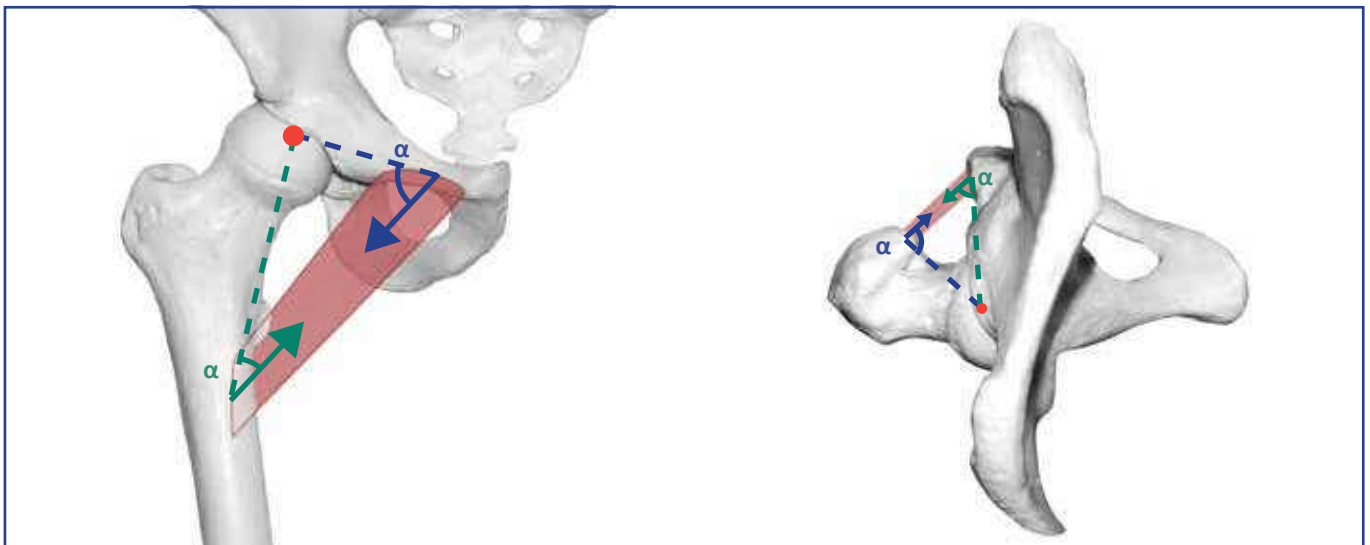
This picture represents the pectineus muscle and its lever arms in the A-P axis of the coxofemoral joint, in its neutral position. There is one lever arm for each of the insertion sites. Lever arms are the shortest lines between the point of application of the force (in this case, the muscle insertion) and the joint axis which is being studied.

MUSCLE FORCE ANGLE

The angle of application of a force (force angle or FA) is the angle that is created between the muscle's line of action and the lever (bone).

This angle determines the relationship between the muscle force vector and the direction in which the lever moves. It should be calculated for the different points of the ROM, since it varies along the articular ROM (because of the changing relationship muscle-to-bone).

Muscle force angles exist at both of the attachment/redirection sites. Thus, there are always two angles present, one for each of the joint's segments.

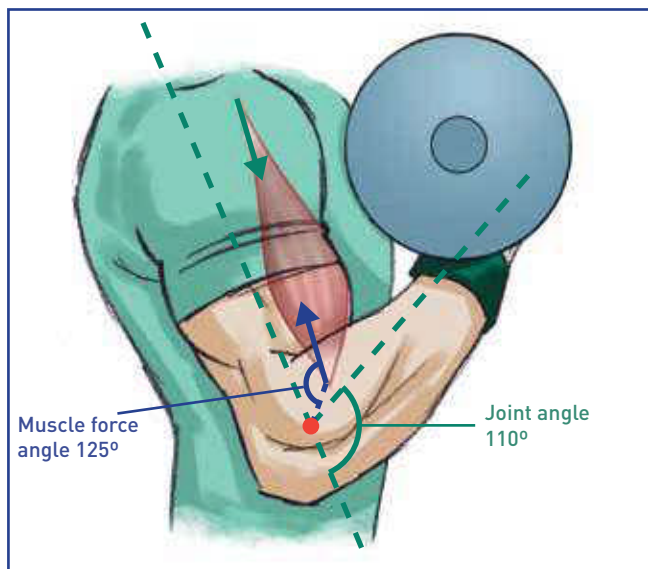


Picture 4.5.8

This picture shows how to analyse the force angles for the pectineus muscle (for A-P axis) and the quadratus femoris (for its S-I axis, from a superior view). They are both being looked at on a neutrally positioned coxofemoral joint.

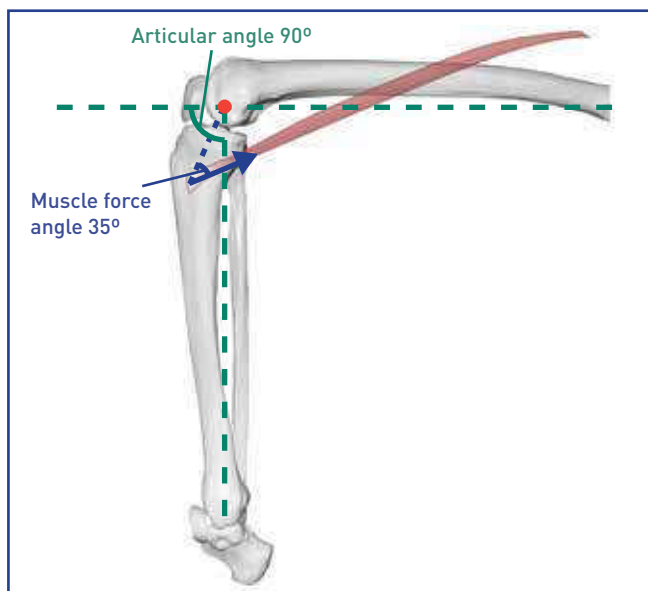
The picture shows how two different force angles are present for each one of the above-mentioned muscles –one angle for each of the insertion sites–. Force angles in the human body are most commonly unlike those explained by traditional seesaws, in that they are usually not the ideal 90° angle, where optimal rotation of the lever would be generated.

There are two concepts that should not be mistaken. One is force angle, and the other is joint or articular angle. Even if the joint is at 90° , it is very likely that the muscle is at a different angle at its attachment site to the bone. And when this is so, not all of the force being produced by the muscle will cause rotation at the axis.



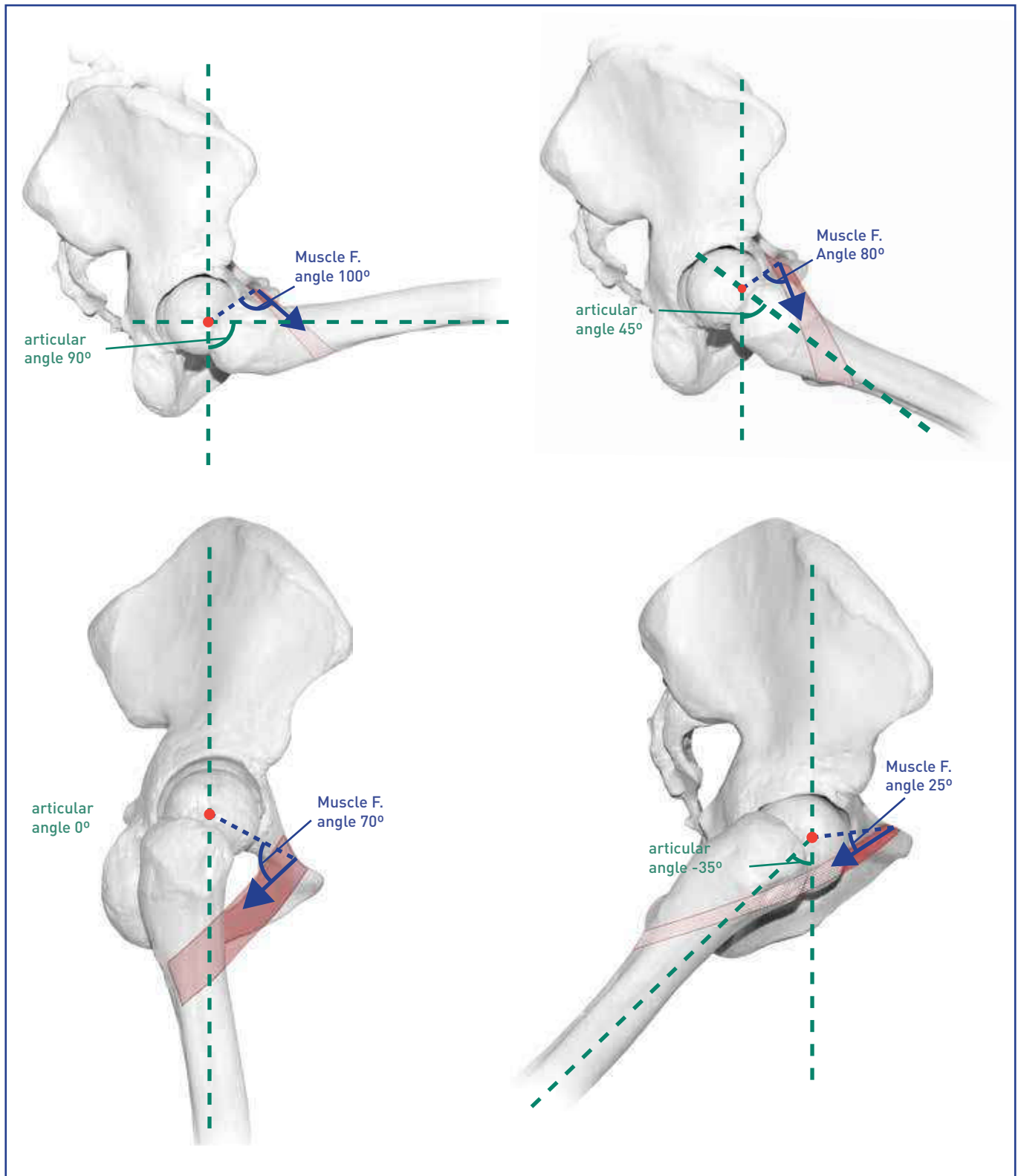
Picture 4.5.9a

This picture shows a sample analysis of the biceps brachii muscle and its force angle, in relationship to the elbow joint, for its L-M axis. The angle in the joint is simultaneously analysed. You can tell how these two angles actually measure different things, so there is no reason for confusion. However, muscle force angles and joint angles frequently get confused, especially when it comes to this specific move. Thinking that the muscle's lever arm runs parallel to the forearm—without taking into consideration the bone's width—is one of the most common mistakes. Also note that the muscle's line of force is not always parallel to the arm.



Picture 4.5.9b

This picture shows the analysis of the sartorius muscle's force angle, in relationship to the L-M axis of the knee joint. The analysis of the articular angle has also been depicted. Equally to what was shown in the previous picture, you can see that both angles measure different things and should, therefore, not be mistaken.



Picture 4.5.10

This picture represents the evolution of the pectineus muscle force angle (to the L-M axis of the hip) as the hip position progresses from 90° of flexion to around 30° of extension. The evolution of this angle throughout the flexion-extension move can be observed and is a consequence of the varying relationship between the pectineus's line of force and the insertion site at the bone. The picture also depicts and analysis of the articular angle, and it once again becomes obvious that they are not the same.

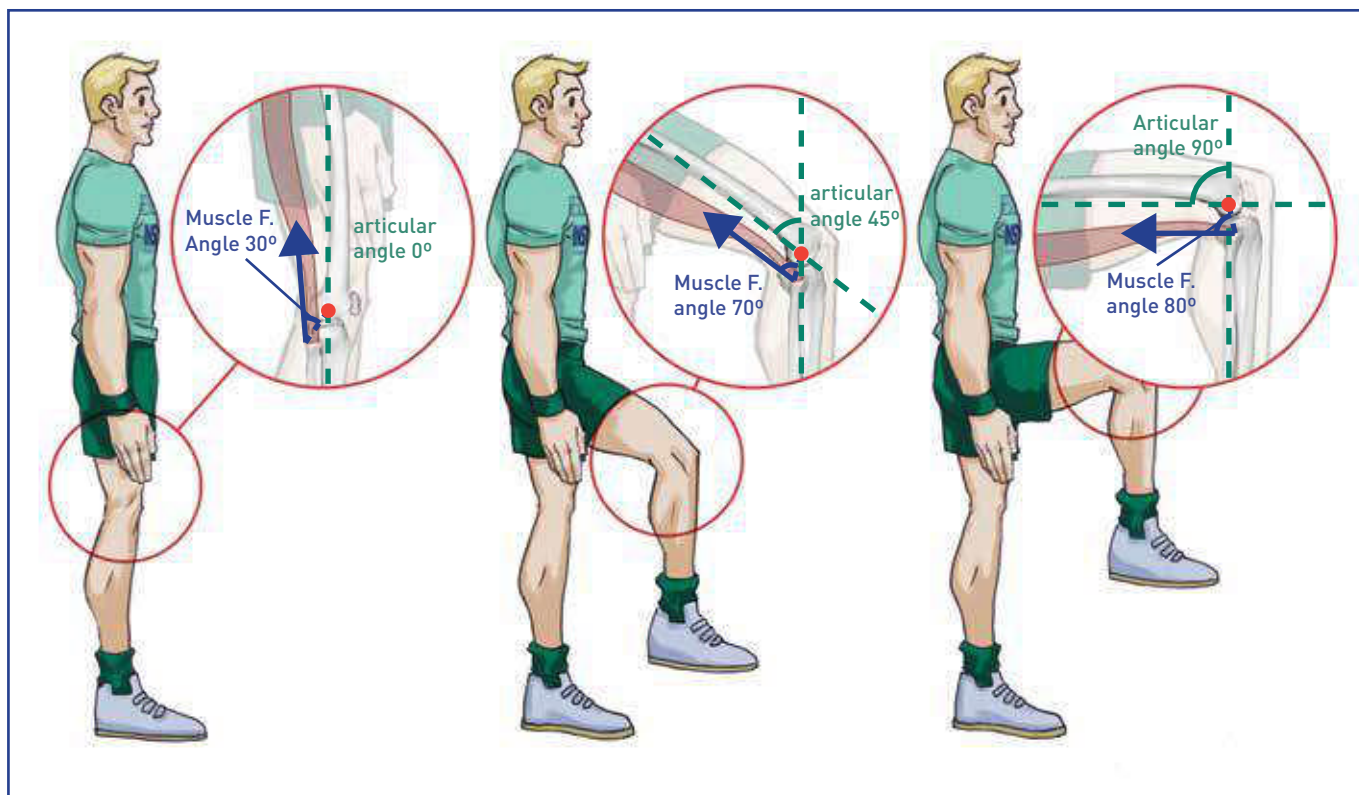
The closer the force angle is to 90°, the more effective the force will be at generating rotational movement on of the segment. Therefore, muscle torque responds to the formula as seen previously regarding lever systems. It is only that the force angle needs to be taken into consideration as a mechanical factor.

$$\text{MUSCLE TORQUE} = (\text{FORCE} * \text{LEVER ARM}) * \sin \alpha$$

As learnt before, the force angle will determine a trigonometric relationship, which will rule over how much

force will actually generate rotation on the lever.

The optimal force angle does not necessarily have to be the strongest point in the ROM (the greatest motor torque). The optimal spot is also subject to the relationship tension-elongation of the whole musculature that is involved in that joint. And to fatigue, as well. For biarticular muscles, things become even a bit more complex. In this case, the tension-elongation relationship will also be dependent upon the positioning of the other joint.



Picture 4.5.11

The evolution of the angle of force for the long head of the biceps femoris is analysed in this picture. To be more specific, it's its relationship to the L-M axis of the knee joint, when moving it from neutral to about 90° of flexion. You can tell how the angle gets closer to 90° as the knee gets flexed, since the relationship between the muscle's line of force and the insertion site changes.

MUSCLE FORCE MOMENT ARM

There are two important factors that determine a muscle's mechanic ability to generate torque: the length (distance) of the lever arm and the force angle.

It is the moment arm that establishes a relationship between the above two mechanical factors. The moment arm determines what proportion of the force applied to a lever actually causes rotation. Because this proportional

measurement establishes a relationship between lever arm (distance) and the sine of the angle (adimensional measurement), we are talking about a unit of distance.

In this way, the muscle's moment arm turns into a graphic representation of the mechanics a muscle holds, and which enables it to generate torque around a given articular axis.

HOW TO CALCULATE A MUSCLE'S MOMENT ARM

Muscle moment arm can be calculated in two ways: either trigonometrically or with a graphical method.

For trigonometric calculations:

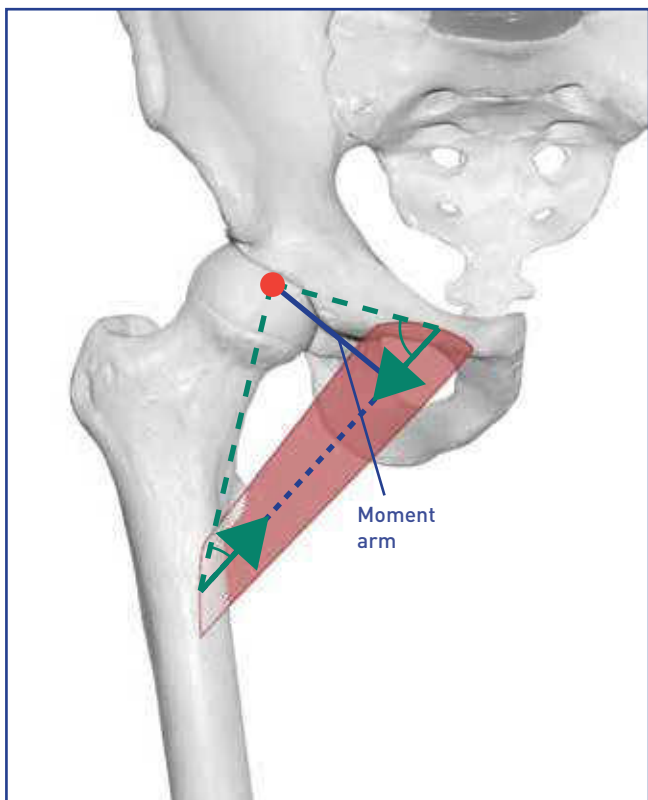
$$\text{MOMENT ARM} = \text{LEVER ARM} * \sin \alpha$$

Logically, this part 'lever arm * sin α ' can be substituted, in this way:

$$\text{MUSCLE TORQUE} = \text{FORCE (amount of tension)} * \text{MOMENT ARM (mechanical traits)}$$

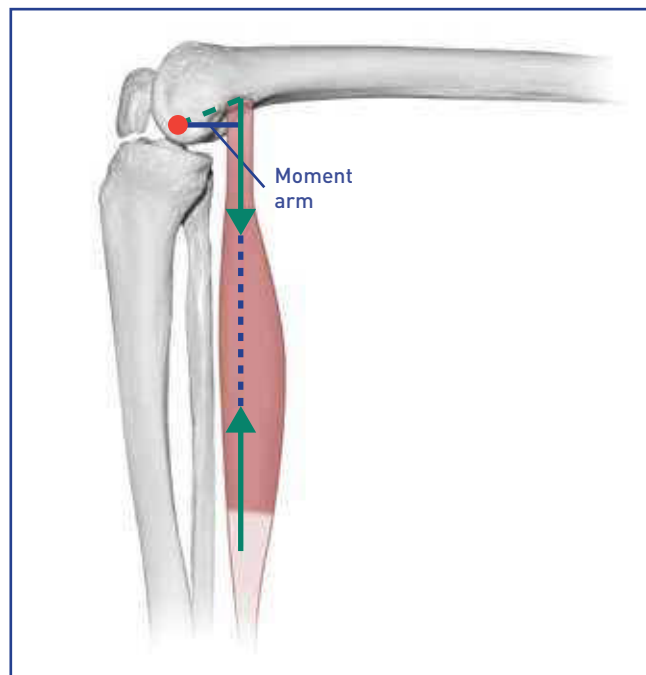
For the graphical method:

Moment arm is that straight line which runs perpendicular to the line of force (line of muscle action), and touches the axis of rotation.



Picture 4.5.12a

This picture shows the analysis for the moment arm of the pectineus muscle, for the A-P axis of the hip joint, in its neutral position. This muscle and hip position have been chosen to easily exemplify in an organised and pedagogic manner how muscle mechanics work. In this picture, you can see how the moment arm constitutes a line that touches the joint axis (interruption) and lies perpendicular to the muscle's line of force.



Picture 4.5.12b

This picture shows the moment arm analysis of the gastrocnemius medialis, for the L-M axis of the knee joint, at 90° of flexion. You can see how the moment arm constitutes a line that touches the axis (interruption) and runs perpendicular to the muscle's line of force.

WHY IS CALCULATING MUSCLE MOMENT ARM IMPORTANT?

Moment arm unites the relationship between the distance and the angle. In this way, it is a fast way to determine a muscle fiber's mechanics to generate force on a given plane.

Data would be incomplete if we only took into consideration the distance factor, for most of the times, the muscle force angle does not equal 90° and thus not all of the force applied onto the lever will generate rotation.

If we only took into consideration the force angle, the resulting information would also be incomplete. The length of the lever arm—which indicates the distance at which the force is being applied onto the system, must be known.

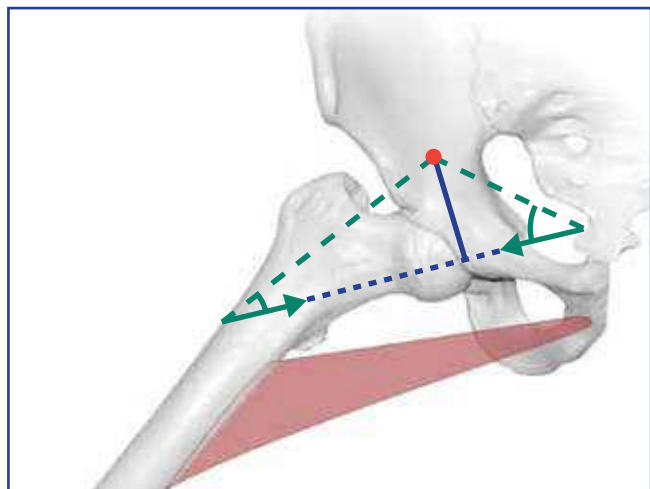
It is a muscle's moment arm that establishes the relationship between both of those characteristics, throughout the ROM and relative to the different articular axes. Because of this, using the graphical method can help to determine, in a fast and simple way, what the mechanics for each of the muscle fibers are, for a given position and articular axis.

ONE SAME MOMENT ARM PER MUSCLE FIBER AND ARTICULAR AXIS

Just as could be appreciated from the above drawings, in the same single muscle, two different lever arms and two different force angles can be found. Each one will have to do with one of the two different existing attachment sites for one given muscle fiber.

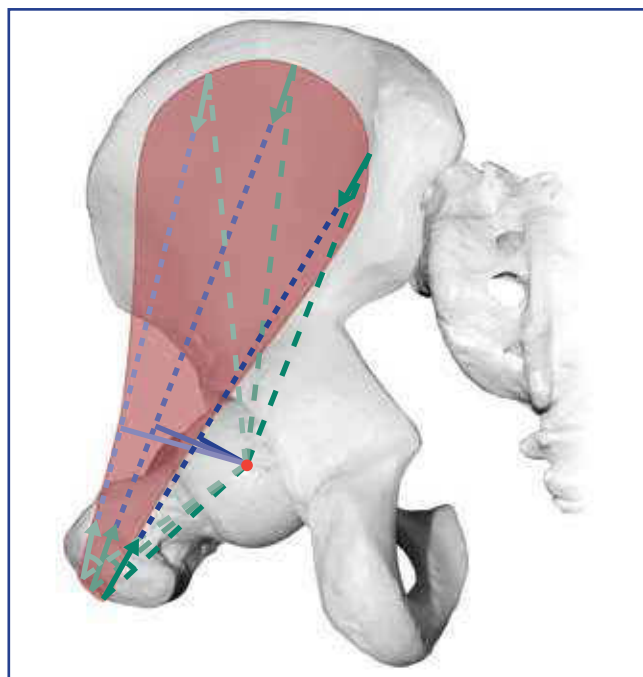
In spite of this, there is only one single line of action for any one given muscle fiber and articular axis. And as a consequence of this, there is only one line that is perpendicular to the force line and which crosses the axis. This last fact makes it clear that only one moment arm will exist, for each fiber (or group of fibers that share one same force line) and articular axis.

By using the algebraic method and doing the appropriate trigonometric calculations for each one of the lever arms for their respective angles, it will become clear that the resulting moment arm measurement always turns out to be the same. Therefore, if one of the attachments has a greater lever arm than the other, the force angle will be proportionally less advantageous (furthest to 90°). At the same time, that attachment with the smallest lever arm, the angle will be proportionally more advantageous (closer to 90°).



Picture 4.5.13a

This picture helps analysing the moment arm for the adductor brevis muscle, in the A-P axis of the hip joint, when the hip is in an abducted position. For these given axis and muscle, only one possible moment arm is possible. It is therefore impossible that any muscle fibre holds greater ability to generate force from one insertion point than from the other, for a given joint and axis. Clearly stated: the adductor brevis muscle is not mechanically more capable of moving the pelvis over the femur, than to move the femur over the pelvis. As previously explained in this book, whether it is one segment or another that moves, depends on which one of them offers less resistance.



Picture 4.5.13b

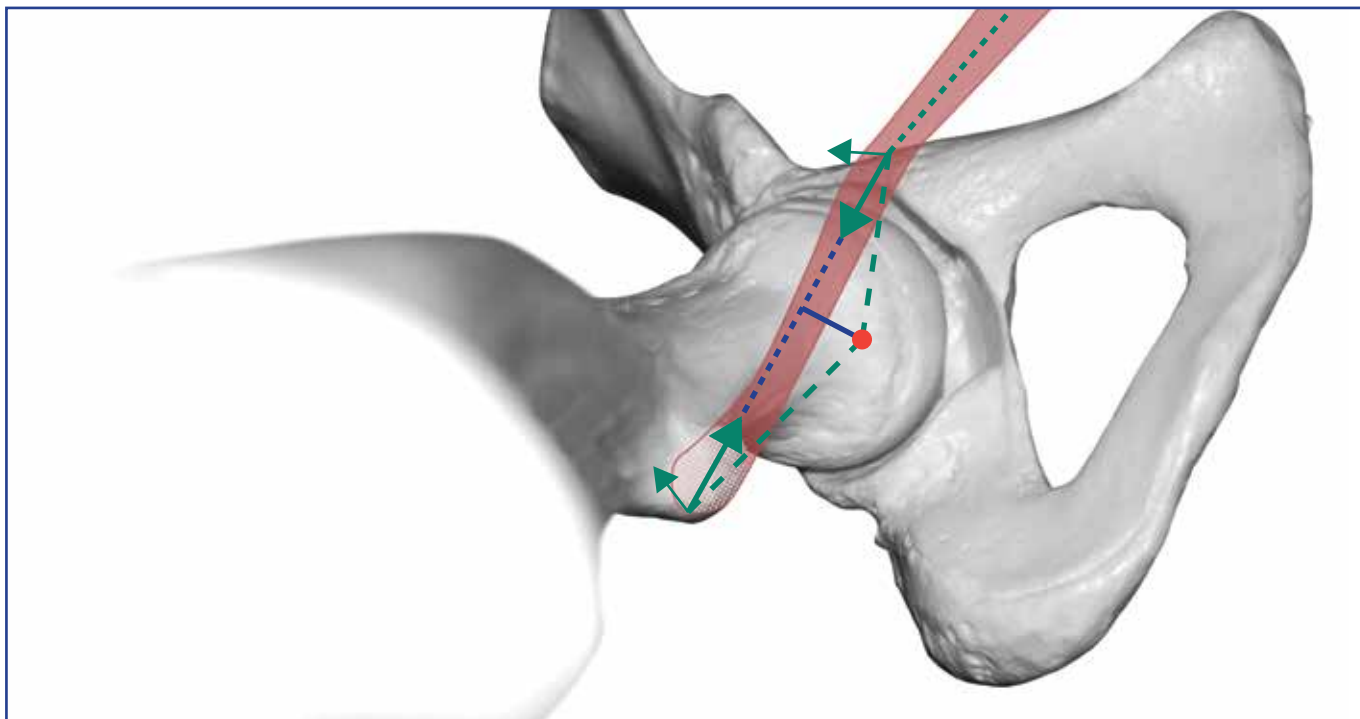
This picture depicts the moment arm of the gluteus medius, for the S-I axis of the hip joint (femur referenced), at 90° of hip flexion. You can see how each muscle fibre and axis hold one single moment arm. You can also see how different fibres hold different lines of force; more specifically, it is the anterior fibres that are mechanically better prone to exert internal rotation. Nevertheless, for this given position, all of the fibres are capable of exerting hip internal rotation. This differs from the neutral positioning of the hip joint, where (also for the S-I axis), the posterior fibres are mechanically capable of generating external rotation of the hip.

LIMITATIONS TO MOMENT ARM

Even though the muscle's moment arm constitutes a clear indication of a fiber's mechanical advantage around a given articular axis, it does not clearly indicate the direction of the rotation it is generating.

This fact causes many professionals to find difficulty in visualizing the lever and the direction of rotation. And it also leads them to believing that they can skip finding out where the lever arm is, and what the direction of force is, as long as they figure out the moment arm.

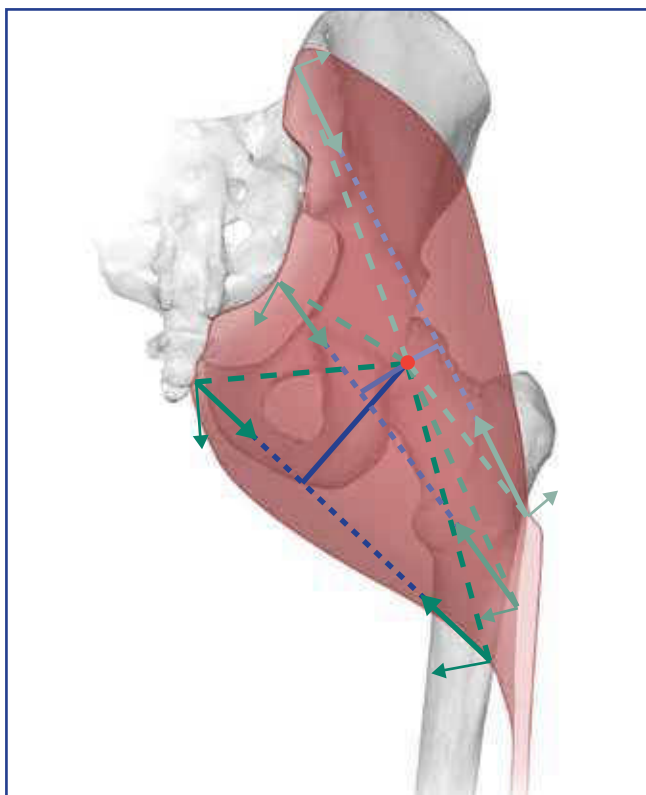
Determining the moment arm for a muscle fiber is, indeed, of the greatest importance. It is also very important to figure out its progression throughout the joint's ROM. But despite of that, we must also be able to visualize the lever arm and the direction of the muscle force if we aim to understand the direction of rotation a specific muscle fiber tries to generate around a given articular axis.



Picture 4.5.14a

This picture helps to analyse the moment arm of the psoas major, for a S-I axis (femur referenced), being the hip slightly abducted. The length of the moment arm gives us a hint about the mechanics.

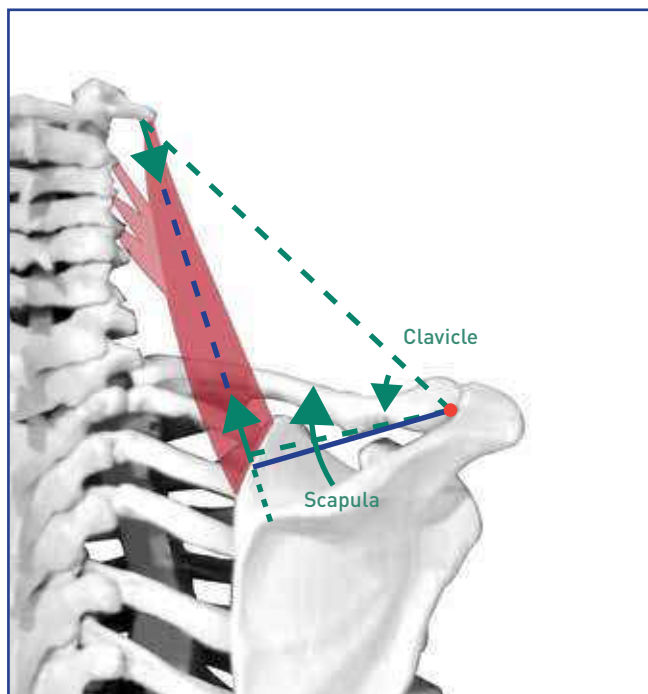
In order to figure out this muscle's intent to generate internal rotation, it is useful to also take a look at the lever arm and the muscle's line of force. Taking all these three factors under consideration helps to better visualize the direction of rotation around the given axis. In this case, and opposite to what is usually stated in bibliography, the psoas major muscle is mechanically capable of generating internal rotation of the hip joint. Later on in this chapter, we'll take a look at the redirectioning effect that applies here.



Picture 4.5.14b

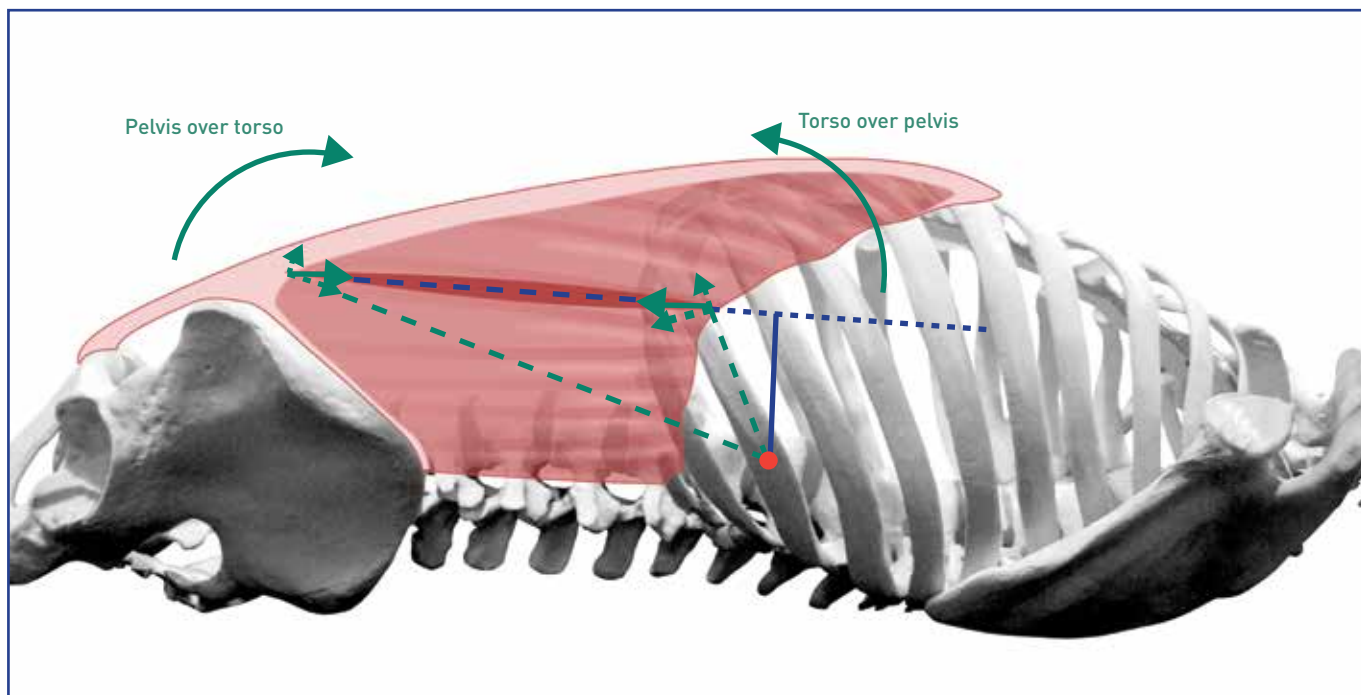
This picture shows the moment arm, lever arms and rotational components for the different fibres that comprise the gluteus maximus. They have been represented for an A-P axis of the hip joint, in its neutral position.

You can see different moment arms, and how the more iliac fibres have opposite directions of rotation to the more coxal fibres. While the superior (iliac) fibres hold mechanics for abduction, the more inferior (coxal) fibres do exactly the opposite. The superior sacral fibres basically have no mechanics for the generation of rotations around this A-P axis of the hip joint.



Picture 4.5.14c

This picture helps analyse the moment arm, lever arm and rotational component of the levator scapulae, around the A-P axis of a neutrally positioned acromioclavicular joint. Both the lever arm and the rotational component aid at the visualization of the direction of rotation generated by muscle torque. In this case, the direction of rotation corresponds to adduction around this A-P axis of the acromioclavicular joint. It is sensible to think that this muscle does also generate torque directly over the cervical vertebrae and around the sternoclavicular joint. However, the picture does not aim to represent all possible torques that this muscle can create, for each of the joints it crosses over.



Picture 4.5.14d

This picture is useful for the analysis of moment arm, lever arm and rotational component of an antero-inferior fibres of the abdomen's external oblique muscle, around the L-M axis of the spine (at a T10-T11 level). It can be seen that this fibre has got mechanics for the flexion of the spine. If a posterior-lateral fibre was to be chosen for study, we would be able to observe that it would have some –though not much– mechanics for spine extension at various vertebral levels.

In the pictures above, you can see how visualizing the lever arm, as well as the direction of the muscle force are key to understanding the direction of rotation the muscle fiber is trying to generate around a given articular axis.

This is the reason why we will next present you with a detailed explanation on:

The relationship between a muscle fiber's line of action and force angle and the rotational and translational components of that given force on the bony lever.

COMPONENTS OF MUSCLE FORCE ON THE LEVER

Torque, or the part of the force that causes rotation on the lever, has to do with the rotational component, which is the part of the force that runs perpendicular to the lever.

This rotational component has the same direction as the lever's direction of movement. This is precisely why it is considered to generate rotation around the articular axis.

The translational component runs along the lever.

Both components are, by definition, perpendicular to each other.

Every muscle force that is applied on a lever generates its own translational and rotational components. This individual components of that muscle force oppose each other, but are not necessarily equal.

As seen before, the only time when there is a lack of translational component is when all of the effort is of rotational character. This situation comes about when the force angle equals exactly 90° .

NOT WHEN THE ARTICULAR ANGLE IS 90° !

Only in those situations where there is no rotational component present, will the force present solely with a translational component. This takes place when there is no moment arm (it's distance equals 0), and thus, the force angle is 0° .

Whenever there are absolutely no rotational forces on a lever or joint, that is called **ROTATIONAL EQUILIBRIUM**. This equilibrium takes place when there are no rotational forces present on a joint, or either when the result for all rotational forces acting on the lever equals zero.

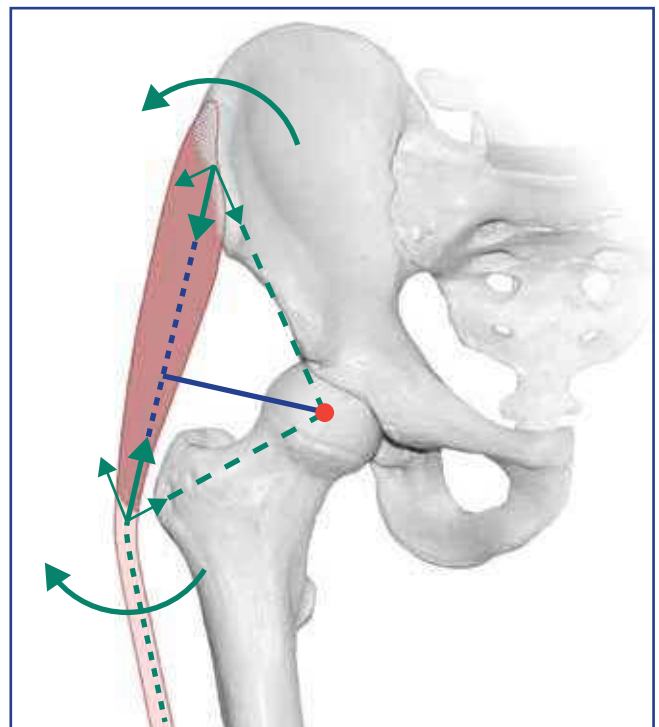
A muscle's rotational component, relative to a specific joint, will be greatest whenever its force angle is greater than 45° and smaller than 135° .

Should the opposite be the case, then the muscle can have more of a translational component.

A muscle presenting with a less optimal force angle, compared to another muscle, will not necessarily show a lesser moment arm. There is one other necessary factor, that being the length of the lever arm (distance), which will help determine the moment arm. A given muscle could have a greater distance from the fulcrum to the point of application of the force.

In summary, two muscles with different force angles could have the same moment arm. Consequently, they could also have the same mechanics to generate rotational forces on a plane, even in cases where their rotational and translational components were different.

But even then, the total amount of torque generated by each of the fibers or fiber bundles depends on several factors that have to do with force (muscle tension). Examples of these are: muscle structure, cross section, main type of fibers, neuromuscular recruitment ability, etc. **This all will be a part of the final, total, equation.**



Picture 4.5.15

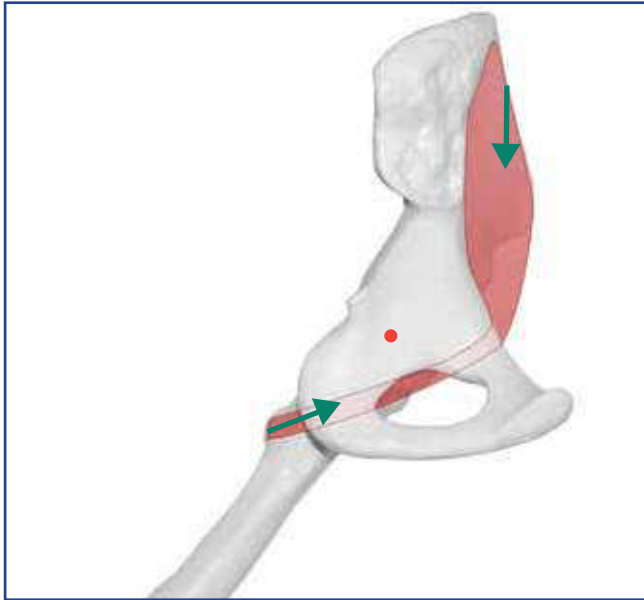
This picture shows the analysis for: moment arm, lever arm, translational and rotational components of the tensor fasciae latae muscle, for the A-P axis of the hip joint, in a neutral position.

The lengths of the different arrows, representing muscle action and its components, do not aim to represent the intensity of tension generated by the muscle. Calculating the amount of force would depend upon several factors. In this picture, arrows merely aim at determining the direction and relative magnitude of the vectors.

ANATOMICAL PULLEYS AND DIRECTION OF THE MUSCLE ACTION LINE

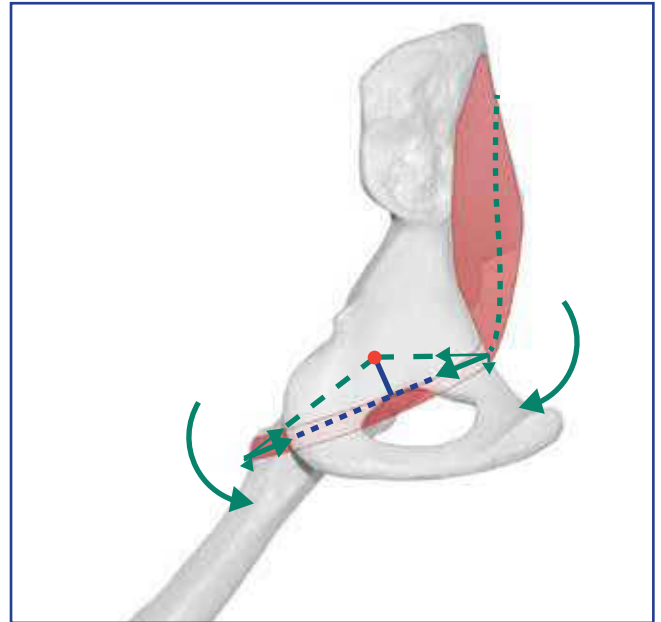
Body structures which alter the pulling direction of any given muscle can be regarded as anatomical pulleys. They create a redirectioning of the muscle's line of force, and can be found in bones, connective tissue or muscle tissue, itself.

As explained in the chapter on lever systems, sometimes a muscle redirectioning takes place. When this is the case, and in order to do the calculations, the tendon's line of force at its attachment site won't be of use any more. Instead, attention should be placed on the spot where the muscle last (directly or indirectly) touches the pully-creating-bone.



Picture 4.5.16a

This picture shows the superior insertion site of the iliac muscle, and how its direction differs to that of the inferior insertion site (at the lesser trochanter). One of the two arrows in the picture represents the generation of force over the L-M axis of the coxofemoral joint, causing extension. Can you guess which one that is?



Picture 4.5.16b

This is what you can do to figure out the line of force that generates torque around the L-M axis of the hip joint:

Take the iliacus muscle, note the direction of its line of force just prior to entering contact with the pelvis, and continue it towards the insertion at the femur (trochanter minor).

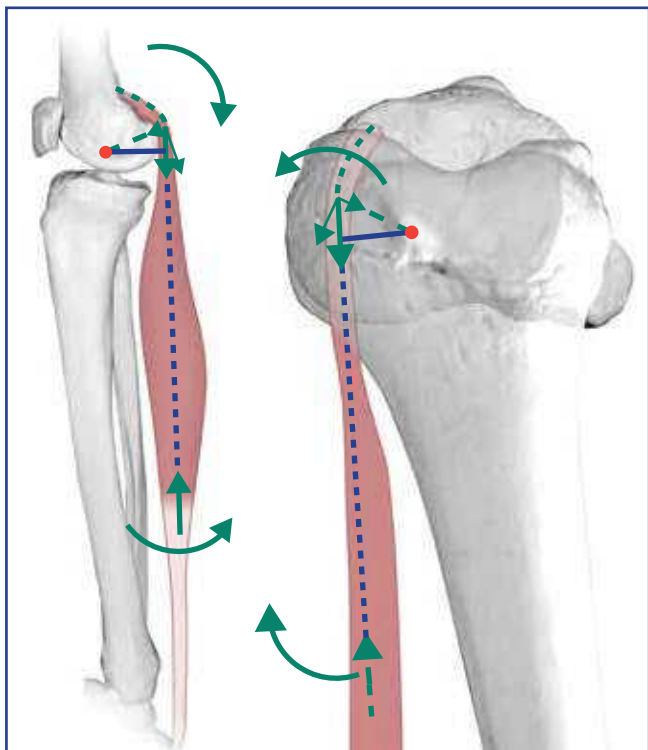
The other line is also important, for tension is transmitted from both of the insertion sites towards the center. However, when trying to determine the mechanics of a specific muscle fibre, the line of force and the convergence must be considered from the point of redirection.

Please pay attention to how, when redirectioning exists, the lever arm does not connect the axis to the insertion site but, instead, it connects it to the point of redirection.

The pictures above show it clearly, how the attachment site does not determine the line of action. Therefore, and in order to do the mechanical calculations, we must consider whether the fibers have been redirectioned.

Muscles do, indeed, create tension from their two attachment sites. However, it is important to bear in mind that only one line of action will be able to create torque. This line of action will be that which connects the two segments that constitute the joint.

In picture 4.5.16a, you can see how the superior line of force connects the pelvis to the pelvis. At the same time, the inferior line connects the pelvis with the femur. In order to calculate the mechanics of the iliac bone, the line to be chosen should be the second one.



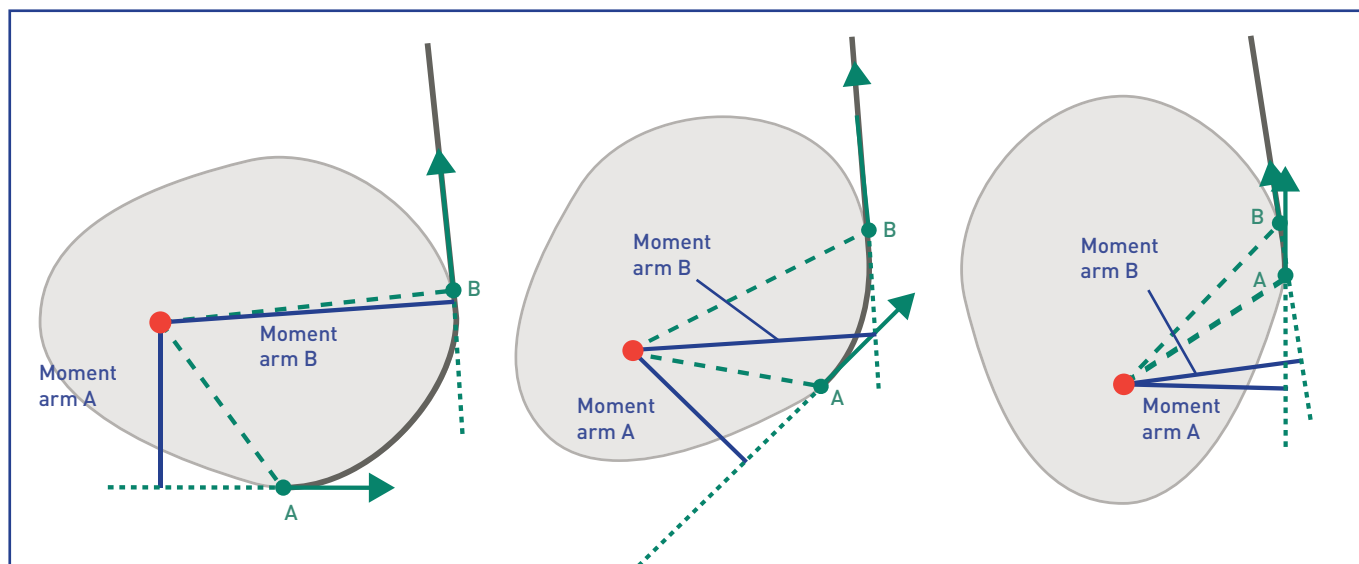
Picture 4.5.17

These two pictures show how, at one of the insertion sites, the force becomes redirected. The picture on the left shows the analysis of the redirectioning the medial gastrocnemius experiments, for the L-M axis of the knee joint, in its neutral position. The picture on the right analyses the redirectioning of the semitendinosus muscle, for the S-I axis of the knee joint, at 90° of flexion.

In the previous chapter, we learned how the direction of any given force affects the mechanics of a given lever system. Let us now consider, as an example, an eccentric cam within a varying resistance machine (the same type one can commonly find in fitness studios). By observing it, one can realize that the distance of the moment arm changes, as the pulley rotates around its axis.

If we were to consider the attachment site to the cable as the point of application of the force, the moment arm would remain the same throughout the rotation. So, it would make no sense to use an eccentric pulley, and a regular, circular pulley could be used, instead.

This makes it clear that, in order to calculate the mechanical variables for any lever system, it will not be the attachment site of the force that should be taken into account. Instead, we should consider the place where the force last contacts the structure.

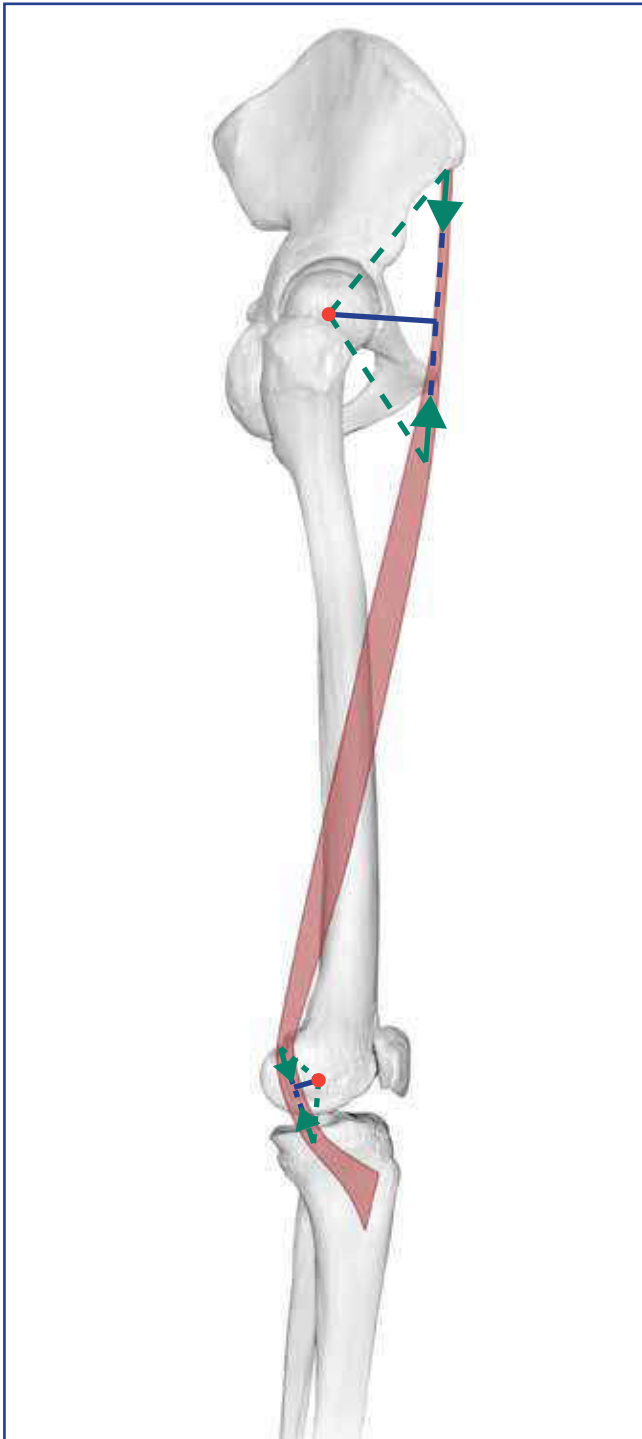


Picture 4.5.18

Moment arm analysis as to: (A) the line of force created at the insertion of a cable in to an eccentric pulley, (B) as to the line of force that's created at the redirectioning.

Clearly, both the contact site and the moment arm remain constant for the analysis of case A, while they change in B. If the insertion site were to be the reference, the use of eccentric pulleys wouldn't make any sense.

Most of the existing literature establishes that it is the attachment site which determines the mechanics of the musculature. An exception to this explanation is sometimes made when talking about the quadriceps muscle. Using the attachment site as the reference could be of use in cases where there is no redirectioning, but it would be incorrect when a redirectioning is present.



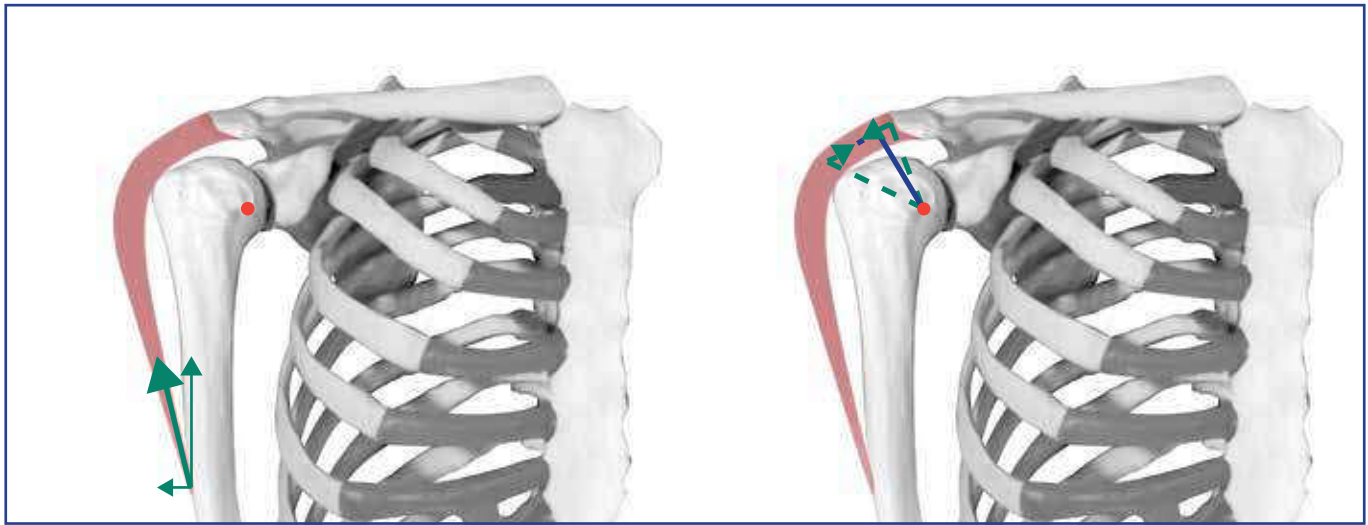
Picture 4.5.19

This is an example of how the sartorius muscle offers a series of lines of force, for the L-M axis of both the hip and the knee joints, in their neutral position. Depending on the resulting moment arms, one given tension of this muscle will generate different rotational torques for the different joints it crosses over. It can also be noted, that the line of force present at knee level and the line of force at the hip are not the same, due to the redirectioning this muscle is subject to on its way from origin to insertion site.

Whenever a muscle crosses over a series of joints, its lines of force should be calculated separately for each one of those joints.

That is to say, one given muscle generates different amounts of torque at each of the joints it affects.

It is important to stress that muscles are not able to choose whether it will generate torque over this one joint or another. As soon as the muscle creates tension, the generated force will affect all of the joint it crosses. And their mechanics must be analyzed separately.

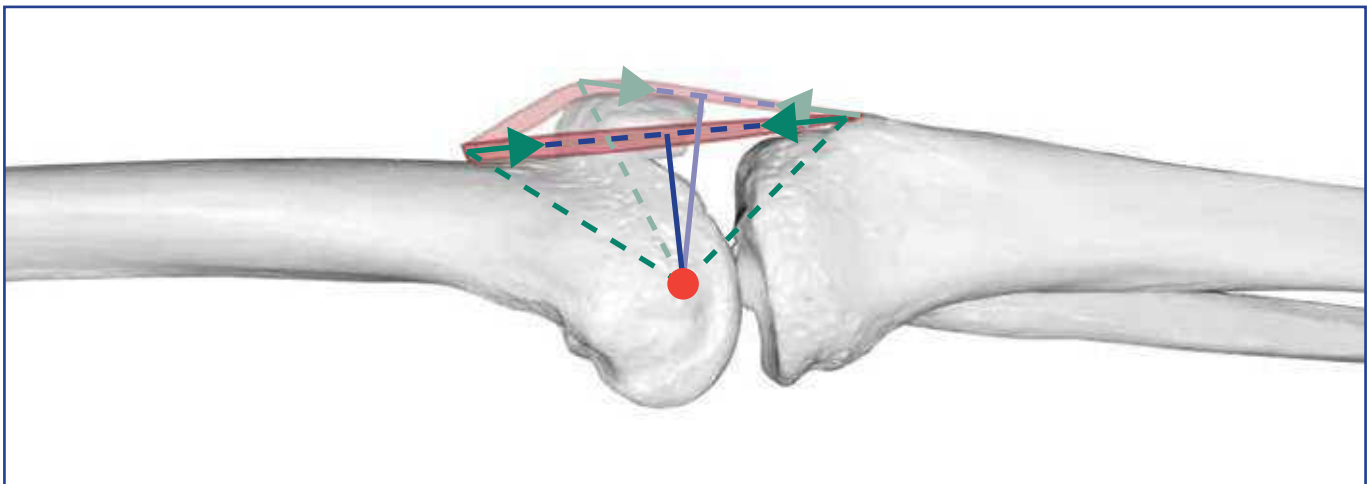


Picture 4.5.20

The example on the left is an extract from a well-known book about joint function and structure. From the picture, it is evident that the redirectioning of the middle fibres of the deltoid muscle, for the L-M axis of the glenohumeral joint, has not been taken into account. The example on the right takes the redirectioning into consideration, for the calculation of the line of force that is responsible for the mechanics.

THE IMPORTANCE OF REDIRECTION AND ANATOMICAL PULLEYS

Many of the redirectionings that take place do so because of what could be called anatomical pulleys. Just like variable resistance machines, anatomical pulleys increase mechanical advantage by increasing the angle of force and, consequently, the moment arm. The following picture shows how important the patella is for the mechanics of the quadriceps muscle, relative to the L-M axis of the knee joint.

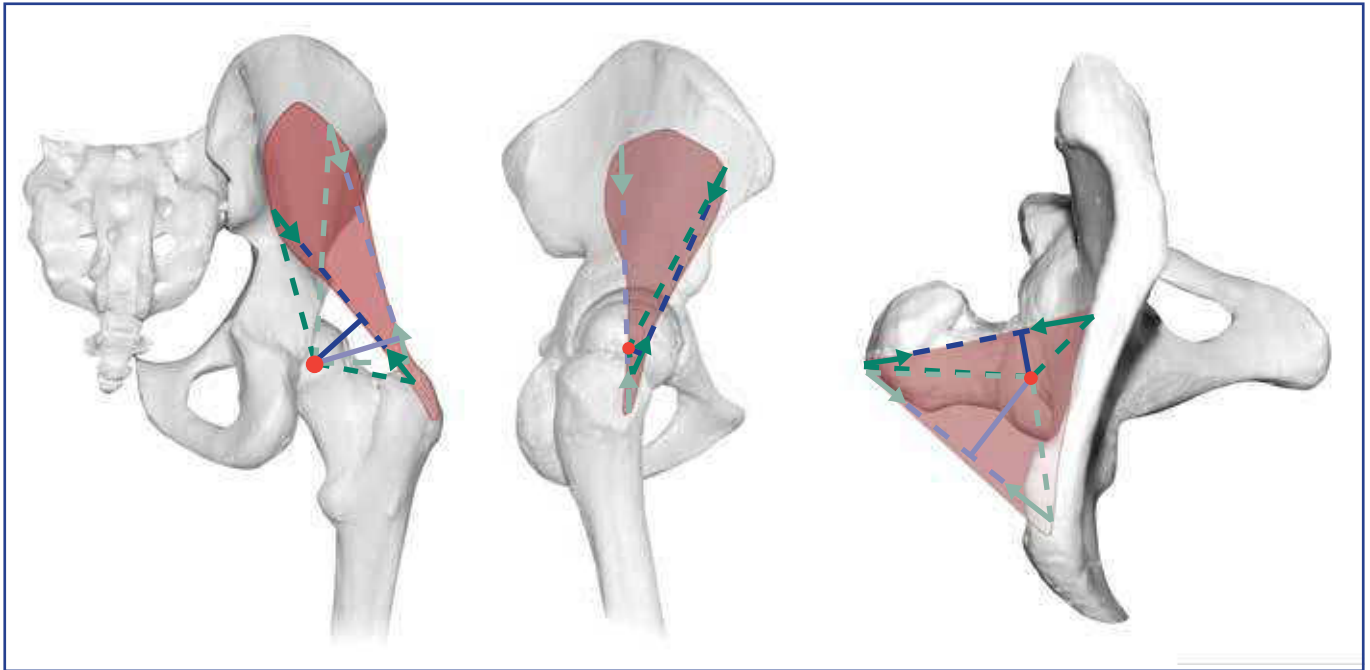


Picture 4.5.21

MUSCLE TORQUE: THREE-DIMENSIONAL versus COPLANAR

Commonly, the direction of the fibers in our muscles do not exactly correspond with any given anatomical plane (sagittal, frontal or transverse). It is therefore characteristic, that the direction in which a given muscle fiber generates rotational force be an oblique plane, relative to the anatomical planes. And not only that, but it also constantly changes throughout the movement of the joints it crosses over.

Sometimes, the plane crosses all three anatomical axes. When this is the case, the musculature generates rotational force in all three dimensions of space (even if the joint does not allow for it, that doesn't mean that any given muscle fiber is not intending to move/control the joint in those given planes).



Picture 4.5.22

This picture represents the mechanics of the gluteus minimus muscle, for the three anatomical axes of the hip joint, in its neutral position. Whenever one of this muscle's fibres contracts, a three-dimensional torque will be created, on the plane corresponding to the lineal force of that fibre. The lineal force will cross over the three anatomical planes, and it will change as the articular position of the hip joint varies. In order to simplify the study process, bi-dimensional analyzes for each of the anatomical planes are usually conducted, which help determine the mechanics for each one of those planes (frontal, transverse and sagittal).

Each muscle fascicle (with the same line of action) has a different direction of force. And not only that, but the direction changes all the time. These facts make muscle actions more complex than what they initially may appear.

For instance, we commonly hear that the biceps brachii is an elbow flexor. The truth is, however, that the contraction of the biceps brachii tries to generate tension and transmit the force on to the lever, which is not only that of elbow flexion. What it actually does is try to generate torque upon all the joints it crosses over directly. And also upon joints it has an indirect relation to, maybe due to connections via connective tissue, bone, capsule. So in all, the biceps brachii tries to create movements of supination, shoulder flexion, protraction of the scapulothoracic joint, etcetera.

MUSCLE FIBERS GENERATE THREE-DIMENSIONAL TORQUE, WHATEVER IT IS THAT OTHER MUSCLE FIBRES ARE DOING

The idea that muscle force generates three-dimensional torque may seem obvious, but it is misunderstood by many exercise professionals.

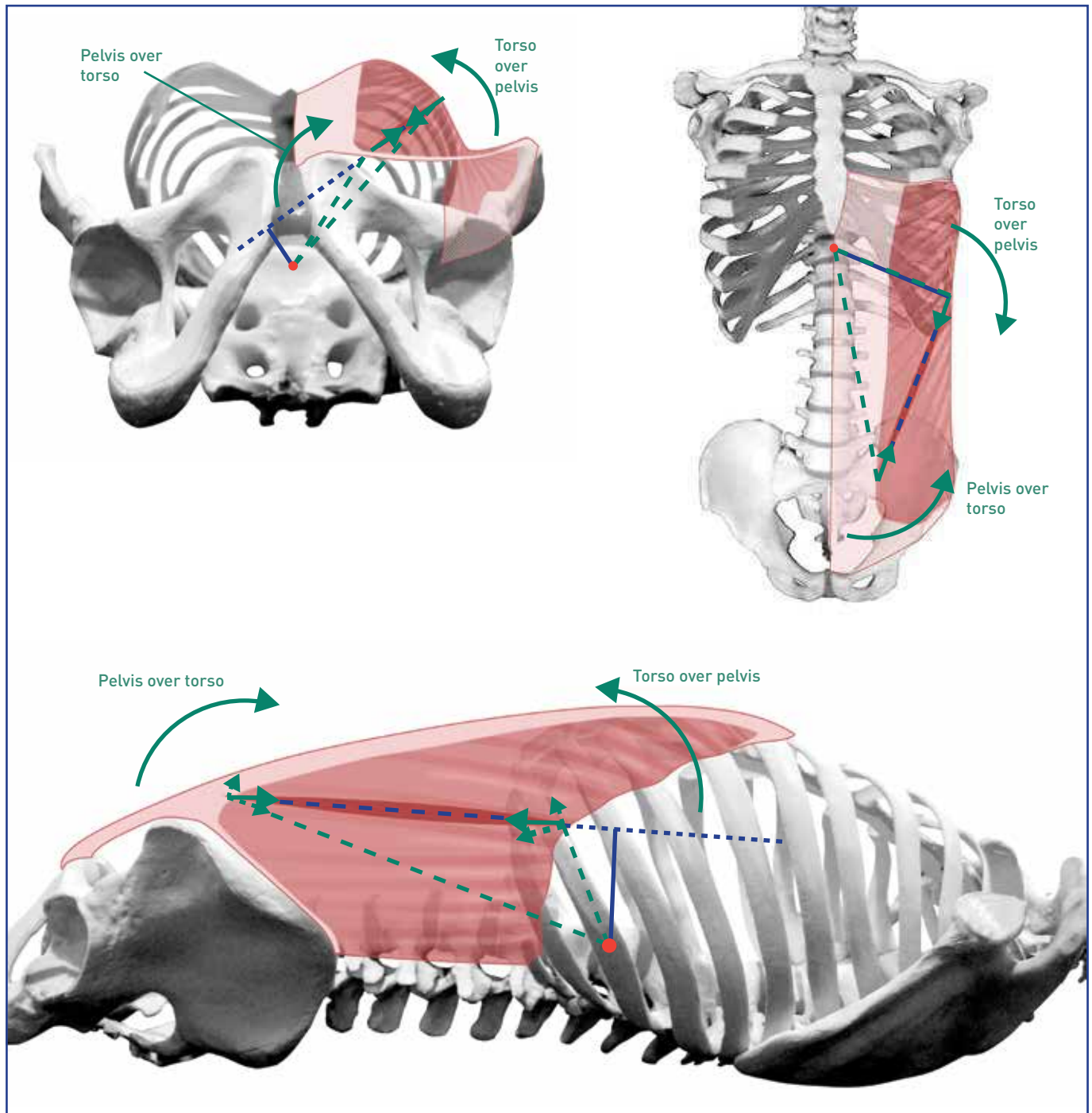
The fact that many books still state that a muscle's actions are different whether it is working unilaterally or bilaterally is a clear example of this. If we look at it from a muscle mechanics perspective, stating something like that is a big mistake.

When a person aims to do a mechanical function analysis of different muscle fibers, they should calculate the three-

dimensional line of force of those fibers. A correct analysis would mean doing bidimensional calculations for each of the different anatomical planes.

example. This is one of those muscles, books usually refer to as having two different actions, depending on whether its fibers are working unilaterally or bilaterally.

Let's use the external oblique muscle of the abdomen as an



Picture 4.5.23

The pictures on the previous page exemplify how the sample muscle fiber has the capacity to generate torque in all three anatomical planes. That is to say, upon its contraction, the fibers is mechanically able to do flexion, as well as rotation and side bending. In fact, every time that specific fiber contracts, it will intend to do exactly the same, regardless of what its counterpart fibers might be doing on the contralateral side, and regardless of whatever other fibers might be generating torque around the same axes. It will try to generate three-dimensional torque, whether it is being counter-acted, or not.

Any given muscle fiber will try to generate the same torque, regardless of it acting unilaterally or bilaterally. Torque is determined by the mechanical traits of that muscle fiber, and its capacity to generate tension (force).

There is, however, a large amount of bibliography stating otherwise. If what it says were true, the mechanics of any given muscle would vary, depending on it acting unilaterally or bilaterally. Think of the gluteus maximus muscle. Since there are two lower limbs in the human body, the gluteus maximus actually has no counterpart around one same hip axis. If we believed that type of bibliography, it would mean that the glute max would no longer be a rotator, whenever the nervous system decided to tense up an internal rotator, in order to counteract the former's three-dimensional torque into external rotation. Following that same line of thought, one would have to say that the tensor fascia latae muscle is a flexor (when the sartorius muscle works simultaneously) or an internal rotator (if it acts on its own). This is inaccurate. And it is also inaccurate to state that an external oblique muscle is a flexor, when it works together with its contralateral counterpart, and lateral flexor when it acts unilaterally.

IMPORTANT

Muscle mechanics calculations can only be performed by analysing what each fiber is able to do, in an isolated manner, around the various joint axes.

What results from that contraction will depend on the relationship between all muscle forces acting upon that given joint, both by muscles crossing over it directly, as well as by those affecting the joint in a more indirect manner. Consequently, whether a muscle is contracting unilaterally or bilaterally does not change the intention of any individual fiber to generate torque.

The fact that a fiber in the left external oblique muscle might be counteracting the capacity of another fiber in the right external oblique muscle does not mean that the latter is not trying to generate the rotation, or the lateral flexion. If that were so, would the right external oblique no longer be a rotator, just because the ipsilateral internal oblique may be also tensing

up and counteracting the rotation?

Let's imagine that a fiber in the external oblique muscle is trying to generate flexion in the spine, and, say, some extensor musculature is counteracting the flexion movement. That doesn't mean that when the external oblique works in a unilateral manner, it loses its ability to generate flexion producing torque around the lateromedial axis.

Think about this question: When flexing the spine forward, what happens if the left external oblique muscle tenses up more than the one on the right side? –Many exercise professionals would reply that this situation would lead to the trunk rotating towards the right, due to the left external oblique generating a larger tension.

What is likely to happen in reality is quite different, though. The truth is, if an individual intends to flex the spine –and as long as the system is able to do so, so not in all cases, the nervous system will find a way to use whatever other musculature available to cover up for that imbalance in the obliques. This is exactly what exemplifies the concept of compensation. And, as a matter of fact, this is much more likely to happen in real life than both obliques contracting exactly with the same intensity.

When an exercise specialist watches a client perform a forward flexion of the spine, can they be sure that all of the musculature, on either side, is working symmetrically?

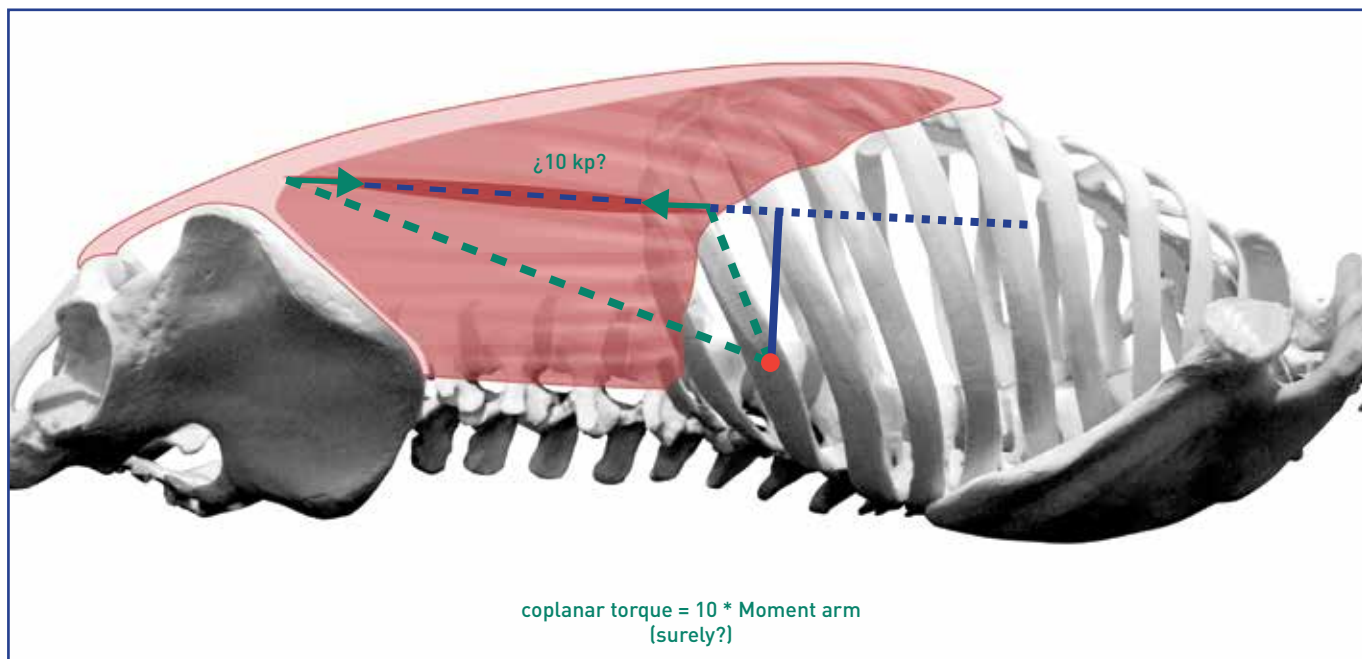
CALCULATING COPLANAR (BIDIMENSIONAL) COMPONENTS FOR A THREE-DIMENSIONAL MUSCLE FORCE

When resistance training, the ability to three-dimensionally generate force shows a new variable, which affects the muscle mechanics relative to a plane of force or to a movement.

No matter whether movement actually takes place or not, taking this new variable into account would imply the need to consider the following: what percentage of the muscle torque lies within the resistance's plane of force? The previous chapter covered how, with non-coplanar forces, whenever not all of the force is in the plane to be analyzed, only a fraction of the three-dimensional-torque-generating-force gets transferred on to the given plane.

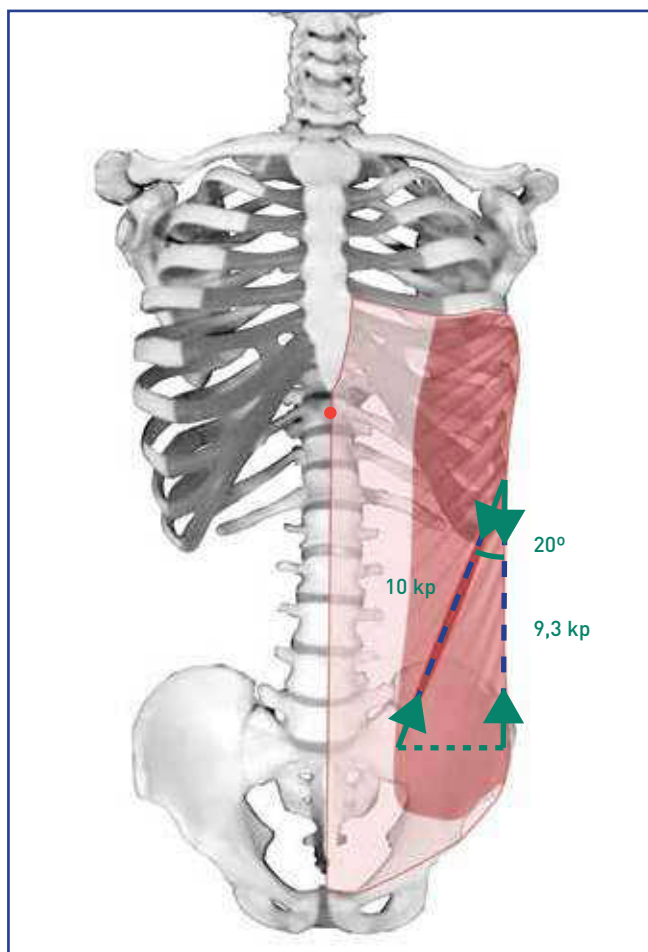
As a consequence, it is that given fiber's (or muscle fascicle sharing that same line of action) torque that should be calculated, but within the plane of force, or plane of movement against resistance that is being analyzed.

Next, we will analyze the mechanics of the external oblique muscle, for the sagittal and frontal planes, as an example.



Picture 4.5.24

Let's consider, for instance, that the amount of tension generated by a muscle were 10kp. Even then, not all of the force generated by this muscle would lie on the sagittal plane, since the line of force crosses over the three planes of space.

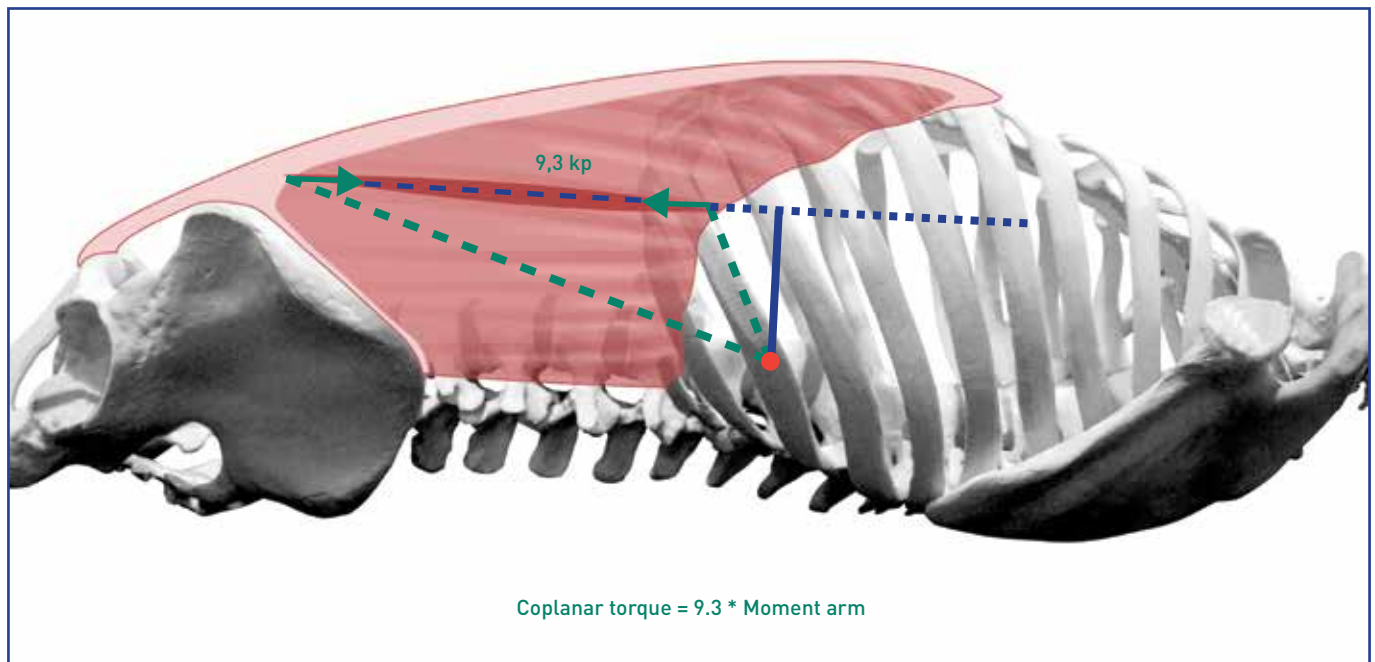


The line of force of this fiber in the oblique muscle does not lie purely on the sagittal plane. Logically, only the percentage of that force what actually lies on that plane will generate a rotational force around the joint. For this specific situation, and for this specific fiber, that rotational force would be called flexion.

Because the line of force does not solely get across one single plane in space, it becomes essential to calculate to what extent it is able to generate rotation, for each of the single axes. In order to do so, different views should be taken into account, for the different axes around which the fiber is able to create rotation. And then, a simple trigonometric calculation will suffice to determine how much of that force actually affects each of the reference planes.

Picture 4.5.25

The portion of force lying within the sagittal plane can be determined by looking at this frontal plane. The portion of coplanar force can be calculated trigonometrically, by using the line of muscle force and the line that lies on the sagittal plane. In particular, the trigonometric calculation would result from multiplying 10kp times $\cos 20^\circ$; totalling 9.3kp.



Picture 4.5.26

Once the portion of force that is on the sagittal plane has been established, torque for flexion can be calculated. This does not mean that this given fibre is unable to generate rotational force around other anatomical axes. If each of the axes were to be subject to study, coplanar components for each specific plane would need to be analysed, in the same manner that has just been shown for the sagittal plane.

IMPORTANT

In spite of it being possible to calculate the coplanar force, fibers will always exert their torque lineally in their own direction. This will most likely not match any anatomical plane, nor any resisted plane of movement. And this fact should be borne in mind.

Taking all of these variables into consideration, it is easy to realize how muscle function is, indeed, a much more complex issue than to merely call a muscle 'a flexor' or 'an extensor'. Things change slightly when all of these mechanical variables are taken into account.

HOW ANGULAR MECHANICS AND MUSCLE LINEAR DEFLECTION RELATE TO EACH OTHER

For any given joint axis, the mechanics for a muscle fiber, and its longitudinal compliance are strictly related.

Angular movement and longitudinal compliance are proportionally related. This means that, if any one fiber has twice as much mechanics than another fiber for a given position, it will also linearly deform twice as much when an angular movement takes place.

The relationship explained above is hard to demonstrate in vivo, since muscle mechanics change constantly throughout the articular movement.

It is important that this relationship is established not only in kinematical terms, but also kinetically. In order to be able to comprehend its kinetics, any one force acting around an interruption (joint axis) should be regarded as a potential generator of rotational movements.

Every time there is a force aiming to generate rotation around an axis, it also has an attempt to linearly deform whatever tissue may have mechanics over that axis. When you read tissue, please read any tissue that has mechanics (tendons, ligaments, capsule, etc.) and not only musculature, even if the latter may play a main role.

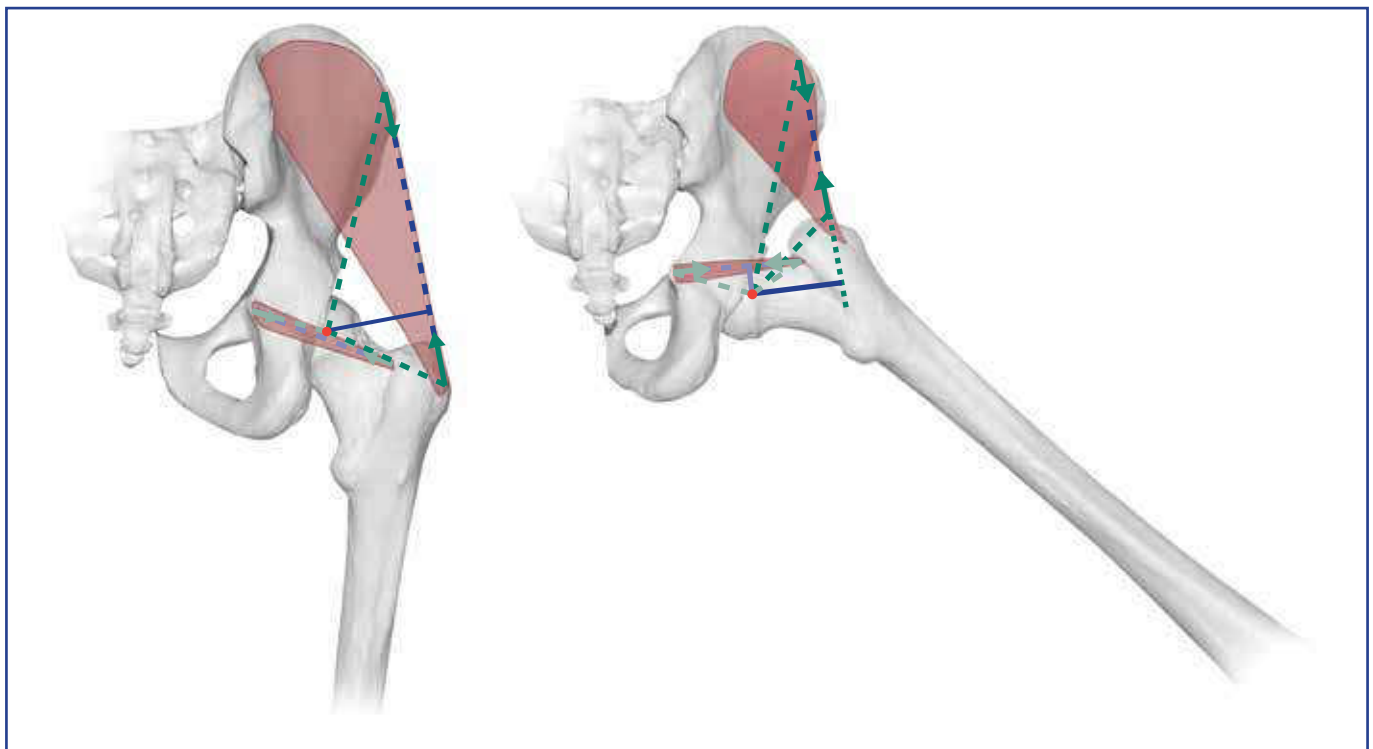
Those tissues having the best mechanics will be the ones receiving the forces trying to cause the greatest linear deformation, in comparison to others. Its mechanoreceptors will be stimulated to the highest intensity (mechanoreceptors are sensitive to linear deformation), in particular those in the muscle spindle and in Golgi tendon organs, but also the receptors within ligaments, fascia, etc.

At the same time, it will be the musculature having the best mechanics for each position –relative to the plane of force, that will receive the greatest strain toward a linear deformation. This shows how, by means of this linear relationship, mechanoreceptors detect forces and inform the nervous system.

Movement may or may not be visible from the outside.

Sometimes nothing will be seen, if for example, the deformation becomes counteracted by the musculature. The above relationship shows that, regardless of what we may see, our system uses its proprioceptive receptors to obtain information about the forces reaching our joints, and not only from resulting articular movement. This is so, even in situations where the deformation experienced by the receptors is invisible to the professional. But even then, the nervous system gets informed and the corresponding muscle response is generated.

All of these mechanisms cause the musculature that is most directly related to the planes of force for each specific scenario to also be the most stimulated, and the most involved in negotiating with the occurring forces.



Picture 4.5.27

This picture shows one given articular ROM of the hip, in the frontal plane. The variation in length of two different muscle fibres has been analysed: a fibre of the superior gemellus muscle and a fibre in the gluteus medius. Due to their mechanical advantage, fibres in the gluteus medius show a noticeably greater deformation. Any given force that acted on the frontal plane would mostly try to cause a deformation on the sensory receptors of the gluteus medius. It would also affect the receptors in the frontal plane associated connective tissue, for the latter are of a mechanical nature and their stimulation depends on linear deformation. Understandably, the proprioceptive system will send signals that will inform about which planes are being physically challenged. And motor responses will be processed, in accordance to the sensory stimuli. Even if an individual were doing a move in the sagittal plane, should he/she receive forces in the frontal plane, their sensory receptors would feel an attempt of deformation (as a consequence of the force), and their central nervous system (CNS) would adjust the muscle response accordingly, relative to their goals (e.g.: keep moving in the sagittal plane and avoid movement in the frontal plane).

SUMMARY: FACTORS UNDERLYING MUSCLE MECHANICS AND ITS JOINT FUNCTION

- Muscles exert force onto both their attachment sites, not only on the moving segment.
- Muscles (or muscle fibers) are unable to produce any more torque on a joint, by choosing one of the attachment sites over another.
- Muscles (or muscle fibers) do not change their individual mechanics, according to them working alone or alongside other muscles (or muscle fibers). And their individual mechanics also remain invariable, whether movement actually occurs or not.
- The line of action is the resultant of all forces reaching one same attachment or redirection site.
- The direction of the muscle force runs from the tendon attachment sites towards the muscle center. Whenever there is a redirectioning, it runs from the redirection site towards the site of direct or indirect contact at the other linkage bone.
- Every time an anatomical pulley is present, the direction of the muscle force is that of the last site of contact with the bony surface. This applies both in cases where there is direct contact, but also when the contact is indirect (via other tissues or structures, similarly to what happens with pulley machines).
- Commonly for multi-joint muscles, the direction of the muscle force is different at each of the joints. The reason for this is that the line of force at each of the attachment sites is usually different, which is also the case with redirectioning pulley systems.
- The moment arm for a muscle line of force is the graphic arising from the two factors that influence muscle mechanics around a given axis: distance of the lever arm and angle of force.
- Any one muscle (or muscle fiber) is responsible for the torque it creates around the joints it crosses over directly.
- Any one muscle (or muscle fiber) also generates joint forces and torque on joints it does not directly cross. This is due to force transmission and valid as long as the system maintains its force unions –rigid union system. Fully understanding this will be one of the major challenges for biomechanists in the upcoming years.
- Due to mechanical restriction chains, muscles of one system joint are able to directly create torque at other joints within the restriction system (you may want to read more about the concept of mechanical restriction chains by Tom Purvis).
- A given muscle generates torque at all of the anatomical axes it may have mechanics on. And because of its layout in space, it will often do so on a plane crossing all three dimensions.
- In order to do coplanar calculations (the most commonly used type in bidimensional calculus), what must be taken into account is the percentage of force actually lying within the plane.

4.6 NEGOTIATING WITH FORCES AND MUSCLE PARTICIPATION

Thoroughly comprehending how important musculature is in the control and movement of joints is key.

The goal of muscles is to guarantee the integrity in joints and motricity and, in that, the continuation of the species. In order to achieve this, the main function of musculature is to generate tension. As simple as that.

The exact manner in which muscles get to participate will depend on the specific requirements from each joint, and for each moment, in order for that joint to become controlled and/or moved around its axes. And therefore,

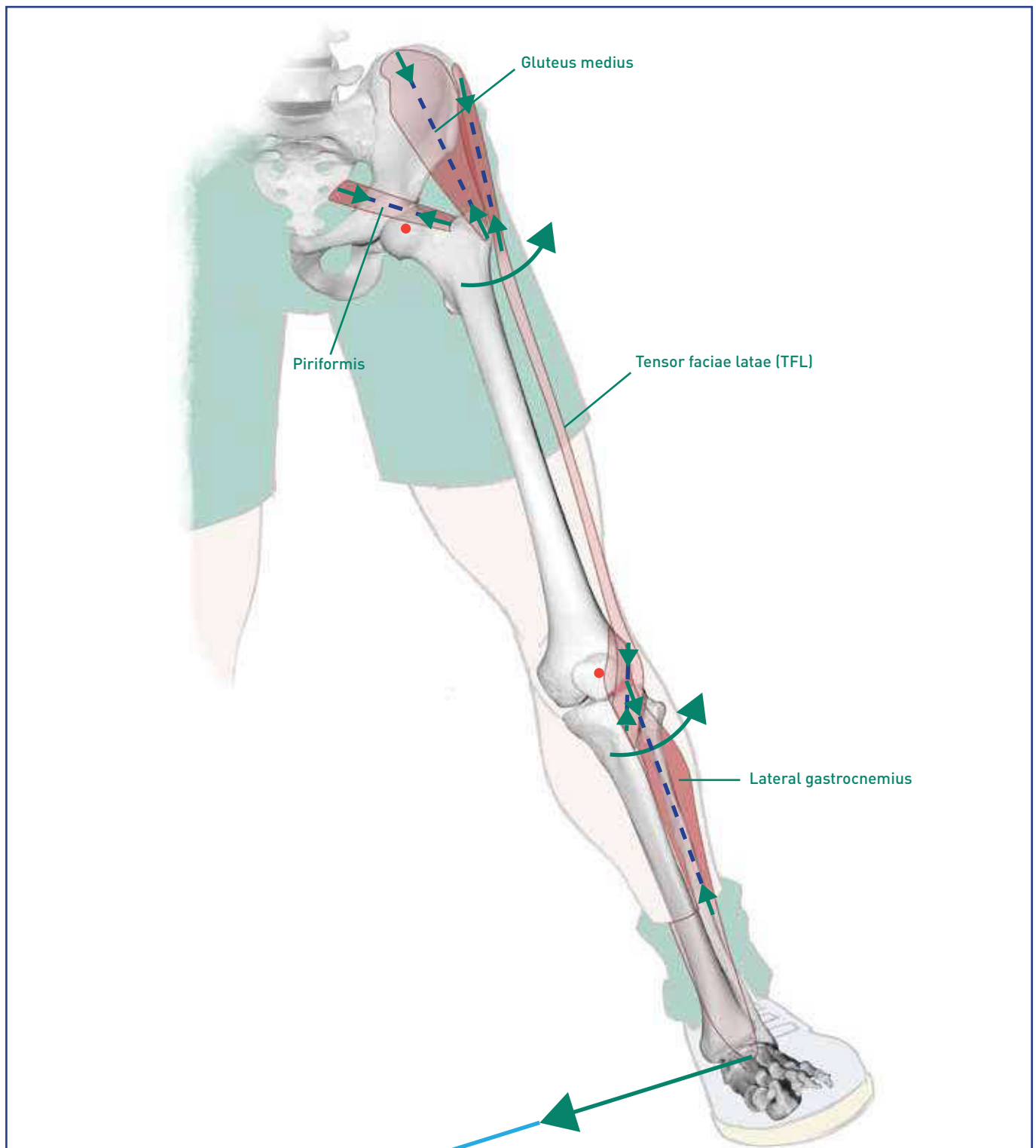
‘PHYSICS IS ONE OF THE MAIN FACTORS THAT WILL DETERMINE THE EVENTUAL MUSCLE RESPONSE’

This means that skeletal muscles become activated, according to the existing forces at the joints, which may need to be controlled or generated. It is the nervous system which controls the muscular response, and it does so depending on the forces acting on the joints.

Unfortunately, muscle participation has become somewhat of a cliché. It has commonly been used in a simplistic manner, and its application to the exercise and strength conditioning world has often been mistaken and inaccurate.

HOW APPLYING FORCES RELATES TO MUSCLE CONTROL

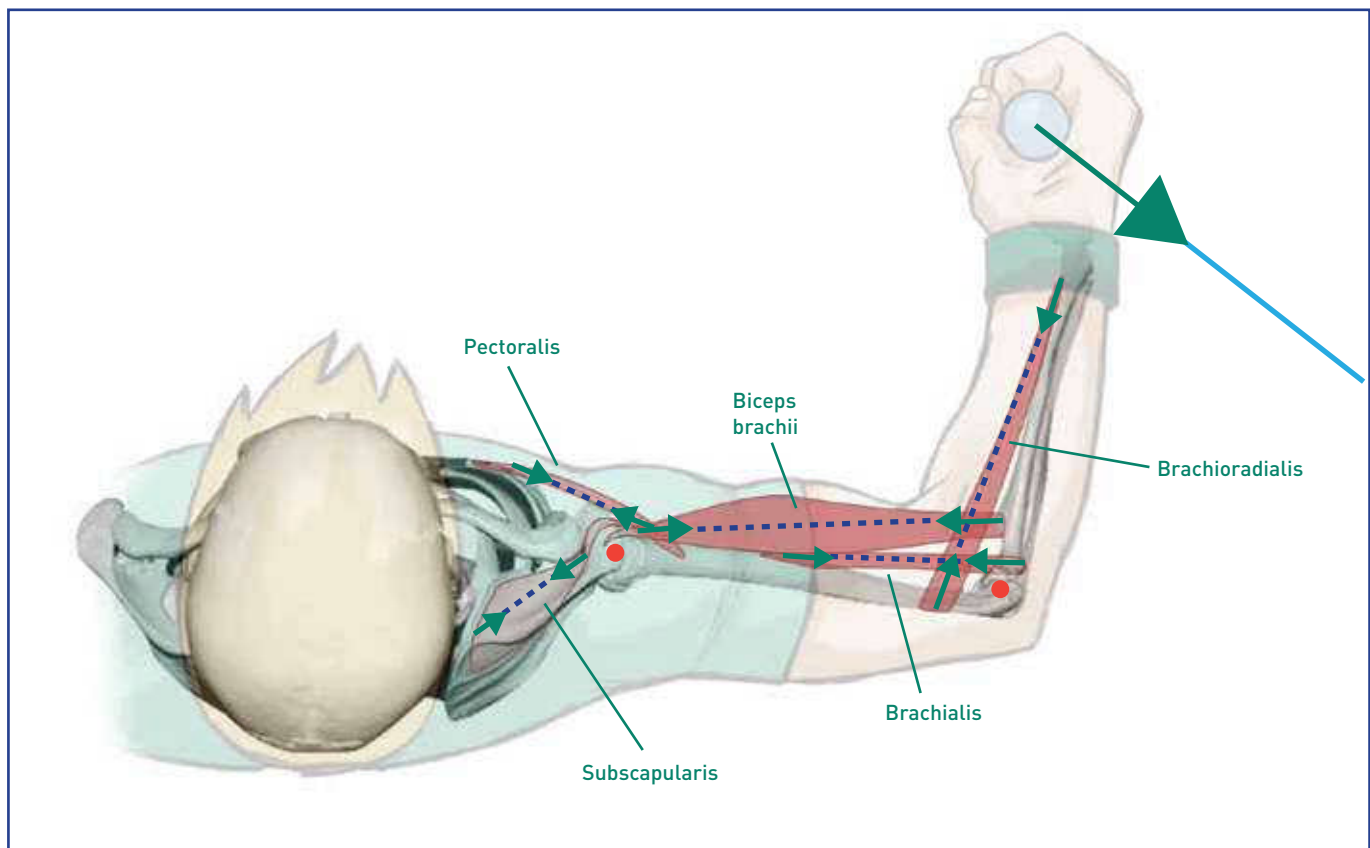
The interaction between forces reaching the human skeleton and forces generated by the body itself is a key issue, if we aim to guarantee human motricity. To put it differently, it is a negotiation that takes place via the articular system. This negotiation takes place at the joints, since it is there where forces reaching the body come to interact. For that reason, it is important that lever systems are well understood, and that this understanding becomes adjusted to the special characteristics of human levers. The latter behave differently to Engineering lever systems and could, therefore, be considered as bio-levers, instead.



Picture 4.6.1

Example of how different musculature is able to interact when having to deal with applied internal/external forces. This scenario represents several muscles that are mechanically able to generate rotational force in hip abduction, namely the TFL, gluteus medius or the piriformis, and in knee abduction (TFL and lateral gastrocnemius).

This picture is a simplification, that should help understand how musculature in general responds to forces. Needless to say, all muscles and connective tissue associated to those joints will play a role, apart from the need of other joints to become stabilized—at least to some degree.



Picture 4.6.2

Example of how different muscles can interact when having to deal with applied internal/external forces. This setting shows various muscles that are mechanically able to generate rotational force on the transverse plane, while doing a press move with a cable. This picture is a simplification that should help understand how muscles respond to forces. Needless to say, all muscles and connective tissues associated to those joints will play a role, apart from the need of other joints to become stabilized –at least to some degree.

If the forces being applied are known (not only the weight stated on the weights, themselves), the dose of rotational force affecting the joints can be controlled, and stimuli can be strategically manipulated. When this is so, it becomes easier to estimate the muscle responses that are required, in order to respond to the strategically generated stimuli, during the training session.

Muscles respond to resistance-generated stimuli, as well as to the attempts to accelerate one own's body mass. As a consequence, it is vital that all professionals understand how to create, manipulate and advance the application of forces. And by forces, it is meant both externally-generated resistances, as well as one own's inertial resistances, derived from the acceleration of one own's body.

TYPES OF LEVERS AND MUSCLE CONTROL

This section will show how a lever being of one or another class is virtually irrelevant when dealing with real exercise mechanics.

What is, indeed, important about lever systems, is being able to detect which forces are generating torque, articular forces, and the relationship between all of them.

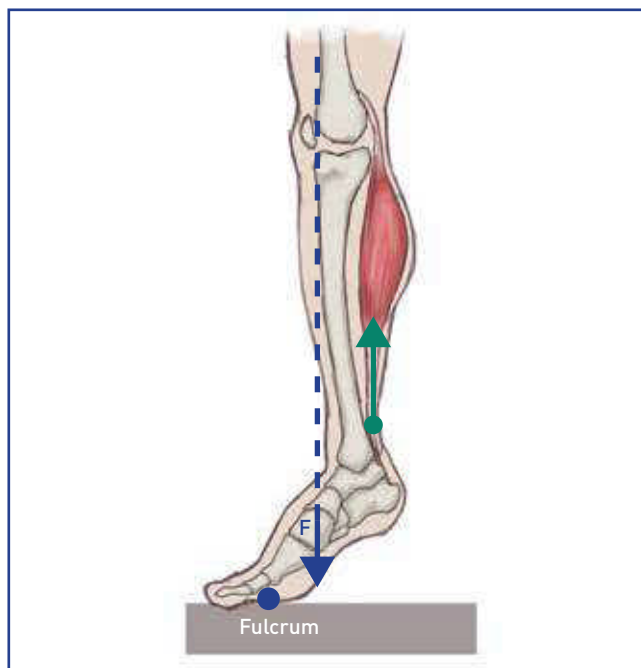
JOINT ROTATIONAL AXES versus EXTERNAL CONTACT AXES

In regards to the analysis of lever classes, the first problem arises from the observation of the typical example for 2nd-class levers (interresistant lever types). See picture 4.6.3.

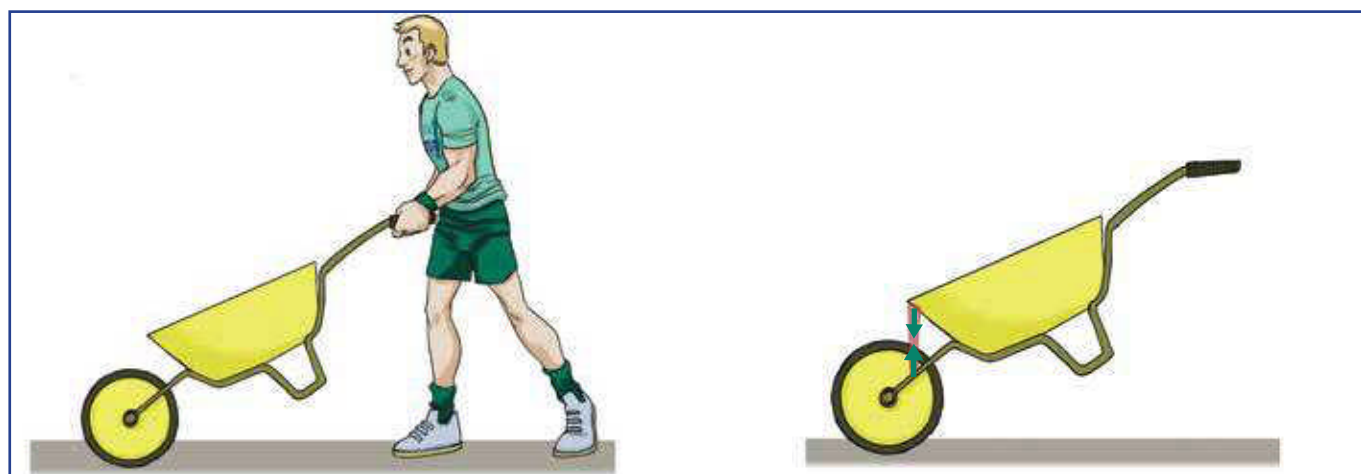
That type of example depicts a standing foot, where the axis is usually depicted by a dot at the point of contact. The resistance is typically depicted at the point which, theoretically, represents the center of mass. And it is the gastrocnemius muscle that plays the role of the power.

In the following lines, arguments will be set forth, to show that the above explanation is invalid, from a Physics stand point.

The problem arises from having placed the axis at the point of contact. As a matter of fact, it wouldn't be incorrect to place the axis there, but it would be appropriate if there were also an external force being applied at yet another point of contact. This, for instance, is what happens when moving a wheelbarrow. In other words, an axis at the point of contact would be an axis that arises from exposing the body to an external force, as is not the case in picture 4.6.3.



Picture 4.6.3



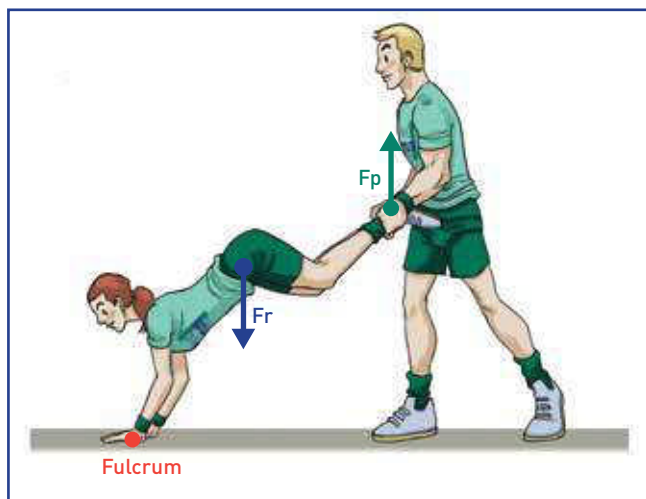
Picture 4.6.4a

The wheelbarrow constitutes a classical example, which presents three different structures, namely the ground, the wheelbarrow and the person that holds it.

In this case, the cart's center of mass lies between the wheel's axis and the force being applied by the person (third body). Because of this, it could be considered a 2nd degree lever.

This example of lever has been frequently compared to the example shown in picture 4.6.3. However, doing so is inappropriate because in the lower leg there is no third body acting upon it. What takes place in the leg would be something like pretending there was some kind of tensor pulling on the wheelbarrow, that originate in the wheelbarrow itself, and not an external force responsible for the creation of a rotational axis.

In the example of the gastrocnemius muscle, as typically depicted in books, there is no external force acting on the leg (body) with which generating a lever system, hypothetically relating to an axis at the point of contact with the foot, would be possible. For that reason, those 'external contact axes' should be taken into consideration, for analysis of muscle levers. Associating internal muscle forces with external contact axes is inaccurate.



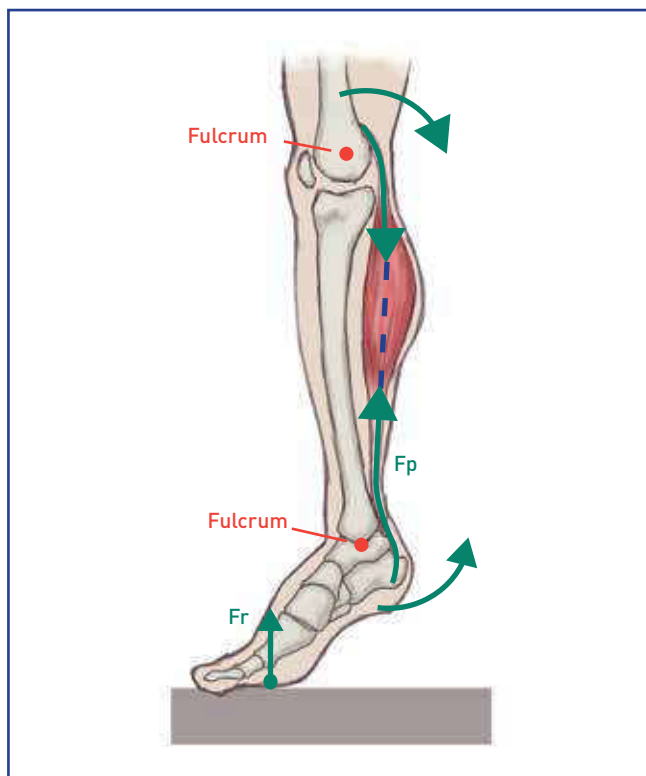
Picture 4.6.4b

This is possibly the closest example of a 2nd degree lever applied to the human body. The 'contact axis' external to the body (the woman with the ground) constitutes the axis. The woman's center of mass constitutes the resistance force. The force the man exerts to hold her constitutes the power. If the goal was to calculate the amount of muscle force, the external contact axis would no longer be considered the fulcrum. Instead, it would be considered the contact force that affects the axes of rotation (joints). And, naturally, joints would go on to be considered the fulcrum.

Muscle forces should be associated with the axes created by the joints. When analysing articular/joint lever systems, points of contact always turn into forces, and not axes.

All of the above implies that the typical example in the literature should be updated and modified. Instead of the usual picture, one should be made where the fulcrum is placed at a joint, and the force corresponds to the reactive created by the contact. Under these new premises, the 2nd-class lever now disappears, and something else becomes the focus: how able is the musculature to exert adequate force within that joint system, and in the right direction, at the point of contact with the ground.

One can logically deduct that, due to the complexity of the joint system, the gastrocnemius muscle would be unable to lift the body up on its own; it wouldn't manage to maintain the integrity of the forefoot and would, eventually, move the tibia backwards. Actually, lifting the body up, and propelling it in the right direction, depends on the orchestration of various muscles.



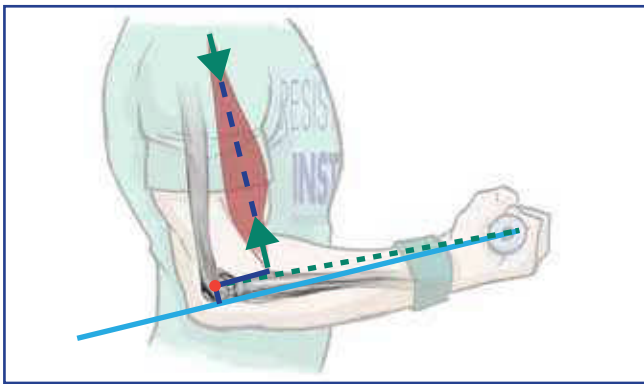
Picture 4.6.5

In a muscle mechanics analysis, the fulcrum (axis) must be placed in the joints to be analyzed. The gastrocnemius muscle is supposed to directly control the ankle and knee. These two will be the axes. The body's point of contact with the ground becomes the force of resistance. It is the body's point of contact with another body (the ground) and therefore is where the force affecting the joints should be applied. By analysing the mechanics of the forces that reach the joints, these levers are no longer a second type lever.

FACTORS MODIFYING THE THEORETICAL MECHANICAL ADVANTAGE OR DISADVANTAGE OF DIFFERENT LEVER CLASSES

Many written texts also commonly determine what a given lever's mechanical advantage or disadvantage is, depending of it being of one lever class or another.

Now, whether a lever has favourable mechanics due to its class only applies under certain circumstances. Unfortunately, it is only those scenarios that are typically taught when learning about lever systems as they apply to the human body. And that only but reinforces the myth that mechanical advantage depends on the class of the lever.



Picture 4.6.6

Even though the cable may show a greater distance to the axis (greater lever arm), the moment arm corresponding to the cable's resistance is less than that of the biceps brachii. In the picture above, it may look as if the biceps brachii had a mechanical disadvantage, but in fact it will require less force than the cable, in order to cause rotation.

It has been stated in this book before, how lever systems are typically taught with the example of a see-saw (or something similar), upon which only forces attacking at a 90°-degree-angle apply, all of which fall within the plane of movement (coplanar forces).

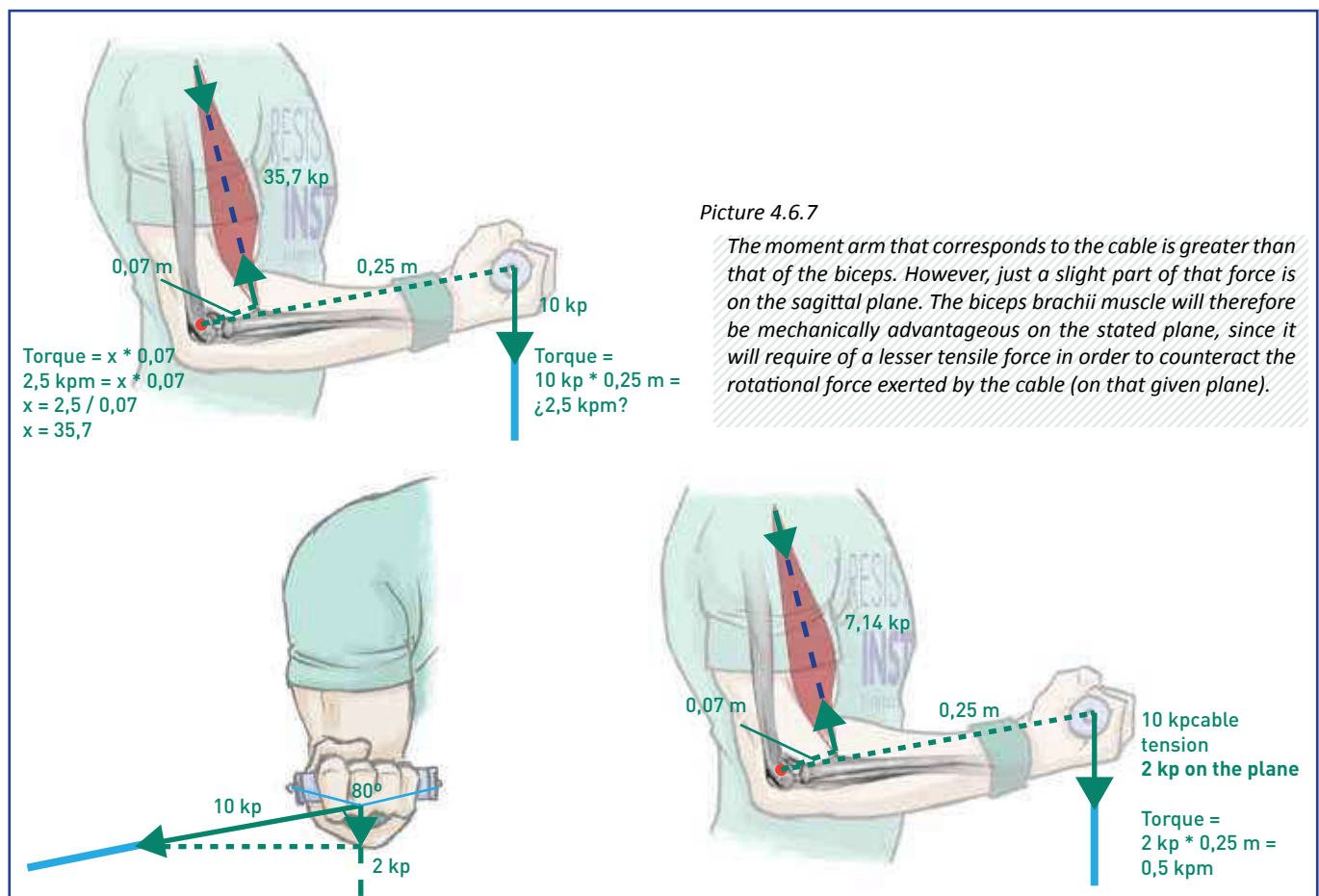
Following is the first aspect which could be disagreed upon: a certain class lever could, in fact, have more distance. But unless its angle of force is optimal, it is possible that its moment arm is smaller, in comparison to a smaller lever arm lever, with a more optimal angle. The reason for that being that it is only the rotational component (the part of the force attacking at 90°) which generates rotation.

Since moment arms take into consideration both the distance and the angle, shouldn't mechanical advantage be determined by the length of the moment arm and not the lever arm?

And if this were so, wouldn't it then be possible that a force at a greater distance to the axis (greater lever arm) would show a lesser mechanic?

There is, yet, another common mistake in the interpretation of lever classes: ignoring the component of the force within the plane being analyzed. As stated before, no matter what the lever arm, or even the moment arm might be, if the force does not lie within a plane that is perpendicular to the axis being analyzed, then only a part of it (or maybe none at all) will have mechanics over the given plane.

Picture 4.6.7 shows how, in spite of a greater distance, a force with a non-coplanar direction can actually have a lesser mechanical advantage for a given plane of analysis. And how, generating torque in that lever scenario would require of a greater force, relative to that given plane.



Picture 4.6.7

The moment arm that corresponds to the cable is greater than that of the biceps. However, just a slight part of that force is on the sagittal plane. The biceps brachii muscle will therefore be mechanically advantageous on the stated plane, since it will require of a lesser tensile force in order to counteract the rotational force exerted by the cable (on that given plane).

The above examples prove how mechanical advantage of an applied force is not necessarily determined by the lever class. And also, how the class of a lever is barely relevant when aiming to determine which forces generate torque.

FORCE AND MOVEMENT: THE ENDLESS CONFUSION

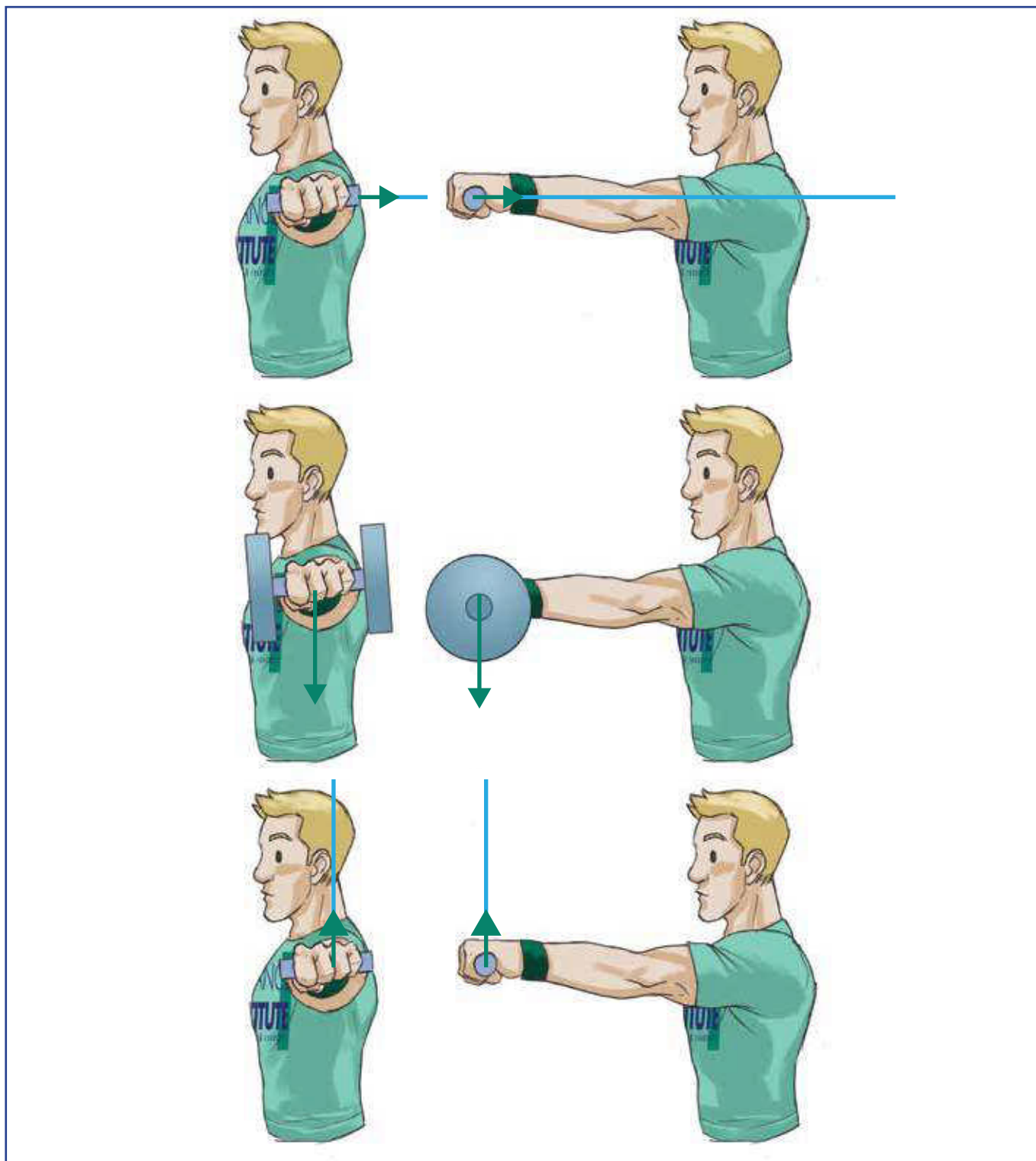
There is this sentence, which seems to have become fashionable in the fitness world. It states that 'your body does not think in terms of muscles, but in terms of movements.' From a Biomechanics perspective, this statement not only holds little meaning, but it can also lead to misunderstandings when analysing the mechanical reality of a resisted exercise.

Our nervous system responds, primarily, to forces, whether movement occurs or not. Needless to say, it is any body's purpose to achieve motricity (movement). However, it should be equally stated that non-movement is intrinsic to motricity, as well. That is to say, the system will choose to

avoid or limit movement, in specific scenarios.

Imagine you are performing some specific movement and then different forces are applied to it. You can imagine how, even if the body wished to think in terms of movement, it is going to use different musculature in order to achieve that movement, in adequate form and with sustained articular integrity.

The picture below clearly represents how, even for one given movement, different muscles will come into action. What is the body thinking about, then?



Picture 4.6.8

The character performs a transverse adduction move with forces acting on different directions. This will generate different strength stimuli, which will lead to different motor responses, even though the individual keeps doing the one same movement.

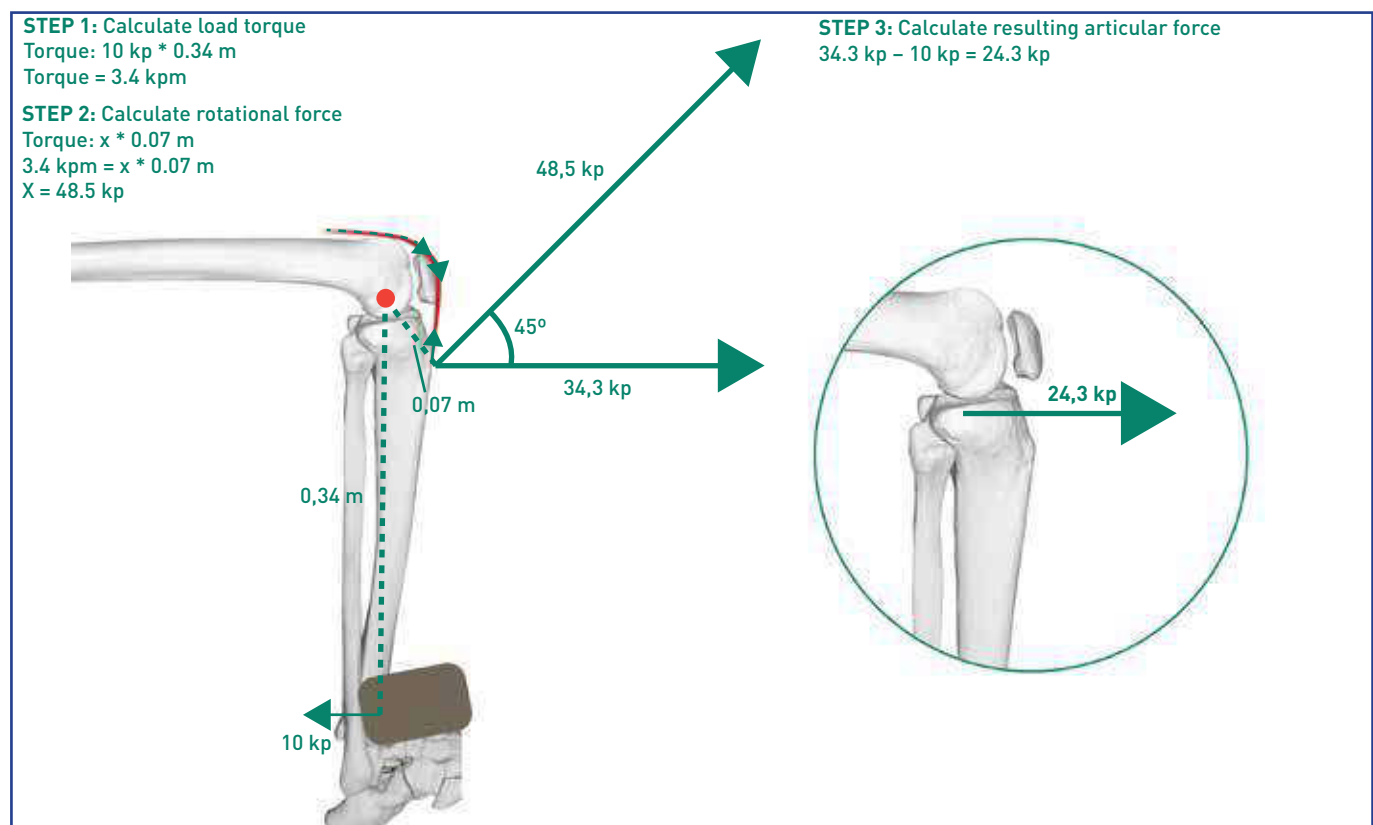
It is forces that determine how a movement will be generated or avoided. And it is forces which create the necessary orchestration for the different scenarios, in spite of there being movement, or not.

ARTICULAR (NON-ROTATIONAL) FORCES AND MUSCLE CONTROL

Because joints are partly structural linkages (shape closure, due to bone geometry), they cannot be merely regarded as biotensegrity linkages, even though the latter plays an important role. With that, we must consider them to be susceptible to three different types of contact forces, just like any other structure would be. These forces are: compression, traction and shear.

The total amount or existing articular/joint force will be obtained by adding up all of the forces that reach that given joint. Those forces can be calculated by creating a

composition of forces that affect the lever both in a parallel and a perpendicular manner. And both due to the load, as well as to the connective and muscle tissue. Needless to say, that is a complicated calculation. Forces are then regarded as positive or negative, depending on the direction they are being applied in. Once they all have been added up, the final direction will be that of the greater module. Whenever the resultant force is parallel to the joint surface, it will create shear for that given ROM position. If, instead, it is perpendicular, that means it creates compression or traction, depending on its direction.



Picture 4.6.9a

This is a simplified calculation of how articular forces may act around the L-M axis of the knee joint. Shear articular forces for this specific scenario equal 24.3 kp (force is parallel to the contact surfaces). This force leads to an anterior displacement of the tibia as a consequence of the force exerted by the quadriceps muscle, not by the resistance. In order to maintain the integrity of the union (joint), this force will need to be counterbalanced by other muscles and connective tissue.

RELATIONSHIP BETWEEN PLACEMENT OF THE LOAD AND JOINT FORCES

Consider these two factors:

- Power's (musculature's) component of rotational force DISTANCE to (joint) axis, and
- Resistance's (external load or own body mass load's)

component of rotational force DISTANCE to (joint) axis.

Both of those factors hold a relationship. That relationship determines that, the greater the distance-to-joint-axis is for the load, in comparison to the power, the greater the articular/joint forces created by the rotational component will be.

STEP 1: Calculate load torque

Torque: $10 \text{ kp} \cdot 0.34 \text{ m}$

Torque = 3.4 kpm

STEP 2: Calculate rotational force

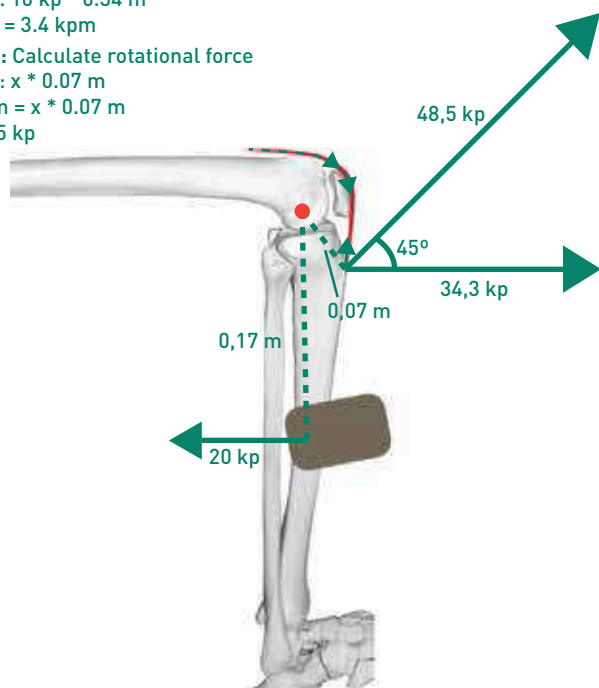
Torque: $x \cdot 0.07 \text{ m}$

$3.4 \text{ kpm} = x \cdot 0.07 \text{ m}$

$x = 48.5 \text{ kp}$

STEP 3: Calculate resulting articular force

$34.3 \text{ kp} - 10 \text{ kp} = 24.3 \text{ kp}$



Picture 4.6.9b

The picture depicts how articular shear forces on the L-M axis of the knee joint vary, when twice as much load is applied at half of the distance. Even though both situations account for equal torque, calculations show that here, the shear force equals 14.3 kp, whereas in the former picture, it was 24.3 kp.

Articular forces depend on the direction of the force being applied onto the structure, as well as on the shape of the joint. Analysing the picture above, you can see how the resistance force creates a compressive force on the tibia and a shear force on the knee joint.

The same picture shows the existing relationship between the magnitude of the load's rotational component (20 kp), relative to the lever arm's distance (0.17 m), and the greater magnitude of the muscle force's rotational component (48.5 kp), relative to its lesser lever arm (0.07 m).

This fact, together with the fact that muscle attachments cannot be non-surgically modified, leads to the following truth: the greater the load's moment arm is, relative to the power's, the more articular forces the rotational component will generate, for any given torque. In picture 4.6.9a, where the load is placed at 0.34m, you can see how the shear forces at the joint equal 24.3 kp towards anterior. Meanwhile, in picture 4.6.9b, where there is twice as much load but placed at half the distance, the articular forces equal 14.3 kp towards anterior. And both scenarios present equal torque for the joint.

The shear force component is created by the quadriceps muscle. This muscle needs to exert more rotational force

than the load, since it is placed at a shorter distance to the axis, and its moment arm is smaller. Contrary to what is often believed in the fitness industry, the anterior shear force is not a consequence of the load, but it is instead due to the muscle power (the quadriceps, in this specific situation).

This implies that, basically every time the quadriceps tenses up in order to generate a rotational force, it will also create an anterior shear force at the knee joint. And this means that it is not only the strengthening machine we use (leg extension), that causes this type of force at the joint.

Pictures 4.6.9a and 4.6.9b are simplified examples, which help grasp what the articular forces are that can arise at the knee, relative to the different forces being applied. It should be borne in mind that this joint is well designed to resist a good amount of these forces, and that is why we have the cruciate ligaments in the knee.

There is yet another important factor, and that is the existing musculature that lies on the opposite side of the articular axis. We are talking muscles, such as the hamstrings, the gracilis, TFL, sartorius, etc. These muscles will co-contract, and are essential in the generation of posterior articular shear forces, which will counteract the anterior shear force created by the quadriceps.

STEP 1: Calculating load Torque

Torque = $20 \text{ kp} \cdot 0.17 \text{ m}$
Torque = 3.4 kpm

STEP 2: Calculating Biceps Femoral Torque

Torque = $10 \text{ kp} \cdot 0.06 \cdot \sin \alpha (115^\circ)$
Torque = 0.54 kpm

STEP 3: Calculating Total Torque

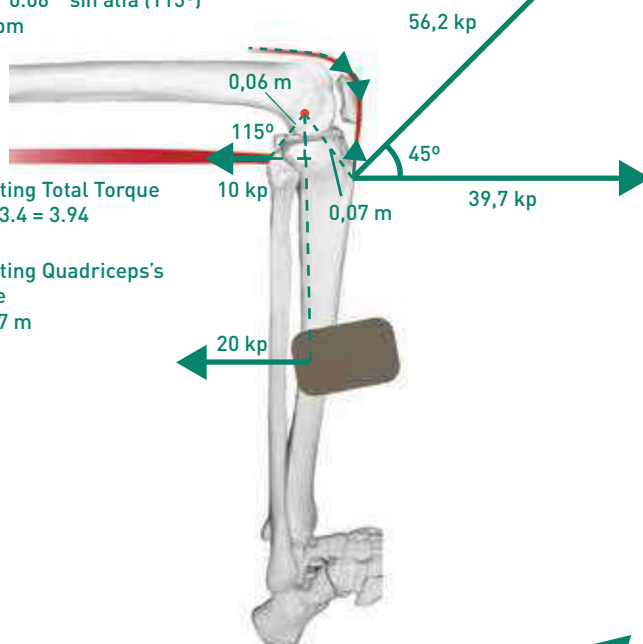
Torque = $0.54 + 3.4 = 3.94$

STEP 4: Calculating Quadriceps's Rotational Force

Torque = $x \cdot 0.07 \text{ m}$
 $3.94 = x \cdot 0.07$
 $x = 56.2$

STEP 5: Calculating Resulting Articular Force

$39.7 - 10 - 20 = 9.7 \text{ kp}$



STEP 1: Calculating load Torque

Torque = $20 \text{ kp} \cdot 0.17 \text{ m}$
Torque = 3.4 kpm

STEP 2: Short head B. Femoris Torque

Torque = $10 \text{ kp} \cdot 0.06 \cdot \sin \alpha (90^\circ)$
Torque = 0.6 kpm

STEP 3: Calculating Total Torque

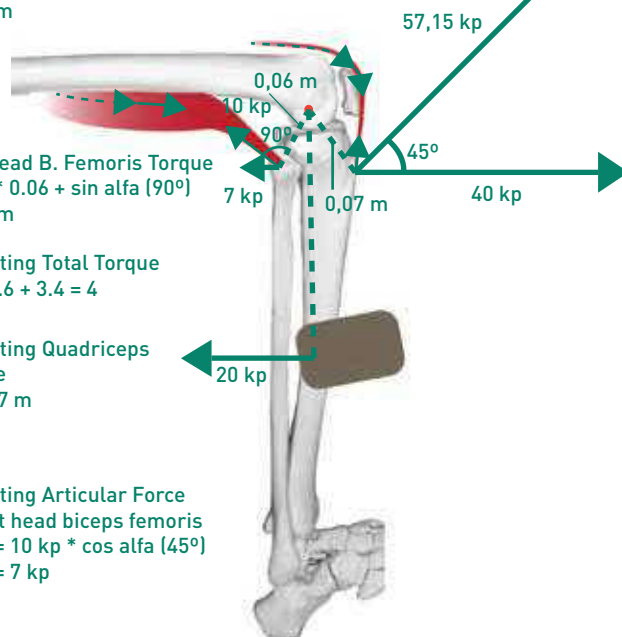
Total torque = $0.6 + 3.4 = 4$

STEP 4: Calculating Quadriceps Rotational Force

Torque = $x \cdot 0.07 \text{ m}$
 $4 = x \cdot 0.07$
 $x = 57.15$

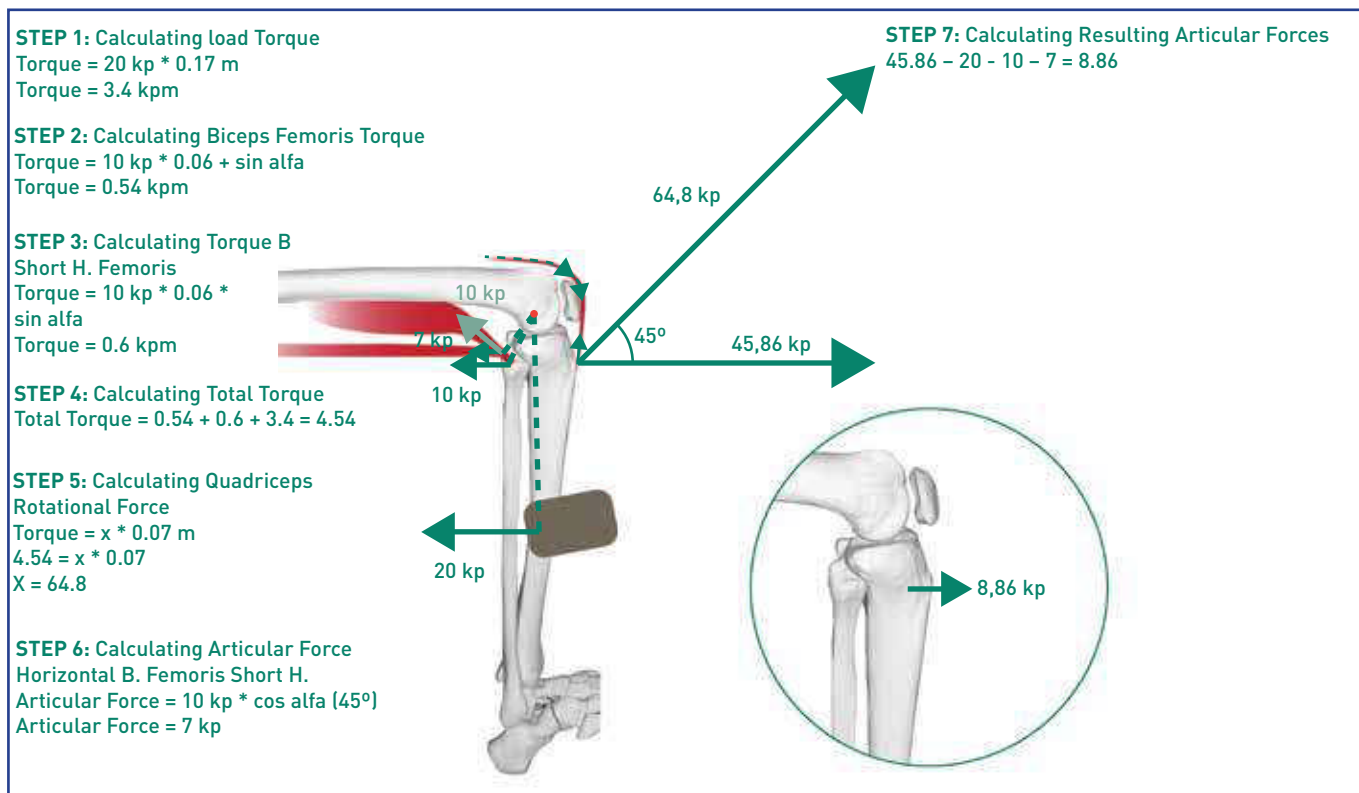
STEP 6: Calculating Resulting Articular Force

$40 - 20 - 7 = 13 \text{ kp}$



Picture 4.6.10

The picture above shows how articular forces become smaller when forces by the different heads of the biceps femoris muscle are added up together. Contraction of the biceps femoris muscle increases torque into flexion which indeed makes the quadriceps have to increase force. By this, total articular forces decrease. This exemplifies the important role co-contraction plays in joint stabilization. Musculature on the opposite side of the joint axis has to contract to some extent, in order to maintain structural integrity. It also helps prevent connective tissue -cruciate ligaments, in particular- from over-working. The values used for biceps femoral force are just an example..



Picture 4.6.11

Any other factor which may increase the difference between positive and negative forces, will logically also modify the articular forces. Examples of those may be the required acceleration, the changes in kinetic energy, etc.

Unfortunately, the term shear is often used with a negative, or malicious meaning. It seems to be frequently used to scare professionals, and convince them that a certain move may be 'bad', because it shear-provoking.

Shear forces should not be necessarily feared. Just like with every other joint, its effects will depend on the its amount and on whether the various structures are prepared to withstand it. Shear forces are present in joints at all times, and joint health depends on these forces being applied adequately. The key lies on comprehending how they work and being able to manipulate the forces, according to the individual's abilities.

Literature quite frequently states that the so-called leg extension machine creates shear forces on the anterior cruciate ligament. The truth is, if the anterior cruciate ligament were to suffer from any force derived from anteriorly directed shear forces in the knee, those would be of a traction type. And due to its inherent structure, traction is exactly what this ligament is most able to resist. In order to actually determine the potential risk-benefit of any one

movement, it would be preferable to be more precise and to determine the specific scenario (amount of force, direction of the force, ROM, affected joints, etc.). And make it also individual-adapted (genetics, phenotype and tissue adaptations, fatigue, injuries, etc.).

Making superfluous statements about the potential risks or benefits of any given movement, its supposed superiority or inferiority, etc is not recommended. By doing so, it is easy to oversimplify reality, where the variables covered in the text above are able to modify results, to a considerable extent.

Quite often, in books and articles, authors explain how a given action, such as the leg extension, is potentially dangerous, due to the shear forces it creates. However, those statement are commonly not accompanied by a thorough explanation on the specific scenarios which should be controlled. From a Biomechanics point of view, any soccer shoot is likely to create greater shear forces than that. To check that, it suffices to do a few biomechanical calculations on forces, relative to the ROM at the knee joint, and the point of application f the ball, and probably that will bring out higher shear forces.

In order to determine the actual potential risk of performing a leg extension exercise, several variables should be taken into account. Those variables have to do with the individual

performing it, the integrity of their cruciate ligaments (have they been injured before?), the load, the acceleration, the point of application of the force (we have already covered how articular forces can be lowered by placing the load somewhere else), the design of the machine, the type of pulley it uses, and whether the individual's musculature actually contracts with an adequate timing (please think about the knee musculature as not just the quadriceps and the hamstrings).

There is yet another concept which becomes frequently associated with the word shear, that being the alteration of the articular cartilage lining in the joints, also known as osteoarthritis. Nevertheless, the main factor leading to that pathology is compression. The latter increases as a function of the intensity of compressive forces and the muscle system's ability to reduce them (by an adequate timing of muscle contraction, which will result in adequate tensegrity, and will eventually reduce the compression at the joint). Once again, it could be of use to insist in this quote by Siff, which was already stated in chapter 1, and is worth remembering:

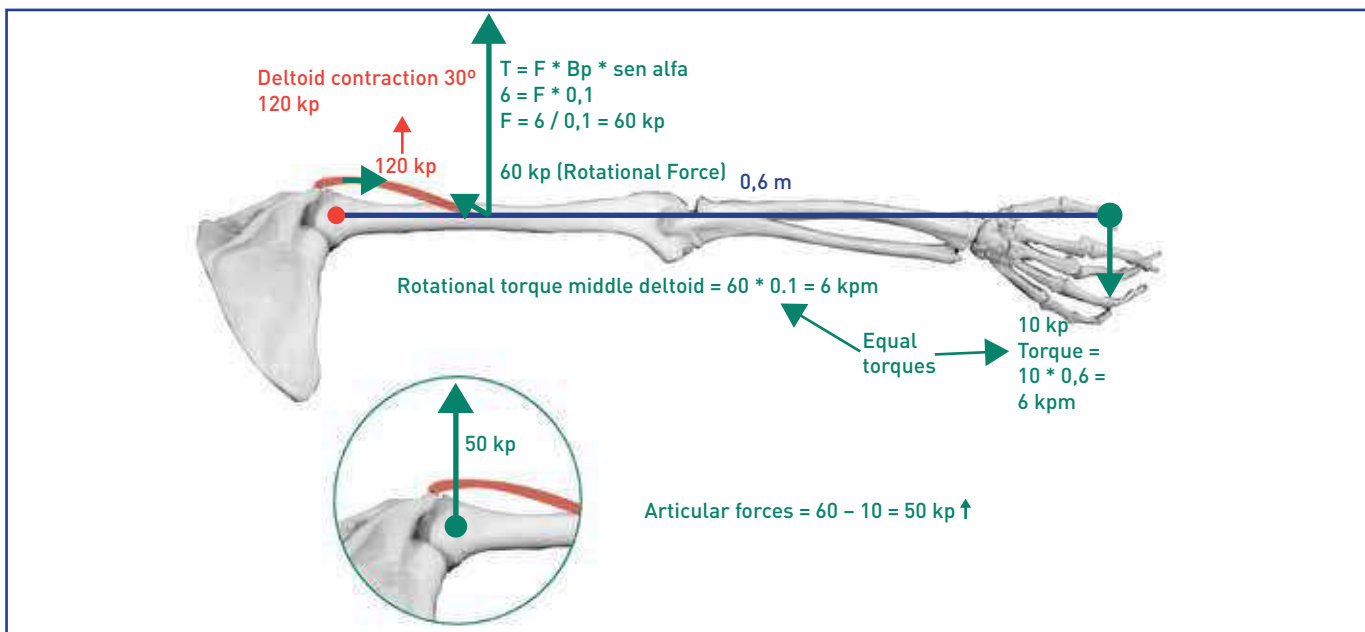
'... we must keep away from making frivolous and unspecific generalizations about the superiority or inferiority, the

safety or the dangers of a given exercise, because every effect of an exercise depends on the situation, the context, the time and the individual.' (Mel Siff)

STRATEGIC MANIPULATION FOR ARTICULAR FORCES DURING EXERCISE

In order to understand how an exercise specialist can strategically manipulate articular forces in exercise, a few examples will be presented. These examples will not take into consideration the exact numbers (which would be the case for structural Engineering), nor the involved tissues. However, they will quite accurately describe what goes on inside the joint, in regards to the articular forces. All of this will lead to a better understanding of the relationship between the mechanics of the point of application of the load, and the mechanics of the muscle force. And the goals of this being explained are:

- To improve the visualization of articular force, in various scenarios.
- To understand the way those articular forces can be manipulated, according to the goals set and the traits of the individual.

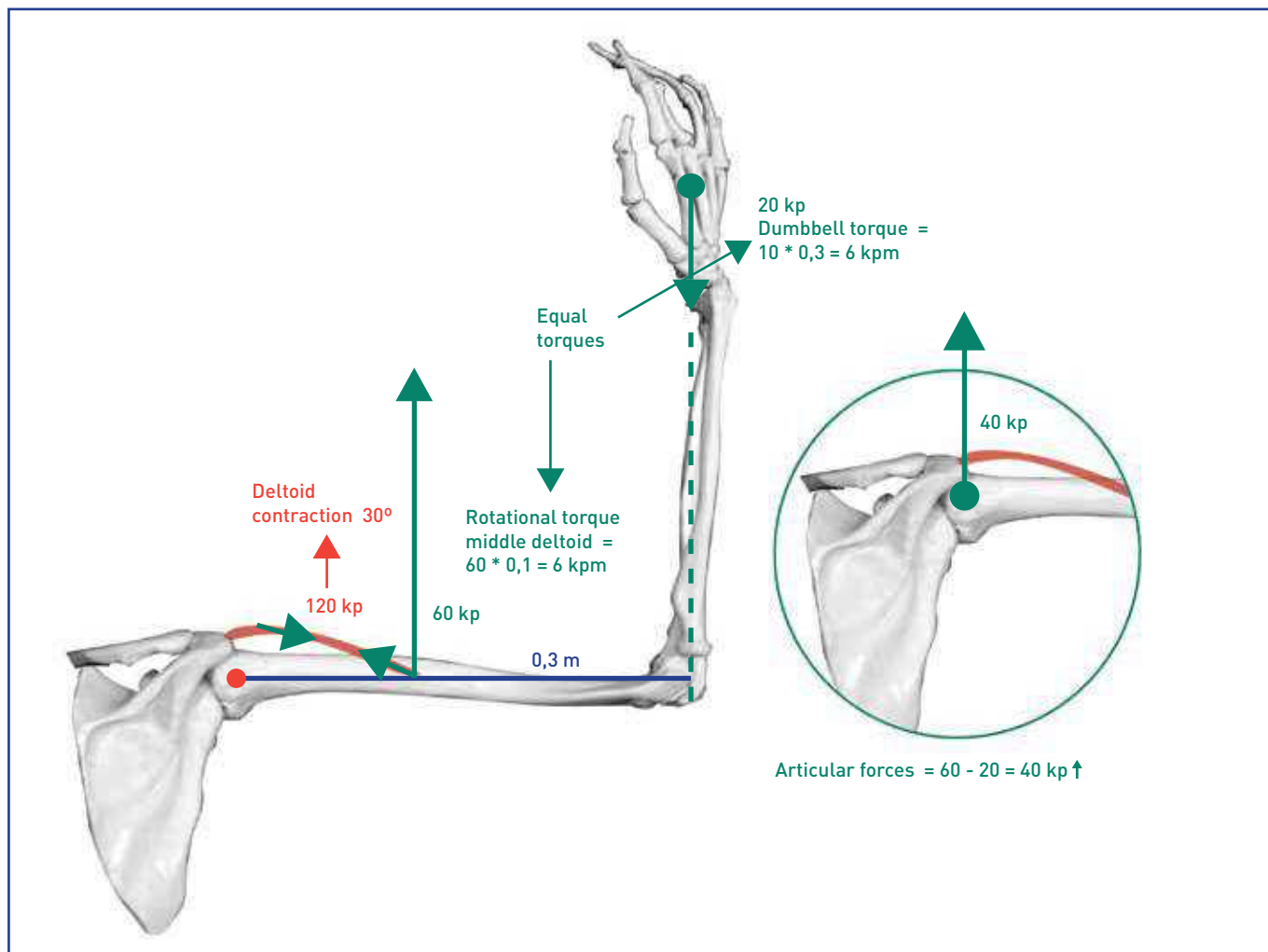


Picture 4.6.12

When the hand holds a force of 10 kp, at 0.6 m of distance from the A-P axis of the GH (glenohumeral) joint, that load will generate a 6 kpm torque. The aim is to calculate the superoinferior articular force in the GH joint.

In order to calculate this force in the joint, all forces taking place in that S-I direction need to be added and subtracted. This picture only depicts some of the forces taking place, in an approximate manner. In order to equal the torque induced by the resistance, the middle head of the deltoid muscle, located at 0.1 m, has to do a rotational force of 60 kp. This will require it to generate 120 kp of muscle tension. Consequently, in order to find out the superoinferior articular forces reaching that given joint, 10 kp must be subtracted (running inferiorly, caused by the load) to the 60 kp caused by the rotational component of the middle head of the deltoid muscle (with a direction towards superior).

The resulting articular forces for this hypothetical scenario: $60 \text{ kp} - 10 \text{ kp} = 50 \text{ kp}$ upwards.



Picture 4.6.13

In the second situation shown, the elbow is flexed at 90° and the external load being applied is the same distance as the elbow, relative to the axis. This will make it necessary to use more external loading in order to generate the same amount of torque as in the previous picture. Comparing to the previous situation, because the load is located at half the distance, twice as much load will be required if we aim to achieve an equal amount of rotational force on the A-P axis of the GH joint. 20 kp are needed in order to achieve a rotational force of 6 kpm. Because we have the same torque at the joint as that in the previous picture, the force the muscle fibre will have to exert is also the same as in the previous picture. Thus, in order to find out the superoinferior articular forces reaching the joint, 20 kp (towards inferior, caused by the load) will have to be subtracted to the 60 kp to superior, caused by the rotational component of the middle head. **$60 \text{ kp} - 20 \text{ kp} = 40 \text{ kp upwards}$.**

Even though the resistance in the joint remains the same, the articular force has been reduced. This fact is a consequence of the relationship between the resistance's mechanics and muscle force.

When the resistance distance decreases, relative to muscle force, articular forces around the rotation axis also decrease. Muscles are usually closer to the joint than loads and so muscles usually show smaller mechanics. Nonetheless, as seen previously, this is not necessarily always so, and it could be otherwise due to the angle and coplanar forces. However, in general, in comparison to resistances, muscles usually have to a much bigger force, in order to be able to generate an equal or greater torque. Most of the times, it is muscle force that directly cause shearing. The placement of the load is what causes the muscle to generate tension in response. Both the muscle and the resistance will induce rotational components. The larger the difference between these two (muscle's and resistance's), the greater the articular force in that direction will be.

Logically, muscles can also generate translational forces. In fact, from the previous example, translational forces originating at the muscle were causing forces with a high compression component into the joint. And because the muscle force angle was small, compression forces were actually higher than shear forces.

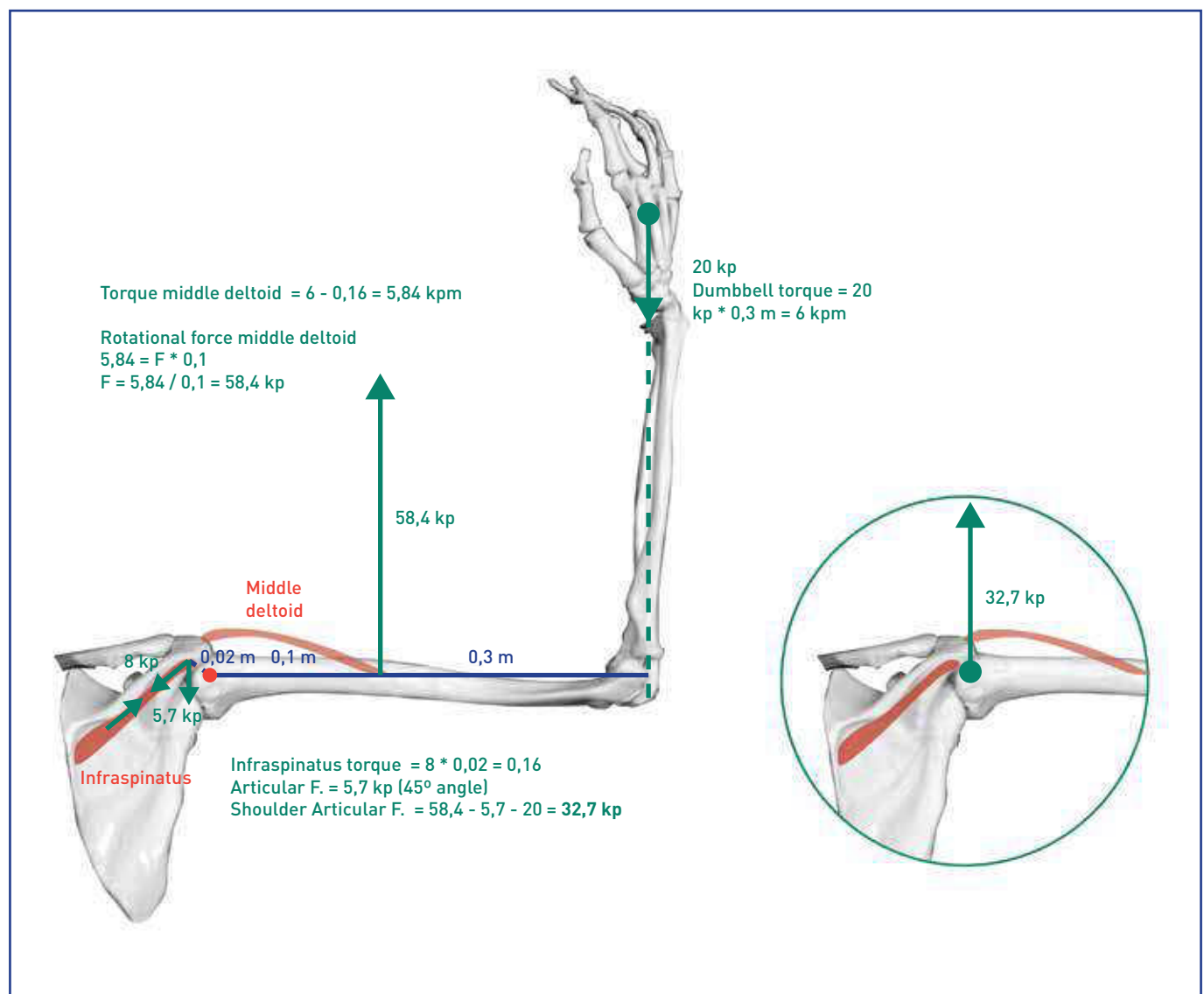
Articular forces, and especially shear forces, are part of the forces reaching our joints. They do not need to be a problem and should not be used to scare exercise professionals.

In the example shown, the geometry of the glenoid fossa (which has a larger superior-inferior surface) helps to control the effect of shear forces in the glenohumeral joint.

That is the reason why the muscles in the rotator cuff play a key role in reducing the potentially negative effects of shear. They are responsible of adding another vertical component, directed downwards, and a translational component, which creates compression. They also help out for shoulder abduction (force couple) and, with that, slightly reduce the

requirements to the deltoid and other abductor muscles.

Articular forces become substantially reduced, thanks to muscles such as the infraspinatus (see picture for an example) helping with abduction of the GH joint (it decreases the amount of force the deltoid and pec major have to exert). Muscles such as the infraspinatus help out by creating forces in the opposite direction, since their torque is generated from the opposite side of the axis. That is to say, they do create abduction of the GH joint, but with joint forces that are opposite to those created by the deltoid.



Picture 4.6.14

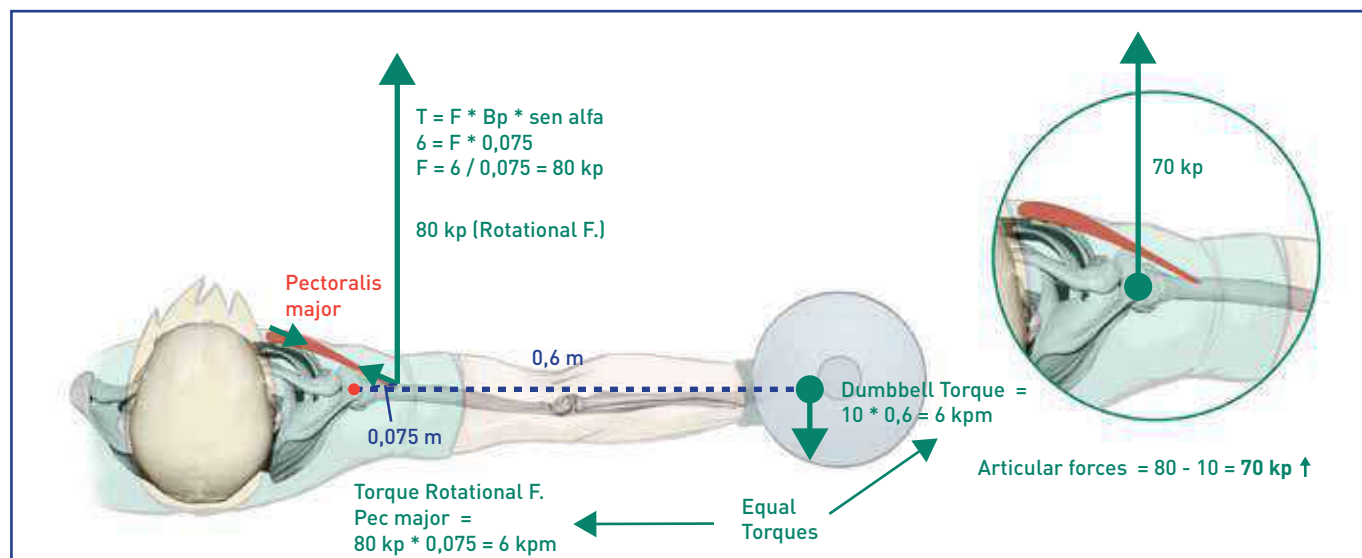
Correct muscle activation of muscles, such as the supraspinatus, noticeably reduces articular forces. In the above examples, the important role of musculature crossing over the GH joint can be noted. If their contraction is correct, it contributes to joint stability, together with the rest of the joint's connective tissues.

If these shear forces were to take place through the joint, in an A-P direction (transverse plane), it would then have to be considered that –for this new direction– the concavity depth of the glenoid fossa and its surface are smaller than for S-I.

The muscles which would negotiate against the resistance for this new plane would be different. The pectoralis major is one of the muscles with the best mechanics on this plane. It attacks close to the axis, which in turn exaggerates the

difference between muscle force and load.

Contrary to what is the case for the knee, the shoulder has no cruciate ligaments, probably to them hypothetically excessively restricting mobility in this joint. This fact makes the shoulder more dependent on the muscle system and, specially, on the rotator cuff musculature (and particularly, in the subscapularis and the supraspinatus).



Picture 4.6.15a

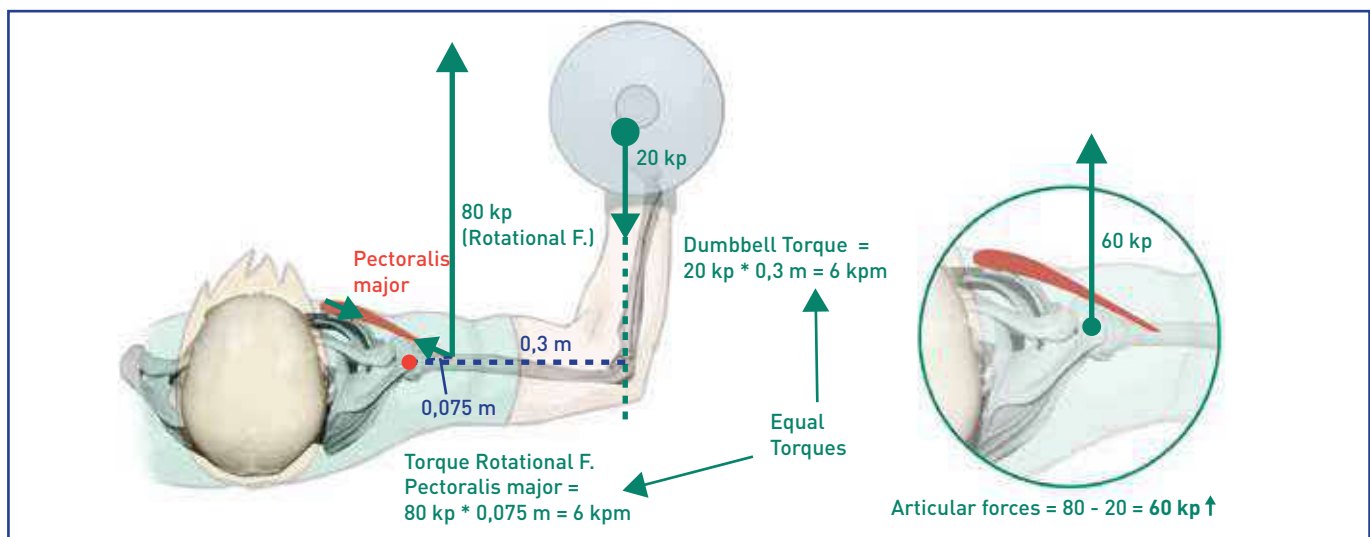
Analysis of articular forces that the pectoralis major muscle could create when having to deal with a dumbbell-generated torque, for a dumbbell placed at 0.6 metres (elbow in neutral).

The articular force around the S-I axis (relative to scapula) equals 6 kpm. The rotational force that the pectoralis has to make in order to overcome a load of 10 kp equals 80 kp. This fact implies an anterior articular force (mostly shear components) of 70 kp.

This articular force needs to be counterbalanced by other musculature and by the joint's connective tissue.

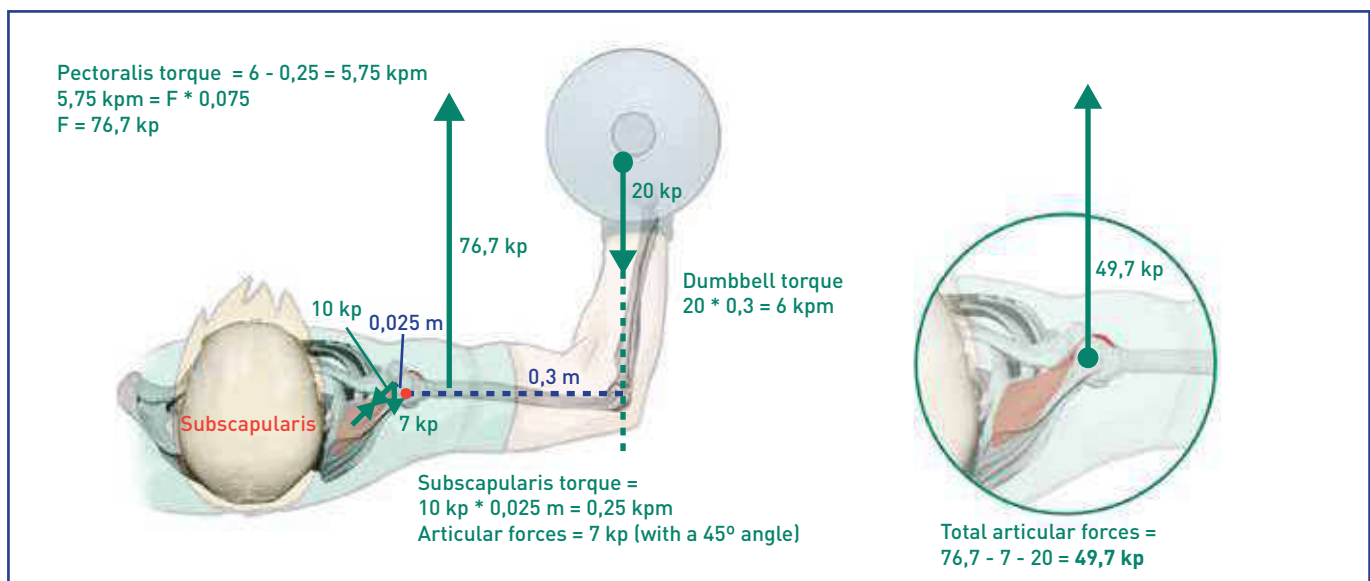
This calculation is merely an example which can help get an idea of how articular forces affect the system, how they can be manipulated and the importance of having musculature activate correctly, in order to be able to maintain articular integrity.

In reality, calculations would be different, since more muscles would be participating. Moreover, things differ from one individual to another, due to their different mechanics and also to the differences in coordination and muscle strength.



Picture 4.6.15b

For the same rotation as in the last picture (6 kpm), articular forces in this example have been brought down to 60 kp by modifying the distance to the point of application of the load. This is due to the influence of the distance joint axis-load placement, relative to the spot where muscle force takes place. For one given rotational force, the smaller the distance between the axis and the applied force, the smaller the amount of articular forces. This fact enables the professional to strategically modify the positioning of the load, in order to manipulate the articular forces, for each specific client.



Picture 4.6.16

This picture represents the role of the subscapularis muscle in the example also shown in picture 4.6.15b. This time, because the subscapularis lies on the opposite side of the axis, compared to the pectoralis major muscle, there is a very positive effect in regard to joint stability. On the one hand, the subscapularis helps negotiate against load-induced-torque. This lowers the necessary amount of force that the pectoralis major has to exert. And it also generates articular force in the direction opposite to the pectoralis. It is similar to what would happen if we were turning a steering wheel, with one hand on one side of the axis, and the other hand on the other. Articular forces decrease noticeable (in this scenario, to less than 50 kp, for a tension of just 10 kp, and a small distance to axis) when the subscapularis tenses up with adequate timing.

This example shows how relevant actions are from muscles that are sometimes regarded as of lesser importance. Unless there is adequately timed contraction of all the musculature involved, many of the supposedly starring muscles would not be able to exert adequate force. It is more likely that the nervous system would avoid their contraction. And it could even be more so. As has been seen in previous pictures about the knee joint, antagonist muscles such as posterior deltoid, infraspinatus, teres minor, etc will play a key role. Even though they do not help with horizontal adduction in the same way the subscapularis does, their correct contraction with the synergists will ensure a decrease in articular forces.

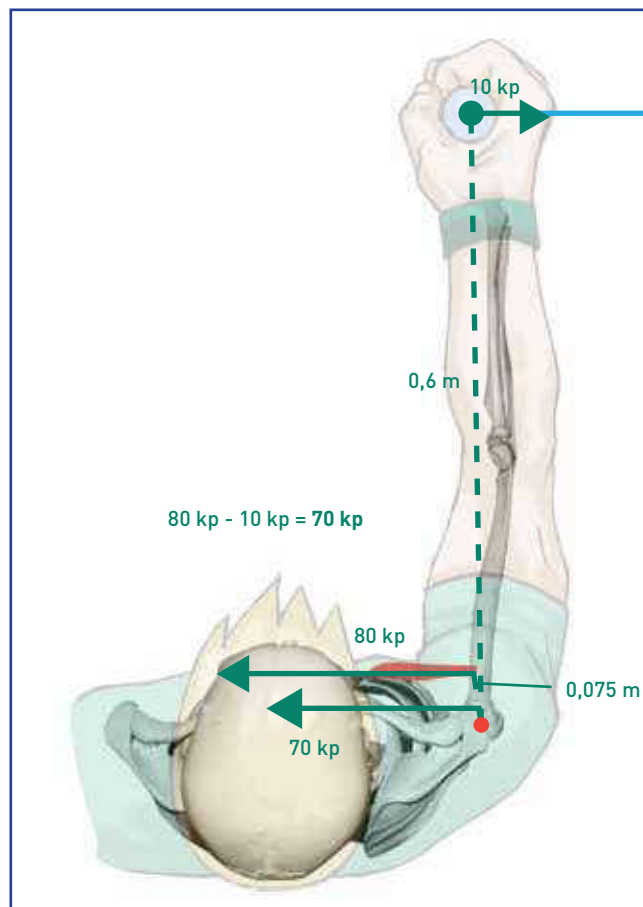
In order to maintain equal torque, using twice as much load in a dumbbell press reduces the amount of shear forces, compared to doing dumbbell chest flies.

It is by no means meant that the so-called chest flies are either good or bad. Readers should keep from making that type of unspecific and inaccurate statements, by now, regarding the superiority or inferiority of any one exercise. What is merely being shared is the effects that articular forces have, depending on the existing distance between the load and the axis. The eventual manipulation of those forces requires that this information is well-comprehended, so that exercises can be adequately selected, scenarios appropriately built (acceleration, joint range, etc.) and all of that by taking into consideration our client's characteristics, progression and individual goals.

The effects derived from the moment of inertia could then be added, as well. That would imply an increase in the given segment's instant acceleration, which will lead to the inertial reactive forces noticeably increasing the shear forces.

It would also be possible to bring the load closer to the axis. This could cause all torques, at whichever joint may be affected, to change, while at the same time decreasing shear forces at the shoulder.

Applying force on to a lever does not necessarily imply that shear stress forces will be generated at the joint. Shear can only be determined by comparing the resulting force with the plane and the shape of the contact surfaces. Picture 4.6.16 depicts a body position where shear forces account for much of the articular forces. Nevertheless, should the individual's position change, so would the articular forces. The reason for that is that the resultant articular force would vary, relative to the articular contact surfaces.



Picture 4.6.17

The rotational component in this picture is the same as in picture 4.6.15a. The difference lies in the plane of the contact surfaces, which has now changed.

This component now causes – to a large extent- compression on the glenoid fossa, since the vector lies close to the perpendicular to the contact surfaces.

4.7 TRADITIONAL KINESIOLOGY AND MUSCLE PARTICIPATION

Kinesiology concerns the study of human body movement. Traditional classifications in Kinesiology can raise some misunderstandings and lead to interpretation difficulties amongst exercise professionals.

The first difficulty arises from just using the term Kinesiology. The latter should not use or include any terms or concepts that have to do with Kinetics (forces present in movement) and should therefore not deal with muscle participation.

Mixing up concepts on muscle participation and joint movement typically leads to wrong interpretations. Oftentimes, the direction of the resistance is assumed, and the difference between movement and forces grow fuzzy.

Certain muscle actions may be taken for granted, in spite of reality very frequently being otherwise. Doing so implies gives rise to misunderstandings and important mistakes, when trying to analyze many of the exercises that use resistances.

WHEN JOINT MOVEMENT IS MISTAKENLY TAKEN FOR MUSCLE PARTICIPATION

The study of the human body is commonly associated with a specific muscle participation, while the direction of the forces being applied does not get specified.

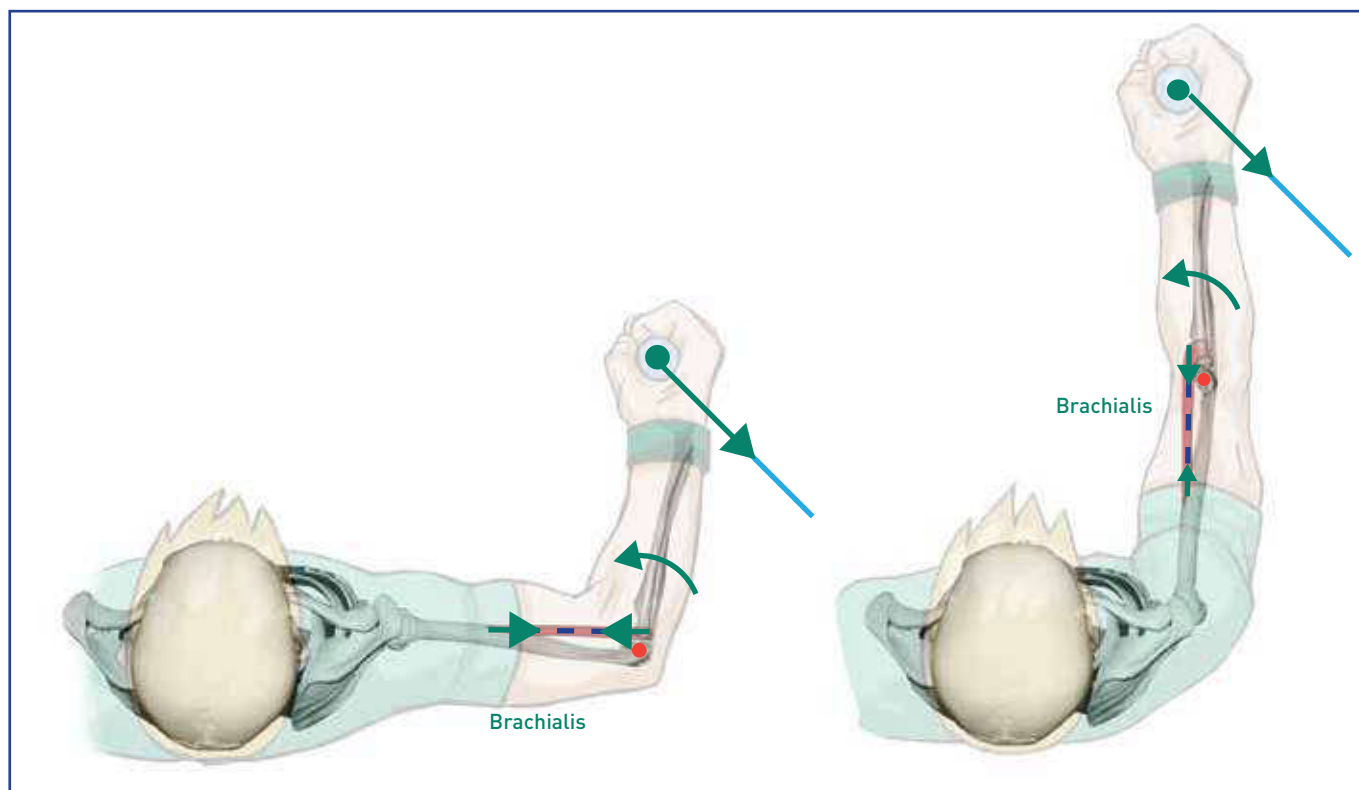
Misinterpreting which muscle is theoretically participating for any given move is way too easy. By looking at the following picture, many people are likely to think that the musculature that is actively resisting the resistance at the elbow joint is the triceps brachii.

One of the most probable reasons why people would assume so is that elbow extension is most commonly mentally associated with the triceps participation. But the direction of the forces being applied, relative to that given movement, are frequently disregarded.

Following the above lines of thought, many exercise professionals would argue that it is the triceps that is working against resistance, since the movement taking place is an extension of the elbow. Thinking in that manner is incorrect. The truth is that, the articular eccentric work of the flexors (the terms articular concentric/eccentric will be further explained later), leads to an elbow extension, which is controlled by the elbow flexors. In other words, the fact that the elbow may be extending does not mean that the triceps is necessarily working against resistance.

How muscles participate in actions does not depend on movement, itself. It depends on the resistance-axis relationship. It is not the movement which guarantees what muscles are doing the work, but instead it is forces which create a need for a specific muscle response.

Talking again about the upcoming picture, forces there generate elbow extension throughout the whole articular range. And that means that the musculature in charge of controlling the elbow must be that, which holds the ability to generate torque into elbow flexion.



Picture 4.7.1

If we analyse those forces reaching the L-M axis of the elbow, it can be appreciated how, for both pictures, the force is inducing extension in the elbow. This implies that the muscles having to negotiate with the resistance will be the flexors of the elbow, all the exercise through. In spite of forces generating elbow extension, the latter is being controlled by the elbow flexors, which put a break to extension. For this example, the brachialis muscle has been chosen.

It has, by no means, been stated that the elbow extensors are not generating tension, also. That would be an absurd, especially because they are doing a co-contraction, and making sure the joint stays stable. What has been stated is that the elbow extensors in this example are not negotiating against the resistance, in a direct manner. Instead, it is the elbow flexors which are doing it. And that is probably the opposite to what most professionals would have thought.

THE REALITY OF MUSCLE PARTICIPATION MECHANICS

MUSCLES GENERATE TENSION AT BOTH OF ITS ATTACHMENT SITES

A given muscle or a given muscle fibre may have the required mechanics to achieve movement in a joint, relative to all its axes and also all the joints it may cross over. But that doesn't mean the result of it tensing up can be predicted, nor can we know what will happen or what could happen at any of those joints.

Picture 4.5.23 ahead can exemplify this, by us looking at a fibre in the external oblique muscle. The fibre has the mechanics to create flexion, contralateral rotation and lateral flexion of the spine. But it tensing up does not mean it will generate all three of those actions. The end movement coming out depends on the orchestration of all

the musculature, as well as on other variables. Such variables could be the amount of kinetic energy that body segment was carrying and/or whether there might be a restriction at any body segment (relative to the external force), which totally or partially prevents it from moving freely around a given joint axis.

As a consequence, there is always more than one muscle controlling an articular axis, and it the following could be stated:

'NO SOLE MUSCLE NEVER WILL EVER RECEIVE ABSOLUTE FOCUS'

Nevertheless, applying resistances on some fibers' specific mechanical plane will, logically, highlight the stimulus on

those muscle fibers.

The above relative focalization is a consequence of two variables: the joint position and the direction of the force, together with the manipulation of the points of support, relative to other joints in the body.

Joints as well as muscles are designed to work together. By doing so, no one single muscle or joint has to specifically resist a load-generated stress, and thus they reduce the risk of injury. This is a way in which the human body is able to work more efficiently and to offer a wide range of movements and planes/axis of force.

In order to determine the muscle participation receiving the greatest focus, the axis (or axes) of motion/force must be identified.

‘MUSCLE PARTICIPATION DEPENDS ON THE MUSCLE-TO-RESISTANCE RELATIONSHIP AND ON THE AXIS, REGARDLESS OF THERE BEING MOVEMENT OR NOT’

For every normal activity, all muscles are basically working to some extent.

The most active muscles will be those that lie directly opposite to the resistance, in the plane of the load.

But other muscles, those which are often even called the antagonists, will be working directly against the forces inside the joint. Previous pictures in the book have helped exemplify how important those muscles not dealing directly against the resistance get to be. It has also been noted that, when the agonist and antagonist musculature simultaneously contract, a phenomenon known as co-contraction occurs.

Co-contraction of muscles around a joint (this has been covered previously, in regards to joint forces) results in an

increase in joint stability, and is actually a way to create synergy. In fact, this co-contraction helps to keep the joint together, it aids with functioning and optimal movement. This is relevant, since those muscles working against the resistance often try to separate both of the joint surfaces from one another.

Naming muscles as ‘agonists’, ‘stabilizers’ or ‘main/starring’ often transmits the idea that that muscle group may be more important than others when trying to generate movement.

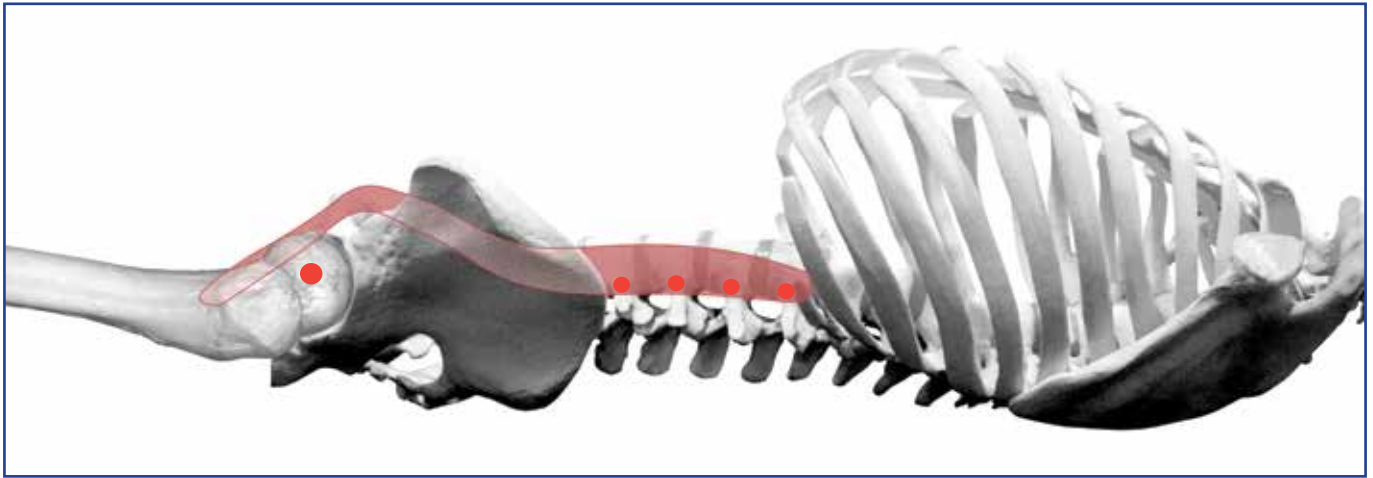
In spite of them ‘not starring’ in the generation of articular torque for a given scenario, in order to generate or to prevent movement, muscles responsible for joint control play a role that is as important as that of muscles acting directly against the resistance. Without those former muscles, many joints would many times not even move.

Having said that, it is also important that no muscle is automatically labelled as stabilizer or mover, per se. Any one muscle may have good mechanics, relative to a given axis, while at the same time having poor mechanics and create articular forces, relative to another axis. Should the latter be the case, that muscle would be acting more as a joint controller. Picture 4.7.2 should help to depict that. In it, you see how the mechanics for the psoas major muscle is analyzed, relative to the L-M and S-I axes of the coxofemoral joint.

It is also feasible that a given muscle may have good mechanics, relative to a given axis and to one of its attachments, and poor mechanics relative to a different joint it crosses over at the other bony attachment. This scenario has been depicted in picture 4.7.3. Here, the mechanics for both attachments of the psoas major muscle is analyzed, relative to the various axes in the hip joint, as well as in the spine.



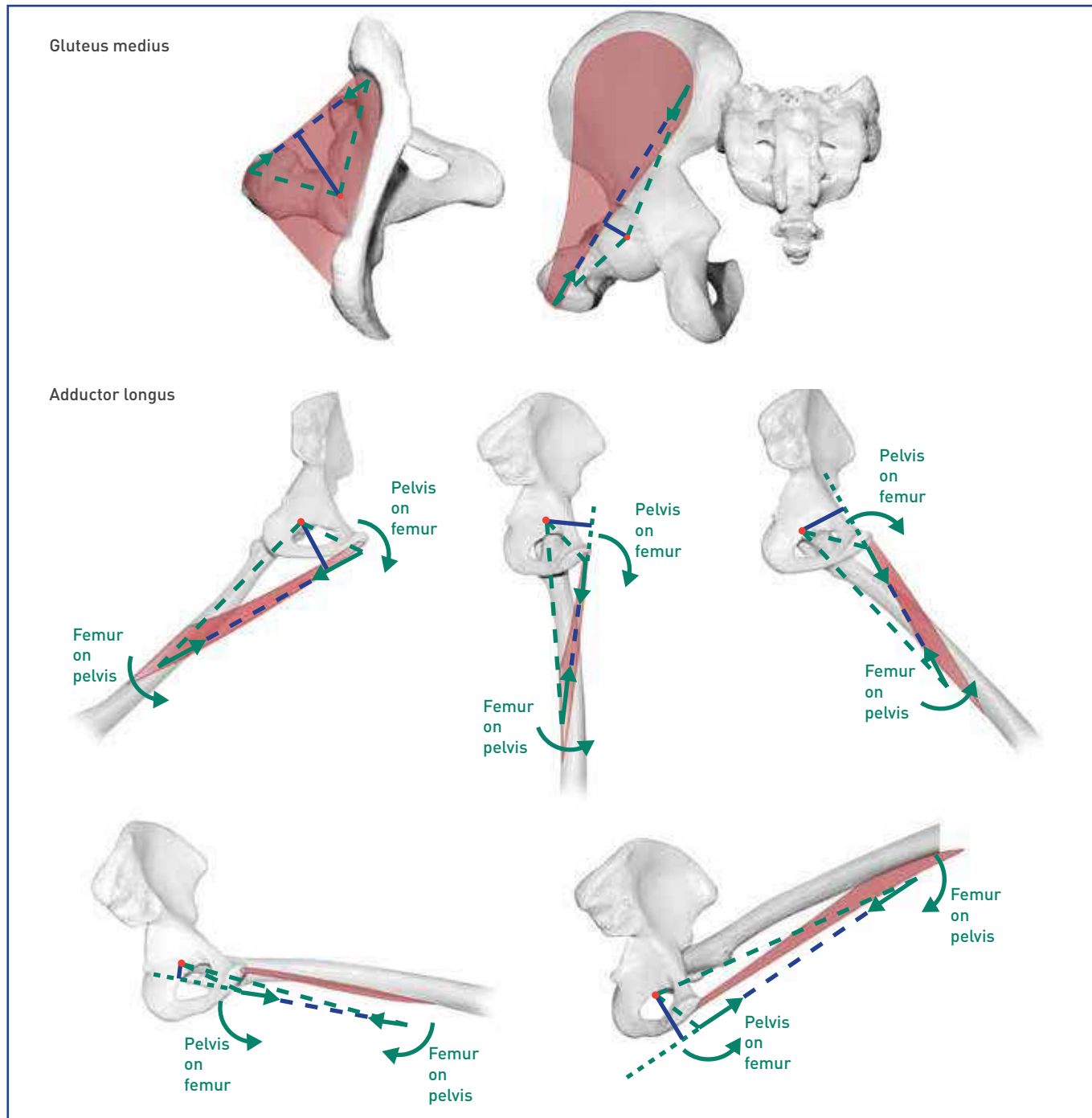
Picture 4.7.2

*Picture 4.7.3*

Muscle function relies on the specific scenario. It is, nevertheless, reasonable to believe that deeper muscles, and those fibers with little torque on given joints, usually serve control purposes, rather than motor purposes on the segment. But they could as well collaborate to moving, to some extent.

WHEN MUSCLE ACTION REVERSES

Reverse actions take place when a muscle's usual role becomes altered by some change in the joint position. This change leads to a modification of the relationships between the muscle, the axis and the resistance. Muscle action reversion is most common at the hip joint and at the shoulder.



Picture 4.7.4

Examples of muscle action inversion. The illustrations show how the posterior fibres of the gluteus medius have the mechanics to enable them to externally rotate the hip in neutral, around the S-I axis (ref. femur). At the same time, if placed at 90° of flexion, these same fibres hold mechanics to internally rotate around the S-I axis.

A similar thing happens with adductor longus muscle. For an extended position, and up to about 60° of hip flexion, its mechanics cause flexion around the L-M axis of the hip. From there on, its mechanics change and now extend the hip around the L-M axis.

4.8 STABILITY AND RESISTANCE TRAINING: NEGOTIATING WITH FORCES

WHAT IS STABILITY?

What is the definition for stability and stabilization, in dictionaries?

STABILITY:

- The strength to stand or endure (e.g. stability of a marriage)
- Safety, firmness (stability of a building, or an investment)
- Resistance to chemical change or to physical disintegration (stability of solids, climate)
- The property of a body that causes it to restore its original balance (stability in a car relies upon its shock absorbers, tires, etc.)

STABILIZATION:

Making stable, steadfast or firm. And to limit variations or fluctuations.

Notice how the terms above have different meanings. The second and fourth definitions may be the most appropriate for the exercise field. Pay attention to the fact that the word static is not mentioned in the text, and nor is 'staticity', which seems to be frequently associated with stability.

From a mechanical perspective, stability has to do with control, which in turn has to do with the concept of force. Stability has to do with the forces that are required in order to prevent other forces from generating changes or damage upon a structure. The goal of stabilizing (which is done via the use of forces) is, thus, to maintain the integrity of any given structure within its functional limits.

STABILIZATION versus ALIGNMENT

Stabilization has to do with forces. Alignment is a matter of body placement. Trainers, therapists, instructors, etc. frequently ask their clients to align the spine.

It then becomes obvious that alignment is nothing but a consequence of stabilization, together with structural geometry. And also that stabilization and the human skeletal shapes, together, will be the ones which determine what position or placement that structure acquires.

In summary, joint alignment is the result of rotational forces adjustments around the various joint axes in the human body, and therefore...

'one must stabilize in order to be able to reach a specific segment placement. And one must also stabilize through movement, and especially, when at the end ranges of functional movement.'

WHAT IS PROPER ALIGNMENT?

The problem with alignment is that it can either be 'x' or 'y', and it would still be alignment! The cue 'align yourself properly' is, therefore, totally unspecific. Even if the desired alignment were achieved, that would not guarantee that there is adequate stabilization, for the given force scenario which might be affecting the structure.

STABILITY AND MOBILITY

STABILITY AND 'STATICITY' ARE NOT SYNONYMS

And consequently, the fact that a structure may be more stable does not imply it being neither more nor less mobile. Mobility and stability are not all that closely related, although in the human body there can be quite a relationship between them. We will cover that later on.

The greater a structure's ability is to control and resist unbalancing forces, the more stable that structure is. That is also what the definition states: restore original balance, avoid rupture, fall, etc. of the given structure. Should a given material, or object lacked stability, that could be associated to the material lacking control against a given force. And similarly, with a low ability to tolerate forces and restore its initial equilibrium. Remember that stability does not equal 'staticity'. And a proof of this is how architects design tall building, conferring them with structural mobility, which in turn gives them higher stability. This can also be seen in vehicles, a car for example, in whether it has shock absorbers or not. Cars are designed to be able to move (both in itself, and from one place to another). A car with no shock absorbers will lack the ability to control and resist the forces created by bumps, which will in turn decrease the vehicle's stability. It must not be ignored that many structures (buildings, cars, etc.) need to be ready to move, withstand displacement or change from one place to another, at least to a certain extent.

WHAT HAPPENS WITH JOINTS AND THE HUMAN BODY?

Joints are, by definition, structures that allow movement between body segments. What this implies is that, should the terms stability and 'staticity' be somehow bound, a joint would never be able to be stable, since it is necessarily related to movement.

Unfortunately, there are still a number of professionals who believe that a joint is more stable, the less movement it has, and consequently fixate it by means of external elements, such as plates, screws, casts, etc.) When such a procedure is performed, joints don't become more stable, but are rather eliminated, since there is no longer movement. The truth is, this procedure can contribute to the generation of instabilities at other close-by joints (to the fixed one, that is). These new instabilities could be a result of increased requirements at the distal joints and will increase their risk of suffering from mechanical disfunction. Fixating a joint by using plates, screws, casts, or the such is not necessarily a bad procedure. It should, however, be used only in extreme situations, depending on the context and the individual. Different joints have different structures,

both at a bone level, as well as ligament-wise, but also in their fibrocartilages, capsule, tendons, etc. This causes each single joint to show a distinct structural support.

What is structural support in a joint?

It is the joint's stability, and is granted by its passive structures, such as the bones, the ligaments, capsule, tendons, fascia and muscle connective tissue. This structural support provides the passive components in the joint with an ability to absorb forces (to a certain extent) and will also fence in what we know as articular or joint range of motion. It is due to the structural support that we find joints with little to no anatomical mobility (e.g. cranial sutures, sacroiliac joints, pubic symphysis, etc.) or joints with larger mobility, such as the glenohumeral joint.

The engineering problem that the human body faces is how to provide structures with mobility and at the same time be able to control it. This mobility-to-stability relationship is a great challenge that has to do with our joints. There are many examples where the human body has had to sacrifice structural support in order to offer joints greater mobility.

The body has managed to find very witty solutions, as is the case in the spinal column, the shoulder girdle and other. It has done so by using systems and arthrokinematics chains, which distribute the amount of movement in the joint system, while maintaining adequate structural stability.

But even with that, it is the controlling function of the musculature that plays the most important role in joint stability.

Once the joint is granted a certain anatomical ROM, due to its structure, and therefore has its characteristic structural support, the human body needs to complement that joint with stability. And this is done via the musculatures, in an active way, and under the control of the nervous system (both voluntarily and involuntarily).

In this book, we will call functional stability to that sum of passive and active stabilizing structures. It would be even more precise to say that structural support does not, by itself, lead to functional stability. Muscles participation is an absolute requirement, in order to achieve final stability of the skeletal system. If we had no muscles, joints would lose their integrity and there would be virtually nothing (ligaments, capsule, etc.) that would be able to maintain integrity. This is an important fact, in order to understand the concept of tensegrity).

What is joint ‘functional stability’?

It is the ability of the neuromuscular system, together with the connective and bony tissues, to control a joint. That is, the ability to generate forces that counteract other disrupting forces and maintain the structure’s equilibrium).

And that for every position in the joint’s anatomical ROM. It is the human skeletal system’s true stability, for everyday activities, for sports, etc.

Who controls functional stability?

A synchronized action of the articular and muscular connective tissue (passively) and the contractile tissue (actively).

As stated above, every joint is built differently and has distinct active and passive stabilization abilities. However, all joints require of muscle control, in order to be able to provide adequate functional stability and control, throughout the anatomical ROM.

What happens when the musculature doesn’t activate well and loses its ability to control the joint?

In that situation, the system would decrease its ability to functionally stabilize a certain position (or several) within the anatomical ROM.

When the above is the case, the situation could lead to other muscles having to upregulate the work (compensate), which would increase the wearing off in the connective tissue around the joint. This wearing off, and with time, potentially increases the risk of suffering from abnormal joint degeneration, tendinitis of the compensatory musculature, degenerative osteoarthritis, etc. As soon as the nervous system detects this lack of muscle control, for the specific positions, it will protect itself and take action. And what it will do is try to prevent joints from acquiring unstable positions.

Our bodies make witty choices, and usually tries to achieve greater mobility at adjacent joints in that chain. But this, in turn, increases wear and decreases functionality.

And how does it do so?

It manages by doing what we could call a ‘protective muscle response’ , which is a consequence of a faulty inhibitory response. The nervous system, via a complex path, regulates tension in other muscles which hold a biomechanical relationship to the instability. This is why Biomechanics is often integrated with the nervous system, and why studying these topics can be called **Neuromechanics**.

Does this mean that the lesser the functional stability, the more limited the movement will be?

It does not necessarily have to be that way but the truth is, that reduced movement is very likely to happen. The nervous system will protect all vulnerable areas and prevent joints from reaching them. Or at least, it will try to do so

How do alignment and joint stability related to each other?

Alignment talks about placement, position. That has been said before. It is therefore a consequence of stabilization (control and strength) of the neuromuscular system, of the genetic code, and the history of use/load of that given osteoarticular structure, primarily.

It is not really the positioning that matter, but rather the amount of control (adequate balance of forces) present for each of the various functional positions.

How to place the spine, even while lying supine (where the superior-inferior forces basically don't apply) is a common instruction by professionals of the health and exercise field. It sometimes feels like this were the most important cue. But they rarely determine whether the muscles that are supposed to stabilize and control those vertebrae are adequately contracting, whereas this is what will in fact determine the degree of functionality in the spine.

Something similar happens with the scapulae. When performing overhead presses or pulls, some professionals find that the scapulothoracic articulation is functionally more stable, the less it moves. It is important to understand that should the SH (scapulohumeral) rhythm not be appropriate, functional stability in that joint will not (for the most part) exist. And the reason for that is that functional stability relies, considerably, on adequate activation and proprioception by the muscles in the spine, the scapulae and the glenohumeral joint.

Following is yet another example of how alignment and stability are often misinterpreted in the exercise field:

'There are still professional who state that 'the spine should not move while performing he exercise. It is to remain neutral, because it is more stable there'. Or 'extending the spine is a contraindication.'

Many of those professionals may not have thought that, if that were so, the body would have developed a fixed, non-articulated structure (something like a broom stick). Such a structure, just like in the example about the care with no shock-absorbers, would probably be less stable. It would not be as prepared to absorb forces and to maintain balance and would generate a considerable limitation in human motion.

It is therefore important to highlight, once again, that placement or positioning is not the problem, at least not as long as they are within the individual's functional limits. The key is stability, for each given scenario, relative to the various positions of the individual.

The reason why exercises should not be lightheadedly regarded as contraindicated is this

'we must keep away from making frivolous and unspecific generalizations about the superiority or inferiority, the safety or the dangers of a given exercise, because every effect of an exercise depends on the situation, the context, the time and the individual.' (Mel Siff)

If a joint shows reduced ROM, due to excessive muscle tightness for example, does that mean the joint is more stable?

The answer is no, even though it may sound contradictory. A lack of mobility can be indicative of the nervous system having identified instability. That is to say, it can simply be the best solution the system has been able to find, in order to maintain that structure stable. Excess muscle tightness is an indicator, a symptom, that something is not working correctly. The cause could be an imbalance of certain muscles. The nervous system might be identifying that as instability, and consequently limiting mobility.

One must not forget, however, that lack of mobility can be due to a series of causes, not having to do with muscle inhibition. It is important to also understand the following: bone degeneration, structural damage, osteophytes, connective tissue fibrosis, surgical procedures leading to structural changes, diseases, viruses, etc. And it can also be due to a combination of the above.

But there is no doubt that tight muscles are often stretched, by using a number of different techniques, without previously analysing what could be the causes leading to that situation. And without assessing muscles, and in so making sure that there are no neuromuscular imbalances. Proceeding that way may not be the best option.

JOINT STABILITY AND ARTHROKINETIC RELATIONSHIP

From a mechanical perspective, there is one single body, that is made of different pieces. And these pieces articulate with one-another, in order to carry out integrated functions.

In that, it is just logical to believe that the nervous system (the system in charge of controlling the whole body) does not act analytically, when performing any given action.

The nervous system integrates the information arriving from each of the pieces. It then interrelates the pieces of information according the mechanical, chemical and other variables. And it finally makes millions of decisions per second.

And for that reason, the whole musculoskeletal system is interrelated, and works in kinetic chains.

Should one single piece not work appropriately, it will affect the system to a greater or lesser extent.

Acknowledging a chain, implies acknowledging the important role that every single piece plays in it.

When any of the links in that chain is not working well, the body will compensate, in order to fulfil the global function. And it will do so even if it has to sacrifice function for a specific piece (specific dysfunction), and the integrity of the system, but it will manage to come from point A to point B.

The neuromuscular system knows of infinite ways to compensate (look for solutions). There are certain relationships and compensations that are common but only rarely does it do the exact same compensation from one individual to another.

Arthrokinetic relationships talk about how forces and joint movements relate to each other. Arthrokinetic relationships are present in the gait phases of pronation and supination, and also in the scapulohumeral rhythm. Those are two

examples of the uncountable relationships the body creates, in order to achieve motricity.

IMPORTANT:

In order to achieve motricity-related goals, should a joint loose movement, another joint will increase it. It should also be taken into account that, whenever a given joint lacks adequate mobility, this will cause an alteration of the forces and the articular surfaces of nearby joints.

This phenomenon can be seen in gait kinematics. For instance, if a hip joint lacks adequate internal rotation, the knee and the foot will experience an alteration of forces for every step that person takes. This means that, whenever a joint is not doing its task correctly, and is not offering adequate stability, the surrounding joints will experiment a change in forces affecting them.

When we study exercise mechanics, this relationship must be acknowledged because any time the system loses stability in a given joint, there is increased risk of experiencing pain, wear or injury in areas that are distant.

And since everything is interconnected, we must acknowledge the importance of specific function in each joint, within the human articular chain. No one joint should be regarded as more important than another. Importance is relative and relies wherever instability exists.

MUSCLE FUNCTION AND JOINT CONTROL

4.9 MUSCLE CONTROL

It has been stated that the 'battle field' for forces is the joints, and that the latter behave, primarily, as lever systems. What is meant by this is that joint stability depends on how well the lever system is able to maintain its integrity, while at the same time allowing for its movement around its axis or axes.

It must be understood that the human articular lever system is a very complex one, and it does not work like traditional lever systems do. In the human body, biotensegrity and force unions are accountable for a number of differences.

The fulcrum (axis) is not usually fixed, but rather frequently it experiences small changes. As a consequence, articular movement is not always purely rotational. In order to maintain adequate axes of instant rotation throughout articular movement, a large number of active stabilizers is often required.

Fulcrums are frequently maintained, or even generated, by the musculature itself, or the ligaments.

If it weren't for muscles, the axis which allows for movement would become unstable, and the remaining joint axis stabilizers (bone, ligaments, capsule, fibrocartilage, etc.) could be damaged.

It is the nervous system that controls all of the above, in a synchronized and complex manner

Understandably, muscle participation is a really complex issue. Sometimes, our system does not activate the musculature that actually would have the greatest ability to generate torque around an axis of movement. And it can simply be because the system has got information about an attachment site to that muscle not offering adequate stability.

Here is an example: hip flexors (for instance, the rectus anterior in the quadriceps), often do not activate correctly, due to the pelvis not being adequately controlled by the muscles in the spine (say, due to a functional deficit in the transverse abdominis muscle).

MUSCLE BALANCE AND IMBALANCE

Muscle balance around a given joint is often surrounded by a number of misconceptions.

In terms of 'amount of force', there should ideally be some kind of ideal anterior/posterior imbalance (that is, should imbalances really exist, since that all depends on the scenario and the individual). This hypothetical imbalance is due to muscles playing different roles in joints; that is some muscles fight against gravity, and bodyweight, and the effects of inertial loads, while others simply move a segment's end portion. Consequently, muscles often have different mechanical abilities. There is a clear example in the relationship between the gastrocs/soleus, relative to the tibialis anterior. Hoping for these muscles to have equal ability to generate tension would be erroneous.

Comparing the strength of the extensor muscles in the knee with that of the flexor muscles in the knee is not very helpful, because:

- Those two actions are not usually assessed under equal circumstances of hip flexion.
- Genetic structure and phenotype will make the extension/flexion ratio different, for each individual.
- If they were, indeed, compared under the same hip flexion conditions, the result would only be applicable to those conditions and should not be considered valid for any other positions.
- Conclusions can only be drawn for the specific scenario the ratios were calculated under. They should not be applied to gait or other scenarios, since the forces reaching the knee joints there would be different.

There is yet another limitation to those calculations, it being the way in which those strength imbalances are determined. Strength training machines in fitness centers are totally not appropriate for these tests, since the figures on the weight plates do not indicate the actual resistance (torque) reaching the joint. Mechanical components and machine design varies from one piece to another and the relationship between them is, therefore, inconsistent.

Using 30 kp on a leg curl machine could create a torque around the joint that differs from the torque resulting from using 30 kp on a leg extension machine. Not only that, but if any of those machines used an eccentric pulley, that would cause the resistance profiles to be different throughout the ROM. Moreover, these machines are not usually of an isokinetic type, so any differences in acceleration throughout the ROM would alter the relationship even more, still.

It is also interesting to analyze those cases of athletes with supposed muscle imbalance, relative to their specific sport. There is the usual example of people talking about an imbalance of the dominant side, relative to the other side, in regards to tennis players' arms. Say we were talking about Rafa Nadal, for example. His dominant side is the left one, so there is a theoretical muscle imbalance with the right. Traditionally, he would be advised to train his right side specifically to even out. But if we look at it more in depth, it could as well be possible that his right side has adequate stability and the left side might be suffering from more

stress and have more imbalances and compensations. But even then, a correct assessment should be made, in order to determine exactly in what plane those imbalances are present, and under which scenarios an athlete proves to be unstable.

Preferrably, that imbalance of Rafa's should be considered aesthetic. It is rather the result of adaptations than an actual functional imbalance. Honestly, functional imbalances would be more likely to take place if his two sides looked equal, especially after so many hours of left-sided training.

What is meant is that the concept of 'aesthetic imbalance' should not be mistaken for a human body's adaptation to forces (stimuli), after years of training. That would be like stating that soccer players have imbalanced upper extremities, because they are not as developed as the lower ones. This page does not suggest, though, that tennis players cannot have an imbalanced non-dominant side, or that soccer players should not train their upper limbs. What is being said is that oversimplification of matters should be avoided, assessment for muscle imbalances is more complex than that, and should not be approached superficially.

POSITIONS OF WEAKNESS

This book mentioned before that, even though muscles may proof strong throughout most of its active range, sometimes there are noticeably points of weakness, for certain joint positions.

In those positions, muscles may be totally unable to activate enough so as to provide adequate amounts of force to control the joint. And that turns them into 'positions of extreme vulnerability'.

This could generate an undesired and potentially dangerous imbalance:

- Around a joint, which would be detrimental to the joint's active stability.
- Within an integrated chain of joint movements and muscle actions, which would increase stress in the joint or for a muscle group in that joint.

MUSCLE FUNCTION AND JOINT CONTROL

PHASES IN MUSCLE CONTRACTION AND JOINT MOVEMENTS

Studying the different phases of muscle contraction and articular movements can be confusing and leads to some misinterpretations.

The first problem arises with the definition of concentric (or myometric), isometric and eccentric (or plyometric) phase.

There are two different ways to define the above phases of contraction and mixing them up usually leads to inappropriate interpretations. That would mean we are bringing together different concepts and is, once again, a proof that sometimes force and movement are not correctly differentiated.

These two conceptual lines, upon which the definitions about contraction phases have been created, have to do with two different references: a muscle reference, or a joint/articular reference.

THE MUSCLE REFERENCE SYSTEM

The phases of muscle contraction talk about the relationship between the muscle contraction and the muscle length.

- **Muscular concentric (myometric):** when musculature contracts and brings origin and insertion closer to each other.
- **Muscular isometric:** when musculature contracts, while the distance between origin and insertion remains doesn't change.
- **Muscular eccentric (plyometric):** when musculature contracts, while origin and insertion separate.

If we chose muscles and their relationship to length, then this is the most reasonable reference to choose. However, it is of little use when the aim is to analyze specific exercises, relative to the direction of rotation caused by the resistance and how that influences the movement in the joint.

Commonly, it is assumed that muscles shorten because

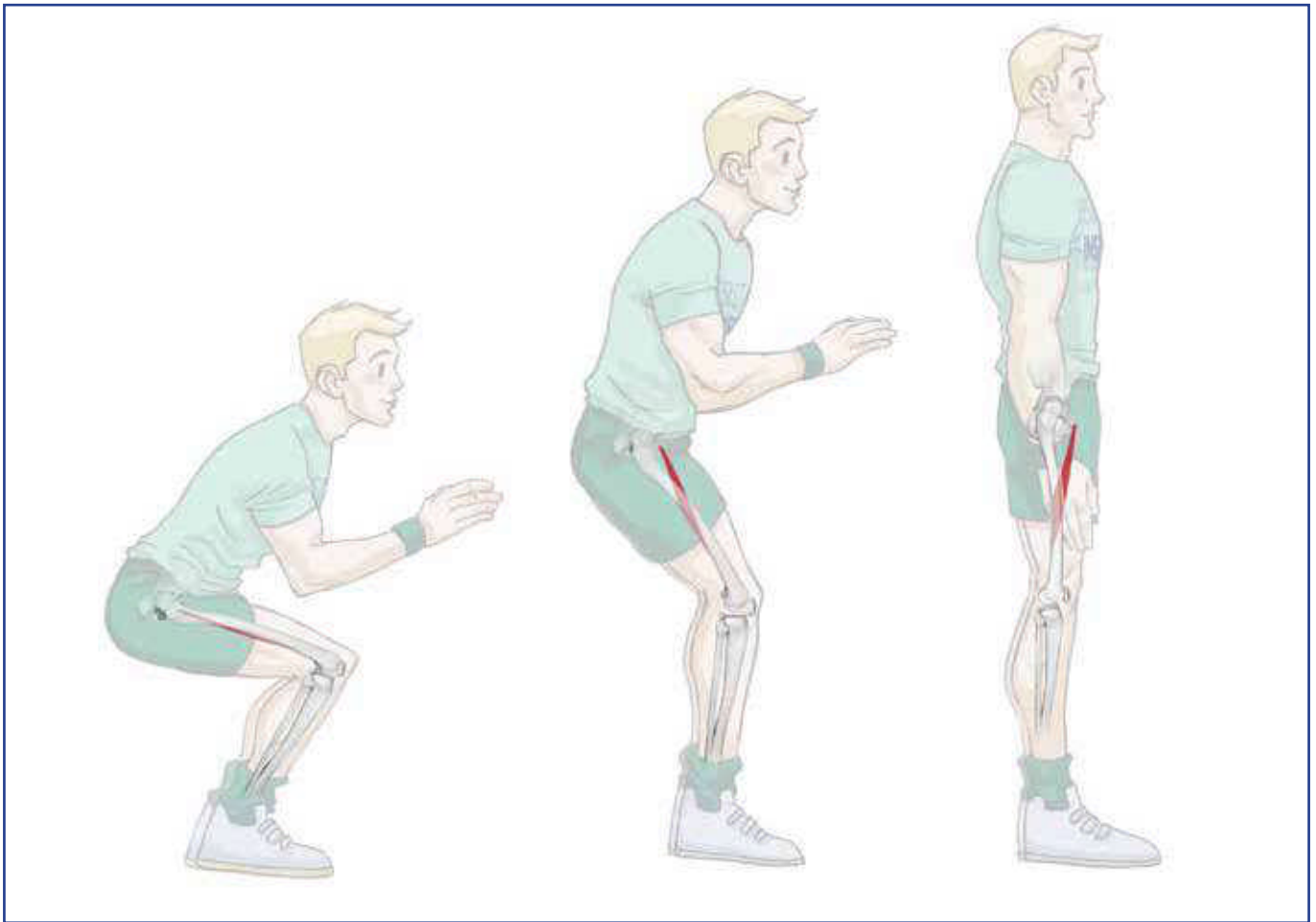
there is a resistance that is trying to lengthen it out. Strictly speaking, this assumption does not strictly derive from the definition, which merely states how a contraction influences the changes in muscle length.

When co-contraction was explained, it was stated that muscles contract at both sides of the axis. This means that, for each action, there may be muscles contracting concentrically on one side of the axis, while on the other side of the axis other sets of muscles may be contracting eccentrically.

Should we choose the muscular reference for every exercise, there would be muscles (muscle fibers) working in various phases. Some fibers may even change phases, along one single movement. And that would be due to the inversion of the muscle action, that we covered previously.

Let's take picture 4.9.1 (showing a squatting movement) as an example and try to describe the lengthening/shortening of the fibers in the adductor longus muscle, while the hip joint is being extended. We can see how from the 100°-110° position up until, approximately, the 60°-70° position of hip flexion, those fibers would shorten. From then on, those fibers would inverse their muscle action and begin to elongate. What this means is that, for a squat movement, and within its ascending phase (hip extension), the adductor longus muscle works concentrically during the first part of the movement and eccentrically during the second part (from 60°-70° until neutral). But, didn't literature state that this was the concentric phase? – Well, this muscle, just as many other adductor muscles, changes its type of contraction within one single phase of articular movement.

Using the muscle-referenced definitions for concentric, isometric, eccentric contractions in this exercise would be complicated. And if used, then things like the specific behaviour of each fiber, or how eventual differences in technique and/or individual can influence the lengthening or shortening of the muscle fibres would have to be taken into account.



Picture 4.9.1

For all of the above, talking about the concentric, isometric or eccentric phases of a squat, while using a muscular reference, is not feasible. Simply because squats are not muscles, and phases would have to be specified, relative to each one of the muscle fibers involved.

Almost no one does this. And so, it is a different reference that is preferred, in order to determine the phases of an exercise, relative to its segmentary motion and the direction of the resistance.

THE ARTICULAR REFERENCE SYSTEM

It deals with the articular movement (direction of rotation around a given joint axis), relative to the direction of the resistance that is posing the external load upon that axis.

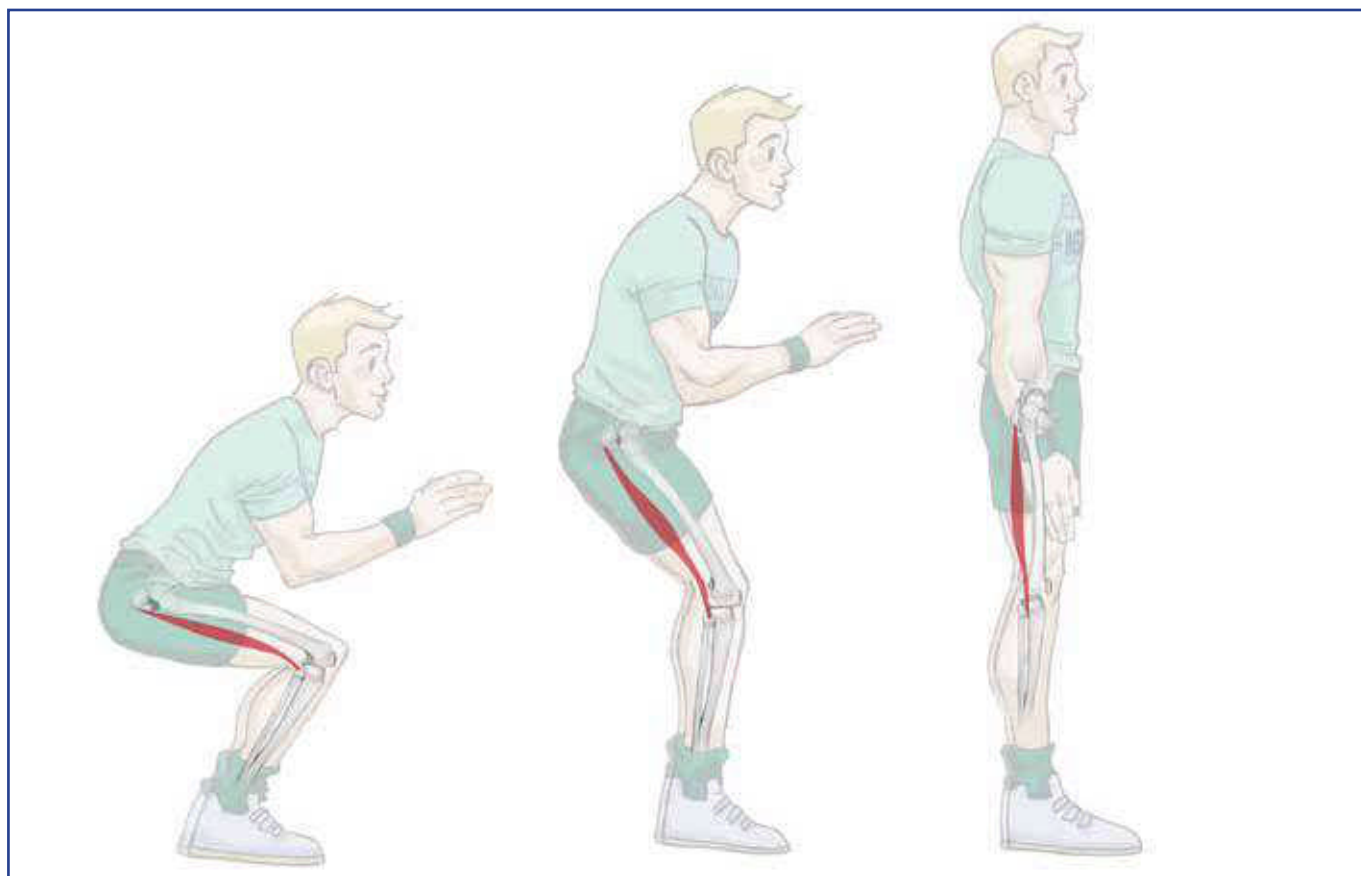
- **Articular concentric:** when the segment moves in the opposite direction to the direction of the resistance. The kinematic direction and the direction of the resistance force are opposite, in the plane.
- **Articular isometric:** when there is no rotational movement going on around the joint axis, and the resistance is counterbalanced by the muscle force.
- **Articular eccentric:** when the direction of the segment and the rotational direction of the resistance are the same. The kinematic direction and the direction of the resistance force are the same, in the plane.

MUSCLE FUNCTION AND JOINT CONTROL

The terms articular concentric/isometric/eccentric have been used, in order to differentiate them from the terms with a muscular reference. It is possible, then, that a muscular concentric action takes place in an articular eccentric phase, or the other way around. It would probably make a lot of sense to change the wording (concentric/isometric/eccentric) for other, not similar, terms. That would avoid misunderstandings. A specific terminology already exists, which describes the articular concentric phase as the positive articular phase. Meanwhile, the articular eccentric phase is the negative phase. But since positive and negative

may lead to thinking in terms of superiority/inferiority, it doesn't seem ideal, either.

In the example above, the one with the squat, it would be possible that during the articular concentric phase (coming up), the biceps femoris muscle worked basically in isometric all the time (since in reality, it shortens relative to the hip joint, and lengthens relative to the knee joint). It becomes just logical that each situation should be studied specifically and that both movement –for each joint– as well as muscle mechanics would have to be determined, for each individual.



Picture 4.9.2

This picture shows how the length of the biceps femoris muscle does not significantly change, even though the knee and hip joints are changing their position.

E-CONCENTRIC

In the past few years, some authors have begun to use a new term, they have called e-concentric.

With this term, some professionals, talk about the scenarios similar to that depicted in picture 4.9.2, where a given muscle may be lengthening at one joint, while shortening at another.

They claim that, in the real functional world, many muscles stretch on one end and shorten on the other, simultaneously. And they use this term, e-concentric, which may well be the result of combining the two concepts: eccentric + concentric).

If we think it thoroughly, exercise specialists will realize that it is impossible to do it at the same time. For any given length, the sarcomere will either elongate or shorten, or maintain its length. It is impossible for it to do more than one thing at a time

If we take the two ends of an elastic band, and we shorten one of them, and lengthen (to the same extent) the other end, the band will not suffer any changes in length, but instead it will have shifted from one place to another. That is actually similar to what happens with muscle length. It can increase, decrease or it can stay the same, throughout the articular ROM (as seen in the pictures above), but it is impossible for it to do more than one at a time.

CHECKING OVER THE 'CONCENTRIC-ECCENTRIC' MODEL

'In the concentric phase, the force wins over the resistance. In the eccentric phase, it's the opposite way around.'

The most popular definition states that the concentric phase is that where the muscle force wins over the resistance; and in the eccentric phase, it is the resistance which wins over the muscle force.

Definitions using this type of reference (based on joint movements, relative to the direction of the resistance) can easily be encountered, by just searching the net or going through some written publications.

This definition gives away how scarce knowledge is, in regards to exercise mechanics and forces acting on lever systems.

During the articular concentric phase, if the muscle force were to win over the resistance, the latter would not stop accelerating during the articular range, and it would end up dismembering and splitting the joint up. Only one thing needs to be understood here: as soon as there is negative acceleration (the segment is slowing down, though it still has positive velocity), the muscle torque is smaller than the torque being generated by the resistance.

Another typical discussion topic is whether one of the phases is preferable to another

The first thing which should be taken into account is the ability to generate adequate muscle tension, for the different scenarios. The resulting phase is secondary to the ability to generate torque.

If muscles are prone to becoming injured during the eccentric phase, there is no simple solution to it, such as working the eccentric phase to prevent injuries. Could that be due to a 'protective firing response', determined by weakness in some other muscle/s?

At Resistance Institute, we are aware that there is still a long way to go, before the complex issue of exercise mechanics is totally comprehended. But inspite of that, nothing is more exciting that being part of a group, formed by individuals who are passionate about Biomechanics. This makes us want to keep exploring and advancing every day, in our quest to become better, and in our believe that things can change, with effort.

It is still possible for many health professionals to think about forces and to follow the motto: Thinking Mechanics.

Thank you.

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